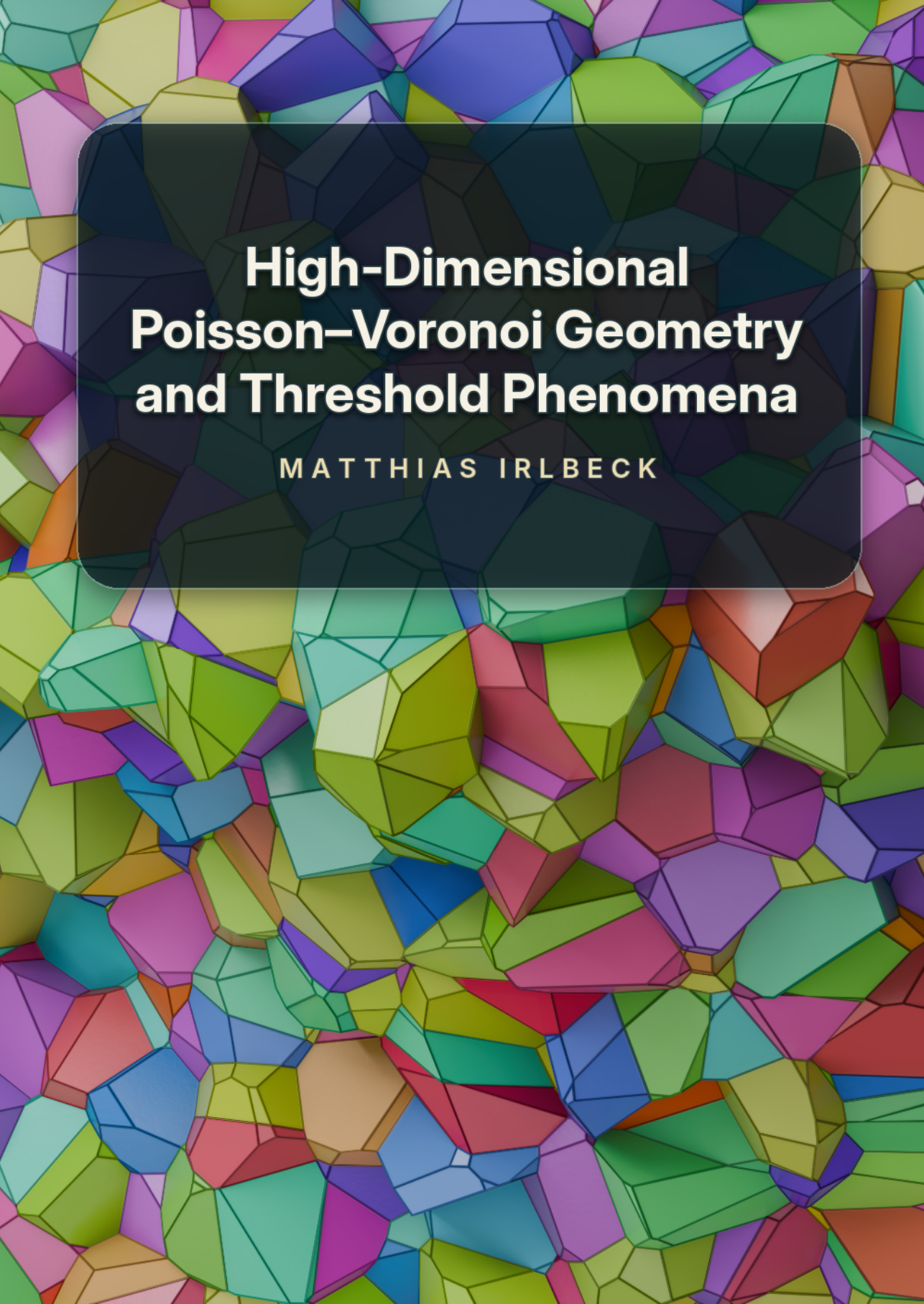




# **High-Dimensional Poisson–Voronoi Geometry and Threshold Phenomena**

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# High-Dimensional Poisson–Voronoi Geometry and Threshold Phenomena

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# CHAPTER 1

## INTRODUCTION

In high-dimensional Euclidean geometry, familiar geometric intuition can be misleading. As the dimension grows, random directions are almost orthogonal, the volume of a ball is concentrated near its boundary, and many random quantities become highly predictable. At the same time, high dimension also creates a large amount of combinatorial complexity: geometric graphs have many possible neighbors, and random point sets can generate polytopes with a complicated face structure.

The three main parts of the thesis follow a similar pattern. We begin with a random point set in a high-dimensional space, either a homogeneous Poisson point process in  $\mathbb{R}^d$  or independent uniform points on the unit sphere  $S^d \subset \mathbb{R}^{d+1}$ . From these points we construct a geometric object, such as a tessellation or a graph, using a local rule. We then introduce another parameter—for example a percolation probability, a distance or angular window, or a scaling of the intensity—and ask when a large-scale property appears. This may be the emergence of an infinite cluster, a change in chromatic number, or a deterministic limiting description of the typical cell. In each case, the main question is to identify the correct threshold and to understand how the geometry of high dimension affects the transition.

*Poisson–Voronoi geometry and percolation.* A homogeneous Poisson point process is a standard model for a random set of points in  $\mathbb{R}^d$ . Informally, it can be thought of as a random cloud of points scattered throughout space. Its Voronoi tessellation divides space by assigning each location to its nearest point. The adjacency structure of the cells gives the Delaunay graph, while imposing stronger proximity conditions leads to subgraphs such as the Gabriel graph. These graphs are geometric and have strong dependencies, but their local behavior, like the degree of a typical vertex, is tied to the geometry of the tessellation. In this setting, we color each cell of the tessellation (or site of the graph) independently with a fixed probability and ask when there can be an infinite connected colored

component. This defines a threshold, the critical probability, and we study the behavior as  $d \rightarrow \infty$ . Although these graphs are not literally trees, the exploration of a cluster near the critical scale becomes increasingly comparable to a branching process. The critical parameter is controlled by a local quantity, namely the degree of a typical vertex. Making this comparison rigorous requires turning geometric information about the Poisson–Voronoi tessellation into quantitative estimates for exploration processes.

*Random Borsuk graph and chromatic thresholds.* The second part moves from  $\mathbb{R}^d$  to the sphere  $S^d$ , and from percolation to graph coloring. A random Borsuk graph is formed by sampling points uniformly on the sphere and connecting pairs of points that are nearly antipodal, with the tolerance determined by a geometric window. We study the asymptotic behavior of the *chromatic number*, that is, the smallest number of colors needed so that adjacent vertices receive different colors. More generally, we study threshold results for  $k$ -colorability. Here the relevant geometry is the geometry of spherical caps and coverings. These geometric quantities determine where the coloring thresholds occur. Unlike the other two parts of the thesis, the results in this chapter are obtained for each fixed dimension.

*The typical Poisson–Voronoi cell and high-dimensional shape laws.* The third part returns to the Poisson–Voronoi tessellation and studies the shape of the typical cell in high dimension. Here shape is understood through classical geometric functionals, such as the inradius, outradius, diameter, mean width, and face structure. These quantities are studied under the natural scaling that keeps the cell at a comparable size as the dimension grows. A main feature of the high-dimensional picture is that different aspects of the cell live on different scales. Typical supporting hyperplanes occur at one radius scale, while extreme points occur at another, and both scales concentrate sharply. However, this concentration does not mean that the cell becomes nearly spherical in a strong metric sense. The polyhedral structure remains important, and it can be measured through face numbers and the geometry of typical facets.

The arguments used throughout the thesis combine tools from stochastic geometry, high-dimensional probability, and random graph theory. On the stochastic geometry side, we use standard properties of Poisson processes, including stationarity, independence, mapping and superposition principles, and the Mecke formula. On the high-dimensional side, we rely on concentration estimates on the sphere and in  $\mathbb{R}^d$ , with careful attention to how the estimates depend on the dimension. To pass from local estimates to global results, we use methods suited to threshold behavior, including exploration process comparisons in percolation and sharp threshold results for monotone properties in random graphs. High dimension plays a central role in this. It often simplifies the problem by revealing the main parameters and by placing the relevant probabilities in regimes where explicit estimates become possible.

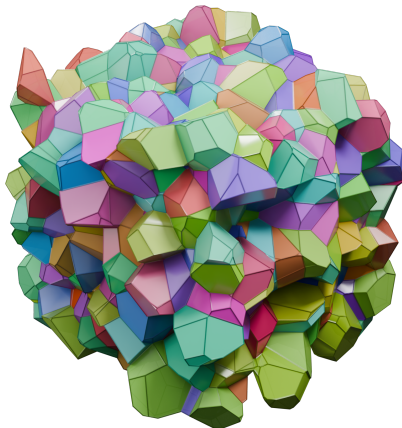


Figure 1.1: Visualization of the Poisson–Voronoi tessellation for  $d = 3$ . All cells that are intersecting a cube are shown.

We now present the main theorems and topics of each chapter in greater detail. Each chapter begins with a short preliminaries section that introduces additional notation and places the results in context.

## 1.1 The Poisson–Voronoi tessellation and the typical cell

We let  $\mathcal{Z} \subset \mathbb{R}^d$  be a *Poisson point process* with constant intensity  $\lambda > 0$ . That is, a random set of points such that  $|A \cap \mathcal{Z}|$  for finite volume (and measurable)  $A \subset \mathbb{R}^d$  follows a Poisson distribution with parameter  $\lambda \text{vol}(A)$  and the number of points in disjoint areas are independent from another. (Here and throughout,  $\text{vol}(\cdot)$  denotes the  $d$ -dimensional Lebesgue measure). Every point of  $\mathcal{Z}$  now gets its unique *Voronoi cell* associated to it. The Voronoi cell of a point  $z \in \mathcal{Z}$  is defined as

$$\mathcal{V}(z) := \{x \in \mathbb{R}^d : \|x - z\| \leq \|x - z'\| \text{ for all } z' \in \mathcal{Z}\}.$$

Thus, the cell consists of all the area that is closer to  $z$  than to all the other points of  $\mathcal{Z}$ . The Voronoi cells  $\{\mathcal{V}(z) : z \in \mathcal{Z}\}$  constitute a dissection of  $\mathbb{R}^d$ , called the *Poisson–Voronoi tessellation*. A standard fact states that, almost surely, every Voronoi cell  $\mathcal{V}(z)$  is a convex polytope with  $z$  in its interior. The Poisson–Voronoi tessellation is one of the central models in stochastic geometry. It is studied in connection with many different applications and has a long history going back at

least to the work of Meijering [78] in the early fifties. For an overview, see the monographs [89, 92] and the references therein.

We will be considering the *typical cell*, which is the Voronoi cell of the origin  $\underline{0}$  in the Voronoi tessellation for  $\mathcal{Z} \cup \{\underline{0}\}$ , the Poisson point process with the origin added in. Throughout, we will denote the typical cell by  $\mathcal{V}_{\text{typ}}$ . Again,  $\mathcal{V}_{\text{typ}}$  is almost surely a polytope with the origin in its interior.

The significance of  $\mathcal{V}_{\text{typ}}$  is that, as the name “typical cell” suggests, it describes the average behavior of the cells of the Poisson–Voronoi tessellation. For instance, if we take the fraction of all Poisson points  $z \in \mathcal{Z}$  inside the ball  $B(\underline{0}, r)$  around the origin of radius  $r$  for which  $\mathcal{V}(z)$  has precisely  $k$  vertices then, as  $r \rightarrow \infty$ , this fraction converges almost surely to  $\mathbb{P}(\mathcal{V}_{\text{typ}} \text{ has } k \text{ vertices})$ . (Here and in the rest  $B(x, r)$  denotes the  $d$ -dimensional open ball with center  $x$  and radius  $r$ .) More generally, for  $h$  a translation invariant and appropriately measurable function from the space of polytopes into  $\mathbb{R}_+$ , we have

$$\lim_{r \rightarrow \infty} \frac{\sum_{z \in \mathcal{Z} \cap B(\underline{0}, r)} h(\mathcal{V}(z))}{|\mathcal{Z} \cap B(\underline{0}, r)|} = \mathbb{E}h(\mathcal{V}_{\text{typ}}) \quad \text{a.s.} \quad (1.1)$$

(See e.g. [27] and the references therein.)

As the Poisson–Voronoi tessellation is a classical subject in stochastic geometry, its typical cell has been studied quite extensively. The behavior of the typical cell as the dimension grows is however still relatively unexplored. In the corresponding chapter we will study the behavior of some parameters of  $\mathcal{V}_{\text{typ}}$  related to its shape and size, as the dimension grows. Namely, we consider the inradius, outradius, diameter, width and mean width, denoted by  $\text{inr}(\cdot)$ ,  $\text{outr}(\cdot)$ ,  $\text{diam}(\cdot)$ ,  $\text{width}(\cdot)$ ,  $\text{meanw}(\cdot)$ , respectively. (Detailed definitions can be found in Section 5.1.) We denote by  $B := B(\underline{0}, 1)$  the  $d$ -dimensional unit ball. We obtain the following convergence results.

**Theorem 1.1.** *For any  $\lambda = \lambda(d) > 0$  we have*

$$\begin{aligned} \frac{\text{outr}(\mathcal{V}_{\text{typ}})}{\sqrt[d]{\lambda \cdot \text{vol}(B)}} &\xrightarrow[d \rightarrow \infty]{\mathbb{P}} 1, & \frac{\text{inr}(\mathcal{V}_{\text{typ}})}{\sqrt[d]{\lambda \cdot \text{vol}(B)}} &\xrightarrow[d \rightarrow \infty]{\mathbb{P}} \frac{1}{2}, \\ \frac{\text{diam}(\mathcal{V}_{\text{typ}})}{\sqrt[d]{\lambda \cdot \text{vol}(B)}} &\xrightarrow[d \rightarrow \infty]{\mathbb{P}} 2, & \frac{\text{meanw}(\mathcal{V}_{\text{typ}})}{\sqrt[d]{\lambda \cdot \text{vol}(B)}} &\xrightarrow[d \rightarrow \infty]{\mathbb{P}} 2. \end{aligned}$$

To clarify, we emphasize that in the above theorem the intensity  $\lambda$  is allowed to vary with the dimension. Applying Stirling’s approximation to the Gamma function for the expression of  $\text{vol}(B)$  (see e.g. (2.1) later), it is easily seen that in the case when  $\lambda > 0$  is a fixed constant, the denominators in Theorem 1.1 can be replaced by  $\sqrt{2\pi e/d}$ .

We will show that the vast majority of the faces of constant co-dimension have microscopic diameter. We let  $f_k(P)$  denote the number of  $k$ -faces of the polytope  $P$  and  $\mathcal{F}_k(P)$  is the corresponding set of  $k$ -faces of  $P$ .

**Theorem 1.2.** *For every fixed  $k \in \mathbb{N}$  and every  $\lambda = \lambda(d) > 0$ , we have*

$$\frac{1}{f_{d-k}(\mathcal{V}_{\text{typ}})} \cdot \frac{\sum_{F \in \mathcal{F}_{d-k}(\mathcal{V}_{\text{typ}})} \text{diam}(F)}{\sqrt[d]{\lambda \cdot \text{vol}(B)}} \xrightarrow[d \rightarrow \infty]{\mathbb{P}} 0.$$

(For clarity, we remark that this last theorem implies that all but a negligible proportion of the faces of co-dimension  $k$  have diameter  $o\left(\sqrt[d]{\lambda \text{vol}(B)}\right)$ .)

Some observations that we shall make in the course of the proofs are of independent interest. The first of these estimates the number of faces of fixed co-dimension  $k$ . The (expected) number of faces of  $\mathcal{V}_{\text{typ}}$  is a theme that has been considered a fair bit in the literature (see e.g. [55, 58]), but as far as we know the following result is new.

**Theorem 1.3.** *For every fixed  $k \in \mathbb{N}$  and every  $\lambda = \lambda(d) > 0$ , we have*

$$\sqrt[d]{f_{d-k}(\mathcal{V}_{\text{typ}})} \xrightarrow[d \rightarrow \infty]{\mathbb{P}} \frac{(k+1)^{(k+1)/2}}{k^{k/2}}.$$

We also obtained results on the width and the geometry of faces of the typical cell  $\mathcal{V}_{\text{typ}}$ . Also, our results imply that the Hausdorff distance between  $\mathcal{V}_{\text{typ}}$  and any ball is large (i.e. at least of the same order as the diameter of  $\mathcal{V}_{\text{typ}}$ ). In order to not overload the introduction with too much on this topic, we will present the additional results at the beginning of [Chapter 5](#), where we will also give all the proofs of the theorems that are stated above.

## 1.2 Poisson–Voronoi percolation

Percolation theory studies certain random graphs and was first introduced by Broadbent and Hammersley in 1957 [24] to give a mathematical model to describe how a liquid spreads in a porous medium. Similar to most of the topics discussed here, percolation theory has the advantage that its underlying stochastic model is easy to define and often straightforward to visualize. Over the past decades, a rich theory has been developed, however key questions remain open [34]. Usually one starts with an infinite graph that satisfies some reasonable properties and one then opens (equivalently keeps or colors) each vertex with probability  $p$  independently from another. This will then create a random subgraph of sites that are open and we speak of *Bernoulli site percolation*. One could also the analogous procedure by considering the edges and keeping each edge with probability  $p$ . The graph that is by far studied the most is  $\mathbb{Z}^d$  with nearest neighbor connections but recently also different models (like the one we study here) have become more popular.

For *Poisson–Voronoi percolation* we start with the exact same setup as before. Thus, we let  $\mathcal{Z}$  be a Poisson point process in  $\mathbb{R}^d$  with constant intensity  $\lambda > 0$

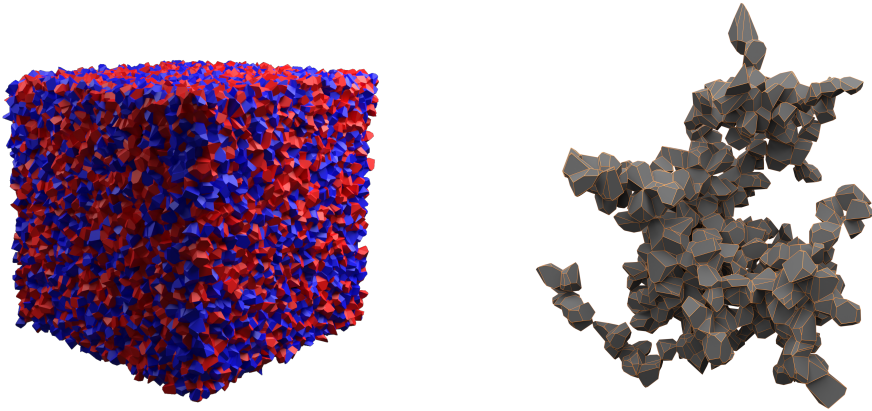


Figure 1.2: Part of the Poisson–Voronoi tessellation that intersects a cube with two random colors, and a cluster for  $d = 3$ .

and consider the Poisson–Voronoi tessellation. We will color each cell *black* with probability  $p$ , independently of one another and say that *percolation* occurs if an unbounded connected black component exists. We are interested in studying the threshold above which an infinite component can occur with positive probability and define the *critical probability*  $p_c(d)$  as

$$p_c(d) := \inf\{p : \mathbb{P}_p(\text{percolation}) > 0\}.$$

It is known that the phase transition is nontrivial in the sense that  $0 < p_c(d) < 1$  and additionally for  $p > p_c$  the infinite component is unique, see Chapter 8.3 in [20]. It can be shown that two cells are adjacent if and only if they share a  $(d - 1)$ -dimensional face [79]. Equivalently, we could also consider the Delaunay graph/triangulation on  $\mathcal{Z}$ , where  $\mathcal{Z}$  are the sites of the graph and two points are connected by an edge if their corresponding Voronoi cells share a face. If we perform ordinary site percolation with parameter  $p$ , each vertex will be black independently with probability  $p$  and percolation occurs if there exists an infinite cluster of black points.

We will also be able to show our main result for site percolation on the Gabriel graph, which is a subgraph of the Delaunay graph. Here we connect  $x, y \in \mathcal{Z}$  by an edge if the *Gabriel ball* of  $x$  and  $y$  contains no point of  $\mathcal{Z}$ . The Gabriel ball  $B_{Gab}(x, y)$  of  $x$  and  $y$  is defined as

$$B_{Gab}(x, y) := B((x + y)/2, \|x - y\|/2),$$

where for  $z \in \mathbb{R}^d$  and  $r \geq 0$  we write  $B(z, r)$  for the open ball around  $z$  with radius  $r$ . Thus it is the smallest ball such that  $x$  and  $y$  are on its boundary. It is immediate if  $B_{Gab}(x, y) \cap \mathcal{Z} = \emptyset$ , then  $\mathcal{V}(x)$  and  $\mathcal{V}(y)$  are adjacent in the Voronoi tessellation of  $\mathcal{Z}$ . As for the Delaunay graph, each site is independently colored black with probability  $p$  and percolation is occurring if there exists an infinite connected component of black points. Then, we can also define  $p_c(d)$  in the identical way for this model.

We note that the occurrence of an infinite component is independent of the intensity parameter  $\lambda$ , since if we consider a homogeneous Poisson point process with intensity  $\lambda'$ , the *mapping theorem* says that the dilation  $x \mapsto (\lambda'/\lambda)^{1/d}x$  will transform this into a process with intensity  $\lambda$  (this is also presented as [Lemma 2.3](#) later). Further, scaling each Poisson point by the same value will not change the combinatorial structure. However, the probability of percolation will depend crucially on the specific dimension. The main result is the asymptotic behavior of the critical probability for large  $d$  in both the Voronoi and Gabriel model.

**Theorem 1.4.** *For Poisson–Voronoi percolation as  $d \rightarrow \infty$*

$$p_c(d) = \frac{e}{d2^d}(1 + o(1)).$$

**Theorem 1.5.** *For Poisson–Gabriel percolation as  $d \rightarrow \infty$*

$$p_c(d) = \frac{1}{2^d}(1 + o(1)).$$

In [Section 3.2.2](#) we see that the limit can be interpreted as the reciprocal of an average or *typical* degree. The following gives the asymptotics in the Voronoi case. For the Gabriel model it is way easier to compute the corresponding value of  $2^d$  for all  $d$  and we will give the argument in [Section 3.2.2](#). We remind ourselves from the last section that  $f_{d-1}(\mathcal{V}_{\text{typ}})$  is the number of  $(d-1)$ -dimensional faces of the typical cell, or in other words (a.s.) the number of adjacent cells.

**Theorem 1.6.** *For the Poisson–Voronoi tessellation and as  $d \rightarrow \infty$*

$$\mathbb{E}f_{d-1}(\mathcal{V}_{\text{typ}}) = \frac{d2^d}{e}(1 + o(1)).$$

The asymptotic results for  $p_c$  also hold for bond percolation with some minor changes to our proofs. However, since site percolation is the natural equivalent to the coloring of Voronoi cells, we do only focus on that. The results corresponding to this section will be proven in [Chapter 3](#).

## 1.3 Chromatic number of the random Borsuk graph

The *Borsuk graph* can be viewed as the opposite of a random geometric graph. The idea is to connect points of the  $d$ -dimensional sphere  $S^d$  that is embedded

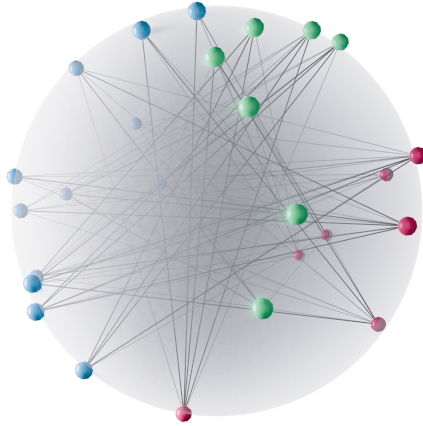


Figure 1.3: Simulation of the random Borsuk graph on  $S^2$  with  $n = 32$ ,  $\alpha = 1$  and  $\chi = 3$ .

into  $\mathbb{R}^{d+1}$  if they are far away from another. More formally, we connect  $u, v \in S^d$  if and only if  $\text{dist}(u, v) > \pi - \alpha$ , where  $\text{dist}(u, v)$  is the length of the shortest curve between  $u$  and  $v$  that stays on the sphere (equivalently the angle between  $u$  and  $v$ ) and  $\alpha > 0$  the parameter of the model. Typically, the parameter  $\alpha$  is chosen small, so that points that are joined by an edge are “nearly antipodal”. The Borsuk graph was introduced in 1967 by Erdős and Hajnal [37] as an explicit example of a graph with a large chromatic number and without short odd cycles. In particular, Erdős and Hajnal showed there exists an  $\alpha_0 = \alpha_0(d)$  such that the Borsuk graph has chromatic number exactly  $d + 2$  for all  $0 < \alpha < \alpha_0$ . More recently, Kahle and Martínez-Figueroa [62] introduced the *random Borsuk graph*, which we will denote by  $G(n, \alpha)$ . It has  $n$  vertices  $X_1, \dots, X_n \in S^d$ , chosen i.i.d. uniformly at random, and an edge  $X_i X_j$  if and only if  $\text{dist}(X_i, X_j) > \pi - \alpha$ . Recently the random Borsuk graph featured in a breakthrough result giving an exponential lower bound for asymmetric Ramsey numbers by Ma et al. [71]. This setup implies that the average degree scales with  $n\alpha^d$ . The results of Kahle and Martínez-Figueroa show that the random Borsuk graph transitions from being  $(d + 1)$ -colorable to needing  $\geq d + 2$  colors in the regime where the average degree is of order  $\ln n$ . To be precise, they have shown that there exist constants  $c_L, c_U$  satisfying

$$\lim_{n \rightarrow \infty} \mathbb{P}(\chi(G(n, \alpha)) \geq d + 2) = \begin{cases} 1 & \text{if } \alpha > c_U \cdot (\ln n/n)^{1/d}, \\ 0 & \text{if } \alpha < c_L \cdot (\ln n/n)^{1/d}. \end{cases} \quad (1.2)$$

Our first result shows that the random Borsuk graph achieves chromatic number  $\geq d + 1$  much “sooner”, namely when the average degree is constant.

**Theorem 1.7.** *For  $d \geq 2$ , there is a constant  $c = c(d) > 0$  such that:*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left( \chi \left( G(n, c \cdot n^{-1/d}) \right) \geq d + 1 \right) = 1.$$

The regime when the average degree is constant is sometimes called the “thermodynamic regime” in the random (geometric) graphs literature. We will show that the switch from being  $k - 1$  colorable to needing at least  $k$  colors occurs in the thermodynamic regime, for all  $3 \leq k \leq d + 1$ . In fact, for  $k = 3$  we can be quite precise:

**Theorem 1.8.** *For  $d \geq 2$ , there is a constant  $c_3 = c_3(d) > 0$  such that for all fixed  $\varepsilon > 0$  and all sequences  $\alpha = \alpha(n)$  it holds that*

$$\lim_{n \rightarrow \infty} \mathbb{P}(\chi(G(n, \alpha)) \geq 3) = \begin{cases} 1 & \text{if } \alpha \geq (c_3 + \varepsilon) \cdot n^{-1/d}, \\ 0 & \text{if } \alpha \leq (c_3 - \varepsilon) \cdot n^{-1/d}. \end{cases}$$

The proof provides an expression for  $c_3$  in terms of the critical intensity for *continuum AB percolation*.

Since having chromatic number one is the same as having no edges, the switch from chromatic number one to at least two occurs well before the thermodynamic regime. For completeness we offer the following precise result, that has a relatively straightforward proof (which we give in Section 4.6).

**Theorem 1.9.** *We have*

$$\lim_{n \rightarrow \infty} \mathbb{P}(\chi(G(n, \alpha)) \geq 2) = \begin{cases} 1 & \text{if } n^2 \alpha^d \rightarrow \infty, \\ 1 - e^{-c \cdot \nu} & \text{if } n^2 \alpha^d \rightarrow \nu \in (0, \infty), \\ 0 & \text{if } n^2 \alpha^d \rightarrow 0, \end{cases}$$

where  $c := \Gamma((d + 3)/2) / (2 \cdot (d + 1) \cdot \sqrt{\pi} \cdot \Gamma((d + 2)/2))$ .

The behavior exhibited by Theorem 1.8 is a prime example of what is called a *sharp threshold* in the random graphs community, and Theorem 1.9 corresponds to what is called a *coarse threshold*. If  $\mathcal{P}$  is a monotone graph property (formally a family of graphs closed under isomorphism and under the addition of edges) then we say  $\beta = \beta(n)$  is a *threshold* for  $\mathcal{P}$  if  $\lim_{n \rightarrow \infty} \mathbb{P}(G(n, \alpha) \in \mathcal{P}) = 1$  for all sequences  $\alpha = \alpha(n)$  satisfying  $\alpha \gg \beta$  and  $\lim_{n \rightarrow \infty} \mathbb{P}(G(n, \alpha) \in \mathcal{P}) = 0$  whenever  $\alpha \ll \beta$ . It is a *sharp threshold* if  $\mathbb{P}(G(n, \alpha) \in \mathcal{P}) \rightarrow 1$  whenever  $\alpha \geq (1 + \varepsilon)\beta$  and  $\mathbb{P}(G(n, \alpha) \in \mathcal{P}) \rightarrow 0$  whenever  $\alpha \leq (1 - \varepsilon)\beta$  for all  $\varepsilon > 0$  (but where we keep  $\varepsilon$  fixed as we take the limit  $n \rightarrow \infty$ ). A *coarse threshold* is simply a threshold which is not sharp. The situation in (1.2) is sometimes called a *semi-sharp threshold*.

For  $k = 2, \dots, d + 1$ , the threshold for  $k$ -colorability is also semi-sharp, as a consequence of Theorem 1.7. The semi-sharpness of these thresholds leaves open the possibility that they are in fact sharp. We conjecture that this is the case. (See Section 4.7 for more discussion and more open problems.) We have not been able to prove our conjecture, but we can offer the following weaker result.

**Theorem 1.10.** *For  $d \geq 2$ , there is a set  $N \subseteq \mathbb{N}$  of density one and sequences  $\alpha_k = \alpha_k(n)$  for  $k = 4, \dots, d + 2$ , such that for every fixed  $\varepsilon > 0$ , every sequence  $\alpha = \alpha(n)$  and every increasing sequence  $n_1, n_2, \dots \in N$ :*

$$\lim_{i \rightarrow \infty} \mathbb{P}(\chi(G(n_i, \alpha)) \geq k) = \begin{cases} 1 & \text{if } \alpha > (1 + \varepsilon)\alpha_k, \\ 0 & \text{if } \alpha < (1 - \varepsilon)\alpha_k. \end{cases}$$

Here  $N$  having density one means that

$$\lim_{n \rightarrow \infty} \frac{|N \cap [n]|}{n} = 1,$$

where for a positive integer  $n \in \mathbb{N}$  we write  $[n] := \{1, \dots, n\}$ . One could paraphrase the result as “the property of having chromatic number  $\geq k$  has a sharp threshold for *almost all*  $n$ ”. We will present our arguments regarding the chromatic number thresholds in [Chapter 4](#).

# CHAPTER 2

## PRELIMINARIES

As in many mathematical texts, we will have to introduce a fair amount of notation and background. This chapter collects the definitions, conventions, and notation that are used throughout multiple chapters. Each subsequent chapter also begins with its own short preliminaries section, covering the additional setup that is specific to the results proved there.

We will write  $B(x, r) := \{y \in \mathbb{R}^d : \|y - x\| < r\}$  for the open  $d$ -dimensional ball with center  $x$  and radius  $r$  and  $\underline{0} := (0, \dots, 0) \in \mathbb{R}^d$ . We will use  $\kappa_d$  to denote the  $d$ -dimensional volume of the  $d$ -dimensional unit ball  $B := B(\underline{0}, 1)$ . That is,

$$\kappa_d := \text{vol}(B) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}, \quad (2.1)$$

where the stated equality is a classical result (see e.g. [88], Corollary 15.15, for a proof). We are going to use standard terminology to denote distributions. For example,  $\text{Po}(\mu)$  is the Poisson distribution with intensity  $\mu > 0$ ;  $\text{Bi}(n, p)$  stands for the Binomial distribution with parameters  $n \in \mathbb{N}$  and  $p \in [0, 1]$ , and  $\text{Unif}(A)$  denotes the uniform distribution on the set  $A$ . A proof for the following Chernoff type bound can be found as Lemmas 1.1 and 1.2 in [85].

**Lemma 2.1** (Chernoff bound). *Let  $\mu > 0$  and  $X \sim \text{Po}(\mu)$  or  $X \sim \text{Bi}(n, p)$  with  $np = \mu$ . If  $x \geq \mu$  it holds that*

$$\mathbb{P}(X \geq x) \leq e^{-\mu H(x/\mu)}$$

*and when  $x \leq \mu$*

$$\mathbb{P}(X \leq x) \leq e^{-\mu H(x/\mu)}$$

*for  $H(a) := 1 - a + a \ln(a)$  and  $H(0) = 1$ .*

## 2.1 Poisson point process

Technically speaking, a point process is a random locally finite counting measure on  $\mathbb{R}^d$ . However, it is often more intuitive to simply imagine them as a random set of points. A Poisson point process (PPP)  $\mathcal{Z} \subset \mathbb{R}^d$  with intensity measure  $\Lambda$  is characterized by:

- (i) for pairwise disjoint Borel sets  $A_1, \dots, A_\ell$ , the random variables  $|\mathcal{Z} \cap A_1|, \dots, |\mathcal{Z} \cap A_\ell|$  are independent; and
- (ii) for any Borel set  $A$  with  $\Lambda(A) < \infty$ ,  $|\mathcal{Z} \cap A|$  is Poisson with mean  $\Lambda(A)$ .

If  $\Lambda(dx) = g(x)dx$  for some nonnegative measurable  $g$ , we call  $g$  the intensity density (with respect to Lebesgue measure). As most of the well known results that are presented here, a proof the following two useful properties can be found in [65].

**Lemma 2.2** (Superposition). *Let  $\mathcal{Z}_1$  and  $\mathcal{Z}_2$  be independent Poisson point processes on  $\mathbb{R}^d$  with intensity measures  $\Lambda_1$  and  $\Lambda_2$ . Then,  $\mathcal{Z} := \mathcal{Z}_1 \cup \mathcal{Z}_2$  is a Poisson point process with intensity measure  $\Lambda_1 + \Lambda_2$ .*

**Lemma 2.3** (Mapping theorem). *Let  $\mathcal{Z}$  be a PPP on  $\mathbb{R}^d$  with intensity measure  $\Lambda$  and let  $T : \mathbb{R}^d \rightarrow \mathbb{R}^m$  be measurable. Assume: (i)  $\Lambda(T^{-1}(B)) < \infty$  for every bounded Borel set  $B \subset \mathbb{R}^m$ ; and (ii)  $\Lambda(T^{-1}(\{y\})) = 0$  for all  $y \in \mathbb{R}^m$ . Then  $T(\mathcal{Z})$  is a PPP on  $\mathbb{R}^m$  with intensity measure  $\Lambda \circ T^{-1}$ .*

If the density  $g(x) = \lambda > 0$  is constant, we call  $\mathcal{Z}$  a homogeneous Poisson point process, or just say it has constant intensity  $\lambda$ . When working with our PPP, the lemma below, in combination with the results we state in the next section, will make it possible for us to explicitly compute probabilities that deal with spatial properties. A proof can be found at Thm. 3.2.2 of [89] or Section 2.4 in [65].

**Lemma 2.4.** *Let  $\mathcal{Z}$  be a homogeneous Poisson point process. Further, let  $B \subseteq A \subset \mathbb{R}^d$  be measurable sets with  $0 < \text{vol}(A) < \infty$  and  $n, k \in \mathbb{N}_0$ . If  $U_1, \dots, U_n$  are independent and uniformly chosen from  $A$  it holds that*

$$\mathbb{P}(|\mathcal{Z} \cap B| = k \mid |\mathcal{Z} \cap A| = n) = \mathbb{P}\left(\sum_{i=1}^n \mathbb{1}_{\{U_i \in B\}} = k\right).$$

Expectations that have the form of a sum that ranges over  $m$ -tuples of the PPP can be computed elegantly with the Mecke formula. In the literature other names for this result are used, like multivariate Mecke formula, Slivnyak–Mecke formula or Campbell–Mecke formula. Let  $\mathcal{M}$  be the set of all counting measures on  $\mathbb{R}^d$ . We equip  $\mathcal{M}$  with the smallest  $\sigma$ -algebra such that, for each Borel-measurable  $A$ , the map

$$\begin{aligned} \Phi_A : \mathcal{M} &\rightarrow \mathbb{R} \cup \{\infty\} \\ \eta &\mapsto \eta(A) \end{aligned}$$

is measurable. Further details can be found in Chapter 3 of [89]. The version that works well for us is from Corollary 3.2.3 in [89].

**Lemma 2.5** (Mecke formula). *Let  $\mathcal{Z}$  be a Poisson point process in  $\mathbb{R}^d$  with constant intensity  $\lambda > 0$ . If for some  $m \in \mathbb{N}$ ,  $f : \mathcal{M} \times (\mathbb{R}^d)^m \rightarrow \mathbb{R}$  is a nonnegative measurable function, we have*

$$\mathbb{E} \sum_{\substack{(x_1, \dots, x_m) \in \mathcal{Z}^m \\ x_i \neq x_j \text{ for } i \neq j}} f(\mathcal{Z}; x_1, \dots, x_m) = \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} \mathbb{E} f(\mathcal{Z} \cup \{x_1, \dots, x_m\}; x_1, \dots, x_m) \lambda^m dx_1 \dots dx_m.$$

## 2.2 Measure concentration

The spherical measure distribution on the  $d-1$  dimensional sphere  $S^{d-1}$  has some useful properties that will play a role in multiple of our arguments. Analogous statements will also hold for balls. Informally speaking, for large dimensions, the mass of the sphere is concentrated around any equator. This is often made rigorous by considering a uniform random variable on the sphere. The measure concentration has major implications on the angle between a uniform random variable and a fixed point. See also Chapter 14 in [72]. The following bound is part (b) of Lemma 2.1 in [23], rephrased in terms of angles rather than spherical caps. For three points  $a, b, c \in \mathbb{R}^d$ , we will write  $\angle abc$  for the angle between  $a - b$  and  $c - b$ .

**Lemma 2.6** ([23]). *Let  $U$  be chosen uniformly at random on the unit sphere  $S^{d-1}$  in  $\mathbb{R}^d$ . For any  $v \in \mathbb{R}^d \setminus \{0\}$  and  $0 < \alpha < \arccos(\sqrt{2/d})$  we have*

$$\frac{\sin^{d-1} \alpha}{6 \cdot \cos \alpha \cdot \sqrt{d}} \leq \mathbb{P}(\angle U \underline{0} v < \alpha) \leq \frac{\sin^{d-1} \alpha}{2 \cdot \cos \alpha \cdot \sqrt{d}}.$$

For  $\delta > 0$ , we also introduce the notation  $(\pi/2 \pm \delta)$  for the open interval  $(\pi/2 - \delta, \pi/2 + \delta)$ . The next bound is a consequence of the lemma above, phrased in a form that is more useful most of the time. It claims that the angle between a uniform random variable on the sphere and a fixed point is roughly  $\pi/2$ , exactly what one would expect if the mass is indeed concentrated around any equator. Different proofs for it can be found for example at Theorem 3.4.5 in [100] or in greater generality at Theorem 14.1.1 in [72].

**Lemma 2.7.** *For all  $0 < \delta < \pi/2$  there is a  $c = c(\delta) > 0$  and  $d_0 = d_0(\delta)$  such that for  $U$  chosen uniformly on  $S^{d-1}$ , all  $d \geq d_0$  and  $y \in \mathbb{R}^d \setminus \{0\}$*

$$\mathbb{P}(\angle U \underline{0} y \notin (\pi/2 \pm \delta)) \leq e^{-cd}.$$

In our proofs, it will also be essential to establish upper bounds on the volume of intersections between balls or the region inside a ball that has a certain angle to

a fixed point. Again, the following could also be derived from the angle properties we just stated. Since it can also be shown directly rather quickly, we give a proof for completeness.

**Lemma 2.8.** *For distinct  $x, y \in \mathbb{R}^d$  there is a  $c = c(\|x - y\|) > 0$  with*

$$\text{vol}(B(x, 1) \cap B(y, 1)) \leq e^{-cd} \text{vol}(B(\underline{0}, 1)).$$

*Proof.* If  $u := \|x - y\| \geq 2$  then  $B(x, 1) \cap B(y, 1) = \emptyset$ . Thus let  $0 < u < 2$ . By translation invariance we may assume  $x = ue_1/2$  and  $y = -ue_1/2$ , where  $e_1 = (1, 0, \dots, 0) \in \mathbb{R}^d$ . Fix  $z \in B(x, 1) \cap B(y, 1)$  and write  $z = (z_1, \hat{z})$  with  $\hat{z} \in \mathbb{R}^{d-1}$ . Then

$$(z_1 - \frac{u}{2})^2 + \|\hat{z}\|^2 \leq 1, \quad (z_1 + \frac{u}{2})^2 + \|\hat{z}\|^2 \leq 1.$$

Adding these two inequalities gives

$$2z_1^2 + \frac{u^2}{2} + 2\|\hat{z}\|^2 \leq 2,$$

hence

$$z_1^2 + \|\hat{z}\|^2 \leq 1 - \frac{u^2}{4}.$$

Thus  $\|z\|^2 \leq 1 - \frac{u^2}{4}$ , so

$$B(x, 1) \cap B(y, 1) \subset B\left(\underline{0}, \sqrt{1 - \frac{u^2}{4}}\right).$$

By monotonicity of volume and the scaling of Euclidean balls,

$$\text{vol}(B(x, 1) \cap B(y, 1)) \leq \left(1 - \frac{u^2}{4}\right)^{d/2} \text{vol}(B(\underline{0}, 1)),$$

and the claimed bound follows.  $\square$

We remark that we will apply the result above for distances  $\|x - y\|$  that are independent of the dimension and by scaling it can be easily adapted for any radius of the balls.

## 2.3 Percolation theory

Not only is one chapter primarily dealing with percolation, we are also going to use some tools from this area in our proofs. In two chapters we are going to couple an exploration algorithm with a percolation model to show that with high probability certain “good” events occur. Here we present some classical results that we will use.

Given a graph  $G$  (most of the time connected, infinite, transitive with bounded degree), we keep every site with probability  $p$ , independently from another. This will create a random subgraph of  $G$  and we call this (Bernoulli) *site percolation*. If we perform the same procedure with the edges, we speak of *bond percolation*. The primary question people ask themselves is whether or not an infinite connected component exists in said random subgraph. If this happens, we say that *percolation* occurs. Similarly as done in the introduction, one can define the *critical probability* as

$$p_c(G) := \inf\{p : \mathbb{P}_p(\text{percolation}) > 0\}.$$

A key characteristic of the model is, that the behavior is vastly different for  $p < p_c$  (*subcritical*) and  $p > p_c$  (*supercritical*).

For the upper bound on  $p_c$  in the Voronoi/Gabriel setting, we are going to construct a coupling with supercritical site percolation on  $\mathbb{Z}^2$ . We will use the next result to show this. For a proof see [45], Thm. 1.33.

**Lemma 2.9.** *The cluster of  $(0, 0)$  in site percolation on  $\mathbb{Z}^2$  is infinite with positive probability for  $p = 15/16$ .*

We are going to make use of two percolation results for the 2-colourability threshold in the random Borsuk graph Chapter 4. One in the supercritical regime and the other subcritical. The following is well known, but, since we did not find a convenient reference, we provide a proof for completeness.

**Lemma 2.10.** *Let  $C_m(\underline{0})$  be the cluster of  $\underline{0}$  in bond percolation of  $\mathbb{Z}^d$  restricted to  $\Lambda_m := \{-m, \dots, m\}^d$ . Then, for all  $0 < \varepsilon < 1/10$  we can choose  $p = p(\varepsilon, d)$  large enough such that for all large enough  $m \geq m_0(\varepsilon, d, p)$*

$$\mathbb{P}_p(|C_m(\underline{0})| > (1 - \varepsilon)|\Lambda_m|) > 1 - \varepsilon.$$

*Proof of Lemma 2.10.* We let  $\rho := \varepsilon^2/(4d)$  and consider bond percolation on  $\mathbb{Z}^2$  with the usual dual graph definition. We choose  $p$  so close to 1 that with  $q := 1 - p$  we have

$$\alpha(q) := \mathbb{P}(\text{there is a dual open circuit surrounding } (0, 0)) \leq \rho.$$

This is possible because of the famous Peierls contour argument that gives  $\alpha(q) \leq 4 \sum_{\ell \geq 4} (3q)^\ell$  by counting self avoiding paths of length  $\ell$ . Fix some  $\beta > 0$  such that  $1 + \beta \log(3q) < 0$ . For each  $m$  set  $r := \lceil \beta \log m \rceil$  and

$$\Delta_m := \{x \in \Lambda_m : \inf_{y \in \partial \Lambda_m} \|x - y\| < r\},$$

which yields  $\Delta_m = O(rm^{d-1}) = o(|\Lambda_m|)$ , so for all large  $m$  it holds  $|\Delta_m| \leq \rho |\Lambda_m|$ . Now fix any 2 coordinate plane  $\Pi \subset \Lambda_m$  and consider connectivity inside  $\Pi$  using only edges of  $\Pi$ . If  $u, v \in \Pi \setminus \Delta_m$  and  $u$  is not connected to  $v$  inside  $\Pi$ , then one

of the following three states must occur: there is a dual open circuit surrounding  $v$ , or there is a dual open circuit surrounding  $u$ , or there is a dual open path attached to the dual boundary of  $\Pi$  that reaches depth at least  $r$  away from the boundary. The first two possibilities have probability  $\leq \alpha(q) \leq \rho$  each. For the third we can again bound the self avoiding paths that yields constants  $C_1, C_2 > 0$  with

$$\begin{aligned} \mathbb{P}(\text{there is dual boundary path reaching depth } \geq r) &\leq C_1 m \sum_{\ell \geq r} (3q)^\ell \leq C_2 m (3q)^r \\ &\leq C_2 m^{1+\beta \log(3q)}. \end{aligned}$$

Thus, as the right hand side is less than  $\rho$  for all large enough  $m$ , we have

$$\mathbb{P}(u \text{ is not connected to } v \text{ in } \Pi) \leq 3\rho.$$

Now fix some  $z \in \Lambda_m \setminus \Delta_m$ . Note that by fixing two coordinates in each step, we can move from 0 to  $z$  by traversing  $d-1$  planes. By symmetry of the planes, the error probability that inside a plane the connection to the specific coordinates of  $z$  does not occur is upper bounded by  $3\rho$ . This yields  $\mathbb{P}(0 \not\leftrightarrow z) \leq 3\rho(d-1) \leq 3d\rho$ , where the connection is meant to be inside  $\Lambda_m$ . Defining  $D_m := |\Lambda_m \setminus C_m(\underline{0})|$  we can compute

$$\mathbb{E}D_m = \sum_{z \in \Lambda_m} \mathbb{P}(0 \not\leftrightarrow z) \leq |\Delta_m| + \sum_{z \in \Lambda_m \setminus \Delta_m} 3d\rho \leq \rho|\Lambda_m| + 3d\rho|\Lambda_m| \leq \varepsilon^2|\Lambda_m|.$$

Therefore by Markov

$$\mathbb{P}(D_m \geq \varepsilon|\Lambda_m|) \leq \frac{\mathbb{E}D_m}{\varepsilon|\Lambda_m|} \leq \varepsilon,$$

which is equivalent to our statement.  $\square$

A proof for the following subcritical exponential decay result can be found at Theorem 10 in Section 4.5 of [20]. It shows that the component sizes decay exponentially fast in the subcritical regime.

**Lemma 2.11.** *Let  $G$  be a connected, infinite, transitive, locally-finite graph with  $|B_r(x)| \leq r^{(\log r)/100}$  for all  $x \in G$ , where  $B_r(x)$  is the ball with radius  $r$  w.r.t. the graph distance around  $x$ . Then, if we consider site percolation on  $G$  with  $p < p_c(G)$  there exists a constant  $c = c(p, G) > 0$  such that for all  $n \in \mathbb{N}$*

$$\mathbb{P}_p(|C_x| \geq n) \leq e^{-cn},$$

where  $C_x$  is the connected component of  $x$ .

# CHAPTER 3

## POISSON–VORONOI PERCOLATION IN HIGH DIMENSIONS

This chapter is based on joint work with Zakhar Kabluchko and Tobias Müller. At the time of writing, it is in the final stages of polishing before release as a preprint.

### 3.1 Outline

#### 3.1.1 Sketch of the proof

We show Theorem 1.4 and 1.5 by proving the corresponding upper and lower bound. The proof of the lower bound is inspired by the work on Voronoi percolation in hyperbolic space [48]. Penrose employed a comparable approach to establish an upper bound on  $p_c$ , as detailed in [84]. The intuition behind the proof strategies is similar to that of other high-dimensional percolation models. The idea is that in high dimensions there is a greater level of independence, meaning that the probability of forming loops inside a cluster is low and thus locally clusters can be compared to trees. Likewise, connected components can be viewed (in some sense) as a branching random walk if a Euclidean embedding is required. It is well known that a Galton–Watson process with offspring distribution  $X$  satisfying  $\mathbb{P}(X = 1) < 1$  and  $\mathbb{E}X \leq 1$  will die out almost surely and will survive with a positive probability if  $\mathbb{E}X > 1$  (see e.g. Section 4.3.4 of [36]). This implies that on the  $k$ -regular tree  $T_k$  it holds  $p_c(T_k) = 1/(k - 1)$ . It is therefore also not surprising that  $p_c(\mathbb{Z}^d) = (1 + o(1))/2d$ , since every vertex of  $\mathbb{Z}^d$  has degree  $2d$ .

However, in our setting the degree of vertices are random and even unbounded. We will see that the critical probability behaves like the reciprocal of the average or typical degree (more details in Section 3.2.2), aligning with the results for other high dimensional percolation models.

The high dimensionality will have a major influence on the geometric structure and we will describe this here informally. More rigorous details are given in Section 3.2.3. Neighbors in the Delaunay/Gabriel graph of a point will mostly lie in an annulus around it. Together with our choice of  $\lambda$ , this will yield that black neighbors of a fixed point are at distance of roughly 2 from it. The reason for this is that for shorter distances, we will only rarely encounter black points for our choice of  $p$ . We note that for each connection between two fixed points, there must exist a ball with them on its boundary that is empty of other Poisson points. It will be quite unlikely that this happens when the points are at great distance, since then the ball has to be large. Further, the neighbors of a point will have an angle of roughly  $\pi/2$  to each other and also to other Poisson points that are in proximity. We will utilize this to show a large level of independence of different edges for large enough dimensions. A priori we would have to consider the whole space  $\mathbb{R}^d$  in order to check for connections. However, the structure we have just described will imply it suffices to consider only a bounded area to capture most possible neighbors. We will use this observation for both the lower and upper bound on  $p_c$ .

### Outline of the upper bound

For this direction we will fix an arbitrary  $\mu > 1$  and show that for all large enough  $d$  a subset of a fixed black component is infinite with a positive probability, given that  $p = \mu e d^{-1} 2^{-d}$  (resp.  $p = \mu 2^{-d}$ ). This value will play the role as the some sort of expected number of neighbors. We are going to construct a coupling of an exploration algorithm of this subset with site percolation on  $\mathbb{Z}^2$ . The algorithm will not have to look for all possible connections, since if an infinite component exists with positive probability for a subset, it will already imply the above statement. The idea is to project the points explored in each iteration down to two dimensions and open a site in  $\mathbb{Z}^2$  if the exploration produced a ‘beneficial’ result for this iteration. We will show that the exploration procedure can be coupled with a branching random walk up to a fixed generation. The benefit of the branching random walk is that connections happen independently, while the status of a connection of two points in the Delaunay/Gabriel graph is generally not independent. When we scale the first two coordinates of such a neighbor by  $\sqrt{d}$ , they will be normally distributed as we increase the dimension. Further, the branching random walk will be supercritical for all large enough  $d$ . These properties also hold more generally for neighbors of points that we find in our exploration algorithm at a later iteration step, since the considered area in each step of the algorithm has a small intersection with the previously explored area

for large  $d$ . Thus, the interference of the previous construction with the current iteration step is small. There will be some truncations that we have to perform, and part of the work will involve showing that these have little impact for large enough  $d$ . Additionally, we will not be able to perform the algorithm in one go, since our coupling will only work until a fixed generation. Thus we will glue multiple exploration algorithms that all run for a finite time together, one for each site of  $\mathbb{Z}^2$ . By utilizing the coupling with site percolation on  $\mathbb{Z}^2$ , we are then able to show that an infinite component exists with positive probability.

### Outline of the lower bound

For the lower bound, we will show that for all  $0 < \mu < 1$  we can choose  $d$  large enough such that percolation does not occur (a.s.) when we pick  $p = \mu ed^{-1}2^{-d}$  (resp.  $p = \mu 2^{-d}$ ). It immediately follows that the existence of an infinite component implies there has to be an infinite path consisting of black Poisson points with adjacent Voronoi cells (resp. Gabriel neighbors). We will show that the probability of the occurrence of such a path is 0 for all large enough  $d$  and the choice of  $p$ . We are more generous in how we define a path compared to the usual sense. This means that in our path not all neighbors are necessary Voronoi neighbors. However, we will show that if the ‘more generous’ paths of infinite length do not exist (a.s.), they also do not exist in the original sense. We will work with this new definition in order to only check Poisson points inside a bounded area to confirm or deny a connection. As outlined above, in large dimensions, neighbors of a typical point will mostly lie in an annulus around it and they will have an angle close to  $\pi/2$  to each other. Even though we expect that distinct connections inside a path have a small probability of interfering with each other, the chances are still nonzero. This would introduce dependencies that we aim to avoid in our derivation. We will introduce two types of edges in such a path, good and bad, where good edges will have properties that are nice to work with. We can now split each black path into multiple *chunks*  $P_1, P_2, \dots$ , where all edges of each part  $P_i$ , except the last edge, is good (singular bad edges are also allowed). Thus we will ‘restart’ every time we encounter a bad edge. These parts will be defined in a way such that the area that determines  $P_i$  can only intersect the region of  $P_{i-1}$  or  $P_{i+1}$  and therefore we do not have any dependence to edges in  $P_j$  for  $|i - j| > 1$ . We then make extensive use of the Mecke formula (2.5) to compute upper bounds on the expected number of different types of chunks. This will then be used to conclude that for all  $0 < \mu < 1$  and large enough  $d$ , the expected number of chunks is small and we do not percolate almost surely.

#### 3.1.2 Related work

Looking at different graphs, the precise value of  $p_c$  is only known in rare cases, and the study of how the model behaves at criticality ( $p = p_c$ ) is of central interest in

percolation theory. Most prominent is the case  $\mathbb{Z}^d$  together with bond percolation where edges connect nearest neighbors, i.e.  $x$  is connected to  $y$  if  $\|x - y\| = 1$ . Here Kesten showed that for  $d = 2$  it holds  $p_c(\mathbb{Z}^2) = 1/2$  [63]. Also the asymptotics of  $p_c(\mathbb{Z}^d)$  for large dimensions are known. We have that  $p_c(\mathbb{Z}^d) = (1 + o(1))/2d$  as  $d \rightarrow \infty$ , which holds for both site and bond percolation. It was independently shown by different authors and proofs for it can be found in [8, 18, 43, 50, 64]. A review of high dimensional percolation on  $\mathbb{Z}^d$  is given in [53]. For more general textbooks that deal with percolation we can refer to [45] or [20], where the latter includes a chapter on Poisson–Voronoi percolation.

Poisson–Voronoi percolation has received significant attention in recent years and is, next to Boolean percolation, a central continuum percolation model. It exhibits a large level of dependence. Given any two points in  $\mathbb{R}^d$ , there is always a positive probability that they lie in the same Voronoi cell and thus if one is black, so is the other. The model was first introduced for first passage percolation by Vahidi-Asl and Wierman [96]. The initial rigorous results on the version we consider here were obtained by Benjamini and Schramm [13], who proved that in dimensions 2 and 3 the crossing probabilities are asymptotically invariant under smooth conformal changes of the metric as the Poisson intensity  $\lambda \rightarrow \infty$ , and by Zvavitch [103], who proved  $\mathbb{P}_{1/2}(\text{percolation}) = 0$  for  $d = 2$ , hence  $p_c(2) \geq 1/2$ . The bound in the other direction, establishing  $p_c(2) = 1/2$ , was given by Bollobás and Riordan [19]. Alternative proofs for this result can be found in [3, 35]. The main benefit of studying the model in the planar case is the self-duality for  $p = 1/2$ , that also helped Tassion in showing the RSW analogue for Poisson–Voronoi percolation [94]. There has been notable interest in noise sensitivity for  $d = 2$  and  $p = p_c = 1/2$ . The model is noise sensitive in the sense that given the Voronoi configuration without any coloring gives asymptotically no information about the crossing probability of a box [3–5], see also the work of Vanneuville [97–99].

Results for  $d \geq 3$  are less frequent. Exponential decay in the subcritical regime was shown by Duminil-Copin, Raoufi and Tassion [35]. Sharpness for  $p > p_c$  was established recently by Dembin and Severo [30]. Exact rigorous results on  $p_c(d)$  for  $d \neq 2$  are unknown. Yet, the bounds  $2^{-d}(9d \log d)^{-1} \leq p_c(d) \leq C2^{-d}\sqrt{d} \log d$  for a constant  $C > 0$  are known to hold for all large enough  $d$  [12]. Theorem 1.4 is an improvement on this. Similar bounds on  $p_c$  also hold for bond percolation on the Poisson–Voronoi configuration [10]. We are not aware of rigorous results on  $p_c$  for Poisson–Gabriel percolation, apart from the fact that it is non-trivial [15].

The setting of Poisson–Voronoi percolation also carries over naturally to spaces that are not  $\mathbb{R}^d$ . For results concerning the hyperbolic plane, see [14, 29, 48, 49] or for more general spaces [26, 44].

## 3.2 Notation, preliminaries and expected degree

### 3.2.1 Probabilistic auxiliary lemmas

For each  $y \in \mathbb{R}^d$  and  $b \in \mathbb{R}$ , the set  $\{x \in \mathbb{R}^d : \langle x, y \rangle \geq b\}$  is said to be a *half-space*. If  $b = 0$ , it is a *linear half-space*.

**Lemma 3.1** (Wendel's formula [101]). *Let  $X_1, \dots, X_n$  be i.i.d. uniformly random points on  $S^{d-1}$ . Then*

$$\mathbb{P}(\text{there is a linear half-space containing all } X_1, \dots, X_n) = \mathbb{P}(\text{Bi}(n-1, 1/2) \leq d-1).$$

The result of Wendel also holds if the points are taken uniformly from the ball  $B(\mathbf{0}, r)$  for any  $r > 0$ , since if we scale points by arbitrary positive scalars, they will still lie in the same linear half-space as before. The next result is sometimes referred to as the Poincaré–Maxwell–Borel lemma (see Lemma 1.2 in [67] or [33]). The total variation distance of two random variables (or distributions) is defined as

$$\|X - Y\|_{\text{TV}} := \sup_A |\mathbb{P}(X \in A) - \mathbb{P}(Y \in A)|,$$

where the supremum is over all measurable sets  $A$ .

**Lemma 3.2** ([91]). *Let  $U = (U_1, \dots, U_d)$  be uniformly distributed over  $S^{d-1}$  and  $m \leq d$  fixed. Then  $\sqrt{d}(U_1, \dots, U_m)$  converges as  $d \rightarrow \infty$  in total variation distance to  $(X_1, \dots, X_m)$ , where the  $X_i$  are i.i.d. with  $X_i \sim \mathcal{N}(0, 1)$ .*

The following result on branching processes can be found in greater generality in [51], Thm. 5.1.

**Lemma 3.3.** *Consider a Galton–Watson process starting from a single point of generation 0 with a  $\text{Po}(\nu)$  offspring distribution for  $\nu \geq 1$ . Then, the second moment of the number of particles at generation  $k$  is at most  $(k+1)\nu^{2k+1}$ .*

### 3.2.2 Expected typical degree

We remind ourselves of the setup that we use here. We let  $\mathcal{Z}$  be a Poisson point process (PPP) in  $\mathbb{R}^d$  with constant intensity  $\lambda > 0$ . We color each point of  $\mathcal{Z}$  black with probability  $p$ , independently of one another, and otherwise they are said to be white. We set  $\mathcal{V}(z)$  (i.e. the Voronoi cell of  $z$ ) black if  $z \in \mathcal{Z}$  is black. We augment the set of points  $\mathcal{Z}$  with a deterministic black reference point  $\mathbf{0} := (0, \dots, 0) \in \mathbb{R}^d$  as our *origin*, implying that the cell  $\mathcal{V}(\mathbf{0})$  is always considered to be black. We will also use the phrase *typical point* for  $\mathbf{0}$ . Additionally,  $\mathcal{Z}_b$  is the set of all black points of  $\mathcal{Z}$  and  $\mathcal{Z}_w$  are all white points. Thus,  $\mathcal{Z} = \mathcal{Z}_b \cup \mathcal{Z}_w$  and the two sets of colored points are independent PPPs with intensity  $\lambda p$  and  $\lambda(1-p)$ , respectively. We will use the notation  $\mathbb{P} = \mathbb{P}_p$  for the law of  $(\mathcal{Z}_b, \mathcal{Z}_w)$ .

As we have pointed out, the global structure of the graph is independent of the intensity parameter. Therefore, we can fix  $\lambda = \text{vol}(B(\underline{0}, 1))^{-1}$  without changing the probability of the occurrence of an infinite cluster. The value of  $\lambda$  will also not influence the expected *typical degree*. The typical degree  $D$  is the degree of  $\underline{0}$  on the Delaunay graph w.r.t.  $\mathcal{Z} \cup \{\underline{0}\}$ . Equivalently, it is the number of neighboring cells of  $\mathcal{V}(\underline{0})$  in the Poisson–Voronoi tessellation  $\text{Vor}_d(\mathcal{Z} \cup \{\underline{0}\})$  of  $\mathcal{Z} \cup \{\underline{0}\}$ . Thus, if we write  $\underline{0} \sim z$  should the Voronoi cells of  $\underline{0}$  and  $z \in \mathcal{Z}$  be adjacent with respect to  $\mathcal{Z} \cup \{\underline{0}\}$ , we can express this as

$$D := \sum_{z \in \mathcal{Z}} \mathbb{1}\{\underline{0} \sim z\}.$$

Since we obtain results for both models, we can also write  $\underline{0} \sim z$  to say that  $\underline{0}$  is connected to  $z$  in the Gabriel graph  $\text{Gab}_d(\mathcal{Z} \cup \{\underline{0}\})$  of  $\mathcal{Z} \cup \{\underline{0}\}$  and the arguments presented here will go through. If  $w \in \mathcal{Z}$  we will use the identical notation  $w \sim z$  to say that  $w$  is a neighbor of  $z$  w.r.t. the Delaunay/Gabriel graph on  $\mathcal{Z}$ . In the same sense, if  $w, z \in \mathbb{R}^d$ , we write  $w \sim z$  if  $w$  and  $z$  are adjacent in the graph formed on the augmented set  $\mathcal{Z} \cup \{w, z\}$ .

Computing an expectation of a random variable with the above structure can be done using the Mecke formula, as given in [Lemma 2.5](#). As a warm up we will first consider the way easier Gabriel model. A related result to the typical degree for the Gabriel graph was first obtained by Matula and Sokal [74]. They showed that when placing  $n$  points uniformly in the unit square in two dimensions, the expected degree of a vertex of the Gabriel graph tends to 4 as  $n$  increases. This was generalized for arbitrary dimensions and more general distributions by Devroye [32]. We were not able to find a clean reference for our specific result, so we give a short proof here.

**Lemma 3.4.** *For the Poisson–Gabriel graph and all  $d \geq 1$*

$$\mathbb{E}D = 2^d.$$

*Proof.* The probability of a fixed point  $z \in \mathbb{R}^d$  being connected to  $\underline{0}$  in the Poisson–Gabriel graph on  $\mathcal{Z} \cup \{\underline{0}, z\}$  is equal to the probability that there are no points of  $\mathcal{Z}$  in  $B_{\text{Gab}}(\underline{0}, z)$ , yielding

$$\begin{aligned} \mathbb{P}(\underline{0} \sim z \text{ for } \mathcal{Z} \cup \{\underline{0}, z\}) &= \mathbb{P}(|\mathcal{Z} \cap B_{\text{Gab}}(\underline{0}, z)| = 0) = e^{-\lambda \text{vol}(B_{\text{Gab}}(\underline{0}, z))} = e^{-\kappa_d \lambda \frac{\|z\|^d}{2^d}} \\ &= e^{-\frac{\|z\|^d}{2^d}}. \end{aligned}$$

Since  $\mathcal{Z}$  is a Poisson point process with intensity  $\lambda$ , we can use the Mecke formula and switch to polar coordinates (see e.g. [88], Cor. 15.14) to get with  $\lambda = 1/\kappa_d$

and substituting  $x = r^d 2^{-d}$

$$\begin{aligned} \mathbb{E}D &= \int_{\mathbb{R}^d} \mathbb{P}(\underline{0} \sim z \text{ for } \mathcal{Z} \cup \{\underline{0}, z\}) \lambda \, dz = \int_{\mathbb{R}^d} e^{-\frac{\|z\|^d}{2^d}} \lambda \, dz \\ &= \int_0^\infty e^{-r^d 2^{-d}} dr^{d-1} \kappa_d \lambda \, dr = 2^d \int_0^\infty e^{-x} dx = 2^d. \end{aligned}$$

□

We now turn our head to the way more challenging Voronoi setting, so the proof of [Theorem 1.6](#). The only external input for the proof is the following known expected face number formula from the literature (Theorem 2.1 inside [\[58\]](#)).

**Lemma 3.5** ([\[58\]](#)).

$$\mathbb{E}D = \mathbb{E}f_{d-1}(\mathcal{V}_{typ}) = d^{d+1} \frac{\sqrt{\pi} \Gamma(d/2)}{\pi \Gamma((d+1)/2)} I_d, \quad (3.1)$$

where

$$I_d := \frac{1}{i} \int_{\pi/2-i\infty}^{\pi/2+i\infty} G_d(w) \, dw, \quad G_d(w) := \frac{F_d(w)^{d-1}}{(\sin w)^{d^2+1}}, \quad F_d(w) := \int_0^w (\sin y)^{d-1} \, dy.$$

The integral defining  $F_d$  is path independent since  $(\sin y)^{d-1}$  is entire. The argument has four steps. First we shift the contour from the line  $\operatorname{Re}(w) = \pi/2$  to the saddle line  $\operatorname{Re}(w) = \pi/4$ . Second we prove a uniform endpoint expansion for  $F_d(w)$ . Third we evaluate the integral on the saddle line, proving both the local Gaussian approximation and the estimates excluding the rest of the contour. Finally we multiply by the explicit prefactor in [\(3.1\)](#).

The saddle point for the facet case here is  $\pi/4$ . A quick way to see why this is the right point is to insert the first order approximation  $F_d(w) \approx (\sin w)^d / (d \cos w)$  into  $G_d$ . Most powers of  $\sin w$  then cancel, and the exponentially relevant part is, up to subexponential factors,

$$\exp\{d\Phi(w)\}, \text{ with } \Phi(w) := -\log \cos w - \log \sin w.$$

On the real interval  $(0, \pi/2)$ , this satisfies

$$\Phi'(w) = \tan w - \cot w,$$

so  $\Phi'(w) = 0$  gives  $\tan^2 w = 1$ , namely  $w = \pi/4$ . Along the vertical line through this point, the quadratic term has negative real part; this is the Gaussian decay that appears below. The required approximation is proved in [Lemma 3.7](#).

**Lemma 3.6.** *For every  $d \geq 2$ ,*

$$I_d = \frac{1}{i} \int_{\pi/4-i\infty}^{\pi/4+i\infty} G_d(w) \, dw. \quad (3.2)$$

*Proof.* We first check where the integrand could have singularities. The function  $(\sin w)^{d-1}$  is entire, so its antiderivative  $F_d$  is entire. Hence the only possible singularities of  $G_d$  are the zeros of  $\sin w$ , that is the points  $n\pi$ ,  $n \in \mathbb{Z}$ . The closed strip

$$\pi/4 \leq \operatorname{Re}(w) \leq \pi/2$$

contains no such zero. Thus  $G_d$  is holomorphic in a neighbourhood of this closed strip.

Cauchy's theorem can be applied immediately to finite rectangles. The only point needing justification is the limit as the height of the rectangle tends to infinity. We prove the required decay. Write  $w = u + iv$ . The identity

$$|\sin(u + iv)|^2 = \sin^2 u + \sinh^2 v$$

shows that, uniformly for  $u \in [\pi/4, \pi/2]$ ,

$$|\sin(u + iv)| \asymp e^{|v|}, \quad |v| \rightarrow \infty. \quad (3.3)$$

For  $F_d$ , use the straight segment from 0 to  $w$ :

$$F_d(w) = w \int_0^1 (\sin(tw))^{d-1} dt.$$

Since  $u$  stays in a bounded interval, the elementary estimate  $|\sin(tu + itv)| \leq C e^{t|v|} \leq C e^{|v|}$  gives, for fixed  $d$ ,

$$|F_d(u + iv)| \leq C_d(1 + |v|)e^{(d-1)|v|}. \quad (3.4)$$

Here the constant may depend on  $d$ , which is harmless in this lemma because we are shifting the contour for each fixed  $d$ .

Combining (3.3) and (3.4), we get

$$\begin{aligned} |G_d(u + iv)| &\leq C_d(1 + |v|)^{d-1} \exp\{((d-1)^2 - (d^2 + 1))|v|\} \\ &= C_d(1 + |v|)^{d-1} e^{-2d|v|}, \end{aligned}$$

uniformly for  $u \in [\pi/4, \pi/2]$ , as  $|v| \rightarrow \infty$ . The exponential decay implies that the integrals over horizontal segments at heights  $\pm R$  tend to zero as  $R \rightarrow \infty$ .

Indeed, let  $\mathcal{R}_R$  be the rectangle with vertices  $\pi/4 \pm iR$  and  $\pi/2 \pm iR$ . Cauchy's theorem gives  $\int_{\partial \mathcal{R}_R} G_d(w) dw = 0$ . Writing the two vertical sides with upward orientation, this identity says

$$\int_{\pi/2 - iR}^{\pi/2 + iR} G_d(w) dw - \int_{\pi/4 - iR}^{\pi/4 + iR} G_d(w) dw = -H_R,$$

where  $H_R$  is the sum of the two horizontal integrals. The quantity  $H_R$  tends to zero by the decay just proved. Letting  $R \rightarrow \infty$  proves (3.2).  $\square$

We shall use the following elementary endpoint expansion. Its role is similar to a one sided Laplace method: for  $w$  in the right half of the strip, the function  $t \mapsto |\sin(tw)|$  is largest at the endpoint  $t = 1$ , so the integral  $\int_0^w (\sin y)^{d-1} dy$  is controlled by the endpoint  $y = w$ .

**Lemma 3.7.** *Let  $K$  be a compact subset of  $\{w \in \mathbb{C} : 0 < \operatorname{Re}(w) < \pi/2\}$ . Then, uniformly for  $w \in K$ ,*

$$F_d(w) = \frac{(\sin w)^d}{d \cos w} \left( 1 - \frac{\tan^2 w}{d} + O_K(d^{-2}) \right), \quad d \rightarrow \infty. \quad (3.5)$$

*Proof.* We split the proof into two parts. First we prove a rough but uniform bound showing that the endpoint  $y = w$  dominates the integral. Then we construct an approximate antiderivative whose derivative agrees with  $(\sin y)^{d-1}$  up to a relative error of order  $d^{-2}$ .

For  $|\operatorname{Re}(w)| < \pi/2$  and  $w \neq 0$ , the function  $t \mapsto |\sin(tw)|$  is strictly increasing on  $[0, 1]$ . Indeed, if  $w = a + ib$ , then

$$|\sin(tw)|^2 = \frac{1}{2} \{ \cosh(2tb) - \cos(2ta) \}.$$

The term  $\cosh(2tb)$  is nondecreasing in  $t \geq 0$ , and  $-\cos(2ta)$  is nondecreasing for  $|a| < \pi/2$ ; at least one monotonicity is strict unless  $w = 0$ .

We need a quantitative form of this endpoint dominance, uniform for  $w \in K$ . For  $0 < t \leq 1$  it holds that

$$\frac{d}{dt} \log |\sin(tw)| = \operatorname{Re}\{w \cot(tw)\}.$$

If  $w = a + ib$ , then a direct computation gives

$$\operatorname{Re}\{w \cot(tw)\} = \frac{a \sin(2at) + b \sinh(2bt)}{\cosh(2bt) - \cos(2at)}. \quad (3.6)$$

For  $w \in K$ , we have  $a = \operatorname{Re}(w) > 0$ . Hence the numerator in (3.6) is positive for every  $t > 0$  since  $a \sin(2at) > 0$  and  $b \sinh(2bt) \geq 0$ . The denominator is also positive. Moreover, as  $t \downarrow 0$ ,

$$w \cot(tw) = \frac{1}{t} + O_K(t),$$

so the expression in (3.6) tends to  $+\infty$  uniformly near  $t = 0$ . By compactness on the remaining range of  $t$ , there is a constant  $c_K > 0$  such that

$$\operatorname{Re}\{w \cot(tw)\} \geq c_K, \quad w \in K, \quad 0 < t \leq 1.$$

Integrating this derivative from  $t$  to 1, we obtain

$$\log |\sin w| - \log |\sin(tw)| = \int_t^1 \operatorname{Re}\{w \cot(sw)\} ds \geq c_K(1-t).$$

Equivalently,

$$|\sin(tw)| \leq |\sin w| e^{-c_K(1-t)}, \quad w \in K, \quad 0 \leq t \leq 1.$$

At  $t = 0$ , the inequality is trivial because  $\sin(0) = 0$ . Integrating along the segment  $y = tw$ , we therefore get

$$\begin{aligned} |F_d(w)| &\leq |w| \int_0^1 |\sin(tw)|^{d-1} dt \\ &\leq |w| |\sin w|^{d-1} \int_0^1 e^{-c_K(d-1)(1-t)} dt \leq C_K \frac{|\sin w|^{d-1}}{d}, \quad w \in K. \end{aligned} \quad (3.7)$$

This is only a rough estimate, but it is the estimate that will control the error after the approximate antiderivative is introduced.

We now define

$$A_d(y) := \frac{(\sin y)^d}{d \cos y} \left( 1 - \frac{\tan^2 y}{d} \right).$$

The choice of  $A_d$  is designed so that its derivative has no error term of relative order  $d^{-1}$ . To see this explicitly, write  $T = \tan y$ . Since

$$\frac{d}{dy} \frac{(\sin y)^d}{\cos y} = (\sin y)^{d-1} (d + T^2), \quad \frac{d}{dy} T^2 = 2T \sec^2 y,$$

a direct differentiation gives

$$A'_d(y) = (\sin y)^{d-1} \left( 1 - \frac{\tan^4 y + 2 \tan^2 y \sec^2 y}{d^2} \right). \quad (3.8)$$

The cancellation of the order  $d^{-1}$  terms in this formula is the reason for including the correction  $-\tan^2 y/d$  in  $A_d$ .

Let

$$K_0 := \{tw : w \in K, 0 \leq t \leq 1\}.$$

This set is compact and  $\cos y \neq 0$  on  $K_0$ . Thus  $\tan y$  and  $\sec y$  are bounded on  $K_0$ , and (3.8) can be written as

$$A'_d(y) = (\sin y)^{d-1} (1 + R_d(y)), \quad \text{with } \sup_{y \in K_0} |R_d(y)| \leq C_K d^{-2}.$$

Since  $A_d(0) = 0$ , integration along the segment from 0 to  $w$  gives

$$F_d(w) = A_d(w) - \int_0^w R_d(y) (\sin y)^{d-1} dy.$$

The error integral is estimated in absolute value along the same segment  $y = tw$ :

$$\left| \int_0^w R_d(y) (\sin y)^{d-1} dy \right| \leq C_K d^{-2} |w| \int_0^1 |\sin(tw)|^{d-1} dt = O_K \left( d^{-2} \frac{|\sin w|^{d-1}}{d} \right),$$

where the last step is exactly (3.7). Since  $|\cos w|/|\sin w|$  is bounded on  $K$ , this is

$$O_K \left( d^{-2} \left| \frac{(\sin w)^d}{d \cos w} \right| \right).$$

Consequently

$$F_d(w) = A_d(w) + \frac{(\sin w)^d}{d \cos w} O_K(d^{-2}),$$

uniformly for  $w \in K$ . Substituting the definition of  $A_d(w)$  gives (3.5).  $\square$

We now evaluate  $I_d$  by the saddle-point method. We put

$$a := \frac{\pi}{4}, \quad H_d := e^{-1} 2^d d^{-(d-1)}.$$

The number  $H_d$  is the height scale of the integrand at the saddle. Indeed, Lemma 3.8 below shows that  $G_d(a + iz/\sqrt{d})$  is asymptotic to  $H_d e^{-2z^2}$  in the window  $|t| = |z|/\sqrt{d}$ . By Lemma 3.6, and with  $w = a + it$ ,

$$I_d = \int_{-\infty}^{\infty} G_d(a + it) dt = \frac{1}{\sqrt{d}} \int_{-\infty}^{\infty} G_d \left( a + \frac{iz}{\sqrt{d}} \right) dz.$$

The first equality follows because  $dw = i dt$  on the vertical contour. The second equality is the rescaling  $t = z/\sqrt{d}$ .

**Lemma 3.8.** *For every fixed  $L > 0$ , uniformly for  $|z| \leq L$  and as  $d \rightarrow \infty$*

$$G_d \left( a + \frac{iz}{\sqrt{d}} \right) \sim H_d e^{-2z^2}. \quad (3.9)$$

*Proof.* Apply Lemma 3.7 uniformly in a fixed compact neighbourhood of  $a$ . For  $|z| \leq L$  and  $w = a + iz/\sqrt{d}$ ,

$$F_d(w) = \frac{(\sin w)^d}{d \cos w} \left( 1 - \frac{\tan^2 w}{d} + O_L(d^{-2}) \right).$$

Substituting this into  $G_d(w) = F_d(w)^{d-1}/(\sin w)^{d^2+1}$ , the powers of  $\sin w$  simplify as

$$d(d-1) - (d^2+1) = -(d+1).$$

Therefore

$$G_d(w) = d^{-(d-1)} \left( 1 - \frac{\tan^2 w}{d} + O_L(d^{-2}) \right)^{d-1} \frac{1}{\cos^{d-1} w \sin^{d+1} w}. \quad (3.10)$$

The first factor in parentheses gives the constant  $e^{-1}$ . Indeed,  $\tan^2 w = 1 + O_L(d^{-1/2})$ . The quantity inside the logarithm is therefore  $1 + O_L(d^{-1})$ , so for all large  $d$  it lies in a fixed small neighbourhood of 1, where the complex logarithm is unique. Hence

$$\begin{aligned} (d-1) \log \left( 1 - \frac{\tan^2 w}{d} + O_L(d^{-2}) \right) &= (d-1) \left( -\frac{\tan^2 w}{d} + O_L(d^{-2}) \right) \\ &= -1 + O_L(d^{-1/2}). \end{aligned}$$

Thus the parenthetical factor in (3.10), raised to the power  $d-1$ , is  $e^{-1}(1+o(1))$ , uniformly for  $|z| \leq L$ .

It remains to evaluate the trigonometric denominator. On the vertical saddle line this can be done by an exact identity, avoiding any logarithmic branch choices for  $\sin$  and  $\cos$ . Since

$$\sin w \cos w = \frac{1}{2} \sin(2w) = \frac{1}{2} \sin \left( \frac{\pi}{2} + \frac{2iz}{\sqrt{d}} \right) = \frac{1}{2} \cosh \left( \frac{2z}{\sqrt{d}} \right),$$

we have

$$\frac{1}{\cos^{d-1} w \sin^{d+1} w} = \frac{\cos w}{\sin w} \frac{1}{(\sin w \cos w)^d} = \cot w \left( \frac{2}{\cosh(2z/\sqrt{d})} \right)^d. \quad (3.11)$$

Uniformly for  $|z| \leq L$ ,  $\cot(a + iz/\sqrt{d}) = 1 + O_L(d^{-1/2})$ . Also,  $\log \cosh u = u^2/2 + O(u^4)$  as  $u \rightarrow 0$ , and therefore

$$d \log \cosh \left( \frac{2z}{\sqrt{d}} \right) = 2z^2 + O_L(d^{-1}).$$

It follows from (3.11) that

$$\frac{1}{\cos^{d-1} w \sin^{d+1} w} = 2^d e^{-2z^2} (1 + o(1)),$$

uniformly for  $|z| \leq L$ . Combining this with (3.10) proves (3.9).  $\square$

The preceding lemma only controls a bounded  $z$ -window, equivalently a  $t$ -window of size  $d^{-1/2}$  around the saddle. To pass from this local asymptotic to the full contour integral, we need estimates showing that the rest of the contour is negligible. The key simplification on the saddle line  $w = a + it$  is that

$$|\sin(a + it)|^2 = |\cos(a + it)|^2 = \frac{1}{2} + \sinh^2 t. \quad (3.12)$$

Away from  $t = 0$ , this common absolute value is larger than  $2^{-1/2}$ , and the large power in the denominator suppresses the integrand exponentially in  $d$ .

**Lemma 3.9.** *There are constants  $\delta > 0$ ,  $c > 0$ ,  $C > 0$  such that, for all sufficiently large  $d$  and all  $|z| \leq \delta\sqrt{d}$ ,*

$$\left| G_d \left( a + \frac{iz}{\sqrt{d}} \right) \right| \leq CH_d e^{-cz^2}. \quad (3.13)$$

Moreover, for every fixed  $T > \delta$ ,

$$\int_{\delta \leq |t| \leq T} |G_d(a + it)| dt = o \left( \frac{H_d}{\sqrt{d}} \right). \quad (3.14)$$

Finally, if  $T$  is chosen sufficiently large, then

$$\int_{|t| \geq T} |G_d(a + it)| dt = o \left( \frac{H_d}{\sqrt{d}} \right). \quad (3.15)$$

*Proof.* We start with a compact part of the saddle line, say  $|t| \leq T$  with  $T$  fixed. Lemma 3.7, applied uniformly on the compact set  $\{a + it : |t| \leq T\}$ , gives

$$F_d(a + it) = \frac{\sin^d(a + it)}{d \cos(a + it)} (1 + O_T(d^{-1})), \quad |t| \leq T.$$

The relative error is  $1 + O_T(d^{-1})$ , not merely bounded by some constant. Consequently, after raising it to the power  $d - 1$ , it remains bounded by a constant depending only on  $T$ . Using (3.12), we obtain for  $|t| \leq T$

$$|G_d(a + it)| \leq C_T d^{-(d-1)} \left( \frac{1}{2} + \sinh^2 t \right)^{-d} = C_T 2^d d^{-(d-1)} (1 + 2 \sinh^2 t)^{-d}. \quad (3.16)$$

This is the main estimate on bounded pieces of the contour. It says that the ratio of  $|G_d(a + it)|$  to its saddle height is controlled by  $(1 + 2 \sinh^2 t)^{-d}$ .

Choose  $\delta > 0$  small. Since  $\log(1 + 2 \sinh^2 t) = 2t^2 + O(t^4)$  as  $t \rightarrow 0$ , there is  $c_1 > 0$  such that

$$\log(1 + 2 \sinh^2 t) \geq c_1 t^2, \quad |t| \leq \delta.$$

Putting  $t = z/\sqrt{d}$  in (3.16) gives

$$|G_d(a + iz/\sqrt{d})| \leq C 2^d d^{-(d-1)} e^{-c_1 z^2} \leq CH_d e^{-cz^2},$$

after changing the constants. This proves (3.13).

If  $\delta \leq |t| \leq T$ , then  $1 + 2 \sinh^2 t \geq 1 + 2 \sinh^2 \delta > 1$ . Thus there is  $q = q(\delta) < 1$  such that  $(1 + 2 \sinh^2 t)^{-d} \leq q^d$  on this region. From (3.16),

$$\int_{\delta \leq |t| \leq T} |G_d(a + it)| dt \leq C_T 2^d d^{-(d-1)} q^d.$$

Dividing by  $H_d/\sqrt{d} = e^{-1}2^d d^{-(d-1)}/\sqrt{d}$  gives  $O_T(\sqrt{d}q^d)$ , which tends to 0. This proves (3.14).

It remains to control the unbounded part of the contour. The compact end-point expansion is no longer available uniformly as  $|t| \rightarrow \infty$ , so we use a separate crude estimate. We prove it for  $t > 0$ ; the case  $t < 0$  follows by complex conjugation, because  $F_d(\bar{w}) = \overline{F_d(w)}$  and  $|\sin(\bar{w})| = |\sin w|$ . Fix some  $T_0 > 1$ . For  $t \geq T_0$ , write

$$F_d(a + it) = F_d(a) + i \int_0^t (\sin(a + is))^{d-1} ds.$$

Let

$$r(s) := |\sin(a + is)| = \left( \frac{1}{2} + \sinh^2 s \right)^{1/2}.$$

For  $s \geq 1$ ,

$$(\log r)'(s) = \frac{\sinh s \cosh s}{\frac{1}{2} + \sinh^2 s}$$

is positive and bounded below by some constant  $m > 0$ . Hence, for  $1 \leq s \leq t$ ,

$$r(s) \leq r(t)e^{-m(t-s)}.$$

Therefore

$$\int_1^t r(s)^{d-1} ds \leq r(t)^{d-1} \int_1^t e^{-m(d-1)(t-s)} ds \leq \frac{1}{m(d-1)} r(t)^{d-1}.$$

The bounded initial part  $[0, 1]$ , together with  $F_d(a)$ , is  $O(r(1)^{d-1})$ . Since  $r(T_0) > r(1)$ , the ratio  $(r(1)/r(T_0))^{d-1}$  is exponentially small, and therefore this contribution is absorbed into  $Cd^{-1}r(t)^{d-1}$ , uniformly for  $t \geq T_0$  and all large  $d$ . Thus

$$|F_d(a + it)| \leq Cd^{-1}r(t)^{d-1}, \quad t \geq T_0.$$

Using again  $|\sin(a + it)| = r(t)$ , we get

$$|G_d(a + it)| \leq \left( \frac{C}{d} \right)^{d-1} r(t)^{(d-1)^2 - (d^2+1)} = \left( \frac{C}{d} \right)^{d-1} r(t)^{-2d}.$$

Since  $r(t)^2 \geq c_2 e^{2t}$  for  $t \geq T_0$ , it follows that, for every  $T \geq T_0$ ,

$$\int_T^\infty |G_d(a + it)| dt \leq d^{-(d-1)} (C_3 e^{-2T})^d,$$

where  $C_3$  is independent of  $T$  and  $d$ . Choose  $T \geq T_0$  so large that  $C_3 e^{-2T} < 1$ . After dividing by  $H_d/\sqrt{d}$ , the last bound is

$$O \left( \sqrt{d} \left( \frac{C_3 e^{-2T}}{2} \right)^d \right) = o(1).$$

Together with the identical estimate for negative  $t$ , this proves (3.15).  $\square$

We can now complete the saddle-point evaluation and get the asymptotics of the expected number of neighboring Voronoi cells of the typical cell.

*Proof of Theorem 1.6.* Choose  $T$  large enough so that (3.15) holds, and keep  $\delta$  as in Lemma 3.9. Let  $L > 0$  be fixed. By Lemma 3.8,

$$\frac{1}{\sqrt{d}} \int_{-L}^L G_d \left( a + \frac{iz}{\sqrt{d}} \right) dz \sim \frac{H_d}{\sqrt{d}} \int_{-L}^L e^{-2z^2} dz.$$

For the part  $L < |z| \leq \delta\sqrt{d}$ , the Gaussian majorant gives the uniform estimate

$$\limsup_{d \rightarrow \infty} \frac{\sqrt{d}}{H_d} \left| \frac{1}{\sqrt{d}} \int_{L < |z| \leq \delta\sqrt{d}} G_d \left( a + \frac{iz}{\sqrt{d}} \right) dz \right| \leq C \int_{|z| > L} e^{-cz^2} dz,$$

and the right-hand side tends to 0 as  $L \rightarrow \infty$ . The remaining parts of the contour are  $\delta \leq |t| \leq T$  and  $|t| \geq T$ ; they are negligible by (3.14) and (3.15). Thus, letting first  $d \rightarrow \infty$  and then  $L \rightarrow \infty$ ,

$$I_d \sim \frac{H_d}{\sqrt{d}} \int_{-\infty}^{\infty} e^{-2z^2} dz = \frac{2^d}{e^{d^{d-1}}} \sqrt{\frac{\pi}{2d}}. \quad (3.17)$$

Finally, using Stirlings approximation yields  $\Gamma(x+a)/\Gamma(x+b) \sim x^{a-b}$  with  $x = d/2$ ,  $a = 0$ , and  $b = 1/2$ , we get

$$\frac{\Gamma(d/2)}{\Gamma((d+1)/2)} \sim \left( \frac{d}{2} \right)^{-1/2}. \quad (3.18)$$

Substituting (3.17) and (3.18) into (3.1), we obtain

$$\mathbb{E}f_{d-1}(\mathcal{V}_{\text{typ}}) \sim d^{d+1} \frac{\sqrt{\pi}}{\pi} \left( \frac{d}{2} \right)^{-1/2} \frac{2^d}{e^{d^{d-1}}} \sqrt{\frac{\pi}{2d}} = e^{-1} d 2^d.$$

In the last line, the gamma-ratio contribution and the Gaussian contribution combine to give the factor

$$\frac{\sqrt{\pi}}{\pi} \left( \frac{d}{2} \right)^{-1/2} \sqrt{\frac{\pi}{2d}} = \frac{1}{d},$$

leaving  $d^{d+1}/d^{d-1} \cdot d^{-1} = d$ . This proves the desired expectation asymptotic.  $\square$

### 3.2.3 Truncating the neighbors

The results of this subsection serve as the cornerstones of our proofs. They are absolutely central for everything that follows. We show that for large dimensions, most neighbors of the typical point  $\underline{0}$  will lie inside the annulus  $A(\underline{0})$ , where

$$A(x) := \{y \in \mathbb{R}^d : r_\ell < \|x - y\| < r_u\},$$

for

$$r_\ell := 2d^{-1000/d} \quad \text{and} \quad r_u := 2d^{1000/d}.$$

We note that with these choices, both  $r_\ell$  and  $r_u$  converge to 2, as  $d \rightarrow \infty$ . Throughout we let  $\delta_1 > 0$  be a fixed small constant to be chosen appropriately in the proof of the upper and lower bound. For  $x, y \in \mathbb{R}^d$  we define the *certificate* of  $x$  and  $y$  as

$$\text{cert}(x, y) := \begin{cases} B((x+y)/2, 1 + \delta_1) & \text{if } x \in B(y, r_u) \text{ and} \\ B_{Gab}(x, y) & \text{otherwise.} \end{cases}$$

It will be crucial for us that the certificate captures most area we have to consider in order to check for connections, and we will make this rigorous in the succeeding lemmas. For the Gabriel model this follows by definition, since then for each fixed  $\delta_1$  and large enough  $d$  we have  $B_{Gab}(x, y) \subseteq \text{cert}(x, y)$ .

If  $z, w \in \mathbb{R}^d$ , we define a *Delaunay ball* for  $w \sim z$  to be any open ball with  $w, z$  on the boundary and no Poisson point of  $\mathcal{Z}$  inside. The points  $w$  and  $z$  are adjacent in the Delaunay graph on  $\mathcal{Z} \cup \{w, z\}$  (or the corresponding cells are neighbors in the Voronoi tessellation) if and only if there exists a Delaunay ball. For  $z \in A(w)$ , we are now going to define two events and show that one is a subset of the other, given the right choices of parameters. The first is

$$L(w, z) := \{w \sim z \text{ in } \text{Vor}_d(\mathcal{Z} \cup \{w, z\})\} \\ \cap \{\text{there is no Delaunay ball for } w \sim z \text{ contained in } \text{cert}(w, z)\}.$$

For any ball  $B(x_1, r)$ , and hyperplane  $H = \{x \in \mathbb{R}^d : \langle x, y \rangle = b\}$  with  $x_1 \in H$ , we call  $B(x_1, r) \cap \{x \in \mathbb{R}^d : \langle x, y \rangle \geq b\}$  a *half-ball*. We will say that the half-ball has center  $x_1$  with radius  $r$ . For a small constant  $\delta_2 = \delta_2(\delta_1) > 0$ , to be determined later, we let the second event be

$$J(w, z) := \left\{ \begin{array}{l} \text{there exists a half-ball with radius } 1 + \delta_2 \text{ contained in } \text{cert}(w, z) \\ \text{that is disjoint from } \mathcal{Z} \end{array} \right\}.$$

The succeeding auxiliary lemma will be useful to establish the existence of a half-ball that is empty of Poisson points.

**Lemma 3.10.** *For all  $\delta_1 > 0$  there exists a  $\delta_2 = \delta_2(\delta_1) > 0$  and  $d_0 = d_0(\delta_1)$  such that for all  $d \geq d_0$  and  $p_1, p_2 \in \mathbb{R}^d$  with  $p_1 \in A(p_2)$  the following holds. If  $B \subset \mathbb{R}^d$  is a ball such that  $p_1, p_2 \in \partial B$  and  $B \not\subseteq \text{cert}(p_1, p_2)$ , there exists a half-ball  $H$  of radius  $1 + \delta_2$  with  $H \subseteq B \cap \text{cert}(p_1, p_2)$ .*

*Proof.* Without loss of generality we let  $\delta_1 < 1$ . Applying a suitable isometry we can assume  $p_1 = (-s, 0, \dots, 0)$  and  $p_2 = (s, 0, \dots, 0)$  for  $r_\ell/2 \leq s \leq r_u/2$  and  $B = B(c, r)$  with  $c = (0, h, 0, \dots, 0)$  for some  $h \geq 0$  and  $r > 0$  such that  $B$  satisfies the requirements of the statement. See also Figure 3.1. We show that setting  $H := B(c', 1 + \delta_2) \cap \{x \in \mathbb{R}^d : x_2 \geq h'\}$  with  $c' = (0, h', 0, \dots, 0)$ ,  $h' = \delta_1/2$  and  $\delta_2 = \delta_1^2/24$  will result in the desired half-ball. We see that for all  $x \in H$  it holds

$$\|x\| \leq \|c'\| + \|x - c'\| \leq h' + 1 + \delta_2 = 1 + \delta_1/2 + \delta_1^2/24 \leq 1 + \delta_1,$$

and thus  $H \subseteq B(0, 1 + \delta_1) = \text{cert}(p_1, p_2)$ . It remains to show that  $H \subseteq B$ . Since  $B \not\subseteq \text{cert}(p_1, p_2)$  it is necessary that  $h + r \geq 1 + \delta_1$ . Additionally, by the requirements on  $B$ , we obtain  $r = \sqrt{s^2 + h^2} = \sqrt{1 + h^2} + o(1)$  as  $d \rightarrow \infty$ . Combining these two facts we get  $h + \sqrt{1 + h^2} + o(1) \geq 1 + \delta_1$  and thus  $h > \delta_1/2 = h'$  for all large enough  $d$ . For  $x \in H$  we can then bound for these dimensions

$$\begin{aligned} \|x - c\|^2 &= \|x - c'\|^2 + \|c' - c\|^2 + 2\langle x - c', c' - c \rangle \\ &\leq (1 + \delta_2)^2 + (h' - h)^2 + 2(x_2 - h')(h' - h) \\ &\leq 1 + 2\delta_2 + \delta_2^2 - h'^2 + h^2 \leq 1 + 3\delta_2 - \delta_1^2/4 + h^2 = 1 + h^2 - \delta_1^2/8. \end{aligned}$$

We are done if the right hand side is less than  $r^2$  for all large enough  $d$ . However, this is easily seen to be true since  $\delta_1 > 0$  and  $r^2 = s^2 + h^2 = 1 + h^2 + o(1)$  as  $d \rightarrow \infty$ . □

It is easier to work with the event  $J(w, z)$ , as it suffices to know the points of  $\mathcal{Z}$  on a fixed bounded area.

**Corollary 3.11.** *For all  $\delta_1 > 0$  there is a  $\delta_2 = \delta_2(\delta_1) > 0$  and  $d_0 = d_0(\delta_1)$  such that for all  $d \geq d_0$  and  $z \in A(w)$  it holds that  $L(w, z) \subseteq J(w, z)$ .*

*Proof.* If  $w$  and  $z$  are neighbors in the Voronoi tessellation  $\text{Vor}_d(\mathcal{Z} \cup \{w, z\})$ , there exist a ball  $B$  with  $w, z \in \partial B$  and  $B \cap \mathcal{Z} = \emptyset$ . The result now follows from Lemma 3.10 and the definition of  $L(w, z)$  and  $J(w, z)$ . □

**Lemma 3.12.** *There is a constant  $c > 0$  and dimension  $d_0$  such that for all  $r \geq r_u/2$  and  $d \geq d_0$*

$$\mathbb{P}(\text{there exists a half-ball of radius } r \text{ and center } \underline{0} \text{ that is disjoint with } \mathcal{Z}) \leq e^{-cr^d}.$$



we define

$$Q_\varepsilon := \left\{ \left[ \frac{i_1 \varepsilon}{\sqrt{d}}, \frac{(i_1 + 1) \varepsilon}{\sqrt{d}} \right) \times \cdots \times \left[ \frac{i_d \varepsilon}{\sqrt{d}}, \frac{(i_d + 1) \varepsilon}{\sqrt{d}} \right) : i_1, \dots, i_d \in \mathbb{Z} \right\}$$

and

$$Q_{\varepsilon, R} := \{q \in Q_\varepsilon : q \subseteq B(\underline{0}, R)\}.$$

We can bound the number of cubes  $N_{\varepsilon, R} := |Q_{\varepsilon, R}|$  inside a ball of radius  $R$  as  $d \rightarrow \infty$  with

$$N_{\varepsilon, R} \leq \frac{\text{vol}(B(\underline{0}, R))}{(\varepsilon/\sqrt{d})^d} = \frac{\kappa_d R^d}{(\varepsilon/\sqrt{d})^d} = \frac{R^d \exp(O(d))}{\varepsilon^d}, \quad (3.19)$$

by Stirling's approximation of the gamma function which implies  $\kappa_d = d^{-d/2} \exp(O(d))$ .

**Lemma 3.13.** *For all  $\delta_2 > 0$  there are constants  $c_1 = c_1(\delta_2) > 0$  and  $c_2 = c_2(\delta_2) > 0$  such that it holds for all  $d \geq d_0(\delta_2)$  and  $z \in A(w)$*

$$\mathbb{P}(J(w, z)) \leq e^{-c_1(1+c_2)^d}.$$

*Proof.* Without loss of generality we assume  $(w + z)/2 = \underline{0}$ . If  $0 < \varepsilon \leq \delta_2/2$  is small enough, the center  $c$  of the half-ball required by  $J(w, z)$  has to lie in some cube  $q \in Q_{\varepsilon, 2(1+\delta_2)}$ . Write this half-ball as  $B(c, 1 + \delta_2) \cap H^+$  where  $H^+ = \{x \in \mathbb{R}^d : \langle x - c, y \rangle \geq 0\}$  is a closed half-space whose boundary hyperplane passes through  $c$  and  $y$  is a unit vector. At least one of the  $2^d$  corners of  $q$  has to be a part of the half-ball. This easy to believe claim can be seen as follows. The cube  $q$  is the convex hull of its  $2^d$  corners  $v_1, \dots, v_{2^d}$ . Suppose, for contradiction, that every corner  $v_i$  lies in the open half-space  $\{x : \langle x - c, y \rangle < 0\}$ . Any point  $x \in q$  can be written as a convex combination  $x = \sum_i \alpha_i v_i$  with  $\alpha_i \geq 0$  and  $\sum_i \alpha_i = 1$ , so

$$\langle x - c, y \rangle = \left\langle \sum_i \alpha_i v_i - c, y \right\rangle = \sum_i \alpha_i \langle v_i - c, y \rangle < 0.$$

Hence  $q$  would be entirely contained in  $\{x : \langle x - c, y \rangle < 0\}$ , which is impossible since  $c \in q$  and  $\langle c - c, y \rangle = 0$ , so our claim follows. As the diameter of  $q$  is  $\varepsilon$  for all dimensions, the center of the half-ball is at most  $\varepsilon$  away from some corner of  $q$ . Let  $v$  be the corner found above. Define  $H_v^+ = \{x : \langle x - v, y \rangle \geq 0\}$ . Then  $H_v^+ \subset H^+$  and, since  $\|v - c\| \leq \varepsilon$ , we have  $B(v, 1 + \delta_2 - \varepsilon) \subset B(c, 1 + \delta_2)$ . Hence

$$B(v, 1 + \delta_2 - \varepsilon) \cap H_v^+ \subset B(c, 1 + \delta_2) \cap H^+,$$

so this new half-ball is also empty of points of  $\mathcal{Z}$ . Thus, there exists a half-ball of radius at least  $1 + \delta_2 - \varepsilon$  centered at a corner of  $q$  that intersects no particle of

$\mathcal{Z}$ . Using  $\varepsilon \leq \delta_2/2$  and by Lemma 3.12 there is now a constant  $c > 0$  such that

$$\begin{aligned} \mathbb{P}(J(w, z)) &= \mathbb{P}\left(\begin{array}{l} \text{there exists a half-ball of radius } 1 + \delta_2 \\ \text{contained in } \text{cert}(w, z) \text{ that is empty of } \mathcal{Z} \end{array}\right) \\ &\leq \mathbb{P}\left(\begin{array}{l} \text{there exists a half-ball of radius } 1 + \delta_2/2 \text{ centered at} \\ \text{a corner of a cube of } Q_{\varepsilon, 2(1+\delta_2)} \text{ that is empty of } \mathcal{Z} \end{array}\right) \\ &\leq 2^d \cdot N_{\varepsilon, 2(1+\delta_2)} \cdot e^{-c(1+\delta_2/2)^d}. \end{aligned}$$

By inequality (3.19) and for  $c_2 = \delta_2/2$ , we obtain our desired upper bound, for all large enough  $d$ .  $\square$

Let  $\delta_1 > 0$  be arbitrary and  $\delta_2 = \delta_2(\delta_1) > 0$  be as given by Corollary 3.11. In the Voronoi model, for  $w, z \in \mathbb{R}^d$ , we will define an event that depends on  $z$  and  $w$  (and  $\delta_1, \delta_2$ ) as

$$\{w \hat{\sim} z\} := \begin{cases} \text{always satisfied, if} & z \in B(w, r_\ell), \\ \left\{ \begin{array}{l} \exists \text{ a Delaunay ball for } w \sim z \text{ contained in } \text{cert}(w, z) \\ \cup J(w, z), \end{array} \right\} & z \in A(w), \\ \left\{ \begin{array}{l} \exists \text{ a hyperplane through } (w+z)/2 \\ \text{and no point of } \mathcal{Z} \cap B_{Gab}(w, z) \text{ on one side of it}, \end{array} \right\} & \text{otherwise.} \end{cases}$$

In the Gabriel case  $w \hat{\sim} z$  will be identical to  $w \sim z$  for all  $w, z \in \mathbb{R}^d$ .

**Lemma 3.14.** *There is a constant  $c > 0$  and  $d_0$  such that for all  $d \geq d_0$  and all  $z \notin B(w, r_u)$*

$$\mathbb{P}(w \sim z \text{ for } \mathcal{Z} \cup \{w, z\}) \leq \mathbb{P}(w \hat{\sim} z \text{ for } \mathcal{Z} \cup \{w, z\}) \leq e^{-c\|w-z\|^d/2^d}.$$

*Proof.* In the Gabriel model this holds with equality and  $c = 1$ , thus it is only left to show for the Voronoi case. We can assume w.l.o.g. that  $(w+z)/2 = \underline{0}$ . The first inequality follows, since if  $w \sim z$  occurs, there exists  $r > 0$  and  $x \in \mathbb{R}^d$  such that  $B(x, r) \cap \mathcal{Z} = \emptyset$  with  $w, z \in \partial B(x, r)$ . Then, there also has to exist a half-ball of radius  $\|w-z\|/2$  inside  $B_{Gab}(w, z)$  that is free of  $\mathcal{Z}$ . See also Figure 3.2. It can be seen more rigorously by letting  $H = B_{Gab}(w, z) \cap \{y \in \mathbb{R}^d : \langle y, x \rangle \geq 0\}$  and computing for  $y \in H$  by using  $r^2 = \|x\|^2 + \|z\|^2$  and  $\|y\| \leq \|z\|$

$$\|y-x\|^2 = \|x\|^2 + \|y\|^2 - 2\langle y, x \rangle = r^2 - \|z\|^2 + \|y\|^2 - 2\langle y, x \rangle \leq r^2.$$

This establishes the first inequality. To show the second, we use Lemma 3.12 to get a constant  $c > 0$  with

$$\begin{aligned} \mathbb{P}(w \hat{\sim} z \text{ for } \mathcal{Z} \cup \{w, z\}) &= \mathbb{P}\left(\begin{array}{l} \exists \text{ a half-ball of radius } \|w-z\|/2 \\ \text{and center } \underline{0} \text{ that is disjoint with } \mathcal{Z} \end{array}\right) \\ &\leq e^{-c\|w-z\|^d/2^d}. \end{aligned}$$

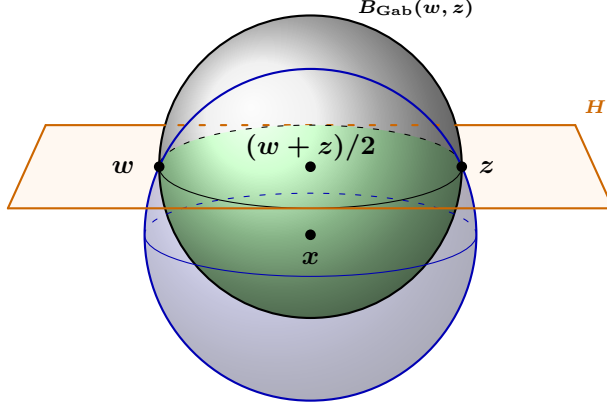


Figure 3.2: The Delaunay ball that is free of  $\mathcal{Z}$  is colored blue. The green area inside the black Gabriel ball represents a half-ball of radius  $\|w - z\|/2$ , defined by  $H$ .

□

**Remark 3.15.** By the proof above and Corollary 3.11 we can choose  $d$  large enough such that for all  $w, z \in \mathbb{R}^d$  it holds  $\{w \sim z\} \subseteq \{w \hat{\sim} z\}$ .

We define for  $w, z \in \mathbb{R}^d$

$$q(w, z) := \mathbb{P}(\text{there is a Delaunay ball for } w \sim z \text{ contained in } \text{cert}(w, z))$$

and  $q(w, z) := \mathbb{P}(w \sim z \text{ for } \mathcal{Z} \cup \{w, z\}) = \exp(-\|w - z\|^d/2^d)$  should we consider the Gabriel graph. The following lemma shows that only adding Voronoi neighbors  $z \in A(\underline{0})$  that have a Delaunay ball that is contained in  $\text{cert}(\underline{0}, z)$  will capture almost all black neighbors of the typical point  $\underline{0}$  when we increase  $d$ .

**Lemma 3.16.** For all  $\mu, \delta_1 > 0$ ,  $w \in \mathbb{R}^d$  and  $p = \mu/\mathbb{E}D$  it holds

$$\mu = \lim_{d \rightarrow \infty} \int_{B(w, r_u)} \mathbb{P}(w \hat{\sim} z \text{ for } \mathcal{Z} \cup \{w, z\}) \lambda p \, dz = \lim_{d \rightarrow \infty} \int_{A(w)} q(w, z) \lambda p \, dz.$$

*Proof.* First, it holds by the Mecke formula, the definition of  $D$  and translation invariance that

$$\mu = \int_{\mathbb{R}^d} \mathbb{P}(w \sim z \text{ for } \mathcal{Z} \cup \{w, z\}) \lambda p \, dz.$$

By Lemma 1.6 and 3.4 we can let  $d_0$  be large enough such that  $\mathbb{E}D \geq 2^d$  for all  $d \geq d_0$ . First, it holds that

$$\int_{B(w, r_\ell)} \lambda p \, dz = \lambda p \, \text{vol}(B(\underline{0}, r_\ell)) \leq p 2^d d^{-1000} \leq \mu d^{-1000},$$

and the right hand side is  $o(1)$  as  $d \rightarrow \infty$ . By Corollary 3.11, Remark 3.15 and bounding for  $z \in A(w)$  we get the inequalities

$$\begin{aligned} \mathbb{P}(w \sim z \text{ for } \mathcal{Z} \cup \{w, z\}) &\leq \mathbb{P}(w \hat{\sim} z \text{ for } \mathcal{Z} \cup \{w, z\}) \leq q(w, z) + \mathbb{P}(J(w, z)) \\ &\leq \mathbb{P}(w \sim z \text{ for } \mathcal{Z} \cup \{w, z\}) + \mathbb{P}(J(w, z)). \end{aligned}$$

Afterwards, we can use Lemma 3.13 to obtain constants  $c_1, c_2 > 0$  that only depend on  $\delta_2 = \delta_2(\delta_1)$  with

$$\int_{A(w)} \mathbb{P}(J(w, z)) \lambda p \, dz \leq p \lambda \operatorname{vol}(A(\underline{0})) e^{-c_1(1+c_2)^d} \leq p 2^d d^{1000} e^{-c_1(1+c_2)^d},$$

and the right hand side decreases to 0 as  $d \rightarrow \infty$ . For  $z \notin B(w, r_u)$  we use Lemma 3.14 to get a constant  $c > 0$  such that for  $d$  sufficiently large

$$\begin{aligned} \int_{\mathbb{R}^d \setminus B(w, r_u)} \mathbb{P}(w \sim z \text{ for } \mathcal{Z} \cup \{w, z\}) \lambda p \, dz &\leq \int_{\mathbb{R}^d \setminus B(\underline{0}, r_u)} e^{-c\|z\|^d/2^d} \lambda p \, dz \\ &= p \int_{r_u}^{\infty} e^{-cr^d 2^{-d}} dr^{d-1} \kappa_d \lambda \, dr = p 2^d \int_{d^{1000}}^{\infty} e^{-cy} dy = p 2^d e^{-cd^{1000}} c^{-1}, \end{aligned}$$

where we switched to polar coordinates and substituted  $y = r^d 2^{-d}$ . The right hand side is again  $o(1)$ , as  $d \rightarrow \infty$ .  $\square$

### 3.3 Upper bound on $p_c$

The main goal of this section is to establish one direction of the equation in Theorem 1.4 and 1.5.

**Theorem 3.17.** *For all  $\mu > 1$  with  $p = p(d) = \mu/\mathbb{E}D$  there is a  $d_0 = d_0(\mu)$  such that for all  $d \geq d_0$  and both the Poisson–Voronoi and Poisson–Gabriel percolation models*

$$\mathbb{P}_p(\text{percolation}) > 0.$$

The proof will be divided into two sections. In Section 3.3.1 we will describe an exploration algorithm and in Section 3.3.2 we will couple this algorithm with a branching random walk. Throughout this section we always assume that  $p = \mu/\mathbb{E}D$  for  $\mu > 1$ . We can without loss of generality upper bound  $\mu \leq 2$ , since if the Theorem holds for  $\mu = 2$ , it will follow for all  $\mu > 2$  by a standard coupling argument (since coloring more points black makes percolation easier). We make this choice to simplify some of the proofs, as we are able to upper bound  $\mu$  by 2. The proofs apply to both models, since we will only have to use that  $p$  is of order  $2^{-d}$  and the result of Lemma 3.16. We will also work with the definitions we had for  $r_u, r_\ell$  and  $A(x)$  on page 32.

### 3.3.1 Exploration algorithm

We aim to construct a coupling between an exploration algorithm, which explores a subset of a cluster of  $\text{Vor}_d(\mathcal{Z})$  or  $\text{Gab}_d(\mathcal{Z})$ , augmented with a finite number of deterministic points, and a site percolation model in  $\mathbb{Z}^2$ , as outlined in Section 3.1.1. The goal is to show that the explored subset is infinite with a positive probability. For each site  $(i, j) \in \mathbb{Z}^2$  we will run an exploration algorithm whenever there is an open site adjacent to it and we will declare  $(i, j)$  open if the algorithm produces a beneficial outcome for us. During the algorithm's definition, we introduce several parameters. We will later show that we can choose them in such a way that sites in  $\mathbb{Z}^2$  are open with a probability close to 1, given a previous history of algorithm runs. The core idea is to successively reveal the Poisson point process in certain regions of  $\mathbb{R}^d$  and determine Voronoi/Gabriel neighbors that meet specific conditions. We emphasize that the underlying Poisson point process  $\mathcal{Z}_b$  (and  $\mathcal{Z}_w$ ) differs between the Voronoi and Gabriel model, since the parameter  $p$  that determines the probability of a point being black is different. However, we are able to define the algorithm for both models in one go, highlighting the minor difference when necessary. We will introduce five parameters to fine-tune the algorithm's routine. The precise values of them will be chosen later on. First, let  $N, k \in \mathbb{N}$  be fixed large numbers and  $\delta_3 > 0$  be a small constant ( $\delta_1$  stays as the parameter of  $\text{cert}(x, y)$  and  $\delta_2$  that was used in the previous section will play a role in the lower bound again). The value of  $k$  will determine the maximum graph distance one algorithm run can explore and  $N$  helps us fine-tune the number of initial points. The small constant  $\delta_3$  will be used to require that neighbors  $x_1, x_2$  of a point  $u$  have an angle of  $\angle x_1 u x_2 \in (\pi/2 \pm \delta_3)$ . Further, set  $M = M(d) := d^{2000}$  and let  $U > 0$  to be determined in the course of the proof. Not all choices of parameters will produce something useful. For each fixed  $\delta_1 > 0$  it will be essential for us to only consider sufficiently large dimensions such that  $B_{\text{Gab}}(x, y) \subseteq \text{cert}(x, y)$ .

We are going to split the exploration algorithm into multiple parts. The first procedure,  $\text{FindNei}(u, G, W, E, V_1)$ , aims to explore/find a subset of Voronoi/Gabriel neighbors of a fixed point  $u \in \mathbb{R}^d$  that all lie in  $A(u)$ . The input  $G = (V, \mathcal{E})$  will be a rooted forest in  $\mathbb{R}^d$ , which represents the structure of the subgraph of the Voronoi/Gabriel graph we have explored up to this point. The finite set of points  $W \subset \mathbb{R}^d$  will consist of all elements of  $\mathcal{Z}$  that we have revealed over all algorithm iterations so far. The measurable set  $E \subset \mathbb{R}^d$  will capture the total region where we have revealed  $\mathcal{Z}$ . The vertices of  $G$  are partitioned into two sets  $V = V_1 \uplus V_2$  of 'unexplored' ( $V_1$ ) and 'explored' vertices ( $V_2$ ). The partition will keep track of what points were already used for exploring its neighbors in a previous algorithm iteration. Any such tuple of inputs, which also fulfill the following conditions, will be called *admissible* for  $\text{FindNei}(u, G, W, E, V_1)$ .

(C1) It holds that  $u \in V_1$  and all edges in  $G$  have a length between  $r_\ell$  and  $r_u$ .

(C2) We have  $V \subseteq W$  and

$$W \cap \left( \bigcup_{v \in V} B(v, 0.9) \setminus \{v\} \right) = \emptyset.$$

(C3) For all  $x \in \mathbb{R}^d$  we have  $|W \cap B(x, 100)| \leq U$ .

(C4) There exists  $\widehat{W} \subset W^2$  with  $w_1 \in A(w_2)$  for all  $(w_1, w_2) \in \widehat{W}$  and

$$E \subseteq \left( \bigcup_{v \in V_2} B(v, r_u) \right) \cup \left( \bigcup_{(w_1, w_2) \in \widehat{W}} \text{cert}(w_1, w_2) \right) \cup \left( \bigcup_{w \in W} B(w, 0.9) \right). \quad (3.20)$$

Further, for all  $(w_1, w_2) \in \widehat{W}$  and  $u \in V$  it holds  $\|(w_1 + w_2)/2 - u\| \geq 0.1$ .

The conditions will be used in our proofs to show that previous iterations will have a low influence on the current one, since the area  $E$  has only a small intersection with the region we consider for the current step.

For  $\text{FindNei}(u, G, W, E, V_1)$ , we will reveal  $\mathcal{Z}$  (or  $\mathcal{Z}_b$ ) inside balls  $B(x, r)$  from some  $x \in \mathbb{R}^d, r > 0$  and it will be crucial to not find too many points. To achieve this we will reveal the points successively. If in the definition of the algorithm we say that we reveal  $\mathcal{Z}$  (resp.  $\mathcal{Z}_b$ ) inside  $B(x, r) \setminus E$  until finding  $n$  points, we reveal all  $\mathcal{Z}$  (resp.  $\mathcal{Z}_b$ ) inside  $B(x, r') \setminus E$  by letting  $r'$  increase from  $r' = 0$  until we either find  $n$  points in  $B(x, r') \setminus E$ , or until  $r' = r$ . If we find  $n$  points before reaching  $r' = r$ , we instantly stop exploring further. By doing this we make sure that there are no more than  $n$  points being revealed inside  $B(x, r)$  (almost surely, since for each fixed  $r'$  there won't be multiple particles on the sphere  $r'S^{d-1}$ ). It is crucial to do this, so we can make sure that (C3) is satisfied.

We are ready to describe the procedure of one  $\text{FindNei}(u, G, W, E, V_1)$  iteration for an admissible  $(u, G, W, E, V_1)$ .

**Algorithm 3.18** ( $\text{FindNei}(u, G, W, E, V_1)$ ). The algorithm will either return a set of points (possibly empty) or ‘failure’. First, we will reveal  $\mathcal{Z}$  in the *safety area*  $S(u) := B(u, 0.9) \setminus (E \cup \{u\})$  until we find one point. If we reveal one, we stop the algorithm and declare failure. Else, we successively reveal

$$(\mathcal{Z}_b \cap B(u, r_u)) \setminus (E \cup B(u, r_\ell))$$

until finding  $M$  points of  $\mathcal{Z}_b$ . Should there be  $M$  points, we declare failure and stop the algorithm. Otherwise, we have now determined a set  $X_u$  of less than  $M$  black points. We then check whether the angles of these points are permitted. We declare failure if there are two distinct  $x, y \in X_u$  with  $\angle xuy \notin (\pi/2 \pm \delta_3)$  or

$\angle xuz \notin (\pi/2 \pm \delta_3)$  for some  $z \in B(u, 100) \cap W \setminus \{u\}$  and  $x \in X_u$ . Should the angles be as we desired, we reveal

$$\mathcal{Z} \cap S(x) \setminus E$$

for all  $x \in X_u$  until finding one point. If there is any  $x$  with  $(W \cup \mathcal{Z}) \cap S(x) \neq \emptyset$ , we instantly stop the algorithm and return failure. Otherwise, we reveal for all  $x \in X_u$

$$\mathcal{Z} \cap \text{cert}(u, x) \setminus E,$$

until finding  $2(1 + \delta_1)^d$  many points of  $\mathcal{Z}$  inside one of those certificates. If we find  $2(1 + \delta_1)^d$  points inside  $\text{cert}(u, x)$  for some  $x \in X_u$ , we stop exploring and return failure. Now we come to the only part of our algorithm that behaves differently for the Voronoi and Gabriel model. Should there exist a ball  $B_x \subset \text{cert}(u, x)$  with  $x, u \in \partial B_x$  and  $(W \cup \mathcal{Z}) \cap B_x = \emptyset$ , we call  $x$  a *Voronoi child* of  $u$ . For the Gabriel setting, we check whether  $(W \cup \mathcal{Z}) \cap B_{Gab}(u, x) = \emptyset$  and call such points *Gabriel children* of  $u$ . This concludes the definition of  $\text{FindNei}(u, G, W, E, V_1)$ . See also Figure 3.3.

For the coupling with the two dimensional percolation model, we will use the projection map  $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^2$  defined by

$$\pi(x_1, \dots, x_d) := \sqrt{d}(x_1, x_2).$$

We will also use the notation

$$S_{i,j} := [i - 1/2, i + 1/2) \times [j - 1/2, j + 1/2)$$

for a square centered at  $(i, j)$  with side length 1. The algorithm  $\text{Decl}((i, j), G, W, E, V_1)$  will determine whether we declare the site  $(i, j)$  open or closed in our site percolation model on  $\mathbb{Z}^2$ . For that, it will try to explore new points that are at graph distance  $k$  from some initial  $N$  points of  $G$ . The inputs  $G = (V, \mathcal{E})$ ,  $W$  and  $E$  have the identical role as before. We will require that these inputs satisfy all conditions **(C1-4)** (except  $u \in V_1$ ) and additionally

$$\textbf{(C5)} \quad |\{v \in V_1 : \pi(v) \in S_{i,j}\}| \geq N.$$

This new condition is necessary to have enough initial (unexplored) points for our algorithm. Later it will be clear that these points indeed always exist when we want to start a new algorithm run.

**Algorithm 3.19** ( $\text{Decl}((i, j), G, W, E, V_1)$ ). In the beginning we will fix  $N$  arbitrary points of  $V_1$  such that if projected with  $\pi(\cdot)$  they lie in  $S_{i,j}$ , that exist by **(C5)**, and denote them as  $K$ . The algorithm is going to iterate through the following procedure. We select an arbitrary  $u \in V_1$  subject to either  $u \in K$  or  $u$  having some  $x \in K$  on a path between  $u$  and a root such that

$$\min_{y \in K} d_G(y, u) < k,$$

where  $d_G(u, v)$  denotes the graph distance between  $u$  and  $v$  (the number of edges on a shortest path between  $u$  and  $v$ ). Next, we are going to run **FindNei** ( $u, G, W, E, V_1$ ). If the algorithm returns failure, we will declare  $(i, j)$  to be closed and stop the procedure. Otherwise, we connect all found Voronoi/Gabriel children to  $u$  by an edge, updating the forest  $G$ . The Voronoi/Gabriel children of  $u$  will receive label 1, making them part of  $V_1$ . Regardless of whether the algorithm stops and declares the site closed, we add the total region where  $\mathcal{Z}$  or  $\mathcal{Z}_b$  was revealed to  $E$  and all Poisson points we found to  $W$ . We then relabel  $u$  by removing it from  $V_1$  and adding it to  $V_2$ . Afterwards, if the site was not determined to be closed yet, we will try to repeat this process for the next candidate for  $u$ . That is, we again select an arbitrary  $u \in V_1$  subject to either  $u \in K$  or  $u$  having some  $x \in K$  on a path between  $u$  and a root with  $\min_{y \in K} d_G(y, u) < k$  and perform the procedure outlined above as long as such a point still exists. If eventually we do not find a node  $u$  that fulfills the required conditions, we let  $V(k)$  denote all points of  $V_1$  that are at exactly graph distance  $k$  from  $K$ . If the site has not already been closed we declare  $(i, j)$  open if

$$|\pi(V(k)) \cap S_{i', j'}| \geq N \text{ for all } (i', j') \sim (i, j), \quad (3.21)$$

and otherwise declare the site closed. (The symbol ‘ $\sim$ ’ for points of  $\mathbb{Z}^2$  refers to adjacency in the nearest neighbor lattice  $\mathbb{Z}^2$ .) This concludes one iteration of  $\text{Decl}((i, j), G, W, E, V_1)$ .

The first algorithm run that we perform is for site  $(0, 0)$  with the inputs

$$V_1 = W := \{x \in \mathbb{R}^d : \exists 3 \leq i \leq N + 2 \text{ with } x_i = 1 \text{ and } x_j = 0 \text{ for all } j \neq i\}, \quad (3.22)$$

and  $V_2 = \mathcal{E} = E = \emptyset$ . The initial points of  $V_1$  will be our roots. The roots will stay unchanged. Note it is immediate that with these choices we fulfill all the necessary requirements of our inputs, since  $|V_1| = N$  and  $\pi(V_1) = \{(0, 0)\}$ . We will then iteratively start an exploration for  $(i, j) \neq (0, 0)$  if there is an adjacent site that is already open and if the status of  $(i, j)$  is still unexplored (i.e. no algorithm has been performed for it yet). The input  $(G, W, E, V_1)$  to  $\text{Decl}((i, j), G, W, E, V_1)$  for site  $(i, j)$  will be the current version that our algorithm constructed from the initial inputs.

It is clear by the definition of the procedure that **(C1)** will continue to hold after all steps of the algorithm. Furthermore, when starting a run of  $\text{Decl}((i, j), G, W, E, V_1)$  for a new site, condition **(C5)** will be fulfilled, as we require a neighboring site to be open and this in turn means that (3.21) has to be satisfied. We see that we cannot ‘destroy’ **(C5)** if we choose to first explore another site, as those algorithm runs will never relabel those points. Condition **(C2)** will continue to hold, since we will always check whether the safety area of a point is empty of all points of  $W$  before we add it as a site to  $G$ . However, there could be a problem

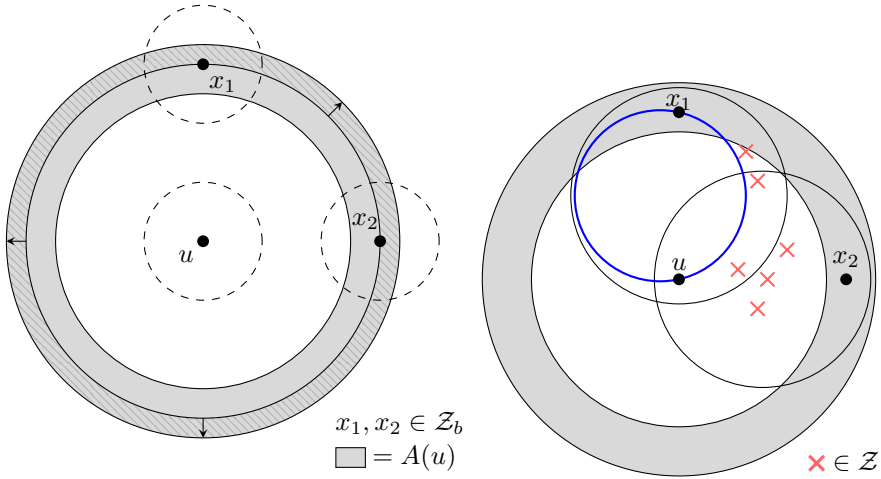


Figure 3.3: Visualization of  $\text{FindNei}(u, G, W, E, V_1)$ . The shaded area on the left side corresponds to a part where we currently reveal  $\mathcal{Z}_b$ . The dotted circles are our safety areas. By construction  $\angle x_1 u x_2 \in (\pi/2 \pm \delta_3)$ , otherwise we would stop exploring. On the right figure the ball with center  $(x_1 + u)/2$  (resp.  $(x_2 + u)/2$ ) is the certificate  $\text{cert}(u, x_1)$  (resp.  $\text{cert}(u, x_2)$ ). Since the blue ball does not contain any particle, it is a Delaunay ball and we connect the Voronoi child  $x_1$  to  $u$ . On the other hand such a ball does not exist in  $\text{cert}(u, x_2)$ , thus  $x_2$  won't be part of our forest.

for the initial points of (3.22) that the algorithm for site  $(0, 0)$  might use. If  $u$  is a root and at the beginning of a run of  $\text{FindNei}(u, G, W, E, V_1)$  the safety area of  $u$  is not empty, then (C2) is violated, as  $u \in V$ . If this occurs, we would instantly close the site  $(0, 0)$  and we would not proceed further. For all other sites this issue is not existent, as all points that serve as an input already had their safety area checked before hand.

The fact that we can choose  $U$  such that (C3) remains intact is not evident from the definition and will be shown in the next lemma.

**Lemma 3.20.** *There is a  $d_0$  and  $U = 10^6(1 + \delta_1)^d NM^{2k}k^2d$  such that for all  $d \geq d_0$  and  $\delta_1 < 1$  the following holds almost surely. If we proceed as outlined above, condition (C3) will continue to hold after each iteration of  $\text{FindNei}(\cdot)$  during  $\text{Dec1}((i, j), G, W, E, V_1)$ .*

*Proof.* Let  $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$  be such that  $\pi(x) \in S_{i,j}$ . We have for all  $y \in B(x, 100)$

$$\|\pi(y) - \pi(x)\| \leq \sqrt{d}\|y - x\| \leq 100\sqrt{d}.$$

If  $d$  is large enough such that  $r_u \leq 3$  and  $\delta_1 < 1$ , each  $\text{FindNei}(u, G, W, E, V_1)$  will only reveal points that lie in  $B(u, 4)$  (for any input). Since for all  $(i', j')$  the algorithm  $\text{Dec1}((i, j), G, W, E, V_1)$  will explore a maximum of graph distance  $k$  away from the initial points by iterating through multiple  $\text{FindNei}(\cdot)$  runs, we will only reveal Poisson points that are closer than  $4k$  to a point of the initial points of  $\text{Dec1}(\cdot)$ . Thus, for all revealed  $y$  during  $\text{Dec1}((i, j), G, W, E, V_1)$  it has to hold that  $\|\pi(y) - \pi(v)\| \leq \sqrt{d}\|y - v\| \leq 4k\sqrt{d}$  for a  $v \in \mathbb{R}^d$  that was chosen as an initial point (and thus  $\pi(v) \in S_{i',j'}$ ). Moreover, both squares  $S_{i,j}$  and  $S_{i',j'}$  have a side length 1 and so a diagonal of  $\sqrt{2}$ . That in turn implies if an algorithm for  $(i', j')$  added some  $y \in B(x, 100)$  to  $W$ , the site  $(i', j')$  has to be at distance of no more than  $100\sqrt{d} + 4k\sqrt{d} + 2\sqrt{2}$  from  $\pi(x)$ . See also Figure 3.4. We conclude that there are fewer than

$$(100\sqrt{d} + 4k\sqrt{d} + 2\sqrt{2})^2 \leq (100 + 4 + 2\sqrt{2})^2 k^2 d \leq (110k)^2 d$$

possibilities for  $(i', j')$  with said property. Each run of the algorithm  $\text{Dec1}((i, j), G, W, E, V_1)$  adds less than  $NM^{k+1}$  black points to  $G$  and every time we check the safety area of one point it can reveal at most one additional point of  $\mathcal{Z}$  (a.s.). To determine whether a point is a Voronoi/Gabriel child, we will never reveal more than  $2(1 + \delta_1)^d$  particles of  $\mathcal{Z}$  (a.s.). Thus, each algorithm run  $\text{Dec1}(\cdot)$  is revealing less than  $4(1 + \delta_1)^d NM^{k+1}$  points of  $\mathcal{Z}$  (a.s.). Since the set  $W$  consists of all points we have ever revealed plus the initial  $N$  points for site  $(0, 0)$ , there cannot be more than

$$4(1 + \delta_1)^d NM^{k+1} (110k)^2 d + N \leq 10^6(1 + \delta_1)^d NM^{2k}k^2d$$

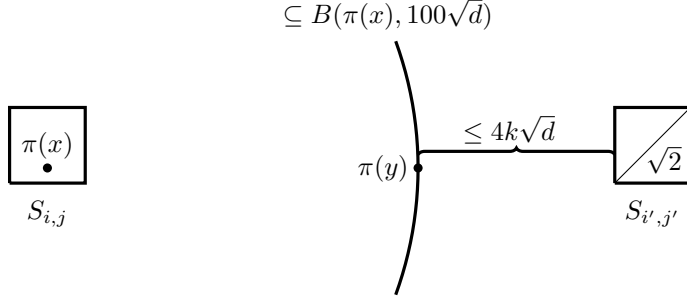


Figure 3.4: Visualization of the maximum distance that we can travel away from a site in the two-dimensional projection.

points that we have already revealed in  $B(x, 100)$  and by choosing  $U$  to be the right hand side above, we can finish the proof.  $\square$

We are now ready to choose our first parameter of the algorithm. From now on we will always work with the  $U$  given by the lemma above. The fact that condition **(C4)** won't break at some point is also not immediate and we show it in the following.

**Lemma 3.21.** *If we choose  $\delta_3$  small and  $d$  large enough, condition **(C4)** will continue to hold after updating  $G, W, E$  and  $V_1$  during  $\text{Dec1}((i, j), G, W, E, V_1)$ .*

*Proof.* If after each run of  $\text{FindNei}(u, G, W, E, V_1)$  where the points of  $X_u$  satisfied the angle condition we add  $(u, x)$  to  $\widehat{W}$  for all  $x \in X_u$ , property (3.20) continues to hold. This follows from the definition of  $\text{FindNei}(u, G, W, E, V_1)$ , since we only reveal annuli, certificates and safety areas. Further, we only reveal certificates if the angle condition was satisfied. For the distance requirement, note that by the cosine rule and the angle condition we can bound for  $v_1 \in B(u, 90) \cap V$  and all newly added  $(x, u) \in \widehat{W}$

$$\begin{aligned} \|(u+x)/2 - v_1\|^2 &\geq \|(u+x)/2 - u\|^2 + \|u - v_1\|^2 \\ &\quad - 2\|(u+x)/2 - u\|\|u - v_1\|\cos(\pi/2 - \delta_3) \\ &\geq r_\ell^2/4 - 200r_u \cos(\pi/2 - \delta_3), \end{aligned}$$

by using  $x \in A(u)$ . If  $v_1 \notin B(u, 90)$ , there is nothing to show. For all explored Voronoi/Gabriel children  $v_2$  during  $\text{FindNei}(u, G, W, E, V_1)$ , and thus new points of  $V$ , we get with the same argument

$$\|(w_1 + w_2)/2 - v_2\|^2 \geq r_\ell^2/4 - 200r_u \cos(\pi/2 - \delta_3),$$

for all  $(w_1, w_2) \in \widehat{W}$  with  $w_1, w_2 \in B(u, 90)$ . Choosing  $d$  large and  $\delta_3$  small enough concludes the proof.  $\square$

We have now ensured that **(C1-5)** will not be violated when iterating over multiple algorithm runs as outlined above. The upcoming lemma shows that Theorem 3.17 follows, if we can choose the parameters in a way such that the algorithm opens a site with a sufficiently large probability. In the remaining part of the section we will then show that we can indeed choose them like that.

**Lemma 3.22.** *Assume that for all  $1 < \mu \leq 2$  and  $\varepsilon > 0$  we can choose  $k = k(\mu)$ ,  $N = N(\mu, \varepsilon)$ ,  $\delta_1, \delta_3$  and  $d_0(\varepsilon, \mu, k, N, \delta_1, \delta_3)$  such that we have for all  $d \geq d_0$  and all admissible inputs*

$$\mathbb{P}(\text{Decl}((i, j), G, W, E, V_1) \text{ declares } (i, j) \text{ open}) > 1 - \varepsilon.$$

*Then, it follows that  $\mathbb{P}_p(\text{percolation}) > 0$  for all  $d \geq d_0$ .*

*Proof.* Let  $\varepsilon = 1/16$ . We start by running  $\text{Decl}((0, 0), G, W, E, V_1)$  with the inputs as defined in (3.22). If the algorithm declares  $(0, 0)$  open, we throw an independent coin with probability of success

$$0 < \frac{15}{16} \cdot \mathbb{P}(\text{Decl}((0, 0), G, W, E, V_1) \text{ declares } (0, 0) \text{ open})^{-1} < 1.$$

If the coin shows success, we set  $(0, 0)$  to be open in the  $\mathbb{Z}^2$  site percolation model and closed otherwise. This implies that  $(0, 0)$  is open with probability  $15/16$ . If we opened  $(0, 0)$ , we then pick an arbitrary neighboring site  $(i, j) \sim (0, 0)$  and apply the same procedure as for  $(0, 0)$  by running  $\text{Decl}((i, j), G, W, E, V_1)$  with the updated inputs. By throwing an adapted coin, site  $(i, j)$  will again be open with probability  $15/16$ . We iterate this routine as long as there are unexplored sites of  $\mathbb{Z}^2$  adjacent to a site that was determined to be open. Thus,  $(0, 0)$  will be open with probability  $15/16$  and all other sites of  $\mathbb{Z}^2$  are open with a probability of  $15/16$ , conditioned on the fact that a neighbor is open.

If the cluster of the origin  $(0, 0)$  in our constructed  $\mathbb{Z}^2$  percolation model is infinite, we can perform infinitely many algorithm runs for a realization of  $\mathcal{Z} = \mathcal{Z}_w \cup \mathcal{Z}_b$ . Note that this implies at least one point from (3.22) must be inside an infinite black component if we add them as typical/deterministic points to  $\mathcal{Z}$ . Since we had  $N$  initial points and all degrees of the constructed graph  $G$  are bounded, the infinite cluster is also present without them. If this occurs with a non zero probability, it would result in  $\mathbb{P}_p(\text{percolation}) > 0$ . If we consider ordinary site percolation on  $\mathbb{Z}^2$  where sites are open with probability  $p'$ , the origin is open with probability  $p'$ . All other sites are an element of the cluster of the origin with probability  $p'$ , if we condition on the fact that there is a neighbor that is an element of said cluster. We can finish the proof by Lemma 2.9, since we have constructed a coupling with the cluster of the origin with  $p' = 15/16$ .  $\square$

There are four different reasons that can result in failure of  $\text{FindNei}(u, G, W, E, V_1)$ . We could:

- (i) reveal more than  $M = M(d)$  black points in  $A(u)$  while determining the set  $X_u$ , or
- (ii) after we successfully found  $X_u$ , there is at least one angle in question that is not sufficiently close to  $\pi/2$ , or
- (iii) find an element of  $Z \cup W$  in some safety area  $S(\cdot)$  that we required to be empty, or
- (iv) reveal  $2(1 + \delta_1)^d$  points inside a certificate while checking if we have a Voronoi/Gabriel connection.

The following lemmas show that for sufficiently large dimension, we will only rarely stop the algorithm early as a result of the four cases above. To be exact, the probability will decay exponentially as we increase  $d$ . For the first case, let  $F_1 = F_1(u, E)$  be the event where we reveal more than  $M(d) = d^{2000}$  black points in the annulus  $A(u)$  during **FindNei**( $u, G, W, E, V_1$ ).

**Lemma 3.23.** *If  $d$  is large enough it holds for all  $u \in \mathbb{R}^d$ ,  $E \subset \mathbb{R}^d$  and  $1 < \mu \leq 2$*

$$\mathbb{P}(F_1(u, E)) \leq e^{-d}.$$

*Proof.* By Lemma 1.6 and 3.4 we can choose  $d$  large enough such that  $\mathbb{E}D \geq 2^d$ . We can then use the Chernoff bound from Lemma 2.1 to say by using  $\mu \leq 2$

$$\begin{aligned} \mathbb{P}(F_1(u, E)) &= \mathbb{P}(|Z_b \cap A(u) \setminus E| \geq M) \leq \mathbb{P}(|Z_b \cap B(\underline{0}, r_u)| \geq M) \\ &= \mathbb{P}(\text{Po}(pr_u^d) \geq d^{2000}) \leq \mathbb{P}(\text{Po}(\mu d^{1000}) \geq d^{2000}) \leq e^{-d^{1000}H(d^{999})} \leq e^{-d}, \end{aligned}$$

where  $H(a) = 1 - a + a \ln(a)$ .  $\square$

Let now  $F_2 = F_2(u, W, E)$  be the event that during **FindNei**( $u, G, W, E, V_1$ ) the angles are not as we desired them to be. That is, we find a  $x \in Z_b \cap A(u) \setminus E$  with  $\angle yux \notin (\pi/2 \pm \delta_3)$  for some  $y \in X_u \cup (W \cap B(u, 100)) \setminus \{u\}$ .

**Lemma 3.24.** *For all  $\delta_3 > 0, k \in \mathbb{N}$  and  $N \in \mathbb{N}$  we can choose  $\delta_1 = \delta_1(\delta_3, k, N) > 0$  small enough such that there is a  $c = c(\delta_1, \delta_3, k, N) > 0$  so for all admissible inputs, large enough  $d$  and  $1 < \mu \leq 2$*

$$\mathbb{P}(F_2(u, W, E)) < e^{-cd}.$$

*Proof.* The angle test will check the angle to two different types of points. The first ones are the particles of  $X_u$ , i.e. other candidates for a connection that will be a subset of the black points of  $A(u)$ . We will write  $Z := Z_b \cap B(u, r_u)$ . It holds for all  $u \in \mathbb{R}^d$  and  $E \subset \mathbb{R}^d$

$$\begin{aligned} &\mathbb{P}(\exists x, y \in Z \setminus E : \angle xuy \notin (\pi/2 \pm \delta_3)) \\ &\leq \mathbb{P}(\exists x, y \in Z : \angle x\underline{0}y \notin (\pi/2 \pm \delta_3)) \\ &\leq \mathbb{P}(\exists x, y \in Z : \angle x\underline{0}y \notin (\pi/2 \pm \delta_3) \mid |Z| \leq M) + \mathbb{P}(|Z| > M) \\ &\leq \mathbb{P}(\exists x, y \in Z : \angle x\underline{0}y \notin (\pi/2 \pm \delta_3) \mid |Z| = M) + \mathbb{P}(|Z| > M). \end{aligned}$$

We now let  $d$  be large enough such that  $\mathbb{E}D \geq 2^d$  (Lemma 1.6 and 3.4). By Lemma 2.1 and since  $M = d^{2000}$  we can now bound the second term by using  $\mu \leq 2$

$$\mathbb{P}(|\mathcal{Z}_b \cap B(\underline{0}, r_u)| > M) \leq \mathbb{P}(\text{Po}(\mu d^{1000}) > d^{2000}) \leq e^{-d^{1000}H(d^{999})} \leq e^{-d}.$$

By Lemma 2.4 and Lemma 2.7 there exists a constant  $c_1 = c_1(\delta_3) > 0$  such that for large enough  $d$

$$\mathbb{P}(\exists x, y \in \mathcal{Z}_b \cap B(\underline{0}, r_u) : \angle xy \notin (\pi/2 \pm \delta_3) \mid |\mathcal{Z}_b \cap B(\underline{0}, r_u)| = M) \leq M^2 e^{-c_1 d}.$$

Since  $M$  was a fixed polynomial we can now choose  $d$  large enough such that for some  $c_2 = c_2(\delta_3) > 0$

$$M^2 e^{-c_1 d} + e^{-d} \leq e^{-c_2 d}.$$

The other points we consider when checking the angles are those we have collected previously that are in proximity to  $u$ , so elements of  $W \cap B(u, 100) \setminus \{u\}$ . By Lemma 3.20 and **(C3)** we get  $|W \cap B(u, 100)| \leq U$ . We can use Lemma 2.7 to follow that the total area that has to be free of black points in order to pass all angle tests with points of  $W \cap B(u, 100) \setminus \{u\}$  is less than

$$U e^{-c_3 d} \text{vol}(B(\underline{0}, r_u)) \leq 10^7 (1 + \delta_1)^d N M^{2k} k^2 d e^{-c_3 d} \text{vol}(B(\underline{0}, r_u)),$$

for a constant  $c_3 = c_3(\delta_3) > 0$ . Since  $c_3$  depends only on  $\delta_3$ , we can now choose  $\delta_1$  small enough and  $d$  large to obtain a constant  $c_4 = c_4(k, N, \delta_1, \delta_3) > 0$  with

$$10^7 (1 + \delta_1)^d N M^{2k} k^2 d e^{-c_3 d} \text{vol}(B(\underline{0}, r_u)) \leq e^{-c_4 d} \text{vol}(B(\underline{0}, r_u)).$$

Thus, the probability that we will find a black point  $x \in X_u$  and  $w \in B(u, 100) \cap W \setminus \{u\}$  with  $\angle xuw \notin (\pi/2 \pm \delta_3)$  is bounded from above by

$$\begin{aligned} 1 - \exp(-\lambda p e^{-c_4 d} \text{vol}(B(\underline{0}, r_u))) &\leq 1 - \exp(-\mu e^{-c_4 d} d^{1000}) \leq 1 - \exp(-2e^{-c_4 d} d^{1000}) \\ &\leq e^{-c_5 d}, \end{aligned}$$

for some  $c_5 = c_5(\delta_1, \delta_3, k, N) > 0$  and all large enough  $d$ .  $\square$

In the following, we are going to write  $F_3 = F_3(u, W, E)$  for the event of a safety area violation occurring for  $\text{FindNei}(u, G, W, E, V_1)$ . That is, the event of failure because at some point in the algorithm there is a point of  $\mathcal{Z} \cup W$  in some safety area that we check.

**Lemma 3.25.** *There is a  $c > 0$  such that for all large enough  $d$ , small enough  $\delta_3$  and all admissible inputs*

$$\mathbb{P}(F_3(u, W, E)) < e^{-cd}.$$

*Proof.* For all  $x \in X_u$ , we know by the definition of our algorithm that all  $w \in W \cap B(u, 100)$  with  $w \neq u$  satisfy  $\angle wux \in (\pi/2 \pm \delta_3)$ , should we reveal the safety areas for them. Further, all these  $w$  are at distance of more than 0.9 from  $u$ , since  $S(u)$  had to be empty of  $W$  in the first place by **(C2)**. Thus, with the cosine rule we estimate

$$\|w - x\|^2 \geq 0.9^2 + r_\ell^2 - 2r_u 100 \cos(\pi/2 - \delta_3),$$

and the lower bound is larger than  $0.9^2$ , given that  $\delta_3$  is small and  $d$  large enough. If  $w = u$  it holds that  $\|w - x\|^2 \geq r_\ell^2 \geq 0.9^2$ . Thus, there cannot be a point in  $S(x)$  that we have revealed in an earlier step (i.e. points of  $W$ ). On the other hand, we may have revealed some area  $E$  in  $S(x)$  that is empty of Poisson points. By definition, in each **FindNei**( $u, G, W, E, V_1$ ) run we check no more than  $M + 1$  safety areas. This implies the area that is revealed and checked to be vacant of all  $\mathcal{Z}$  has a volume of at most  $(M + 1) \text{vol}(S(\underline{0}))$ . We follow, the probability of not closing the site by a safety area violation is lower bounded by

$$\exp(-\lambda(M + 1) \text{vol}(S(\underline{0}))) = \exp(-(M + 1)0.9^d) \geq 1 - e^{-cd},$$

where the inequality holds for some  $c > 0$  and  $d$  large enough.  $\square$

For the last case of an early closure of a site we are going to write  $F_4 = F_4(u, E)$  for the event of revealing  $2(1 + \delta_1)^d$  many points of  $\mathcal{Z}$  in a certificate while verifying a Voronoi/Gabriel connection during **FindNei**( $u, G, W, E, V_1$ ).

**Lemma 3.26.** *For all  $\delta_1 > 0$  there is a  $c = c(\delta_1) > 0$  such that for all large enough  $d \geq d_0(\delta_1)$ ,  $u \in \mathbb{R}^d$  and  $E \subset \mathbb{R}^d$*

$$\mathbb{P}(F_4(u, E)) \leq e^{-cd}.$$

*Proof.* By construction, we will never reveal more than  $M = d^{2000}$  certificates during **FindNei**( $\cdot$ ). Thus by Lemma 2.1

$$\mathbb{P}(F_4(u, E)) \leq M \mathbb{P}(|B(\underline{0}, 1 + \delta_1) \cap \mathcal{Z}| \geq 2(1 + \delta_1)^d) \leq M e^{-(1 + \delta_1)^d H(2)} \leq e^{-cd},$$

for a constant  $c = c(\delta_1) > 0$  and all large enough  $d \geq d_0(\delta_1)$ .  $\square$

We defined one run of **FindNei**( $u, G, W, E, V_1$ ) with the goal to have as little interference as possible with previous steps. The following lemma shows there cannot be any points of  $W$  that lie inside a relevant certificate. In the Gabriel model this might instantly block the connection and it would reduce the probability of a connection in the Voronoi case. This implies when we have found the set  $X_u$ , it is a priori not clear if a point cannot be connected to  $u$ , because of information that we have collected previously. Conversely, the region  $E$  within the certificate is confirmed to be free of particles  $\mathcal{Z}$ . However, this will ‘help us’, as it makes connections more likely.

**Lemma 3.27.** *We can choose  $\delta_1, \delta_3$  small and  $d$  large enough such that the following holds. If during  $\text{FindNei}(u, G, W, E, V_1)$  the points  $X_u$  pass the angle test, there cannot be a point of  $W$  inside any  $\text{cert}(x, u)$  for all  $x \in X_u$ .*

*Proof.* Assume we are determining the Voronoi/Gabriel children of some point  $u$  and we have passed the angle test. Let  $x \in X_u$  be a candidate for a connection and  $w \in W \cap B(u, 100) \setminus \{u\}$ . Since it holds that  $\angle xuw \in (\pi/2 \pm \delta_3)$  and  $w \notin S(u)$  we have by the cosine rule

$$\|(u+x)/2 - w\|^2 \geq 0.9^2 + (r_\ell/2)^2 - 200r_u \cos(\pi/2 - \delta_3).$$

Choosing  $\delta_1, \delta_3$  small and  $d$  large enough, we obtain  $w \notin \text{cert}(x, u)$ , since the distance is lower bounded by a number larger than  $1 + \delta_1$ .  $\square$

The area of  $E \cap B(u, r_u)$  will not be considered during a run of  $\text{FindNei}(u, G, W, E, V_1)$ . The following lemma shows we would only rarely find black points there.

**Lemma 3.28.** *For all  $k, N \in \mathbb{N}$  and small enough  $\delta_1 > 0$  there is  $c = c(\delta_1, k, N) > 0$  such that for all admissible  $(u, G, W, E, V_1)$ ,  $1 < \mu \leq 2$  and large enough  $d$*

$$\lambda p \text{vol}(E \cap B(u, r_u)) < e^{-cd}.$$

*Proof.* First, we remind ourselves it holds by **(C4)** that

$$E \subseteq \left( \bigcup_{v \in V_2} B(v, r_u) \right) \bigcup \left( \bigcup_{(w_1, w_2) \in \widehat{W}} \text{cert}(w_1, w_2) \right) \bigcup \left( \bigcup_{w \in W} B(w, 0.9) \right)$$

for some  $\widehat{W} \subset W^2$ . Additionally, by **(C2)** we get  $\|u - v\| \geq 0.9$  for all  $v \in V_2$ . Therefore, by Lemma 2.8 there is a constant  $c_1 > 0$  such that for all  $v \in V_2$

$$\text{vol}(B(u, r_u) \cap B(v, r_u)) \leq e^{-c_1 d} \text{vol}(B(\underline{0}, r_u)).$$

Condition **(C4)** says that the center of  $\text{cert}(w_1, w_2)$  is at distance of at least 0.1 from  $u$  for all  $(w_1, w_2) \in \widehat{W}$ . In combination with Lemma 2.8 this yields a constant  $c_2 > 0$  with

$$\text{vol}(B(u, r_u) \cap \text{cert}(w_1, w_2)) \leq e^{-c_2 d} \text{vol}(B(\underline{0}, r_u)),$$

for all large enough  $d$ . We also get the trivial bound of

$$\text{vol}(B(u, r_u) \cap B(w, 0.9)) \leq \kappa_d 0.9^d.$$

Note by Lemma 3.20 and **(C3)** there are less than  $U$  points of  $W$  in  $B(u, 100)$ . Thus in total, if we let  $d$  be large enough such that  $\mathbb{E}D \geq 2^d$

$$\begin{aligned} \lambda p \text{vol}(E \cap B(u, r_u)) &\leq \lambda p ((Ue^{-c_1 d} + U^2 e^{-c_2 d}) \text{vol}(B(\underline{0}, r_u)) + U \kappa_d 0.9^d) \\ &\leq \mu U^2 (e^{-c_1 d} + e^{-c_2 d} + 0.9^d) d^{1000} \\ &\leq 10^{13} (1 + \delta_1)^{2d} N^2 M^{4k} k^4 (e^{-c_1 d} + e^{-c_2 d} + 0.9^d) d^{1002}. \end{aligned}$$

Since  $c_1, c_2$  are independent from both  $d$  and  $\delta_1$  we can choose  $d$  large enough and  $\delta_1$  small such that

$$10^{13}(1 + \delta_1)^{2d} N^2 M^{4k} k^4 (e^{-c_1 d} + e^{-c_2 d} + 0.9^d) d^{1002} \leq e^{-c_3 d},$$

for a constant  $c_3 = c_3(\delta_1, k, N) > 0$ .  $\square$

### 3.3.2 Coupling with a branching random walk

In this section we will compare the distribution of the projected Voronoi/Gabriel children in our exploration to a branching random walk (BRW). Our goal is to show that the assumptions of Lemma 3.22 are indeed satisfied. A branching random walk is a spatial version of a Galton-Watson process. We start with a deterministic set of points  $Z_0 \subset \mathbb{R}^d$  of generation 0. Then, each particle will produce offspring independently of each other according to some common point process. The union of all these children will be the points of  $Z_1$ . Iterating this procedure will result in the BRW  $(Z_n)_{n \in \mathbb{N}_0}$ . For the definition of our BRW we recall that for  $w \in \mathbb{R}^d$  and  $z \in A(w)$  in the Voronoi model

$$q(w, z) = \mathbb{P}(\text{there is a Delaunay ball for } w \sim z \text{ contained in } \text{cert}(w, z))$$

and  $q(w, z) = \mathbb{P}(w \sim z \text{ for } \mathcal{Z} \cup \{w, z\})$  in the Gabriel setting. Let  $x \in \mathbb{R}^d$  be arbitrary. In the BRW  $(Y_n(x))_{n \in \mathbb{N}_0}$  we will start at the initial point  $x = Y_0(x)$ . Then, we let  $\mathcal{Z}(x)$  be a PPP in  $A(x)$  with intensity  $\lambda p$ , independent of everything else. Each  $z \in \mathcal{Z}(x)$  is now independently said to be a child of  $x$  with probability  $q(x, z)$ . The union of all these children are the particles of  $Y_1(x)$ . Performing this procedure for each  $y \in Y_1(x)$ , instead of  $x$ , will result in the points of  $Y_2(x)$ , and yields by iterating the branching random walk  $(Y_n(x))_{n \in \mathbb{N}_0}$ . Note, that each time we sample offspring we will use a new PPP.

**Lemma 3.29.** *For all  $k, N \in \mathbb{N}$  and if  $\delta_1, \delta_3$  are small enough, there is a  $c = c(\delta_1, \delta_3, k, N) > 0$  such that for all admissible inputs,  $1 < \mu \leq 2$  and sufficiently large  $d$  we can couple the exploration algorithm  $\text{FindNei}(u, G, W, E, V_1)$  with  $Y_1(u)$  to obtain*

$$\mathbb{P} \left( \begin{array}{l} Y_1(u) \text{ is a subset of the children} \\ \text{determined by } \text{FindNei}(u, G, W, E, V_1) \end{array} \right) > 1 - e^{-cd}.$$

*Proof.* The point process we are going to use for the children of  $u$  in the BRW will be  $\mathcal{Z}_b$  in  $A(u) \setminus E$  (so exactly the same PPP as for the exploration algorithm!) and for  $A(u) \cap E$  let  $\hat{\mathcal{Z}}_b(u)$  be a Poisson point process with intensity  $\lambda p$ , independent from everything else. Then, each  $z \in \hat{\mathcal{Z}}_b(u)$  is set to be a child of  $u$  in the BRW with probability  $q(u, z)$ , independently from another. It is clear that all points of  $\hat{\mathcal{Z}}_b(u)$  can never be Voronoi/Gabriel children of  $u$  by the definition of

the algorithm. However, if we choose  $\delta_1, \delta_3$  small enough, we have by Lemma 3.28 that  $\{|\widehat{\mathcal{Z}}_b(u)| > 0\}$  occurs with probability less than

$$1 - \exp(-\lambda p \operatorname{vol}(E \cap B(u, r_u))) \leq 1 - \exp(-e^{-c_1 d}) \leq e^{-c_2 d}$$

for constants  $c_1 = c_1(\delta_1, k, N) > 0, c_2 = c_2(\delta_1, k, N) > 0$  and large enough  $d$ . For  $x \in \mathcal{Z}_b \cap A(u) \setminus E$ , so the remaining candidates for children in the BRW, we define

$$I(x) := \operatorname{cert}(u, x) \bigcap \bigcup_{\substack{y \in \mathcal{Z}_b \cap A(u) \setminus E \\ y \neq x}} \operatorname{cert}(u, y).$$

Thus  $I(x)$  is the area of  $\operatorname{cert}(u, x)$  that intersects at least one other certificate  $\operatorname{cert}(u, y)$ . For each  $x \in \mathcal{Z}_b \cap A(u) \setminus E$  let now  $\widehat{\mathcal{Z}}(x)$  be a new and independent PPP with intensity  $\lambda$  in  $I(x) \cup E$ . If there exists a ball inside  $\operatorname{cert}(u, x)$  with  $u$  and  $x$  on its boundary that is free of all  $\widehat{\mathcal{Z}}(x) \cup (\mathcal{Z} \setminus (I(x) \cup E))$ , we declare  $x$  to be a descendant of  $u$  in the BRW for the Voronoi model. We see this yields the right offspring distribution since  $(\widehat{\mathcal{Z}}(x) \cup (\mathcal{Z} \setminus (I(x) \cup E))) \cap \operatorname{cert}(u, x)$  is a PPP with intensity  $\lambda$  inside  $\operatorname{cert}(u, x)$  and by construction these point processes are independent for different candidates of  $x \in \mathcal{Z}_b \cap A(u) \setminus E$ . Similarly, we set  $x$  to be a descendant in the BRW for the Gabriel model if there is no point of  $\widehat{\mathcal{Z}}(x) \cup (\mathcal{Z} \setminus (I(x) \cup E))$  inside  $B_{Gab}(x, u)$ .

When will a point  $x$ , which we have designated as an offspring in the BRW, not be a Voronoi/Gabriel child of  $u$ ? One reason could be that the algorithm declares failure if either  $F_1, F_2, F_3$  or  $F_4$  occurs. Then, we would connect no points to  $u$ . However, by the Lemmas 3.23 to 3.26 the four stated events that result in failure do not happen with a probability of at least  $1 - e^{-c_6 d}$  for all large enough  $d$  and a constant  $c_6 = c_6(\delta_1, \delta_3, k, N) > 0$ . The other reason could be that for some  $x \in \mathcal{Z}_b \cap A(u) \setminus E$  that is an offspring in the BRW we find a point of  $W \cup \mathcal{Z}$  inside  $I(x) \cup E$  that blocks the connection. By Lemma 3.27 we know that if  $F_1, F_2$  and  $F_3$  do not occur, the already revealed area  $E \cap \operatorname{cert}(u, x)$  is empty of  $W$ , given that  $\delta_1, \delta_3, d^{-1}$  are small enough. The remaining case where the coupling fails to create a subset would be  $\mathcal{Z} \cap I(x) \setminus E \neq \emptyset$ . Note for  $x, y \in A(u)$  with  $\angle xuy \in (\pi/2 \pm \delta_3)$  we can use the cosine rule with Lemma 2.8 to get a constant  $c_3 = c_3(\delta_3) > 0$  such that

$$\operatorname{vol}(\operatorname{cert}(u, x) \cap \operatorname{cert}(u, y)) \leq e^{-c_3 d} \operatorname{vol}(B(\mathbf{0}, 1 + \delta_1)).$$

If  $F_1$  does not occur, we have less than  $M$  points in  $\mathcal{Z}_b \cap A(u) \setminus E$ . Under  $(F_2)^c$  we further get that  $x, y \in \mathcal{Z}_b \cap A(u) \setminus E$  all satisfy  $\angle xuy \in (\pi/2 \pm \delta_3)$ . Therefore, if  $F_1$  and  $F_2$  do not occur, we obtain by choosing  $d$  sufficiently large and  $\delta_1$  small

enough a constant  $c_4 = c_4(\delta_1, \delta_3) > 0$  with

$$\begin{aligned} \lambda \operatorname{vol} \left( \bigcup_{x \in \mathcal{Z}_b \cap A(u) \setminus E} (I(x) \setminus E) \right) &\leq \lambda \cdot \sum_{x \in \mathcal{Z}_b \cap A(u) \setminus E} \operatorname{vol}(I(x)) \\ &\leq \lambda M^2 e^{-c_3 d} \operatorname{vol}(B(\underline{0}, 1 + \delta_1)) \\ &= M^2 e^{-c_3 d} (1 + \delta_1)^d \leq e^{-c_4 d}. \end{aligned}$$

If we assume that  $F_1, F_2$  do not occur, we can use this bound to say that the probability of finding a  $x \in \mathcal{Z}_b \cap A(u) \setminus E$  with  $\mathcal{Z} \cap I(x) \setminus E \neq \emptyset$  (and thus a point that could be a descendant in the constructed BRW but not a child in the algorithm) is upper bounded by

$$1 - \exp(-e^{-c_4 d}) \leq e^{-c_5 d},$$

as always with  $d$  large enough and some constant  $c_5 = c_5(\delta_3, \delta_1) > 0$ . We conclude with the described coupling above, the probability of the children of  $u$  in the BRW not forming a subset of the Voronoi/Gabriel children is less than  $e^{-cd}$  for some constant  $c > 0$ , dependent only on the specific parameter values (for all large  $d$  and sufficiently small  $\delta_1, \delta_3$ ).  $\square$

With a bit more work also an exact coupling up to a fixed generation would have been possible with high probability. However, only having a subset is enough for our needs.

The BRW  $(Y_n(\underline{0}))_{n \in \mathbb{N}_0}$  is indeed supercritical for large enough  $d$ , even though we only add neighbors from  $A(\underline{0})$ .

**Lemma 3.30.** *For all  $\mu > 1$  and large enough  $d \geq d_0(\mu)$  it holds that  $|Y_1(\underline{0})| \sim \operatorname{Po}(\nu)$  for a  $\nu = \nu(\mu) > 1$ .*

*Proof.* The Poisson distribution is immediate since the independence of connections imply that the points of  $Y_1(\underline{0})$  follow a marked PPP with intensity  $q(\underline{0}, \cdot) \lambda p$  (see e.g. Section 5.2 of [65]). The supercritical expectation follows from Lemma 3.16 and the choice of  $p$ .  $\square$

We write  $(Z_n)_{n \in \mathbb{N}_0}$  for the branching random walk in  $\mathbb{R}^d$  where we start with  $N$  particles of generation 0 and each  $x \in Z_0$  will produce an independent BRW  $(Y_n(x))_{n \in \mathbb{N}_0}$ . Then,  $Z_n := \bigcup_{x \in Z_0} Y_n(x)$ . The following lemma is an adaptation from a result of Penrose [83], Lemma 2 and 3. It shows if we start with enough initial points of  $Z_0$  in  $S_{0,0}$  and let the BRW run for a long enough time, we will again find a lot of points in neighboring squares with a large probability.

**Lemma 3.31.** *For all  $\varepsilon > 0$  and  $\mu > 1$  we can choose  $k = k(\mu)$  and  $d \geq d_0(\mu)$  large enough such that there exist  $N = N(\mu, \varepsilon)$  satisfying the following. For all  $Z_0$  with  $|\pi(Z_0) \cap S_{0,0}| = N$  it holds that*

$$\mathbb{P}(|\pi(Z_k) \cap S_{i,j}| \geq N \text{ for all } (i,j) \sim (0,0)) > 1 - \varepsilon.$$

*Proof.* Conditioned on  $\{|Y_1(\underline{0})| \geq 1\}$ , we let  $C$  be a particle of  $Y_1(\underline{0})$  chosen uniformly at random (e.g. by sampling i.i.d. uniform random variables on  $(0, 1)$  as a label for each offspring and choosing the one with the smallest label). We can write  $C = RX$ , where  $R$  is a random variable with support on  $(r_\ell, r_u)$  and  $X$  is chosen uniformly at random from  $S^{d-1}$  and is independent from  $R$ . This follows from the fact that all points of  $Y_1(\underline{0})$  must lie inside  $A(\underline{0})$  and from the rotational invariance of the offspring distribution in our BRW. Let now  $w \in \mathbb{R}^2$  be such that  $w = (w_1, w_2) \in S_{0,0}$  and  $k \geq 1$  is arbitrary. Further, for  $1 \leq i \leq k$ , we write  $C_i = R_i X_i$  for i.i.d. random variables, distributed like  $C = RX$ . Define  $W_k := w + \pi(C_1) + \dots + \pi(C_k)$  and set

$$S := \sum_{i=1}^k R_i^2 \in [kr_\ell^2, kr_u^2], \quad S_d := \sum_{i=1}^k R_i \pi(X_i).$$

Scaling preserves total variation, so for every  $r > 0$ , by Lemma 3.2

$$\|\mathcal{L}(r \pi(X)) - \mathcal{N}(0, r^2 I_2)\|_{\text{TV}} = \Delta_d,$$

with  $\Delta_d \rightarrow 0$  as  $d \rightarrow \infty$ . Using independence of the steps and the fact that total variation is subadditive under products, we obtain for every choice  $(R_i) \in [r_\ell, r_u]^k$ ,

$$\left\| \mathcal{L}((R_i \pi(X_i))_{i=1}^k \mid R_1, \dots, R_k) - \mathcal{L}((R_i G_i)_{i=1}^k \mid R_1, \dots, R_k) \right\|_{\text{TV}} \leq k \Delta_d,$$

where  $G_1, \dots, G_k$  are i.i.d.  $\mathcal{N}(0, I_2)$ , independent of the  $X_i$ . Thus we also obtain

$$\left\| \mathcal{L}(S_d \mid R_1, \dots, R_k) - \mathcal{L}(G \mid R_1, \dots, R_k) \right\|_{\text{TV}} \leq k \Delta_d,$$

where  $G := \sum_{i=1}^k R_i G_i \sim \mathcal{N}(0, S I_2)$ . We now take some  $w = (w_1, w_2) \in S_{0,0} = [-1/2, 1/2]^2$ , so that

$$I_1(w_1) := [\tfrac{1}{2} - w_1, \tfrac{3}{2} - w_1], \quad I_2(w_2) := [-\tfrac{1}{2} - w_2, \tfrac{1}{2} - w_2]$$

and  $S_{1,0} - w = I_1(w_1) \times I_2(w_2)$ . Let  $Z \sim \mathcal{N}(0, S)$ , with  $S \in [kr_\ell^2, kr_u^2]$ . For any interval  $J \subset [-2, 2]$  of length 1,

$$\mathbb{P}(Z \in J) = \int_J \frac{1}{\sqrt{2\pi S}} e^{-\frac{x^2}{2S}} dx \geq \frac{1}{\sqrt{2\pi S}} e^{-\frac{2^2}{2S}} \geq \frac{1}{\sqrt{2\pi k} r_u} e^{-\frac{2^2}{2kr_\ell^2}} \geq \frac{c_0}{\sqrt{k}},$$

for some constant  $c_0 > 0$ . Since  $I_1(w_1), I_2(w_2) \subset [-2, 2]$  and both have length 1, we obtain

$$\mathbb{P}(\mathcal{N}(0, S) \in I_1(w_1)) \geq \frac{c_0}{\sqrt{k}}, \quad \mathbb{P}(\mathcal{N}(0, S) \in I_2(w_2)) \geq \frac{c_0}{\sqrt{k}}.$$

Using independence of the two coordinates of  $\mathcal{N}(0, S I_2)$ ,

$$\mathbb{P}(\mathcal{N}(0, S I_2) \in S_{1,0} - w) = \mathbb{P}(\mathcal{N}(0, S) \in I_1(w_1)) \mathbb{P}(\mathcal{N}(0, S) \in I_2(w_2)) \geq \frac{c_0^2}{k},$$

uniformly in  $w \in S_{0,0}$  and  $S \in [kr_\ell^2, kr_u^2]$ . Combining this with the total variation estimate yields

$$\mathbb{P}(W_k \in S_{1,0} \mid R_1, \dots, R_k) = \mathbb{P}(S_d \in S_{1,0} - w \mid R_1, \dots, R_k) \geq \frac{c_0^2}{k} - k \Delta_d.$$

Choosing  $d$  large enough so that  $k \Delta_d \leq \frac{1}{2} \frac{c_0^2}{k}$  gives

$$\mathbb{P}(W_k \in S_{1,0} \mid R_1, \dots, R_k) \geq \frac{c_2}{k}$$

for some  $c_2 > 0$  independent of  $w$  and  $(R_i)_{i=1}^k$ , and hence also

$$\mathbb{P}(W_k \in S_{1,0}) \geq \frac{c_2}{k}$$

uniformly in  $w \in S_{0,0}$ . We now let  $x \in \mathbb{R}^d$  be such that  $\pi(x) = w$  for  $w \in S_{0,0}$ . Conditioned on  $\{|Y_k(x)| \geq 1\}$  we enumerate the particles  $Y_k(x)$  uniformly at random by  $y_1, \dots, y_{|Y_k(x)|}$  (e.g., by sampling i.i.d. uniform random variables on  $(0, 1)$  as a label for each particle and then order them in a descending manner). The location of each  $\pi(y_i)$  is distributed like  $W_k$ . This follows from the fact that in  $\mathbb{R}^d$  we start at  $x$  in generation 0 and for each generation every step of an offspring to its parent is distributed like  $C$ . Then, we apply the linear projection  $\pi(\cdot)$  to get to two dimensions, which yields  $W_k$  in distribution. For each fixed  $k \geq 1$  and all large enough  $d \geq d_0(\mu, k)$  Lemma 3.30 yields a  $\nu = \nu(\mu) > 1$  with

$$\begin{aligned} \mathbb{E}|\pi(Y_k(x)) \cap S_{1,0}| &= \mathbb{E} \mathbb{E} \left[ \sum_{i=1}^{|Y_k(x)|} \mathbb{1}\{\pi(y_i) \in S_{1,0}\} \mid |Y_k(x)| \right] \\ &= \mathbb{E} \left[ \sum_{i=1}^{|Y_k(x)|} \mathbb{E} \left[ \mathbb{1}\{\pi(y_i) \in S_{1,0}\} \mid |Y_k(x)| \right] \right] \\ &= \mathbb{E} \left[ \sum_{i=1}^{|Y_k(x)|} \mathbb{E} \left[ \mathbb{1}\{W_k \in S_{1,0}\} \right] \right] = \mathbb{E}|Y_k(x)| \mathbb{P}(W_k \in S_{1,0}) \\ &= \nu^k \mathbb{P}(W_k \in S_{1,0}) \geq \nu^k \frac{c_2}{k}. \end{aligned}$$

Using this lower bound, we can choose  $k(\mu)$  and  $d(\mu)$  large enough such that  $\mathbb{E}|\pi(Y_k(x)) \cap S_{1,0}| > 3$  uniformly over all  $x$  with  $\pi(x) \in S_{0,0}$ . By Lemma 3.3 we estimate

$$\text{Var}(|\pi(Y_k(x)) \cap S_{1,0}|) \leq \mathbb{E}|Y_k(x)|^2 \leq (k+1)\nu^{1+2k}.$$

The bounds above were obtained by considering one arbitrary starting point in  $S_{0,0}$  of generation 0. If we now start the process  $(Z_n)_{n \in \mathbb{N}_0}$  with  $N$  points in  $S_{0,0}$ ,

it is the union of  $N$  i.i.d. branching random walks with one starting point each. Since the results above were uniform in  $x$  it implies

$$\mathbb{E}|\pi(Z_k) \cap S_{1,0}| > 3N \quad \text{and} \quad \text{Var}(|\pi(Z_k) \cap S_{1,0}|) \leq N(k+1)\nu^{1+2k}.$$

Utilizing Chebyshev's inequality we can obtain for all  $\varepsilon > 0$

$$\begin{aligned} \mathbb{P}(|\pi(Z_k) \cap S_{1,0}| < N) &\leq \mathbb{P}(|\pi(Z_k) \cap S_{1,0}| < \mathbb{E}|\pi(Z_k) \cap S_{1,0}| - 2N) \\ &\leq \mathbb{P}(|\pi(Z_k) \cap S_{1,0}| - \mathbb{E}|\pi(Z_k) \cap S_{1,0}| > 2N) \leq \frac{\text{Var}(|\pi(Z_k) \cap S_{1,0}|)}{(2N)^2} \\ &\leq \frac{(k+1)\nu^{1+2k}}{4N} < \frac{\varepsilon}{4}, \end{aligned}$$

where the last inequality holds by choice of  $N = N(\varepsilon, \mu)$ . Using the symmetry of our squares the above inequality also holds if we replace  $S_{1,0}$  by  $S_{-1,0}$ ,  $S_{0,1}$  or  $S_{0,-1}$ . So

$$\begin{aligned} \mathbb{P}(|\pi(Z_k) \cap S_{i,j}| \geq N \text{ for all } (i,j) \sim (0,0)) \\ = 1 - \mathbb{P}(|\pi(Z_k) \cap S_{i,j}| < N \text{ for one } (i,j) \sim (0,0)) \\ \geq 1 - 4\mathbb{P}(|\pi(Z_k) \cap S_{1,0}| < N) > 1 - \varepsilon. \end{aligned}$$

□

We are now ready to show that the assumptions from Lemma 3.22 are indeed satisfied.

**Lemma 3.32.** *For all  $1 < \mu \leq 2$  and  $\varepsilon > 0$  we can choose  $k = k(\mu)$ ,  $N = N(\mu, \varepsilon)$ ,  $\delta_1, \delta_3$  and  $d_0(\varepsilon, \mu, k, N, \delta_1, \delta_3)$  such that we have for all  $d \geq d_0$  and all admissible inputs*

$$\mathbb{P}(\text{Decl}((i,j), G, W, E, V_1) \text{ declares } (i,j) \text{ open}) > 1 - \varepsilon.$$

*Proof.* First, we let  $k$  and  $N$  be one given by Lemma 3.31 for  $\varepsilon/2$  and the choice of  $\mu$ . Using Lemma 3.29, we can couple the found points of each exploration step inside  $\text{Decl}((i,j), G, W, E, V_1)$  with  $Y_1(u)$  for some  $u$ , so that the BRW points form a subset of the points found by the algorithm with a probability of more than  $1 - e^{-cd}$  for a constant  $c = c(\delta_1, \delta_3, N, k) > 0$ , should  $\delta_1, \delta_3$  be small enough and  $d$  sufficiently large. The exploration algorithm will start at  $N$  deterministic points and explore until generation  $k$ . We use the initial points, that we denoted by  $K$  in the definition of Algorithm 3.19, as the particles of  $Z_0 = K$ . Thus, if the couplings succeed in each iteration step, the set of explored points at distance  $k$  from the initial  $N$  points must have  $Z_k$  as a subset. We will run  $\text{FindNei}(u, G, W, E, V_1)$  no more than  $NM^k$  times, and so there will be no more than  $NM^k$  couplings that have to succeed. If we write  $C = C(G, W, E, V_1)$  for the event that all couplings

in  $\text{Decl}((i, j), G, W, E, V_1)$  work out, we can let  $d \geq d_0(\varepsilon, \delta_1, \delta_3, k, N)$  be large enough such that

$$\mathbb{P}(C) \geq 1 - NM^k e^{-cd} = 1 - Nd^{2000k} e^{-cd} > 1 - \frac{\varepsilon}{2}.$$

Then we can lower bound for all large enough  $d \geq d_0(\varepsilon, \mu, \delta_1, \delta_3, k, N)$

$$\begin{aligned} & \mathbb{P}(\text{Decl}((i, j), G, W, E, V_1) \text{ declares } (i, j) \text{ open}) \\ & \geq \mathbb{P}(C \cap \{|\pi(Z_k) \cap S_{i', j'}| \geq N \mid \forall (i', j') \sim (i, j)\}) \\ & \geq 1 - \mathbb{P}(C^c) - \mathbb{P}(\{|\pi(Z_k) \cap S_{i', j'}| \geq N \mid \forall (i', j') \sim (i, j)\}^c) > 1 - \varepsilon. \end{aligned}$$

□

*Proof of Theorem 3.17.* Theorem 3.17 follows by combining Lemma 3.22 and 3.32.

□

### 3.4 Lower bound on $p_c$

The main theorem of this section that we will work towards is:

**Theorem 3.33.** *For all  $0 < \mu < 1$  with  $p = p(d) = \mu/\mathbb{E}D$  there is a  $d_0 = d_0(\mu)$  such that for all  $d \geq d_0$  and both the Poisson–Voronoi and Poisson–Gabriel percolation model*

$$\mathbb{P}_p(\text{percolation}) = 0.$$

Like in the previous section, let  $\mathcal{Z} = \mathcal{Z}_b \cup \mathcal{Z}_w$  be a homogeneous PPP in  $\mathbb{R}^d$  with intensity  $\lambda = 1/\kappa_d = \text{vol}(B(\underline{0}, 1))^{-1}$ . We also remind ourselves of  $r_u = 2d^{1000/d}$ . The overall proof strategy follows the ideas of Section 3.3 in [48]. We are going to bound the expectation of the number of certain paths and the following is defining bad and good edges in such a path. In this section we will let  $\delta_1 > 0$  and  $d$  large enough such that we obtain the value  $\delta_2 = \delta_2(\delta_1) > 0$ , given by Corollary 3.11. Like on page 36, for  $z_1 \in A(z_2)$ , we write  $z_1 \hat{\sim} z_2$  if there is a Delaunay ball for  $z_1 \sim z_2$  that is contained in  $\text{cert}(z_1, z_2)$ , or there is a half-ball of radius  $1 + \delta_2$  in  $\text{cert}(z_1, z_2)$  that contains no particle of  $\mathcal{Z}$ . If  $z_2 \notin B(z_1, r_u)$  we set  $z_1 \hat{\sim} z_2$  to be equal to the event where there exists a hyperplane through  $(z_1 + z_2)/2$  with no point of  $\mathcal{Z} \cap B_{\text{Gab}}(z_1, z_2)$  on one side of it. For  $z_1 \in B(z_2, r_\ell)$  the event  $z_1 \hat{\sim} z_2$  is satisfied for any realization of  $\mathcal{Z}$ . In the Gabriel model  $z_1 \hat{\sim} z_2$  is always equal to  $z_1 \sim z_2$ .

**Definition 3.34.** We say a sequence  $x_0, x_1, \dots, x_k \in \mathbb{R}^d$  of distinct points is a pseudopath if  $x_{i-1} \hat{\sim} x_i$  for all  $i \geq 1$ . We call the edge  $x_{i-1}x_i$  *good* if  $x_i \in B(x_{i-1}, r_u)$ . Otherwise, we speak of a *bad* edge. A pseudopath is good if all edges are good, and bad if at least one edge is bad. If the definition holds for  $x_0, x_1, \dots, x_k \in \mathcal{Z}_b \cup \{\underline{0}\}$ , they form a black pseudopath.

In a pseudopath we are slightly more generous in defining what constitutes a connection, compared to the usual sense, as  $\{x_{i-1} \sim x_i\} \subseteq \{x_{i-1} \hat{\sim} x_i\}$ , see also Remark 3.15. Since we will show that percolation does not occur (a.s.), even if we add certain edges, we also obtain the result for the original graph. The major benefit in doing that is that now we only have to consider the bounded region  $\text{cert}(x_{i-1}, x_i)$  in order to determine whether the two points are connected. This would not be the case for ordinary Voronoi connections. For a sequence of points  $P = x_0, x_1, \dots, x_k$  we will use the notation

$$\text{cert}(P) := \bigcup_i \text{cert}(x_{i-1}, x_i).$$

At the moment  $\delta_1 > 0$  is arbitrary but fixed and we will show it can be chosen small enough such that the argument of this section holds true. It will also be crucial to choose  $d$  large enough such that  $B_{Gab}(x, y) \subseteq \text{cert}(x, y)$  for all  $y \in B(x, r_u)$ . Using the notion of good and bad edges we will use the next two definitions to split a path into multiple *chunks*.

**Definition 3.35.** A finite pseudopath  $P = x_0, \dots, x_k$  where the first  $k-1$  edges are good is called a *chunk*. Note that a chunk can be one bad edge and can also consist of only good edges. Further, we say that  $P$  has length  $k$ . The chunk is black if the pseudopath  $P$  is black.

**Definition 3.36.** We call a (infinite or finite) sequence of chunks  $P_1, P_2 \dots$  a linked sequence of chunks if the following conditions hold:

1.  $\text{cert}(P_i) \cap \text{cert}(P_j) = \emptyset$  if  $|i - j| > 1$ .
2. Writing  $P_i = x_0^i, \dots, x_{k_i}^i$  and  $P_{i-1} = x_0^{i-1}, \dots, x_{k_{i-1}}^{i-1}$  we have

$$\text{cert}(P_{i-1}) \cap \text{cert}(x_0^i, x_1^i) \neq \emptyset \quad \text{but} \quad \text{cert}(P_{i-1}) \cap \text{cert}(P_i) \setminus \text{cert}(x_0^i, x_1^i) = \emptyset.$$

3. If  $k_i$  is the length of  $P_i$  it holds that  $k_i \leq d^2$  and if the final edge of  $P_i$  is good we require  $k_i = d^2$ .

If  $x_{k_{i-1}}^{i-1} = x_0^i$ , the link between  $P_{i-1}$  and  $P_i$  is said to be of type  $a$  and otherwise of type  $b$ .

**Lemma 3.37.** *For all large enough  $d$ , almost surely for all  $r > 0$ , there are finitely many pairs  $z_1, z_2 \in \mathcal{Z}_b$  with  $\text{cert}(z_1, z_2) \cap B(\underline{0}, r) \neq \emptyset$  and  $z_1 \hat{\sim} z_2$ .*

*Proof.* For each  $r > 0$  consider the random variable

$$Y_r := |\{(z_1, z_2) \in \mathcal{Z}_b^2 : \text{cert}(z_1, z_2) \cap B(\underline{0}, r) \neq \emptyset, z_1 \hat{\sim} z_2\}|.$$

Then we can express the expectation of  $Y_r$  with the Mecke formula (Lemma 2.5) as

$$\mathbb{E}Y_r = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(z_1 \hat{\sim} z_2 \text{ for } \mathcal{Z} \cup \{z_1, z_2\}) \mathbb{1}_{\{\text{cert}(z_1, z_2) \cap B(\underline{0}, r) \neq \emptyset\}} (\lambda p)^2 dz_2 dz_1. \quad (3.23)$$

Note that if  $z_2 \in B(z_1, r_u)$  in the integral above there can only be an intersection of  $\text{cert}(z_1, z_2)$  and  $B(\underline{0}, r)$  if  $z_1 \in B(\underline{0}, r + r_u + 1 + \delta_1)$ . Thus

$$\begin{aligned} \int_{\mathbb{R}^d} \int_{B(z_1, r_u)} \mathbb{P}(z_1 \hat{\sim} z_2 \text{ for } \mathcal{Z} \cup \{z_1, z_2\}) \mathbb{1}_{\{\text{cert}(z_1, z_2) \cap B(\underline{0}, r) \neq \emptyset\}} (\lambda p)^2 dz_2 dz_1 \\ \leq (\lambda p)^2 \text{vol}(B(\underline{0}, r + r_u + 1 + \delta_1)) \text{vol}(B(\underline{0}, r_u)) < \infty. \end{aligned}$$

By Lemma 3.14 there is a constant  $c > 0$  such that for both models and for all  $z_2 \notin B(z_1, r_u)$  and large enough  $d$

$$\mathbb{P}(z_1 \hat{\sim} z_2 \text{ for } \mathcal{Z} \cup \{z_1, z_2\}) \leq e^{-c\|z_1 - z_2\|^d/2^d}.$$

With this bound we can estimate the remaining part of (3.23) by

$$\begin{aligned} \mathbb{E}Y_r &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d \setminus B(z_1, r_u)} \mathbb{P}(z_1 \hat{\sim} z_2 \text{ for } \mathcal{Z} \cup \{z_1, z_2\}) \mathbb{1}_{\{\text{cert}(z_1, z_2) \cap B(\underline{0}, r) \neq \emptyset\}} (\lambda p)^2 dz_2 dz_1 \\ &\leq \int_{\mathbb{R}^d} \int_{\mathbb{R}^d \setminus B(z_1, r_u)} e^{-c\|z_1 - z_2\|^d/2^d} \mathbb{1}_{\{\text{cert}(z_1, z_2) \cap B(\underline{0}, r) \neq \emptyset\}} (\lambda p)^2 dz_2 dz_1. \end{aligned}$$

If  $u = (z_1 + z_2)/2$  is the center of  $\text{cert}(z_1, z_2) = B_{G_{ab}}(z_1, z_2)$  and  $v = (z_1 - z_2)/2$ , we see in order for  $\text{cert}(z_1, z_2) \cap B(\underline{0}, r) \neq \emptyset$  to hold,  $u \in B(\underline{0}, r + \|v\|)$  is necessary. If we make the change of variables to  $u$  and  $v$  it holds that  $z_1 = u + v$  and  $z_2 = u - v$ . Thus, the Jacobian has the form

$$\begin{pmatrix} I_d & I_d \\ I_d & -I_d \end{pmatrix}. \quad (3.24)$$

The determinant is equal to  $(-2)^d$  and can be determined by subtracting row  $i$  from row  $d + i$  for  $1 \leq i \leq d$  to get an upper triangle matrix. Using Jensen's inequality we obtain  $|(x + y)/2|^d \leq |x|^d/2 + |y|^d/2$  for all  $x, y \in \mathbb{R}^d$ . From Lemma 1.6 and 3.4 we can also let  $d$  be large enough such that  $\mathbb{E}D \geq 2^d$ . Then, we can first make a change of variable to  $u, v$  and afterwards to polar coordinates,

yielding

$$\begin{aligned}
& \int_{\mathbb{R}^d} \int_{\mathbb{R}^d \setminus B(z_1, r_u)} e^{-c\|z_1 - z_2\|^d / 2^d} \mathbf{1}_{\{\text{cert}(z_1, z_2) \cap B(\underline{0}, r) \neq \emptyset\}} (\lambda p)^2 dz_2 dz_1 \\
& \leq \int_{\mathbb{R}^d} \int_{B(\underline{0}, r + \|v\|)} e^{-c\|v\|^d} (\lambda p)^2 2^d du dv \\
& = \int_{\mathbb{R}^d} (r + \|v\|)^d e^{-c\|v\|^d} \lambda p^2 2^d dv = \int_0^\infty (r + t)^d e^{-ct^d} dt^{d-1} p^2 2^d dt \\
& \leq d \int_0^\infty (r^d t^{d-1} + t^{2d-1}) e^{-ct^d} dt = d(r^d (cd)^{-1} \Gamma(1) + c^{-2} d^{-1} \Gamma(2)) < \infty.
\end{aligned}$$

Thus, for all fixed  $r > 0$  the random variable  $Y_r$  is almost surely finite. We can finish the proof by noting that  $\{Y_b < \infty\} \subset \{Y_a < \infty\}$  for  $a < b$  to conclude

$$\mathbb{P}\left(\bigcap_{r>0} \{Y_r < \infty\}\right) = \lim_{r \rightarrow \infty} \mathbb{P}(Y_r < \infty) = 1.$$

□

We aim to show that b) in the upcoming lemma does not hold (a.s.), given the choice of  $p$  and for large enough  $d$ .

**Lemma 3.38.** *Almost surely one of the following holds for all large enough  $d$ :*

- a) *The black cluster of  $\underline{0}$  in the Delaunay/Gabriel graph on  $\mathcal{Z} \cup \{\underline{0}\}$  is finite.*
- b) *There exists an infinite linked sequence of black chunks starting from  $\underline{0}$ .*

*Proof.* Fix a realization of  $\mathcal{Z} = \mathcal{Z}_b \cup \mathcal{Z}_w$ . Assume that the black component of  $\underline{0}$  in the Delaunay/Gabriel graph on  $\mathcal{Z} \cup \{\underline{0}\}$  is infinite. Then, there exists a black infinite path  $P = \underline{0}, x_1, x_2, x_3, \dots$  of distinct  $x_i \in \mathcal{Z}_b$  with  $x_{i-1} \sim x_i$  for all  $i \geq 1$  if we set  $x_0 = \underline{0}$ . By Corollary 3.11 and Remark 3.15 it then also holds that  $x_{i-1} \hat{\sim} x_i$ . If  $t_1 \leq d^2$  is either the first index such that  $x_{t_1-1}x_{t_1}$  is a bad edge (this could be already the first edge) or if  $t_1 = d^2$  and all edges until this point are good, the first steps  $P_1 = \underline{0}, \dots, x_{t_1}$  form a chunk by definition. Now, let  $x_{s_2}$  be the last index of  $P$  with

$$\text{cert}(P_1) \cap \text{cert}(x_{s_2}, x_{s_2+1}) \neq \emptyset.$$

This point  $s_2$  exists and is finite almost surely, because of Lemma 3.37 and since there is some  $r > 0$  with  $\text{cert}(P_1) \subset B(\underline{0}, r)$ . An issue can occur when  $x_{t_1+1}$  lies on the line through  $x_{t_1-1}$  and  $x_{t_1}$ , since then we might never intersect  $\text{cert}(P_1)$  again. However, this does not happen almost surely, since the points are a result of a Poisson point process and for any two fixed points the line through them has Lebesgue measure zero. Let now  $t_2 \leq s_2 + d^2$  be either the first index after  $s_2$

where  $x_{t_2-1}x_{t_2}$  is a bad edge or  $t_2 = s_2 + d^2$  and all edges in  $P$  between  $x_{s_2}x_{s_2+1}$  and  $x_{t_2-1}x_{t_2}$  are good. Then,  $P_2 = x_{s_2}, \dots, x_{t_2}$  is a chunk by construction. Note also that  $P_1, P_2$  is a linked sequence of chunks. The following chunks can be constructed iteratively by the identical procedure. Having already found the black linked chunks  $P_1, P_2, \dots, P_{i-1}$ , let  $s_i$  be the last index such that

$$\text{cert}(P_{i-1}) \cap \text{cert}(x_{s_i}, x_{s_i+1}) \neq \emptyset.$$

By construction we also obtain

$$\text{cert}(P_j) \cap \text{cert}(x_{s_i}, x_{s_i+1}) = \emptyset,$$

for the other  $j < i - 1$ . The index  $s_i$  exists and is finite since we can use Lemma 3.37 and that  $\text{cert}(P_{i-1}) \subset B(\underline{0}, r)$  for some  $r > 0$ . We again let  $t_i \leq s_i + d^2$  be either the first index after  $s_i$  such that  $x_{t_i-1}x_{t_i}$  is a bad edge, or  $t_i = s_i + d^2$  and all the edges in  $P$  between index  $s_i$  and  $t_i$  are good. Then, choosing  $P_i = x_{s_i}, \dots, x_{t_i}$  will result in a chunk such that  $P_1, \dots, P_i$  is a linked sequence of chunks. Since we can continue this procedure for all steps  $i$ , there has to exist a black and infinite linked sequence of chunks starting from  $\underline{0}$  (almost surely).  $\square$

We introduce several types of chunks. If for a chunk  $P = x_0, \dots, x_k$

I: all edges are good, it is of type  $I$ ,

II: if  $r_u + w - 1 \leq \|x_{k-1} - x_k\| < r_u + w$  for some  $w \in \mathbb{N}$ , it is of type  $(II, w)$ .

We will say that an edge is of type  $\tau \in \{I, (II, w) : w \in \mathbb{N}\}$ , if it is the last edge in a type  $\tau$  chunk. It is immediate that these cases cover all possible chunks in a linked sequence of chunks. We write  $C_k^\tau$  for the number of black type  $\tau$  chunks with length  $k$  starting at  $\underline{0}$ .

For  $k \in \mathbb{N}$  we can define  $x_0 := \underline{0}$  and express

$$C_k^I = |\{(x_1, \dots, x_k) \in \mathcal{Z}_b^k : x_{i-1} \hat{\sim} x_i, x_i \in B(x_{i-1}, r_u) \text{ for } 1 \leq i \leq k\}|.$$

We see that if we treat such a good pseudopath as a member in a linked sequence of chunks, it must hold that  $k \leq d^2$ .

**Lemma 3.39.** *Provided  $\delta_1 > 0$  is small enough there exists a constant  $0 < u = u(\mu) < 1$  such that we have for all  $d \geq d_0(\mu)$  and  $k \leq d^2$*

$$\mathbb{E}C_k^I \leq u^k.$$

*Proof.* By the Mecke formula we can write the expectation as

$$\mathbb{E}C_k^I = \int_{B(\underline{0}, r_u)} \dots \int_{B(x_{k-1}, r_u)} \mathbb{P}(x_0 \hat{\sim} x_1, \dots, x_{k-1} \hat{\sim} x_k \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_k\}) (\lambda p)^k dx_k \dots dx_1.$$

We let  $E_k = E_k(d)$  be the set of all  $k$ -tuples  $(x_1, \dots, x_k) \in (\mathbb{R}^d)^k$  with  $\|y_k - y_i\| \geq 0.1$  for all  $i < k$ , where  $y_i = (x_{i-1} + x_i)/2$  is the center of a certificate with  $x_0 = \underline{0}$ . We will split the integral into the part where we integrate over  $E_k$  and the corresponding complementary area. First, when we integrate over  $(\mathbb{R}^d)^k \setminus E_k$ , we can upper bound

$$\begin{aligned}
& \int_{B(\underline{0}, r_u)} \cdots \int_{B(x_{k-1}, r_u)} \mathbb{P}(x_0 \hat{\sim} x_1, \dots, x_{k-1} \hat{\sim} x_k \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_k\}) \\
& \quad \cdot \mathbb{1}\{(\mathbb{R}^d)^k \setminus E_k\} (\lambda p)^k dx_k \dots dx_1 \\
& \leq \sum_{i < k} \int_{B(\underline{0}, r_u)} \cdots \int_{B(x_{k-1}, r_u)} \mathbb{P}(x_0 \hat{\sim} x_1, \dots, x_{k-1} \hat{\sim} x_k \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_k\}) \\
& \quad \cdot \mathbb{1}\{\|y_k - y_i\| < 0.1\} (\lambda p)^k dx_k \dots dx_1 \\
& \leq \sum_{i < k} \int_{B(\underline{0}, r_u)} \cdots \int_{B(x_{k-1}, r_u)} \mathbb{P}(x_0 \hat{\sim} x_1, \dots, x_{k-2} \hat{\sim} x_{k-1} \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_{k-1}\}) \\
& \quad \cdot \mathbb{1}\{x_k \in B(x_i + x_{i-1} - x_{k-1}, 0.2)\} (\lambda p)^k dx_k \dots dx_1 \\
& \leq k \lambda p \text{vol}(B(\underline{0}, 0.2)) \mathbb{E} C_{k-1}^I.
\end{aligned}$$

Since  $k \leq d^2$ ,  $0 < \mu < 1$  and  $\mathbb{E} D \geq 2^d$  for all large enough  $d$  we can further estimate

$$k \lambda p \text{vol}(B(\underline{0}, 0.2)) \mathbb{E} C_{k-1}^I \leq d^{1002} 0.2^d \mathbb{E} C_{k-1}^I \leq 4^{-d} \mathbb{E} C_{k-1}^I,$$

given that  $d$  is sufficiently large. What remains to estimate is the part of the integral where we integrate over  $E_k$ . We now define the area where  $\text{cert}(x_{k-1}, x_k)$  intersects the other certificates as

$$I(x_0, \dots, x_k) := \bigcup_{i < k} \text{cert}(x_{i-1}, x_i) \cap \text{cert}(x_{k-1}, x_k).$$

Since the absence of points of  $\mathcal{Z}$  is always beneficial for connecting fixed points, we obtain

$$\begin{aligned}
& \mathbb{P}(x_0 \hat{\sim} x_1, \dots, x_{k-1} \hat{\sim} x_k \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_k\}) \\
& \leq \mathbb{P}(x_0 \hat{\sim} x_1, \dots, x_{k-1} \hat{\sim} x_k \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_k\} \mid \mathcal{Z} \cap I(x_0, \dots, x_k) = \emptyset).
\end{aligned}$$

We now observe,  $x_{k-1} \hat{\sim} x_k$  only depends on the points in  $\text{cert}(x_{k-1}, x_k)$ , and therefore under  $\{\mathcal{Z} \cap I(x_0, \dots, x_k) = \emptyset\}$  we use this independence to say

$$\begin{aligned}
& \mathbb{P}(x_0 \hat{\sim} x_1, \dots, x_{k-1} \hat{\sim} x_k \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_k\} \mid \mathcal{Z} \cap I(x_0, \dots, x_k) = \emptyset) \\
& = \mathbb{P}(x_0 \hat{\sim} x_1, \dots, x_{k-2} \hat{\sim} x_{k-1} \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_{k-1}\} \mid \mathcal{Z} \cap I(x_0, \dots, x_k) = \emptyset) \\
& \quad \cdot \mathbb{P}(x_{k-1} \hat{\sim} x_k \text{ for } \mathcal{Z} \cup \{x_{k-1}, x_k\} \mid \mathcal{Z} \cap I(x_0, \dots, x_k) = \emptyset),
\end{aligned}$$

where we also used that the absence of points is beneficial for connections. We recall that Lemma 2.8 gives us a constant  $c > 0$  such that for all  $(x_1, \dots, x_k) \in E_k$

$$\text{vol}(I(x_0, \dots, x_k)) \leq k e^{-cd} \text{vol}(B(\underline{0}, 1 + \delta_1)) \leq d^2 e^{-cd} \text{vol}(B(\underline{0}, 1 + \delta_1)),$$

and therefore

$$\mathbb{P}(\mathcal{Z} \cap I(x_0, \dots, x_k) \neq \emptyset) \leq 1 - \exp(-d^2 e^{-cd} (1 + \delta_1)^d).$$

Since  $c > 0$  is a constant, the right hand side is  $o(1)$  as  $d \rightarrow \infty$  for all small enough  $\delta_1 > 0$ . Combining the bounds above we obtain by translation invariance

$$\begin{aligned} & \int_{B(\underline{0}, r_u)} \dots \int_{B(x_{k-1}, r_u)} \mathbb{P}(x_0 \sim x_1, \dots, x_{k-1} \sim x_k \text{ for } \mathcal{Z} \cup \{x_0, \dots, x_k\}) \\ & \quad \cdot \mathbf{1}_{E_k}(\lambda p)^k dx_k \dots dx_1 \\ & \leq (1 + o(1)) \mathbb{E} C_{k-1}^I \int_{B(\underline{0}, r_u)} \mathbb{P}(\underline{0} \sim z \text{ for } \mathcal{Z} \cup \{\underline{0}, z\}) \lambda p dz. \end{aligned}$$

So in total

$$\mathbb{E} C_k^I \leq \mathbb{E} C_{k-1}^I \left( 4^{-d} + (1 + o(1)) \int_{B(\underline{0}, r_u)} \mathbb{P}(\underline{0} \sim z \text{ for } \mathcal{Z} \cup \{\underline{0}, z\}) \lambda p dz \right),$$

and the recursion concludes the proof by Lemma 1.6, 3.4 and 3.16.  $\square$

In the following we are going to write for  $x \in \mathbb{R}^d$  and  $w \in \mathbb{N}$

$$A_w(x) := \{y \in \mathbb{R}^d : r_u + w - 1 \leq \|x - y\| < r_u + w\}$$

for the region where points would be considered for an  $(II, w)$  connection with  $x$ .

**Lemma 3.40.** *There is a constant  $c > 0$  such that for all large enough  $d$  and all  $w \in \mathbb{N}$*

$$\mathbb{E} |\{z \in A_w(\underline{0}) \cap \mathcal{Z}_b : \underline{0} \sim z\}| \leq e^{-c\beta_{w-1}},$$

with  $\beta_{w-1} := (r_u + w - 1)^d / 2^d$ .

*Proof.* By the Mecke formula we can write

$$\mathbb{E} |\{z \in A_w(\underline{0}) \cap \mathcal{Z}_b : \underline{0} \sim z\}| = \int_{A_w(\underline{0})} \mathbb{P}(\underline{0} \sim z \text{ for } \mathcal{Z} \cup \{\underline{0}, z\}) \lambda p dz.$$

Since  $z \notin B(\underline{0}, r_u)$  we obtain by Lemma 3.14 a constant  $0 < c \leq 1$  with

$$\begin{aligned} & \int_{A_w(\underline{0})} \mathbb{P}(\underline{0} \sim z \text{ for } \mathcal{Z} \cup \{\underline{0}, z\}) \lambda p dz \leq \int_{A_w(\underline{0})} e^{-c\|z\|^d / 2^d} \lambda p dz \\ & = \int_{r_u+w-1}^{r_u+w} e^{-cr^d / 2^d} p dr^{d-1} dr \leq \int_{\beta_{w-1}}^{\beta_w} e^{-cz} dz = (e^{-c\beta_{w-1}} - e^{-c\beta_w}) / c, \end{aligned}$$

where we used  $\mathbb{E}D \geq 2^d$  for all large enough  $d$  and made the switch to polar coordinates and used the change of variable  $z = r^d/2^d$ . Choosing  $d$  large enough such that  $-c\beta_0 + \ln(c^{-1}) \leq -c\beta_0/2$  concludes the proof.  $\square$

The following result makes it possible for us to split a chunk of type II and length  $k$  into the part consisting of  $k - 1$  good edges and the final one, to bound the expected number of chunks.

**Lemma 3.41.** *There is a constant  $c > 0$  such that for all large enough  $d$  it holds for all  $w \in \mathbb{N}$  and  $k \leq d^2$*

$$\mathbb{E}C_k^{(II,w)} \leq e^{-c\beta_{w-1}}.$$

*Proof.* We can write by the Mecke formula

$$\mathbb{E}C_k^{(II,w)} = \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} \mathbb{E}g_{C_k^{(II,w)}}(\underline{0}, x_1, \dots, x_k)(\lambda p)^k dx_k \dots dx_1,$$

where  $g_{C_k^{(II,w)}}(\underline{0}, x_1, \dots, x_k)$  is the indicator function that  $\underline{0}, x_1, \dots, x_k$  forms a chunk with type  $(II, w)$  of length  $k$ , with respect to  $\mathcal{Z} \cup \{\underline{0}, x_1, \dots, x_k\}$ . Since the first  $k - 1$  edges are good it holds that  $x_i \in B(x_{i-1}, r_u)$  for all  $i < k$ . By Lemma 3.40 we can thus bound for constants  $c_1, c_2 > 0$  and all large enough  $d$

$$\begin{aligned} \mathbb{E}C_k^{(II,w)} &\leq \int_{\mathbb{R}^d} \cdots \int_{A_w(x_{k-1})} \mathbb{P}(x_{k-1} \hat{\sim} x_k \text{ for } \mathcal{Z} \cup \{x_{k-1}, x_k\}) \\ &\quad \cdot \mathbb{1}_{\{x_i \in B(x_{i-1}, r_u) \text{ for } i < k\}} (\lambda p)^k dx_k \dots dx_1 \\ &\leq (\lambda p)^{k-1} \text{vol}(B(\underline{0}, r_u))^{k-1} e^{-c_1\beta_{w-1}} \leq d^{1000d^2} e^{-c_1\beta_{w-1}} \leq e^{-c_2\beta_{w-1}}. \end{aligned}$$

$\square$

The upcoming result will help us estimate the number of expected links when dealing with a ‘ $b$  link’.

**Lemma 3.42.** *Let  $P = x_0, \dots, x_k$  be a sequence of points in  $\mathbb{R}^d$  such that  $x_i \in B(x_{i-1}, r_u)$  for all  $1 \leq i \leq k$ . Then for*

$$X(P) := |\{(z_1, z_2) \in \mathcal{Z}_b^2 : \text{cert}(z_1, z_2) \cap \text{cert}(P) \neq \emptyset, z_2 \in B(z_1, r_u), z_1 \hat{\sim} z_2\}|$$

*we have for all large enough  $d \geq d_0(\mu)$*

$$\mathbb{E}X(P) \leq k(1 + \delta_1)^d.$$

*Proof.* Define for  $1 \leq i \leq k$

$$X_i(P) := |\{(z_1, z_2) \in \mathcal{Z}_b^2 : \text{cert}(z_1, z_2) \cap \text{cert}(x_{i-1}, x_i) \neq \emptyset, z_2 \in B(z_1, r_u), z_1 \hat{\sim} z_2\}|,$$

to obtain  $X(P) \leq \sum_i X_i(P)$ . Then we have by using the Mecke formula

$$\begin{aligned} \mathbb{E}X_i(P) &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(z_1 \hat{\sim} z_2 \text{ for } \mathcal{Z} \cup \{z_1, z_2\}) \\ &\quad \cdot \mathbb{1}_{\{\text{cert}(z_1, z_2) \cap \text{cert}(x_{i-1}, x_i) \neq \emptyset, z_2 \in B(z_1, r_u)\}} (\lambda p)^2 dz_2 dz_1 \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(z_1 \hat{\sim} z_2 \text{ for } \mathcal{Z} \cup \{z_1, z_2\}) \\ &\quad \cdot \mathbb{1}_{\{(z_1+z_2)/2 \in B(\underline{0}, 2(1+\delta_1)), (z_1-z_2)/2 \in B(\underline{0}, r_u/2)\}} (\lambda p)^2 dz_2 dz_1. \end{aligned}$$

Making the change of variable  $u = (z_1 + z_2)/2$  and  $v = (z_1 - z_2)/2$  (as in (3.24)) yields

$$\begin{aligned} &\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(z_1 \hat{\sim} z_2 \text{ for } \mathcal{Z} \cup \{z_1, z_2\}) \\ &\quad \cdot \mathbb{1}_{\{(z_1+z_2)/2 \in B(\underline{0}, 2(1+\delta_1)), (z_1-z_2)/2 \in B(\underline{0}, r_u/2)\}} (\lambda p)^2 dz_2 dz_1 \\ &= 2^d \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(u+v \hat{\sim} u-v \text{ for } \mathcal{Z} \cup \{u+v, u-v\}) \\ &\quad \cdot \mathbb{1}_{\{u \in B(\underline{0}, 2(1+\delta_1)), v \in B(\underline{0}, r_u/2)\}} (\lambda p)^2 du dv. \end{aligned}$$

By translation invariance we can rewrite

$$\mathbb{P}(u+v \hat{\sim} u-v \text{ for } \mathcal{Z} \cup \{u+v, u-v\}) = \mathbb{P}(\underline{0} \hat{\sim} 2v \text{ for } \mathcal{Z} \cup \{\underline{0}, 2v\})$$

and by using  $\mu < 1$  together with Lemma 3.16 we get for all large enough  $d \geq d_0(\mu)$

$$\begin{aligned} &2^d \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(u+v \hat{\sim} u-v \text{ for } \mathcal{Z} \cup \{u+v, u-v\}) \\ &\quad \cdot \mathbb{1}_{\{u \in B(\underline{0}, 2(1+\delta_1)), v \in B(\underline{0}, r_u/2)\}} (\lambda p)^2 du dv \\ &= 2^d \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(\underline{0} \hat{\sim} 2v \text{ for } \mathcal{Z} \cup \{\underline{0}, 2v\}) \mathbb{1}_{\{u \in B(\underline{0}, 2(1+\delta_1)), v \in B(\underline{0}, r_u/2)\}} (\lambda p)^2 du dv \\ &\leq 2^d (1+\delta_1)^d \int_{\mathbb{R}^d} \mathbb{P}(\underline{0} \hat{\sim} 2v \text{ for } \mathcal{Z} \cup \{\underline{0}, 2v\}) \mathbb{1}_{\{v \in B(\underline{0}, r_u/2)\}} \lambda p dv \\ &\leq (1+\delta_1)^d \int_{B(\underline{0}, r_u)} \mathbb{P}(\underline{0} \hat{\sim} z \text{ for } \mathcal{Z} \cup \{\underline{0}, z\}) \lambda p dz \leq (1+\delta_1)^d. \end{aligned}$$

□

**Lemma 3.43.** *Let  $\delta_1$  be small enough,  $d$  sufficiently large and  $x_1, x_2, y_1, y_2 \in \mathbb{R}^d$ . Further, we let  $w_x \geq 1$  if  $x_1 x_2$  would be an edge of type  $(II, w_x)$  and otherwise set  $w_x = 0$ . The number  $w_y$  is defined identically for  $y_1 y_2$ . If  $\text{cert}(x_1, x_2) \cap \text{cert}(y_1, y_2) \neq \emptyset$  it holds that  $x_1 \in B((y_1 + y_2)/2, 4 + 2w_x + w_y)$ .*

*Proof.* Let  $\delta_1$  be small enough such that  $3(1 + \delta_1) \leq 4$ . The radius of  $\text{cert}(y_1, y_2)$  is  $1 + \delta_1$  for  $w_y = 0$  and otherwise for large enough  $d$  it is less than  $(r_u + w_y)/2 \leq 1 + \delta_1 + w_y$ . The same holds for  $\text{cert}(x_1, x_2)$  with  $w_x$  instead of  $w_y$ . The point  $x_1$  has to be of distance less than  $(r_u + w_x)/2 \leq 1 + \delta_1 + w_x$  from the center of  $\text{cert}(x_1, x_2)$ . Adding up the three distances we conclude that  $x_1$  must be at distance of less than  $3(1 + \delta_1) + 2w_x + w_y \leq 4 + 2w_x + w_y$  from the center of  $\text{cert}(y_1, y_2)$ , given that  $\text{cert}(x_1, x_2) \cap \text{cert}(y_1, y_2) \neq \emptyset$ .  $\square$

We define  $\mathcal{T} := \{I, (II, w) : w \in \mathbb{N}\}$ . For  $n \in \mathbb{N}$  and vectors  $\underline{k} = (k_1, \dots, k_n) \in \mathbb{N}^n$ ,  $\underline{\tau} = (\tau_1, \dots, \tau_n) \in \mathcal{T}^n$ ,  $\underline{t} = (t_1, \dots, t_{n-1}) \in \{0, 1\}^{n-1}$  we let  $S_{\underline{k}, \underline{\tau}, \underline{t}}$  be the number of  $n$  linked black chunks where the first starts from  $\underline{0}$ , the  $i$ th chunk has length  $k_i$  and type  $\tau_i$  and is linked to the next chunk with link type  $a$  for  $t_i = 0$  and  $b$  if  $t_i = 1$ . For  $n = 1$  there is no link as we only have one chunk. The length of the chunks  $\underline{k}$  and the number of  $b$  links will determine how many points  $m = m(\underline{k}, \underline{t})$  form the linked sequence of chunks counted by  $S_{\underline{k}, \underline{\tau}, \underline{t}}$ . We can express the expectation by the Mecke formula as

$$\mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}} = (\lambda p)^m \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} \mathbb{E}g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1, \dots, x_m) dx_1 \dots dx_m, \quad (3.25)$$

where  $g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1, \dots, x_m)$  is the indicator that  $\underline{0}, x_1, \dots, x_m$  is an element counted by  $S_{\underline{k}, \underline{\tau}, \underline{t}}$  with respect to the augmented point process  $\mathcal{Z} \cup \{\underline{0}, x_1, \dots, x_m\}$ . Not all choices of vectors will form an admitted linked sequence of chunks. For example  $\mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}} = 0$  if there exists  $k_i > d^2$  or for  $\tau_i = I$  it holds  $k_i \neq d^2$ . The following two lemmas will establish a recursion for  $S_{\underline{k}, \underline{\tau}, \underline{t}}$ . We will use the notation  $(\underline{k}, k) = (k_1, \dots, k_n, k)$  for  $\underline{k} \in \mathbb{N}^n$  and  $k \in \mathbb{N}$ . To deal with the situation where  $n = 1$  and  $\underline{t} \in \{0, 1\}^{n-1}$  in the upcoming lemma we will let  $(\underline{t}, t) = t$ . With the notation  $w\mathbb{1}_{\{\tau_n = (II, w)\}}$  we want to say that this expression is equal to 0 if  $\tau_n = I$  and otherwise equal to  $w$ , corresponding to the specific  $(II, w)$  type of  $\tau_n$ .

**Lemma 3.44.** *For  $n \in \mathbb{N}$  let  $\underline{k} \in \mathbb{N}^n$ ,  $\underline{\tau} \in \mathcal{T}^n$  and  $\underline{t} \in \{0, 1\}^{n-1}$  be arbitrary. Given that  $\delta_1$  is sufficiently small there is a  $d_0$  such that for all  $d \geq d_0$ ,  $\tau \in \mathcal{T}$ ,  $k \geq 2$  and  $t \in \{0, 1\}$*

$$\mathbb{E}S_{(\underline{k}, k), (\underline{\tau}, \tau), (\underline{t}, t)} \leq (3 + w\mathbb{1}_{\{\tau_n = (II, w)\}})^d \cdot \mathbb{E}C_{k-1}^\tau \cdot \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}}. \quad (3.26)$$

*Proof.* We can assume that  $k \leq d^2$ , since otherwise there exist no such linked sequence of chunks. In the same sense for the case of  $\tau = I$  and  $k \neq d^2$  it holds that the left hand side of (3.26) is equal to 0, thus only  $k = d^2$  is nontrivial in this instance. We are going to use the notation of Definition 3.36 and write  $P_i = x_0^i, \dots, x_{k_i}^i$  for the  $i$ th chunk counted by  $S_{(\underline{k}, k), (\underline{\tau}, \tau), (\underline{t}, t)}$ . With this we get  $x_0^{n+1} = x_{k_n}^n$  if  $t = 0$  and we will integrate over these two points separately should we have a  $b$  link. For a fixed  $n$ , the number of  $b$  links determines the number of points  $m$  that form the linked sequence of chunks  $S_{(\underline{k}, k), (\underline{\tau}, \tau), (\underline{t}, t)}$ . We will separate

the proof into two cases.

**Case 1:**  $\tau_n = \tau = I$ :

First assume that both  $\tau_n, \tau$  are of type  $I$ . We are going to split the integrals of (3.25) into two parts. The first will correspond to the case where  $(x_0^{n+1} + x_1^{n+1})/2$  (the center of  $\text{cert}(x_0^{n+1}, x_1^{n+1})$ ) is at distance of less than 0.1 to at least one other center of a certificate of  $P_n$  or  $P_{n+1}$ . We will write  $y_1, \dots, y_{k_n}$  for an enumeration of the center of certificates of  $P_n$  and  $y_{k_n+1}, \dots, y_{k_n+k-1}$  for the center of certificates of  $x_1^{n+1}, \dots, x_k^{n+1}$  (i.e.  $P_{n+1}$  without the first point). Additionally, we let  $g_{C_{k-1}^\tau}(x_1^{n+1}, \dots, x_k^{n+1})$  be the indicator function that  $x_1^{n+1}, \dots, x_k^{n+1}$  is a chunk of length  $k-1$  with respect to  $\mathcal{Z} \cup \{x_1^{n+1}, \dots, x_k^{n+1}\}$  where the last edge has type  $\tau$ . By the definition of a linked sequence of chunks, all certificates of  $P_{n+1}$ , except for the first edge, do not intersect any certificates of previous chunks. Then we can remove the requirement of  $x_0^{n+1} \approx x_1^{n+1}$  having to occur and use the resulting independence (the remaining certificates are disjoint) to bound the integrand in the first part by

$$\mathbb{E}g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1^1, \dots, x_{k_n}^n) \cdot \mathbb{E}g_{C_{k-1}^\tau}(x_1^{n+1}, \dots, x_k^{n+1}) \\ \cdot \mathbb{1}\{\|(x_0^{n+1} + x_1^{n+1})/2 - y_j\| < 0.1 \text{ for one } y_j, x_1^{n+1} \in B(x_0^{n+1}, r_u)\}.$$

We can now compute the integrals of (3.25) one by one. First, it holds for all fixed  $x_1^{n+1}$  by translation invariance

$$(\lambda p)^{k-1} \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} \mathbb{E}g_{C_{k-1}^\tau}(x_1^{n+1}, \dots, x_k^{n+1}) dx_2^{n+1} \dots dx_k^{n+1} \\ = (\lambda p)^{k-1} \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} \mathbb{E}g_{C_{k-1}^\tau}(0, x_2^{n+1}, \dots, x_k^{n+1}) dx_2^{n+1} \dots dx_k^{n+1} = \mathbb{E}C_{k-1}^\tau.$$

Moreover, we have  $k_n, k \leq d^2$  and thus it holds in the case of  $t = 1$  for all fixed  $P_n$  and  $x_2^{n+1}, \dots, x_k^{n+1}$

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{1}\{\|(x_0^{n+1} + x_1^{n+1})/2 - y_j\| < 0.1 \text{ for one } y_j, x_1^{n+1} \in B(x_0^{n+1}, r_u)\} (\lambda p)^2 dx_0^{n+1} dx_1^{n+1} \\ \leq d^{2000} 0.2^d.$$

For the  $a$  link, i.e.  $t = 0$ , we have one integral less, resulting in the crude upper bound

$$\int_{\mathbb{R}^d} \mathbb{1}\{\|(x_{k_n}^n + x_1^{n+1})/2 - y_j\| < 0.1 \text{ for one } y_j\} \lambda p dx_1^{n+1} \leq 2d^2 0.2^d \leq d^{2000} 0.2^d.$$

Since further

$$(\lambda p)^{m-k-\mathbb{1}\{t=1\}} \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} \mathbb{E}g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1^1, \dots, x_{k_n}^n) dx_1^1 \dots dx_{k_n}^n = \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}},$$

we can bound the first part of the integrals of (3.25) by

$$d^{2000} 0.2^d \mathbb{E} C_{k-1}^\tau \mathbb{E} S_{\underline{k}, \underline{\tau}, \underline{t}}.$$

The remaining segment that we have to deal with is where the center of  $\text{cert}(x_0^{n+1}, x_1^{n+1})$  is at distance of more than 0.1 from all other center of certificates of  $P_n$  and  $P_{n+1}$ . We define

$$I(P_n, P_{n+1}) := \text{cert}(x_0^{n+1}, x_1^{n+1}) \cap (\text{cert}(P_n) \cup \text{cert}(x_1^{n+1}, \dots, x_k^{n+1})). \quad (3.27)$$

We note if this area is free of  $\mathcal{Z}$  it is always beneficial for connections. So we get by conditioning

$$\mathbb{E} g_{(\underline{k}, k), (\underline{\tau}, \tau), (\underline{t}, t)}(x_1^1, \dots, x_k^{n+1}) \leq \mathbb{E} [g_{(\underline{k}, k), (\underline{\tau}, \tau), (\underline{t}, t)}(x_1^1, \dots, x_k^{n+1}) \mid I(P_n, P_{n+1}) \cap \mathcal{Z} = \emptyset].$$

We realize on  $E(P_n, P_{n+1}) := \{I(P_n, P_{n+1}) \cap \mathcal{Z} = \emptyset\}$ , the event that the points  $\underline{0}, x_1^1, \dots, x_k^n$  form the linked sequence of chunks  $P_1, \dots, P_n$ , that  $x_1^{n+1}, \dots, x_k^{n+1}$  create a chunk of length  $k-1$  of type  $\tau$  and that  $x_0^{n+1} \approx x_1^{n+1}$  for  $\mathcal{Z} \cup \{x_0^{n+1}, x_1^{n+1}\}$  are mutually independent (since these three events now all depend on disjoint area). With this observation we estimate the integrand over the second indicator by

$$\begin{aligned} & \mathbb{E} [g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1^1, \dots, x_k^n) \mid E(P_n, P_{n+1})] \cdot \mathbb{E} [g_{C_{k-1}^\tau}(x_1^{n+1}, \dots, x_k^{n+1}) \mid E(P_n, P_{n+1})] \\ & \cdot \mathbb{1}\{\|(x_0^{n+1} + x_1^{n+1})/2 - y_j\| > 0.1 \text{ for all } y_j, \text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset\} \\ & \cdot \mathbb{1}\{x_1^{n+1} \in B(x_0^{n+1}, r_u)\} \cdot \mathbb{P}(x_0^{n+1} \approx x_1^{n+1} \mid E(P_n, P_{n+1})) \\ & \leq \mathbb{E} [g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1^1, \dots, x_k^n)] \cdot \mathbb{E} [g_{C_{k-1}^\tau}(x_1^{n+1}, \dots, x_k^{n+1})] \\ & \cdot \mathbb{1}\{\|(x_0^{n+1} + x_1^{n+1})/2 - y_j\| > 0.1 \text{ for all } y_j, \text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset\} \\ & \cdot \mathbb{1}\{x_1^{n+1} \in B(x_0^{n+1}, r_u)\} \cdot \mathbb{P}(x_0^{n+1} \approx x_1^{n+1}) \cdot \mathbb{P}(E(P_n, P_{n+1}))^{-3}. \end{aligned}$$

It has to hold that  $x_1^{n+1} \in B(x_0^{n+1}, r_u)$ , as for  $k \geq 2$  the first edge is always good. Since  $\tau_n, \tau$  are both of type  $I$  we obtain by Lemma 2.8, if we integrate over  $\{\|(x_0^{n+1} + x_1^{n+1})/2 - y_j\| > 0.1 \text{ for all } y_j\}$ , a constant  $c > 0$  such that

$$\text{vol}(I(P_n, P_{n+1})) \leq 2d^2 e^{-cd} \text{vol}(B(\underline{0}, 1 + \delta_1)),$$

utilizing that all certificates have a radius less than  $1 + \delta_1$  and there are less than  $2d^2$  intersections in question. Thus,  $\mathbb{P}(E(P_n, P_{n+1})) \geq \exp(-2d^2 e^{-cd}(1 + \delta_1)^d)$ . Therefore, we can upper bound the part of (3.25) where we integrate over the area where  $(x_0^{n+1} + x_1^{n+1})/2$  is at distance of more than 0.1 from all other center of certificates of  $P_n$  and  $P_{n+1}$  by

$$\begin{aligned} & (\lambda p)^{m-k-1} \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} \mathbb{E} g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1^1, \dots, x_k^n) \cdot \mathbb{P}(x_0^{n+1} \approx x_1^{n+1}) \\ & \cdot \mathbb{1}\{\text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset, x_1^{n+1} \in B(x_0^{n+1}, r_u)\} dx_1^1 \dots dx_1^{n+1} \\ & \cdot \mathbb{E} C_{k-1}^\tau \cdot \exp(6d^2 e^{-cd}(1 + \delta_1)^d). \end{aligned}$$

By Lemma 3.42 we have for all fixed  $P_n$  in the  $b$  link situation

$$\begin{aligned}
& (\lambda p)^2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(x_0^{n+1} \approx x_1^{n+1}) \\
& \quad \cdot \mathbb{1}\{\text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset, x_1^{n+1} \in B(x_0^{n+1}, r_u)\} dx_0^{n+1} dx_1^{n+1} \\
& \leq d^2(1 + \delta_1)^d.
\end{aligned}$$

This bound also holds for  $t = 0$ , where  $x_0^{n+1} = x_{k_n}^n$ , so we have one integral less. We can see this by Lemma 3.16, since for all fixed  $P_n$  and large enough  $d$

$$\begin{aligned}
& \lambda p \int_{B(x_0^{n+1}, r_u)} \mathbb{P}(x_0^{n+1} \approx x_1^{n+1}) \mathbb{1}\{\text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset\} dx_1^{n+1} \\
& \leq \lambda p \int_{B(\underline{0}, r_u)} \mathbb{P}(\underline{0} \approx x_1^{n+1}) dx_1^{n+1} \leq d^2(1 + \delta_1)^d.
\end{aligned}$$

As before, the remaining integrals yield  $\mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}}$ . If now  $\delta_1$  is small and  $d$  large enough such that  $\exp(6e^{-cd}(1 + \delta_1)^d) \leq 2$  and  $d^{2000}0.2^d \leq (1 + \delta_1)^d$ , we obtain in total for the non  $II$  edge case

$$\begin{aligned}
\mathbb{E}S_{(\underline{k}, k), (\underline{\tau}, \tau), (\underline{t}, t)} & \leq (d^{2000}0.2^d + d^2(1 + \delta_1)^d \exp(6e^{-cd}(1 + \delta_1)^d)) \mathbb{E}C_{k-1}^\tau \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}} \\
& \leq 3d^2(1 + \delta_1)^d \mathbb{E}C_{k-1}^\tau \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}} \\
& \leq 3^d \mathbb{E}C_{k-1}^\tau \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}}.
\end{aligned}$$

**Case 2:** At least one of  $\tau_n$  or  $\tau$  is of type  $II$ :

In this setting we let  $w = 0$  if  $\tau_n$  is equal to  $I$  and otherwise  $w \in \mathbb{N}$  expresses that  $\tau_n = (II, w)$ . Since  $k \geq 2$ , only the last edge of  $P_{n+1}$  can be bad, so we know that the first edge has to be good. Thus  $x_1^{n+1} \in B(x_0^{n+1}, r_u)$  has to hold, otherwise the indicator is zero and the expectation in the integral will be zero too. If we now omit the condition that  $x_0^{n+1} \approx x_1^{n+1}$  has to occur, we utilize the resulting independence and Lemma 3.43 (to deal with the case of  $t = 1$ , for  $t = 0$  we do not get the  $(2 + w)^d$  factor) to upper bound

$$\begin{aligned}
& \mathbb{E}S_{(\underline{k}, k), (\underline{\tau}, \tau), (\underline{t}, t)} \leq \\
& (\lambda p)^{m-k-1} \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} \mathbb{E}g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1^1, \dots, x_{k_n}^n) \cdot \mathbb{1}\{x_1^{n+1} \in B(x_0^{n+1}, r_u)\} \\
& \quad \cdot \mathbb{1}\{\text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset\} dx_1^1 \dots dx_1^{n+1} \cdot \mathbb{E}C_{k-1}^\tau \\
& \leq k_n d^{1000} (2 + w)^d \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}} \mathbb{E}C_{k-1}^\tau \leq (3 + w)^d \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}} \mathbb{E}C_{k-1}^\tau,
\end{aligned}$$

where the last inequality holds for all large enough  $d$ .  $\square$

The previous result dealt with every case of the final chunk for  $\mathbb{E}S_{(\underline{k}, k), (\underline{\tau}, \tau), (\underline{t}, t)}$  except when it is of type  $\tau = II$  and length  $k = 1$ . There, most of the work

involves handling the situation where the second to last chunk also ends with a  $II$  edge, as the volume of the intersection of the two certificates might still be quite large in absolute terms.

**Lemma 3.45.** *For  $n \in \mathbb{N}$  let  $\underline{k} \in \mathbb{N}^n$ ,  $\underline{\tau} \in \mathcal{T}^n$  and  $\underline{t} \in \{0, 1\}^{n-1}$  be arbitrary. If  $\delta_1$  is sufficiently small, there is a  $d_0$  and  $c > 0$  such that for all  $d \geq d_0$ ,  $\tau \in \mathcal{T}$ ,  $z \geq 1$  and  $t \in \{0, 1\}$  the following bounds hold. If  $\tau_n = (II, w)$  with  $w > 3z$*

$$\mathbb{E}S_{(\underline{k}, 1), (\underline{\tau}, (II, z)), (\underline{t}, t)} \leq \exp((r_u + (w - 1)/3)^d / 2^{d-2} - c\beta_{z-1}) \cdot \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}}.$$

*In all other cases for  $\tau_n$  and  $n \geq 2$*

$$\begin{aligned} \mathbb{E}S_{(\underline{k}, 1), (\underline{\tau}, (II, z)), (\underline{t}, t)} &\leq e^{-c\beta_{z-1}} \cdot (2 + v\mathbb{1}_{\{\tau_{n-1} = (II, v)\}})^d \\ &\quad \cdot \mathbb{E}S_{(k_1, \dots, k_{n-1}), (\tau_1, \dots, \tau_{n-1}), (t_1, \dots, t_{n-2})}, \end{aligned}$$

*with  $\beta_{z-1} = (r_u + z - 1)^d / 2^d$  and if  $n = 1$  we obtain*

$$\mathbb{E}S_{(k_1, 1), (\tau_1, (II, z)), t} \leq e^{-c\beta_{z-1}}.$$

*Proof.* We will split the proof into two cases, dependent on the relative length of the last edge of chunk  $n$  to the length of the singular edge of chunk  $n + 1$ . Like in the proof of the previous lemma, we will work with the expectation and  $m = m(\underline{k}, \underline{t}, t)$  given by the Mecke formula in (3.25) and write  $P_i = x_0^i, \dots, x_{k_i}^i$  for the  $i$ th chunk counted by  $S_{(\underline{k}, 1), (\underline{\tau}, (II, z)), (\underline{t}, t)}$ .

**Case 1:**  $\tau_n = (II, w)$  with  $3z < w$ :

First assume that  $\tau_n = (II, w)$  with  $3z < w$  (since  $z \geq 1$  this also implies  $w \geq 4$ ). Here we can bound by using the definition from (3.27) with  $k = 1$

$$\begin{aligned} \text{vol}(I(P_n, P_{n+1})) &\leq \text{vol}(\text{cert}(x_0^{n+1}, x_1^{n+1})) \leq \text{vol}(B(\mathbb{0}, (r_u + z)/2)) \\ &\leq \text{vol}(B(\mathbb{0}, (r_u + (w - 1)/3)/2)). \end{aligned}$$

We also obtain  $\mathbb{P}(E(P_n, P_{n+1})) \geq \exp(-(r_u + (w - 1)/3)^d / 2^d)$  uniformly over all  $P_n$  and  $P_{n+1}$  considered. Under  $E(P_n, P_{n+1})$ , the fact whether  $\mathbb{0}, x_1^1, \dots, x_{k_n}^n$  form the linked sequence of chunks  $P_1, \dots, P_n$  is independent from  $x_0^{n+1} \leadsto x_1^{n+1}$ . This yields the upper bound

$$\begin{aligned} \mathbb{E}S_{(\underline{k}, 1), (\underline{\tau}, (II, z)), (\underline{t}, t)} &\leq \\ &(\lambda p)^m \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} \mathbb{E}g_{\underline{k}, \underline{\tau}, \underline{t}}(x_1^1, \dots, x_{k_n}^n) \mathbb{P}(x_0^{n+1} \leadsto x_1^{n+1}) \\ &\quad \cdot \mathbb{1}_{\{x_1^{n+1} \in A_z(x_0^{n+1}), \text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset\}} dx_1^1 \dots dx_1^{n+1} \\ &\quad \cdot \exp((r_u + (w - 1)/3)^d / 2^{d-1}). \end{aligned}$$

Lemma 3.43 tells us that  $x_0^{n+1}$  has to be at distance of no more than  $4+2z+w$  from the center of a certificate of  $P_n$  in order to achieve  $\text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset$ . There are less than  $d^2$  such centers. If we are dealing with a  $b$  link, Lemma 3.40 yields a constant  $c_1 > 0$  so for all fixed  $P_n = x_0^n, \dots, x_{k_n}^n$  and large enough  $d$

$$\begin{aligned}
& (\lambda p)^2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(x_0^{n+1} \approx x_1^{n+1}) \\
& \quad \cdot \mathbb{1}\{x_1^{n+1} \in A_z(x_0^{n+1}), \text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset\} dx_0^{n+1} dx_1^{n+1} \\
& \leq d^2 (\lambda p)^2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(x_0^{n+1} \approx x_1^{n+1}) \\
& \quad \cdot \mathbb{1}\{x_1^{n+1} \in A_z(x_0^{n+1}), x_0^{n+1} \in B(\underline{0}, 4+2z+w)\} dx_0^{n+1} dx_1^{n+1} \\
& \leq d^2 (2+w+z)^d \int_{\mathbb{R}^d} \mathbb{P}(\underline{0} \approx x_1^{n+1}) \mathbb{1}\{x_1^{n+1} \in A_z(\underline{0})\} \lambda p dx_1^{n+1} \\
& \leq d^2 (2+w+z)^d e^{-c_1 \beta_{z-1}} \leq (2+w)^d (2+z)^d e^{-c_1 \beta_{z-1}}.
\end{aligned}$$

By the same arguments, the bound is also valid in the  $t = 0$  situation where we only integrate over  $x_1^{n+1}$ , as  $x_0^{n+1} = x_{k_n}^n$  would be part of  $P_n$ . The remaining integrals over  $x_1^n, \dots, x_{k_n}^n$  yield  $\mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}}$ . Then, we obtain a constant  $c_2 > 0$  such that for all  $3z < w$  and all large enough  $d \geq d_0$

$$\begin{aligned}
\mathbb{E}S_{(\underline{k}, 1), (\underline{\tau}, (II, z)), (\underline{t}, t)} & \leq \exp((r_u + (w-1)/3)^d / 2^{d-1}) (2+w)^d (2+z)^d e^{-c_1 \beta_{z-1}} \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}} \\
& \leq \exp((r_u + (w-1)/3)^d / 2^{d-2}) e^{-c_2 \beta_{z-1}} \mathbb{E}S_{\underline{k}, \underline{\tau}, \underline{t}}.
\end{aligned}$$

**Case 2:**  $\tau_n$  is either  $I$ , or  $(II, w)$  for a  $w \in \mathbb{N}$  with  $3z \geq w$ :

To avoid having to do the calculations separately we will set  $w = 0$  in the derivations below for the instance of  $\tau_n = I$ . In the same sense for  $n \geq 2$  we let  $v = 0$  if  $\tau_{n-1} = I$  and otherwise  $v$  expresses that  $\tau_{n-1} = (II, v)$ . By definition, the first  $k_n - 1$  edges of  $P_n$  have a length of less than  $r_u$  and the final edge is shorter than  $r_u \leq 2 + 5z$  for  $w = 0$  and  $r_u + w \leq 2 + 5z$  if  $w > 0$  and  $d$  is large enough. We will let  $G(x_0^n, \dots, x_{k_n}^n)$  be the indicator of  $x_{k_n}^n \in B(x_{k_n-1}^n, 2+5z)$  and if  $k_n > 1$  we additionally require that of  $x_i^n \in B(x_{i-1}^n, r_u)$  for  $1 \leq i \leq k_n - 1$ . Since we consider linked sequence of chunks, the connection  $x_0^{n+1} \approx x_1^{n+1}$  is independent from the fact whether  $P_1, \dots, P_{n-1}$  form a linked sequence of chunks of length  $n-1$ , since the corresponding certificates are disjoint. If we omit the requirements regarding the connection having to occur for the chunk  $P_n$ , we can thus use this

independence to bound if  $n \geq 2$

$$\begin{aligned} \mathbb{E}S_{(\underline{k},1),(\underline{\tau},(II,z)),(\underline{t},t)} &\leq \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} \mathbb{E}g_{(k_1,\dots,k_{n-1}),(\tau_1,\dots,\tau_{n-1}),(\underline{t}_1,\dots,\underline{t}_{n-2})}(x_1^1, \dots, x_{k_{n-1}}^{n-1}) \\ &\quad \cdot \mathbb{P}(x_0^{n+1} \leadsto x_1^{n+1}) \cdot \mathbb{E}G(x_0^n, \dots, x_{k_n}^n) \\ &\quad \cdot \mathbb{1}_{\{x_1^{n+1} \in A_z(x_0^{n+1}), \text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset\}} \\ &\quad \cdot \mathbb{1}_{\{\text{cert}(x_0^n, x_1^n) \cap \text{cert}(P_{n-1}) \neq \emptyset\}} (\lambda p)^m dx_1^1 \dots dx_1^{n+1}. \end{aligned}$$

In the following we can assume that  $n \geq 2$  and the bounds we calculate below will carry over to the bound we claimed for the  $n = 1$  case, where we are only dealing with  $P_1$  and  $P_2$ . In this situation there is no chunk  $P_{n-1}$ , so we can write

$$\begin{aligned} \mathbb{E}S_{(k_1,1),(\tau_1,(II,z)),(\underline{t}_1,t)} &\leq \\ &\int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} \mathbb{P}(x_0^2 \leadsto x_1^2) \cdot \mathbb{E}G(\underline{0}, x_1^1, \dots, x_{k_1}^1) \\ &\quad \cdot \mathbb{1}_{\{x_1^2 \in A_z(x_0^2), \text{cert}(x_0^2, x_1^2) \cap \text{cert}(P_1) \neq \emptyset, x_{k_1}^1 \in A_w(x_{k_1-1}^1)\}} (\lambda p)^m dx_1^1 \dots dx_1^2. \end{aligned}$$

We again compute the integrals one by one. By Lemma 3.43 it has to hold that  $x_0^{n+1}$  lies in a ball of radius  $4 + 2z + w$  around a center of a certificate of  $P_n$ . Then, we see for each fixed  $x_1^1, \dots, x_{k_n}^n$ , it holds by Lemma 3.40 in the  $b$  link setting for large enough  $d$

$$\begin{aligned} &\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(x_0^{n+1} \leadsto x_1^{n+1}) \mathbb{1}_{\{x_1^{n+1} \in A_z(x_0^{n+1}), \text{cert}(x_0^{n+1}, x_1^{n+1}) \cap \text{cert}(P_n) \neq \emptyset\}} (\lambda p)^2 dx_0^{n+1} dx_1^{n+1} \\ &\leq d^2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{P}(x_0^{n+1} \leadsto x_1^{n+1}) \mathbb{1}_{\{x_1^{n+1} \in A_z(x_0^{n+1}), x_0^{n+1} \in B(\underline{0}, 4 + 2z + w)\}} (\lambda p)^2 dx_0^{n+1} dx_1^{n+1} \\ &\leq d^2 p \text{vol}(B(\underline{0}, 4 + 2z + w)) e^{-c_1 \beta z - 1} \leq (2 + 5z)^d e^{-c_1 \beta z - 1}, \end{aligned}$$

for a constant  $c_1 > 0$  and estimating  $w \leq 3z$ . The bound also holds for  $t = 0$ , as we only integrate over  $x_1^{n+1}$  there. Again, by Lemma 3.43, we obtain for all fixed  $P_{n-1}$  and  $t = 1$  that

$$\begin{aligned} &\int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} \mathbb{E}G(x_0^n, \dots, x_{k_n}^n) \mathbb{1}_{\{\text{cert}(x_0^n, x_1^n) \cap \text{cert}(P_{n-1}) \neq \emptyset\}} (\lambda p)^{k_{n+1}} dx_{k_n}^n \dots dx_0^n \\ &\leq (\lambda p)^{k_{n+1}} d^2 \text{vol}(B(\underline{0}, 4 + 2w + v)) \text{vol}(B(\underline{0}, r_u))^{k_{n-1}} \text{vol}(B(\underline{0}, 2 + 5z)) \\ &\leq d^2 (2 + v)^d (2 + 5z)^{2d} d^{1000d^2}, \end{aligned}$$

and in the  $a$  link case we again do not integrate over  $x_1^n$ , as it is also part of  $P_{n-1}$ . The remaining integrals over  $x_1^1, \dots, x_{k_{n-1}}^{n-1}$  yield  $\mathbb{E}S_{(k_1,\dots,k_{n-1}),(\tau_1,\dots,\tau_{n-1}),(\underline{t}_1,\dots,\underline{t}_{n-2})}$  and thus in total for some constant  $c_3 > 0$  and all large enough  $d$

$$\begin{aligned} \mathbb{E}S_{(\underline{k},1),(\underline{\tau},(II,z)),(\underline{t},t)} &\leq d^2 (2 + v)^d (2 + 5z)^{3d} d^{1000d^2} e^{-c_1 \beta z - 1} \\ &\quad \cdot \mathbb{E}S_{(k_1,\dots,k_{n-1}),(\tau_1,\dots,\tau_{n-1}),(\underline{t}_1,\dots,\underline{t}_{n-2})} \\ &\leq e^{-c_3 \beta z - 1} (2 + v)^d \mathbb{E}S_{(k_1,\dots,k_{n-1}),(\tau_1,\dots,\tau_{n-1}),(\underline{t}_1,\dots,\underline{t}_{n-2})}. \end{aligned}$$

□

In the following lemma we will sum over all possible chunks and use the recursions from the previous two lemmas to show that  $b)$  of Lemma 3.38 does not happen almost surely for our choice of  $p$  and all large enough dimensions.

**Lemma 3.46.** *We can choose  $\delta_1$  small enough such that for each fixed  $0 < \mu < 1$  there exists  $d_0 = d_0(\mu)$  such that for all  $d \geq d_0$  there is almost surely no infinite linked sequence of black chunks starting from the origin.*

*Proof.* We define  $S_n$  to be the number of  $n$  linked black chunks that start from the origin. Further, for  $\tau \in \mathcal{T}$  we let  $S_n^\tau$  be the number of  $n$  linked black chunks that start from the origin where the last chunk is of type  $\tau$ . This definition yields by using Tonelli's theorem to interchange expectation with the series

$$\mathbb{E}S_n = \mathbb{E}S_n^I + \sum_{z \in \mathbb{N}} \mathbb{E}S_n^{(II,z)} \leq \mathbb{E}S_n^I + \sum_{z \in \mathbb{N}} f_d(z) \mathbb{E}S_n^{(II,z)} =: \Sigma_n,$$

where we define  $f_d(z) := \exp((r_u + (z-1)/3)^d / 2^{d-2})$  for  $z \geq 4$  and  $f_d(z) := (4z)^d$  otherwise. We are going to show that for all large enough  $d$  it holds  $\Sigma_1, \Sigma_2 < \infty$  and  $\Sigma_{n+1} < (\Sigma_n + \Sigma_{n-1})/3$ . Then,  $\Sigma_n$  converges to 0 with  $n \rightarrow \infty$  for those dimensions and so does  $\mathbb{E}S_n$ . As

$$\mathbb{P}(\cap_{n \in \mathbb{N}} \{S_n \geq 1\}) = \lim_{n \rightarrow \infty} \mathbb{P}(S_n \geq 1) = 0,$$

we can conclude the proof and follow that there is almost surely no infinite sequence of linked black chunks starting from the origin for these parameters. First we can write

$$\Sigma_1 = \mathbb{E}C_{d^2}^I + \sum_{\ell_2=1}^{d^2} \sum_{z \in \mathbb{N}} f_d(z) \mathbb{E}C_{\ell_2}^{(II,z)}.$$

By Lemma 3.41 we obtain a constant  $c_1 > 0$  (for large enough  $d$ ) so that for each fixed  $\ell_2$

$$\sum_{z \geq 4} f_d(z) \mathbb{E}C_{\ell_2}^{(II,z)} \leq \sum_{z \geq 4} \exp((r_u + (z-1)/3)^d / 2^{d-2} - c_1 \beta_{z-1}), \quad (3.28)$$

where  $\beta_{z-1} = (r_u + z - 1)^d / 2^d$ . For all  $z \geq 4$  it holds that  $2 < r_u + (z-1)/3 < r_u + z - 1$ , yielding

$$\lim_{d \rightarrow \infty} \exp((r_u + (z-1)/3)^d / 2^{d-2} - c_1 (r_u + z - 1)^d / 2^d) = 0.$$

We can now choose  $d_0$  large enough such that for all  $z \geq 4$  and  $d \geq d_0$  it holds  $4(r_u + (z-1)/3)^d \leq c_1 (r_u + z - 1)^d / 2$ . Then, we estimate for all  $d \geq d_0$  and some

$$c_2 = c_2(d) > 0$$

$$\begin{aligned} \sum_{z \geq 4} \exp((r_u + (z-1)/3)^d / 2^{d-2} - c_1 \beta_{z-1}) &\leq \sum_{z \geq 4} \exp(-c_1 \beta_{z-1}/2) \\ &\leq \sum_{z \geq 4} \exp(-c_1 (z-1)^d / 2^{d+1}) = \sum_{z \geq 4} e^{-c_2 (z-1)^d} < \infty. \end{aligned}$$

We can follow that the r.h.s. of (3.28) is finite and  $o(1)$  as  $d \rightarrow \infty$  by monotone convergence. Since by Lemma 3.39 and 3.41 the remaining part of  $\Sigma_1$  is a finite sum of finite expectations, we conclude  $\Sigma_1 < \infty$ . For  $n \geq 1$  we can take the sum over all possible linked chunks to rewrite

$$\Sigma_{n+1} = \sum_{\substack{\underline{k} \in \mathbb{N}^n, \underline{\tau} \in \mathcal{T}^n, \underline{t} \in \{0,1\}^{n-1} \\ 1 \leq k \leq d^2, t \in \{0,1\}}} \mathbb{E} S_{(\underline{k}, k), (\underline{\tau}, I), (\underline{t}, t)} + \sum_{z \in \mathbb{N}} f_d(z) \mathbb{E} S_{(\underline{k}, k), (\underline{\tau}, (II, z)), (\underline{t}, t)}. \quad (3.29)$$

If in the sums above we ignore the case where  $k = 1$  in the  $II$  final chunk instance we can use Lemma 3.44 together with  $(3+w)^d \leq f_d(w)$  for all  $w \in \mathbb{N}$  and large enough  $d$  to upper bound these sums for all sufficiently small  $\delta_1$  by

$$\Sigma_n \cdot 2 \cdot 3^d \cdot \left( \mathbb{E} C_{d^2}^I + \sum_{k=1}^{d^2} \sum_{z \in \mathbb{N}} f_d(z) \mathbb{E} C_k^{(II, z)} \right),$$

where the factor of 2 arises from the two possibilities of  $t \in \{0,1\}$ . For large enough  $d \geq d_0(\mu)$ , using Lemmas 3.39 and 3.41, we obtain constants  $c_3 > 0$  and  $0 < c_4 = c_4(\mu) < 1$  with

$$\mathbb{E} C_{d^2}^I + \sum_{k=1}^{d^2} \sum_{z \in \mathbb{N}} f_d(z) \mathbb{E} C_k^{(II, z)} \leq c_4^{d^2} + d^2 \sum_{z \in \mathbb{N}} f_d(z) e^{-c_1 \beta_{z-1}}.$$

By applying monotone convergence as we have done it above for  $n = 1$  together with

$$2 \cdot 3^d \cdot d^2 \sum_{z < 4} f_d(z) e^{-c_1 \beta_{z-1}} \leq 100^d e^{-c_1 d^{1000}} = o(1),$$

we can upper bound the part where we do not consider  $k = 1$  with a final  $II$  edge by  $\Sigma_n \cdot o(1)$  as  $d \rightarrow \infty$ . For the summands of (3.29) where we deal with  $k = 1$  and  $\tau_n = (II, w)$  such that  $w > 3z$ , we compute by Lemma 3.45 for a constant  $c_5 > 0$

$$\sum_{\substack{\underline{k} \in \mathbb{N}^n, \underline{\tau} \in \mathcal{T}^n, \underline{t} \in \{0,1\}^{n-1} \\ t \in \{0,1\}, z \in \mathbb{N} \\ \tau_n = (II, w) \text{ for } w > 3z}} f_d(z) \mathbb{E} S_{(\underline{k}, 1), (\underline{\tau}, (II, z)), (\underline{t}, t)} \leq \Sigma_n \cdot \sum_{\substack{z \in \mathbb{N} \\ t \in \{0,1\}}} f_d(z) e^{-c_5 \beta_{z-1}} = \Sigma_n \cdot o(1).$$

The last cases of the sum of (3.29) that we have to deal with is where  $\tau_n$  is of type  $I$ , or  $(II, w)$  for  $w \leq 3z$  and  $k = 1$ . We can estimate similarly as above for  $n \geq 2$  by monotone convergence, Tonelli's theorem and Lemma 3.45 that gives us a constant  $c_5 > 0$  such that

$$\begin{aligned}
& \sum_{\substack{\underline{k} \in \mathbb{N}^n, \underline{\tau} \in \mathcal{T}^n, \underline{t} \in \{0,1\}^{n-1} \\ t \in \{0,1\}, w \leq 3z \text{ if } \tau_n = (II, w)}} \sum_{z \in \mathbb{N}} f_d(z) \mathbb{E} S_{(\underline{k}, 1), (\underline{\tau}, (II, z)), (\underline{t}, t)} \\
&= \sum_{\substack{\underline{k} \in \mathbb{N}^{n-1}, \underline{\tau} \in \mathcal{T}^{n-1} \\ \underline{t} \in \{0,1\}^{n-2}}} \sum_{\substack{1 \leq k_n \leq d^2 \\ t, t_{n-1} \in \{0,1\}}} \sum_{z \in \mathbb{N}} \sum_{\substack{\tau_n \in \mathcal{T} \\ w \leq 3z \text{ if } \tau_n = (II, w)}} f_d(z) \mathbb{E} S_{(\underline{k}, k_n, 1), (\underline{\tau}, \tau_n, (II, z)), (\underline{t}, t_n, t)} \\
&\leq \sum_{\substack{\underline{k} \in \mathbb{N}^{n-1}, \underline{\tau} \in \mathcal{T}^{n-1} \\ \underline{t} \in \{0,1\}^{n-2}}} \sum_{z \in \mathbb{N}} 16d^2 z f_d(z) e^{-c_5 \beta_{z-1}} (2 + v \mathbb{1}_{\{\tau_{n-1} = (II, v)\}})^d \mathbb{E} S_{\underline{k}, \underline{\tau}, \underline{t}} \\
&\leq \Sigma_{n-1} \cdot 16d^2 2^d \sum_{z \in \mathbb{N}} z f_d(z) e^{-c_5 \beta_{z-1}} = \Sigma_{n-1} \cdot o(1),
\end{aligned}$$

where Lemma 3.45 also says that the upper bound carries over to  $n = 1$  if we set  $\Sigma_0 := 1$ . We can thus conclude if  $d_0$  is large enough it holds for all  $d \geq d_0$  and small enough  $\delta_1$  that  $\Sigma_{n+1} \leq (\Sigma_n + \Sigma_{n-1})/3$ , and for  $n = 2$  we have that  $\Sigma_2 < \infty$ . As we have stated at the beginning, this finishes the proof.  $\square$

We are finally ready to show the main result of this section.

*Proof of Theorem 3.33.* Combining Lemma 3.46 and Lemma 3.38, we conclude that for all  $0 < \mu < 1$  we can choose  $d_0(\mu)$  large enough such that for all  $d \geq d_0$  almost surely the black component of  $\underline{0}$  for  $\mathcal{Z} \cup \{\underline{0}\}$  is finite. Thus, by translation invariance, we get for all  $z \in \mathbb{R}^d$  and  $d \geq d_0$

$$g(z) := \mathbb{P}(\text{the black component of } z \text{ for } \mathcal{Z} \cup \{z\} \text{ is infinite}) = 0,$$

where we treat  $z$  as a deterministic black point. If we now define the random variable  $N$  with range  $\{0, \infty\}$  as

$$N := \sum_{z \in \mathcal{Z}_b} \mathbb{1}_{\{\text{the black component of } z \text{ is infinite}\}},$$

we have with the Mecke formula (given  $d$  is sufficiently large)

$$\mathbb{E} N = \int_{\mathbb{R}^d} g(z) \lambda p \, dz = 0.$$

We conclude that  $N = 0$  almost surely and that a.s. there exists no infinite black component.  $\square$

The main result now follows by joining the upper and lower bound, provided by the two sections.

*Proof of Theorem 1.4 and 1.5.* Combining Theorem 3.17 and 3.33 yields our main Theorems of this chapter.  $\square$

### 3.5 Discussion and suggestions for further work

We have shown that both in the Poisson–Voronoi and Poisson–Gabriel percolation models the critical probability  $p_c(d)$  behaves like the inverse of the expected number of neighbors of a typical point as  $d \rightarrow \infty$ . One next natural step would be to show this “*mean-field behavior*” more rigorously. The informal heuristic is that most high-dimensional percolation models (like we have seen here) exhibit a great level of independence: loops in the graph play only a negligible role, different paths are less likely to interfere with each other and clusters “behave” like branching processes, where connections would be formed completely independently from each other. Thus, it is reasonable to assume that for large enough  $d$  a lot of other quantities of interest in Poisson–Voronoi percolation are similar to the analogue of the corresponding branching process. If this is the case, one speaks of mean-field behavior. The standard way of doing that is to show the existence of power law exponents and observe that they are the same as in the branching setting. This was already done for other models, for example the result for bond percolation on  $\mathbb{Z}^d$  [50], and in the continuum for the Random Connection Model (RCM) [54].

We write  $f \asymp g$  if  $c g \leq f \leq C g$  for some uniform constants  $c, C > 0$ . One value that is of interest is the expected size of the black cluster  $C_0$  of the typical cell in Poisson–Voronoi percolation on  $\mathcal{Z} \cup \{\underline{0}\}$ . So,  $C_0$  are all the Poisson points, whose black Voronoi cell is part of the black cluster of  $\underline{0}$ . We say that  $\gamma > 0$  exists if for all  $p < p_c$

$$\mathbb{E}_p |C_0| \asymp (p_c - p)^{-\gamma}.$$

The exponent  $\delta > 0$  should capture the cluster size tail decay at criticality and exists if for all  $n \in \mathbb{N}$

$$\mathbb{P}_{p_c}(|C_0| \geq n) \asymp n^{-1/\delta}.$$

The last exponent we want to study is  $\beta > 0$ , that exists if there is an  $\varepsilon > 0$  such that for all  $p \in (p_c, p_c + \varepsilon)$

$$\mathbb{P}_p(|C_0| = \infty) \asymp (p - p_c)^\beta.$$

It is important to note that the existence of  $\beta$  would imply that  $\mathbb{P}_{p_c}(\text{percolation}) = 0$ , i.e. that there is almost surely no infinite cluster at criticality. The analogue for all dimensions (for example for bond percolation on  $\mathbb{Z}^d$ ) is a major open problem in probability theory and seems to be out of reach with current methods. One indicator that is commonly used to detect mean-field behavior is the *triangle*

*condition.* For that, we will write  $\tau_p(x, y)$  for the probability that the cells of  $x$  and  $y$  are part of the same black cluster, when  $x, y \in \mathbb{R}^d$  are being added as deterministic black points to  $\mathcal{Z}$ . The *triangle diagram* is defined as

$$\Delta_p := \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \tau_p(\underline{0}, x) \tau_p(x, y) \tau_p(y, \underline{0}) \lambda^2 dx dy$$

and the triangle condition is satisfied if  $\Delta_{p_c} < \infty$ . The name comes from the fact that, when the black components are viewed as a graph, this quantity is an upper bound on the expected number of triangles that contain a fixed point. Heuristically speaking, these loops should only appear rarely for large  $d$ . The overall goal would be to show that the triangle condition is satisfied for all large enough dimensions and that this implies the existence of the just defined exponents. It is wise to split this into two parts.

**Conjecture 3.47.** Assume that  $\Delta_{p_c} < \infty$ . Then,  $\delta, \gamma$  and  $\beta$  exist and take on the values  $\delta = 2, \gamma = 1$  and  $\beta = 1$ .

The values for  $\delta, \gamma, \beta$  are exactly the same as in the branching process model, where connections occur fully independent from another. Thus, the result would prove that key quantities like the cluster distribution or the percolation probability have the same structure in both instances, i.e. the models are very similar to another (given  $\Delta_{p_c} < \infty$ ).

The other step would be to show that the assumption made above is actually satisfied for all large enough dimensions.

**Conjecture 3.48.** There exists a  $d_0$  such that for all  $d \geq d_0$  it holds  $\Delta_{p_c} < \infty$ .



## CHAPTER 4

# THRESHOLDS FOR COLOURING THE RANDOM BORSUK GRAPH

This chapter is based on joint work with Álvaro Acitores Montero, Tobias Müller and Matěj Stehlík. It is available as a preprint [80].

### 4.1 Outline

#### 4.1.1 Related work

For context, we give a very brief and very incomplete overview of related results for the Erdős-Rényi random graph and random geometric graph model. The study of thresholds for monotone properties in general has played a central role in literature on the Erdős-Rényi random graph model. Landmarks include the work of Bollobás and Thomason [21] who observed that every monotone property has a threshold, the work of Friedgut and Kalai [41] showed that any monotone property will transition from holding with probability  $o_n(1)$  to probability  $1 - o_n(1)$  inside an interval for the parameter  $p$  of length  $O(1/\ln n)$  and Friedgut's seminal paper [39] that essentially gave a characterization of the graph properties that have a sharp threshold. For more background on thresholds for monotone properties in the Erdős-Rényi model we refer the reader to [40, 87] and the references therein. For random geometric graphs, McColm [75] established an analogue of the result of Bollobás and Thomason, but only in dimension  $d = 1$ . For other dimensions it remains an open problem whether every monotone graph property has a threshold. Goel et al. [42] proved a result analogous to the one of Friedgut and Kalai.

Perhaps even more than monotone properties, the study of the chromatic number of random graphs has had a huge influence on the development of the field. It was already discussed in the founding work of Erdős and Rényi [38]. In

1975 Grimmett and McDiarmid [47] gave upper and lower bounds for the case of constant  $p$  that were a factor two apart, and in a seminal 1988 paper, Bollobás [16] was able to show the probability mass of the chromatic number is concentrated near the lower bound of Grimmett and McDiarmid. More detailed questions on the distribution of the chromatic number for the Erdős-Rényi random graph however remained open even for constant  $p$ , and the research line is actively pursued to this day. A recent highlight is for instance [52].

For  $p = p(n)$  tending to zero with  $n$ , Łuczak [70] showed that the distribution of the chromatic number has almost all of its probability mass concentrated on two consecutive integers  $k, k+1$ , where  $k = k(n, p)$  depends on both  $n$  and  $p$ . Alon and Krivelevich [9] subsequently extended the range of  $p$  where Łuczak's result holds. Achlioptas and Friedgut [1] showed that for each fixed  $k \geq 3$ , the property of being  $k$ -colourable has a sharp threshold in the regime when the average degree is constant, and subsequently Achlioptas and Naor [2] obtained more detailed information on the shape of the threshold and on the value of the aforementioned  $k(n, p)$  in the case when  $p = \text{const} \cdot n^{-1}$ . It is widely believed in the community, but remains unproven in general, that the (sharp) threshold for  $k$ -colourability of the Erdős-Rényi random graph is of the form  $p = \text{const} \cdot n^{-1}$ . In this light, our Conjecture 4.52 seems (even more) natural. For random geometric graphs, the chromatic number was studied in [76, 81, 85]. In particular, Müller showed an analogue of Łuczak's two-point concentration result : if the average degree is not too large, then the probability mass of the chromatic number is concentrated on two consecutive integers. However, in the thermodynamic regime (when the average degree is constant) the value of these two integers tends to infinity with  $n$ , so that the behaviour is rather different from what we observe in the current chapter for the random Borsuk graph, and the mentioned results on the Erdős-Rényi graph. Intuitively speaking, the reason is that the chromatic number for random geometric graphs is "forced" by groups of points that are close together in the ambient space. In particular, in the thermodynamic regime the random geometric graph even has cliques of growing size, while the random Borsuk graph does not have cliques of size  $> 2$ .

### 4.1.2 The structure of the chapter.

In Section 4.1.3 we give an informal description of some of the intuitions and strategies guiding the proofs of our main results. Next, in Section 4.2 we list some notations, definitions and results from the literature that we will be using in the rest of the chapter. Section 4.3 contains the proof of Theorem 1.7, Section 4.4 contains the proof of Theorem 1.10, Section 4.5 contains the proof of Theorem 1.8, and Section 4.6 contains the proof of Theorem 1.9. We include some more discussion and suggest some open problems in Section 4.7.

### 4.1.3 Informal description of the proofs.

In this section, we describe some of the main intuitions and strategies guiding the proofs in an informal, non-rigorous manner in order to aid the reader in digesting the chapter more easily.

**Theorem 1.7.** The strategy for the proof of Theorem 1.7 is to show that, when the average degree is a large constant, we can identify a subgraph of the random Borsuk graph that has chromatic number (at least)  $d+1$ . More specifically, we will exhibit a subgraph that behaves like a Borsuk graph but in dimension  $d-1$  rather than  $d$ . The technique for lower bounding the chromatic number used by Kahle and Martinez-Figueroa involves showing that the spherical caps (balls with respect to  $\text{dist}$ , see Section 4.2.1 for a detailed definition) with opening angle  $\alpha/2$  around the antipodes of the random points cover the sphere, and then apply a version of the Lyusternik-Schnirelmann theorem. See Lemma 4.7 below. A candidate subgraph that we might want to try this technique on is the one induced by all points within a small distance (of order a small constant times  $\alpha$ , say) of the equator  $S^{d-1} \times \{0\}$ . Unfortunately that will not work however, because in the thermodynamic regime, even when the average degree is a large constant, there will be lots of “empty patches” on the sphere with diameter of order  $\alpha$  and larger that are completely devoid of points. In particular the equator will not be covered by the aforementioned spherical caps around antipodes of points.

To get around this, we find a suitable embedding  $\varphi : S^{d-1} \rightarrow S^d$  that stays close to the equator, avoids empty patches, is nearly antipodal (satisfies  $\varphi(-u) \approx -\varphi(u)$ ) and is “locally nearly flat”. To construct this embedding, we find it technically easier to work in  $\mathbb{R}^d$  rather than  $S^d$ . So we apply stereographic projection to obtain a random set of points in  $\mathbb{R}^d$  and construct a map  $h : S^{d-1} \rightarrow (-1, 1)$  such that for every  $u \in S^{d-1}$  one of the random points is within distance  $\varepsilon \cdot \alpha$  of  $\psi(u) := (1 + h(u)) \cdot u$ , and moreover  $h(u) + h(-u) = 0$ , and  $h$  is Lipschitz with an appropriate constant (so that it “looks flat” at the scale  $O(\alpha)$ ). The construction of  $\psi$  proceeds iteratively. Roughly speaking, we order the “bad patches” by scale. We start by avoiding the largest patches. Each iteration we add a small perturbation in order to also avoid patches on the next smaller scale. The perturbations are made in such a way that on a local scale the function  $h$  changes very gradually.

We now consider the subgraph spanned by all (stereographically projected) points within distance  $\varepsilon\alpha$  of the image of  $\psi$ . Because of the careful choice of  $h$ , it turns out that this subgraph itself contains an induced subgraph of the  $(d-1)$ -dimensional Borsuk graph to which the lower bound technique of Kahle and Martinez-Figueroa (stated as Lemma 4.7 below) applies.

**Theorem 1.8.** The main intuition behind the proof of Theorem 1.8 is that locally the random Borsuk graph in the thermodynamic regime behaves like the continuum AB percolation model. This model, first introduced by Iyer and Yo-

geshwaran in [57], is defined by taking two independent, homogeneous Poisson processes  $\mathcal{Z}_A$  and  $\mathcal{Z}_B$  on  $\mathbb{R}^d$ , the first having intensity  $\lambda_A$  and the second  $\lambda_B$ , and joining each point of  $\mathcal{Z}_A$  to any point of  $\mathcal{Z}_B$  within distance  $r$ . (There are no edges with both endpoints in  $\mathcal{Z}_A$  or both endpoints in  $\mathcal{Z}_B$ .) Let us consider a spherical cap  $C$  of opening angle some large constant  $K$  times  $\alpha$ . In the thermodynamic regime, as  $n \rightarrow \infty$ , up to rescaling by  $1/\alpha$ , the geometry of  $C$  behaves like that of a ball of radius  $K$  in  $\mathbb{R}^d$ . And, under the same rescaling, the set  $\mathcal{X}_A$  of all random points that fall inside the cap  $C$  behaves like a constant intensity Poisson process restricted to this ball of radius  $K$ . Now also consider the set of points that fall in  $-C$ . Their antipodes, which we shall denote by  $\mathcal{X}_B$ , lie in  $C$  and, similarly to before, under the appropriate rescaling  $\mathcal{X}_B$  behaves like a constant intensity Poisson process, independent of the one describing  $\mathcal{X}_A$  and of the same intensity. What is more, the subgraph of the Borsuk graph spanned by the points in  $C \cup -C$  is approximately the same as the graph we get if we join a point of the first Poisson point process to all points of the second Poisson point process at distance at most one. That is, the “local picture” of the random Borsuk graph is that of the continuum AB percolation model with  $\lambda_A = \lambda_B$  and  $r = 1$ , restricted to some large ball. Being a bit more careful, one can see that the intensity in fact scales with  $\alpha$  (the larger  $\alpha$ , the larger the intensity). The constant  $c_3$  in Theorem 1.8 corresponds to the *critical value*  $\lambda_c$  of the continuum AB percolation. That is, if  $\lambda_A = \lambda_B < \lambda_c$  all clusters (connected components) of the model are finite, and for  $\lambda_A = \lambda_B > \lambda_c$  there is an infinite cluster – both statements holding with probability one and with respect to the model that is not restricted to a ball but lives on all of  $\mathbb{R}^d$ .

Recall that a graph is bipartite (two-colourable) if and only if there is no odd cycle, and that any odd cycle in the Borsuk graph must have at least  $\Omega(1/\alpha)$  (which is  $\Omega(n^{1/d})$  if we are in the thermodynamic regime) vertices. That the random Borsuk graph is two colourable when we choose  $\alpha$  in such a way that the corresponding continuum AB percolation model is subcritical follows relatively easily once we have established that cluster sizes decay exponentially in the subcritical regime. Even though continuum AB percolation has been considered in a few papers (e.g. [31, 86]) since the work of Iyer and Yogeshwaran, the literature does not provide the subcritical exponential decay result we need. (We work in the specific situation when  $\lambda_A = \lambda_B$ , but the focus of the cited works is on other choices of the parameters.) We end up adapting various arguments from the percolation literature to the continuum AB percolation setting.

Showing that if  $\alpha$  is such that the corresponding continuum AB percolation model is supercritical then the random Borsuk graph contains an odd cycle is (even) more involved. As explained, locally the random Borsuk graph is well-approximated by the continuum AB percolation model. The challenge is to patch together many of these local pictures into a global one where we are sure that a (necessarily macroscopic) odd cycle exists with probability  $1 - o_n(1)$ . In order to

achieve this we adapt the “sprinkling technique” of Grimmett and Marstrand [46], developed for their famous “percolation on slabs” result.

**Theorem 1.9.** As mentioned before, Theorem 1.9 follows via a relatively straightforward argument. We consider the range of  $\alpha$  where the expected number of edges is constant, and use the “method of moments” to show that the number of edges converges in distribution to a Poisson random variable.

**Theorem 1.10.** The proof of Theorem 1.10 leverages a “sharp thresholds for Boolean functions” result from [82], stated as Proposition 4.4 below. We use this result as follows. Rather than choosing precisely  $n$  points uniformly at random on  $S^d$ , we switch to considering a Poisson point process on  $S^d$  with  $\mu = n$  points in expectation. Lemma 4.6 below establishes that it is enough to prove the result for this “Poissonized” version of the model. We choose  $\alpha_k = \alpha_k(n)$  such that the probability that the (Poissonized) random Borsuk graph with parameter  $\mu = n$  and  $\alpha = \alpha_k$  has chromatic number  $\geq k$  is precisely  $1/2$ . We discretize the model, choosing a dissection of  $S^d$  into sets  $C_1, \dots, C_m$  of small diameter, and for  $i = 1, \dots, m$  we take  $X_i$  to be the indicator variable that at least one of the random points falls inside  $C_i$ . Having chosen the sets  $C_i$  sufficiently small, the event that the (Poissonized) random Borsuk graph has chromatic number  $\geq k$  is well-approximated by an increasing event defined in terms of  $X_1, \dots, X_m$ . The aforementioned sharp thresholds result then allows us to conclude that (the discretized version of) the event that the chromatic number of the Poissonized random Borsuk graph is at least  $k$  has a sharp threshold in the parameter  $\mu$  (the expected total number of points of the Poissonized random graph). In slightly more detail : the sharp thresholds result implies that if there were no sharp threshold, then we could (deterministically) include a bounded number of points in specific locations in the vertex set and increase  $\alpha$  very slightly to boost the probability of having chromatic number  $\geq k$  to close to one. Increasing the expected number of points from  $\mu$  to  $(1 + \varepsilon)\mu$  corresponds to adding another Poisson process with  $\varepsilon\mu$  points in expectation that is independent of the original process. Such a Poisson process will contain a set of points that “looks like” the points in specific locations provided by the sharp threshold result.

Finally we use (1.2), respectively Theorems 1.7 and 1.8, to show that the sequence  $\alpha_k = \alpha_k(n)$  does not behave too “wildly”. More specifically, for most  $n$  the ratio of the values  $\alpha_k(n)$  and  $\alpha_k(\lceil(1 + \delta)n\rceil)$  is close to one (but we cannot exclude the possibility that occasionally there are “jumps”). For those values of  $n$  where the ratio is close to one, we are able to transfer the sharp threshold in terms of the expected number of points to one in terms of  $\alpha$ .

## 4.2 Notation and preliminaries

Here we list some notations, definitions and results from the literature that we will use in the proofs.

### 4.2.1 Geometric preliminaries

We remind ourselves that we write  $\text{dist}(u, v)$  for the spherical distance between  $u, v \in S^d$ , so the length of the shortest path between the points that stays on the sphere. The *spherical cap* around  $u \in S^d$  of opening angle  $\alpha$  is:

$$\text{cap}(u, \alpha) := \{x \in S^d : \text{dist}(u, x) < \alpha\}.$$

For all  $u, v \in S^d$  the Euclidean distance and spherical distance are related by the following identity:

$$\|u - v\| = 2 \cdot \sin(\text{dist}(u, v)/2).$$

(As can be seen by considering the triangle with corners  $0, u, v$ .) For convenient future reference, we point out the following consequence:

$$(2/\pi) \cdot \text{dist}(u, v) \leq \|u - v\| \leq \text{dist}(u, v), \quad (4.1)$$

for all  $u, v \in S^d$ . (We use that  $\alpha \geq \sin \alpha \geq (2/\pi) \cdot \alpha$  for  $0 \leq \alpha \leq \pi/2$ .)

We denote by  $\pi : S^d \setminus \{(0, \dots, 0, 1)\} \rightarrow \mathbb{R}^d$  the  $d$ -dimensional stereographic projection from the “north pole”. It is given explicitly by

$$\pi(x_1, \dots, x_{d+1}) := \left( \frac{x_1}{1 - x_{d+1}}, \dots, \frac{x_d}{1 - x_{d+1}} \right).$$

As is easily checked,  $\pi$  is a diffeomorphism (a smooth bijection whose inverse is also smooth). In several of our arguments below, we will want to compare the geodesic distance between two points  $a, b \in S^d$  with the Euclidean metric between their images  $\pi(a), \pi(b) \in \mathbb{R}^d$ . The following bounds suffice for our purposes.

**Lemma 4.1.**

(i) For all  $a, b \in S^d \setminus \{(0, \dots, 0, 1)\}$  we have

$$\|\pi(a) - \pi(b)\| \geq (1/2) \cdot \text{dist}(a, b).$$

(ii) For every  $\varepsilon > 0$  there is a  $\delta = \delta(\varepsilon) > 0$  such that

$$\|\pi(a) - \pi(b)\| \leq (1/2 + \varepsilon) \cdot \text{dist}(a, b),$$

for all  $a, b \in \text{cap}((0, \dots, 0, -1), \delta)$ .

(iii) For all  $a, b \in S^d \cap \{x_{d+1} \leq t\}$  it holds

$$\|\pi(a) - \pi(b)\| \leq \frac{2-t}{(1-t)^2} \text{dist}(a, b).$$

The proof of Lemma 4.1 makes use of the following observation, which we separate out as a lemma.

**Lemma 4.2.** *Let  $f : [0, 1] \rightarrow S^d \setminus \{(0, \dots, 0, 1)\}$  be a continuously differentiable curve, and set  $g := \pi \circ f$ . Then*

$$\|g'(t)\| = \frac{1}{1 - f_{d+1}(t)} \cdot \|f'(t)\|,$$

for all  $0 \leq t \leq 1$ .

*Proof.* Writing  $f(t) = (f_1(t), \dots, f_{d+1}(t))$ ,  $g(t) = (g_1(t), \dots, g_d(t))$ , the chain rule tells us that

$$g'_i(t) = \sum_{j=1}^{d+1} (\partial \pi_i / \partial x_j)_{x=f(t)} \cdot f'_j(t),$$

The partial derivatives of  $\pi$  are given by:

$$(\partial \pi_i / \partial x_j) = \begin{cases} \frac{1}{1 - x_{d+1}} & \text{if } i = j, \\ \frac{x_i}{(1 - x_{d+1})^2} & \text{if } j = d + 1, \\ 0 & \text{otherwise.} \end{cases}$$

(Note  $1 \leq i \leq d$  while  $1 \leq j \leq d + 1$ .) This translates to:

$$g'_i(t) = \frac{f'_i(t)}{1 - f_{d+1}(t)} + \frac{f_i(t) \cdot f'_{d+1}(t)}{(1 - f_{d+1}(t))^2} = \frac{f'_i(t) \cdot (1 - f_{d+1}(t)) + f_i(t) \cdot f'_{d+1}(t)}{(1 - f_{d+1}(t))^2},$$

for all  $0 \leq t \leq 1$  and  $1 \leq i \leq d$ . We can now compute (suppressing the parameter  $t$  everywhere for conciseness):

$$\begin{aligned}
\|g'\|^2 &= \frac{1}{(1-f_{d+1})^4} \cdot \left( \sum_{i=1}^d (f'_i)^2 (1-f_{d+1})^2 + \sum_{i=1}^d 2f'_i f_i f'_{d+1} (1-f_{d+1}) \right. \\
&\quad \left. + \sum_{i=1}^d f_i^2 (f'_{d+1})^2 \right) \\
&= \frac{1}{(1-f_{d+1})^4} \cdot \left( \sum_{i=1}^d (f'_i)^2 (1-f_{d+1})^2 + \left[ \sum_{i=1}^d f_i^2 \right]' f'_{d+1} (1-f_{d+1}) \right. \\
&\quad \left. + \sum_{i=1}^d f_i^2 (f'_{d+1})^2 \right) \\
&= \frac{1}{(1-f_{d+1})^4} \cdot \left( \sum_{i=1}^d (f'_i)^2 (1-f_{d+1})^2 + [1-f_{d+1}^2]' f'_{d+1} (1-f_{d+1}) \right. \\
&\quad \left. + (1-f_{d+1}^2)(f'_{d+1})^2 \right) \\
&= \frac{1}{(1-f_{d+1})^4} \cdot \left( \sum_{i=1}^d (f'_i)^2 (1-f_{d+1})^2 - 2f_{d+1}(f'_{d+1})^2 (1-f_{d+1}) \right. \\
&\quad \left. + (1-f_{d+1}^2)(f'_{d+1})^2 \right) \\
&= \frac{1}{(1-f_{d+1})^4} \cdot \left( \sum_{i=1}^d (f'_i)^2 (1-f_{d+1})^2 + (f'_{d+1})^2 (1-2f_{d+1}+f_{d+1}^2) \right) \\
&= \frac{1}{(1-f_{d+1})^4} \cdot \left( \sum_{i=1}^{d+1} (f'_i)^2 (1-f_{d+1})^2 \right) = \frac{1}{(1-f_{d+1})^2} \cdot \|f'\|^2.
\end{aligned}$$

□

*Proof of Lemma 4.1.* We use the previous lemma. In order to prove part (i), suppose that we chose  $f$  in such a way that  $g$  traces the line-segment between  $\pi(a)$  and  $\pi(b)$ . (That is,  $f(t) := \pi^{-1}((1-t)\pi(a) + t\pi(b))$  so that  $g(t) = (1-t)\pi(a) + t\pi(b)$ .) We have

$$\begin{aligned}
\|\pi(a) - \pi(b)\| &= \text{length}(g) = \int_0^1 \|g'(t)\| \, dt \geq \frac{1}{2} \cdot \int_0^1 \|f'(t)\| \, dt = \frac{1}{2} \cdot \text{length}(f) \\
&\geq \frac{1}{2} \cdot \text{dist}(a, b),
\end{aligned}$$

where in the first inequality we use that  $-1 \leq f_{d+1}(t) < 1$  for all  $t$ , and in the last inequality we use that  $f$  traces some curve connecting  $a, b$  that stays inside  $S^d$  – hence is at least as long as the geodesic. This establishes part (i).

Now suppose that  $a, b \in \text{cap}((0, \dots, 0, -1), \delta)$  for some small  $\delta > 0$ , to be determined. We consider the situation where  $f : [0, 1] \rightarrow S^d \setminus \{(0, \dots, 0, 1)\}$

traces the geodesic from  $a$  to  $b$  (and again  $g := \pi \circ f$ ). We have  $\text{dist}(a, b) < 2\delta$ , so that  $f$  must stay inside  $\text{cap}((0, \dots, 0, -1), 3\delta)$ . In particular  $f_{d+1}(t) < -\cos(3\delta)$  for all  $0 \leq t \leq 1$ . Arguing similarly to before, we find

$$\begin{aligned} \|\pi(a) - \pi(b)\| &\leq \text{length}(g) = \int_0^1 \|g'(t)\| \, dt \leq \left( \frac{1}{1 + \cos(3\delta)} \right) \cdot \int_0^1 \|f'(t)\| \, dt \\ &= \left( \frac{1}{1 + \cos(3\delta)} \right) \cdot \text{length}(f) = \left( \frac{1}{1 + \cos(3\delta)} \right) \cdot \text{dist}(a, b). \end{aligned}$$

Since  $\lim_{\delta \searrow 0} \cos(3\delta) = 1$ , we can and do assume that we chose  $\delta > 0$  small enough in the beginning to ensure  $1/(1 + \cos(3\delta)) < \frac{1}{2} + \varepsilon$ . This concludes the proof of (ii). For (iii) we are going to write  $\hat{x} = (x_1, \dots, x_d)$  for the first  $d$  coordinates to get

$$\pi(x) - \pi(y) = \frac{\hat{x} - \hat{y}}{1 - x_{d+1}} + \hat{y} \left( \frac{1}{1 - x_{d+1}} - \frac{1}{1 - y_{d+1}} \right)$$

and then use the triangle inequality to obtain

$$\|\pi(x) - \pi(y)\| \leq \frac{\|\hat{x} - \hat{y}\|}{1 - x_{d+1}} + \|\hat{y}\| \frac{|x_{d+1} - y_{d+1}|}{(1 - x_{d+1})(1 - y_{d+1})}.$$

Now, if  $x_{d+1}, y_{d+1} \leq t$  and  $\|\hat{y}\| \leq \|y\| \leq 1$  we can further estimate

$$\|\pi(x) - \pi(y)\| \leq \frac{\|\hat{x} - \hat{y}\|}{1 - t} + \frac{\|x - y\|}{(1 - t)^2} \leq \frac{2 - t}{(1 - t)^2} \|x - y\| \leq \frac{2 - t}{(1 - t)^2} \text{dist}(x, y),$$

as the chordal distance between  $x$  and  $y$  is less than the spherical distance.  $\square$

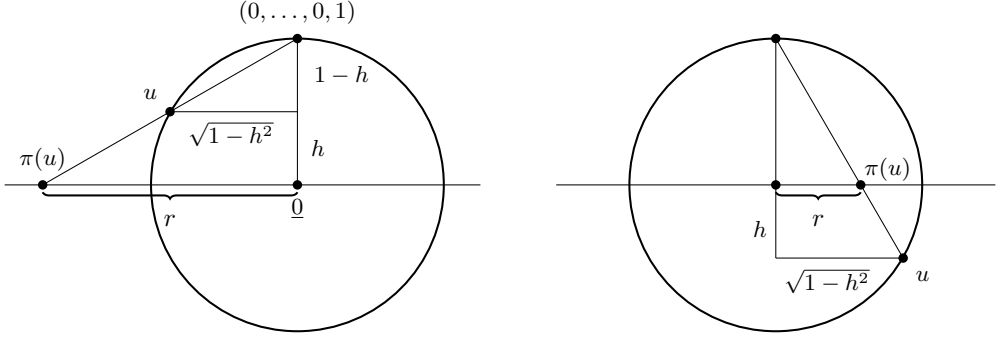
The following result is certainly known in the literature. We have not found a convenient reference with a stand-alone proof, but it can for instance be deduced from Equation 3.7 in [68].

**Lemma 4.3.** *Let  $U \stackrel{d}{=} \text{Unif}(S^d)$  and  $Z := \pi(U)$ . Then the pdf of  $Z$  is given by*

$$f_Z(z) = \frac{1}{(d+1)\kappa_{d+1}} \cdot \left( \frac{2}{1 + \|z\|^2} \right)^d.$$

*Proof of Lemma 4.3.* First, we are going to show that the density of  $H := U_{d+1}$  satisfies

$$f_H(h) = \frac{d\kappa_d}{(d+1)\kappa_{d+1}} (1 - h^2)^{(d-2)/2} \mathbb{1}_{\{-1 \leq h \leq 1\}}.$$

Figure 4.1: Relation between  $H$  and  $R$  for  $r > 1$  (left) and  $r \leq 1$  (right).

For  $Y$  being distributed uniformly on the  $d+1$  dimensional unit ball  $B^{d+1}$  it holds that  $U \stackrel{d}{=} Y/\|Y\|$ . We then have for all  $h \geq 0$

$$\begin{aligned} \mathbb{P}(H \leq h) &= \mathbb{P}(Y \in B^{d+1} \cap \{x_{d+1} \leq h\} \setminus \{\lambda x : x \in B^{d+1} \cap \{x_{d+1} = h\}, 0 \leq \lambda \leq 1\}) \\ &= \frac{1}{\kappa_{d+1}} \left( \int_{-1}^h (1-s^2)^{d/2} \kappa_d ds - \int_0^h \kappa_d (s\sqrt{1-h^2}/h)^d ds \right) \\ &= \frac{\kappa_d}{\kappa_{d+1}} \left( \int_{-1}^h (1-s^2)^{d/2} ds - \frac{1}{d+1} h(1-h^2)^{d/2} \right). \end{aligned}$$

Using this we can obtain the density by differentiating

$$\begin{aligned} f_H(h) &= \frac{d}{dh} \mathbb{P}(H \leq h) \\ &= \frac{\kappa_d}{\kappa_{d+1}} \left( (1-h^2)^{d/2} - \frac{1}{d+1} (1-h^2)^{d/2} + \frac{d}{d+1} h^2 (1-h^2)^{(d-2)/2} \right) \\ &= \frac{d\kappa_d}{(d+1)\kappa_{d+1}} (1-h^2)^{(d-2)/2}. \end{aligned}$$

Our claim for  $f_H(h)$  for arbitrary  $h$  follows by symmetry, as  $f_H(h) = f_H(-h)$ . We can now relate the height  $H$  of  $U$  to the distance from the origin after projecting, namely  $R := \|Z\|$ . From Fig. 4.1

we see that  $1/r = (1-h)/\sqrt{1-h^2} = \sqrt{(1-h)/(1+h)}$ , as the triangle with base  $r$  and height 1 is similar to the one with base  $\sqrt{1-h^2}$  and height  $1-h$ . Rearranging terms yields the relation

$$h(r) = \frac{r^2 - 1}{r^2 + 1}.$$

By symmetry of the sphere it holds that  $f_Z$  only depends on  $R$ , up to measure zero, meaning we can abuse the notation and write  $f_Z(s)$  for  $s \geq 0$ . Thus, we can

compute by making the change to  $d$ -dimensional polar coordinates

$$\mathbb{P}(R \leq r) = \int_0^r f_Z(s) s^{d-1} d\kappa_d ds = \mathbb{P}(H \leq h(r)) = \int_{-1}^{h(r)} \frac{d\kappa_d}{(d+1)\kappa_{d+1}} (1-s^2)^{(d-2)/2} ds.$$

After differentiating we obtain

$$\begin{aligned} f_Z(r) r^{d-1} d\kappa_d &= f_H(h(r)) h'(r) = \frac{d\kappa_d}{(d+1)\kappa_{d+1}} (1-h(r)^2)^{(d-2)/2} \left[ \frac{r^2-1}{r^2+1} \right]' \\ &= \frac{d\kappa_d}{(d+1)\kappa_{d+1}} (1-h(r)^2)^{(d-2)/2} \frac{4r}{(1+r^2)^2}. \end{aligned}$$

Since additionally it holds that

$$1-h(r)^2 = 1 - \left( \frac{r^2-1}{r^2+1} \right)^2 = \frac{4r^2}{(1+r^2)^2},$$

we can rearrange terms to find

$$f_Z(r) = \frac{1}{(d+1)\kappa_{d+1}} \left( \frac{2}{1+r^2} \right)^d,$$

as required.  $\square$

### 4.2.2 Some more ingredients from probability theory

For  $\bar{p} = (p_1, \dots, p_n) \in (0, 1)^n$  the notation  $\mathbb{P}_{\bar{p}}(\cdot)$  will signify the situation where  $X_1, \dots, X_n$  are independent random variables with  $X_i \stackrel{d}{=} \text{Be}(p_i)$ . Observe that, for every  $A \subseteq \{0, 1\}^n$ , the probability  $\mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A]$  can be written as a polynomial in  $p_1, \dots, p_n$ . In particular, this probability is a continuous function of the  $p_i$ -s and the partial derivatives  $\frac{\partial}{\partial p_i} \mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A]$  exist. Recall that a set  $A \subseteq \{0, 1\}^n$  is called an up-set if  $a \in A$  implies that also  $(a_1, \dots, a_{i-1}, 1, a_{i+1}, \dots, a_n) \in A$  for all  $1 \leq i \leq n$ . In other words, if we change a zero coordinate into a one then we do not leave the set  $A$ . We remark that if  $A$  is an up-set then  $\mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A]$  is non-decreasing in each parameter  $p_i$ .

The following result from [82] can be considered as an asymmetric version of Bourgain's powerful sharp threshold result (that appeared in the appendix of Friedgut's influential paper [39]).

**Proposition 4.4** ([82]). *For every  $C > 0$  and  $0 < \varepsilon < 1/2$  there exist  $K = K(C, \varepsilon) \in \mathbb{N}$ ,  $c = c(C, \varepsilon) > 0$  such that the following holds, for every  $n \in \mathbb{N}$  and every up-set  $A \subseteq \{0, 1\}^n$ .*

*If  $\bar{p} \in (0, 1)^n$  is such that  $\mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A] \in (\varepsilon, 1 - \varepsilon)$  and*

$$\sum_{i=1}^n p_i(1-p_i) \cdot \frac{\partial}{\partial p_i} \mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A] \leq C,$$

then there exist indices  $i_1, \dots, i_K \in \{1, \dots, n\}$  such that one of the following holds:

$$(a) \quad \mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A \mid X_{i_1} = \dots = X_{i_K} = 1] \geq \mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A] + c,$$

or

$$(b) \quad \mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A \mid X_{i_1} = \dots = X_{i_K} = 0] \leq \mathbb{P}_{\bar{p}}[(X_1, \dots, X_n) \in A] - c.$$

For  $m \in \mathbb{N}$  and  $x \in \mathbb{R}$  we will write  $(x)_m := x(x-1)\dots(x-m+1)$  for the *falling factorial*. For a proof of the following classical result see e.g. Theorem 1.22 in [17].

**Lemma 4.5.** *Assume that for some fixed  $\mu > 0$  the sequence of random variables  $X_1, X_2, \dots$  satisfy for all  $m \in \mathbb{N}$*

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n)_m = \mu^m.$$

*Then,  $X_n$  converges to  $\text{Po}(\mu)$  in distribution as  $n \rightarrow \infty$ .*

### 4.2.3 Preliminaries on (random) Borsuk graphs

If the dimension plays a role, we will write  $G_d(n, \alpha)$  instead of  $G(n, \alpha)$ . For  $V \subseteq S^d$ , we denote by  $G(V, \alpha)$  the subgraph of the Borsuk graph (with parameter  $\alpha$ ) induced by the vertices  $V$ . Throughout the rest of the chapter, we let  $X_1, X_2, \dots$  be i.i.d. uniform on  $S^d$  and let  $N \stackrel{d}{=} \text{Po}(n)$  be a Poisson distributed random variable, independent of the  $X_i$ -s. We consider the Poissonized Borsuk graph  $G(\mathcal{X}, \alpha)$  where  $\mathcal{X} = \{X_1, \dots, X_N\}$ . It suffices to prove Theorems 1.7, 1.8 and 1.10 for  $G(\mathcal{X}, \alpha)$  instead of  $G(n, \alpha) = G(\{X_1, \dots, X_n\}, \alpha)$  because:

**Lemma 4.6.** *For every  $k \in \mathbb{N}$ , we have*

$$(i) \quad \text{If } \mathbb{P}_n(\chi(G(\mathcal{X}, \alpha)) \geq k) = 1 - o_n(1) \text{ then also } \mathbb{P}(\chi(G(n, \alpha)) \geq k) = 1 - o_n(1),$$

and;

$$(ii) \quad \text{If } \mathbb{P}_n(\chi(G(\mathcal{X}, \alpha)) \geq k) = o_n(1) \text{ then also } \mathbb{P}(\chi(G(n, \alpha)) \geq k) = o_n(1).$$

*Proof.* By obvious monotonicity, we have

$$\begin{aligned} \mathbb{P}_n(\chi(G(\mathcal{X}, \alpha)) \geq k) &= \sum_m \mathbb{P}(\chi(G(\mathcal{X}, \alpha)) \geq k \mid N = m) \mathbb{P}(N = m) \\ &= \sum_m \mathbb{P}(\chi(G(m, \alpha)) \geq k) \mathbb{P}(N = m) \\ &\leq \mathbb{P}(\chi(G(n, \alpha)) \geq k) \cdot \mathbb{P}(N \leq n) + \mathbb{P}(N > n) \\ &= \mathbb{P}(\chi(G(n, \alpha)) \geq k) \cdot (1/2 + o_n(1)) + 1/2 + o_n(1), \end{aligned}$$

where we use  $\mathbb{P}(N \leq n) = 1/2 + o_n(1)$ , which follows for instance from the central limit theorem. The first item follows. For the second item, we similarly have

$$\begin{aligned} \mathbb{P}_n(\chi(G(\mathcal{X}, \alpha)) \geq k) &\geq \sum_{m \geq n} \mathbb{P}(\chi(G(m, \alpha)) \geq k) \mathbb{P}(N = m) \\ &\geq \mathbb{P}(\chi(G(n, \alpha)) \geq k) \cdot \mathbb{P}(N \geq n) \\ &= \mathbb{P}(\chi(G(n, \alpha)) \geq k) \cdot (1/2 + o_n(1)). \end{aligned}$$

□

As a side remark we point out that the lemma and its proof actually hold if we replace “chromatic number at least  $k$ ” with any graph property that is “preserved under taking supergraphs” (i.e. if we arbitrarily add vertices and/or edges to a graph having the property then the resulting graph also has the property). We shall make use of the following observation due to Kahle and Martinez-Figueroa [62].

**Lemma 4.7** ([62]). *If  $\bigcup_{v \in V} \text{cap}(v, \alpha/2) = S^d$  then  $\chi(G(V, \alpha)) \geq d + 2$ .*

This result is not stated as a separate lemma or theorem in [62], but is proved integral to the proof of Theorem 1.1. The proof in [62] is also phrased in terms of a different – but equivalent – definition for the Borsuk graph from the one we employ. Since the computations translating between the two settings take more mental effort than the proof itself and the proof is short and sweet, we include it for the benefit of the reader.

*Proof.* Suppose that  $G(V, \alpha)$  can be coloured with  $d + 1$  colours. Then we can write  $V = V_1 \cup \dots \cup V_{d+1}$  such that there are no edges with both endpoints in the same  $V_i$ . Let us set

$$U_i := \bigcup_{v \in V_i} \text{cap}(v, \alpha/2).$$

By a version of the Lusternik-Schnirelmann theorem for open sets (see e.g. [73], Theorem 2.1.1.) there exist  $1 \leq i \leq d + 1$  and  $u \in S^d$  such that  $u, -u \in U_i$ . By definition of  $U_i$ , there are  $v, w \in V_i$  such that  $u \in \text{cap}(v, \alpha/2)$ ,  $-u \in \text{cap}(w, \alpha/2)$ . The triangle inequality for the geodesic distance gives:

$$\text{dist}(v, w) \geq \text{dist}(u, -u) - \text{dist}(u, v) - \text{dist}(-u, w) > \pi - \alpha,$$

so that  $vw$  must be an edge of  $G(V, \alpha)$ , contradicting the choice of  $V_1, \dots, V_{d+1}$ . It follows there cannot be a colouring of  $G(V, \alpha)$  in  $d + 1$  colours.  $\square$

## 4.3 When having $d$ colors is not enough

Here we will derive the proof of Theorem 1.7. Rather than working directly on the  $d$ -sphere, we find it conceptually easier to work in  $\mathbb{R}^d$ . We let  $\mathcal{X} = \{X_1, \dots, X_N\}$  be as above and set

$$\mathcal{Z} := \pi[\mathcal{X}] = \{Z_1, \dots, Z_N\}.$$

We say a function  $f : S^{d-1} \rightarrow \mathbb{R}$  is (spherically) *odd* if  $f(u) + f(-u) = 0$  for all  $u \in S^{d-1}$ . It is  $\varepsilon$ -*Lipschitz* if  $|f(u) - f(v)| \leq \varepsilon \cdot \|u - v\|$  for all  $u, v \in S^{d-1}$ .

**Lemma 4.8.** *For every  $\varepsilon > 0$  there exists a  $c = c(\varepsilon) > 0$  such that the following holds, setting  $\alpha = c \cdot n^{-1/d}$ . With probability  $1 - o_n(1)$ , there exists a function*

$$h : S^{d-1} \rightarrow \left(-n^{-0.9/d}, n^{-0.9/d}\right),$$

*that is odd,  $\varepsilon$ -Lipschitz and*

$$\mathcal{Z} \cap B((1 + h(u)) \cdot u, \varepsilon \cdot \alpha) \neq \emptyset,$$

*for all  $u \in S^{d-1}$ .*

The lemma and its proof are inspired by a clever proof technique in the recent paper [11] (Sections 3, 4 and 5).

### 4.3.1 The proof of Lemma 4.8.

We fix constants  $\delta = \delta(\varepsilon)$ ,  $K = K(\varepsilon)$  to be chosen more precisely over the course of our arguments. The constant  $\delta$  will be a small positive real and  $K$  will be a large integer. We dissect  $\mathbb{R}^d$  into axis-parallel hypercubes of side-length

$$s_0 := \delta \cdot \alpha,$$

in the obvious way. These cubes are called *level-0 cubes*. A level-1 cube is a hypercube of side length  $s_1 := K \cdot s_0$  that is the union of a group of  $K \times \cdots \times K$  level-0 cubes. For each  $i \geq 1$ , having defined the level  $(i-1)$ -cubes, a level- $i$  cube is a hypercube of side length

$$s_i := K^i \cdot s_{i-1} = K^{i+(i-1)+\cdots+1} \cdot s_0 = K^{i(i+1)/2} \cdot s_0, \quad (4.2)$$

that is the union of a group of  $K^i \times \cdots \times K^i$  level- $(i-1)$  cubes.

Specifically, the set of  $i$ -cubes is

$$\mathcal{Q}_i := \{p + [0, s_i]^d : p \in s_i \mathbb{Z}^d\}.$$

We say a level-0 cube  $q \in \mathcal{Q}_0$  is *good* if  $q \cap \mathcal{Z} \neq \emptyset$ . Otherwise it is *bad*. For each  $i \geq 1$ , each level- $i$  cube  $q \in \mathcal{Q}_i$  is the union of  $(s_i/s_{i-1})^d = K^{di}$  level- $(i-1)$  cubes. The cube  $q \in \mathcal{Q}_i$  is called *good* if at most one of these level- $(i-1)$  cubes is bad. Otherwise, if at least two of these level- $(i-1)$  cubes are bad, then  $q$  is bad.

**Lemma 4.9.** *For every  $K, \delta$  there is a  $c = c(K, \delta)$  such that, setting  $\alpha := c \cdot n^{-1/d}$ ,  $s_0 := \delta \cdot \alpha$ , etc., as above, for every  $i \geq 2 \log \log n$ , with probability  $1 - o_n(1)$ , every level- $i$  cube  $q \in \mathcal{Q}_i$  such that  $q \subseteq B(\underline{0}, 100)$  is good.*

*Proof.* By Lemma 4.3 and the mapping theorem for Poisson processes (see e.g. [65], Section 2.3),  $\mathcal{Z}$  is a Poisson point process on  $\mathbb{R}^d$  with intensity function

$$f(x) := n \cdot \frac{1}{(d+1)\kappa_{d+1}} \cdot \left( \frac{2}{1 + \|x\|^2} \right)^d.$$

It follows that each level-0 cube  $q \in \mathcal{Q}_0$  that is contained in  $B(\underline{0}, 100)$  is bad with probability at most

$$\begin{aligned} \mathbb{P}(q \text{ is bad}) &\leq \exp \left[ -s_0^d \cdot n \cdot \frac{1}{(d+1)\kappa_{d+1}} \cdot \left( \frac{2}{10001} \right)^d \right] \\ &= \exp \left[ -\frac{1}{(d+1)\kappa_{d+1}} \cdot \left( \frac{2}{10001} \right)^d \cdot \delta^d \cdot c^d \right] \\ &=: p_0. \end{aligned} \tag{4.3}$$

We point out that by choosing  $c$  large enough, we can get  $p_0$  to be as close to zero as we like. Applying the union bound, we see that for every  $q \in \mathcal{Q}_1$  that is contained in  $B(\underline{0}, 100)$ , we have

$$\mathbb{P}(q \text{ is bad}) \leq \binom{(s_1/s_0)^d}{2} p_0^2 \leq (s_1/s_0)^{2d} p_0^2 = K^{2d} p_0^2 =: p_1.$$

Repeating the argument, we obtain upper bounds  $p_2, p_3, \dots$  for the probability that a cube  $q \in \mathcal{Q}_i$  that is contained in  $B(\underline{0}, 100)$  is bad, satisfying the recursion:

$$p_i = (s_i/s_{i-1})^{2d} \cdot p_{i-1}^2 = K^{2di} p_{i-1}^2.$$

Iterating (feeding the recursion back into itself) we find

$$p_i = (K^{2d})^{i+2(i-1)+4(i-2)+\dots+2^{i-1}} \cdot p_0^{2^i}.$$

The exponent of  $K^{2d}$  can be written as:

$$\sum_{j=0}^i 2^j \cdot (i-j) = 2^i \sum_{j=0}^i (1/2)^{i-j} \cdot (i-j) \leq 2^i \cdot \left( \sum_{k \geq 0} k 2^{-k} \right) = 2^{i+1}.$$

We see that

$$p_i \leq (K^{4d} \cdot p_0)^{2^i} \leq 2^{-2^i},$$

where the last inequality holds provided we chose  $c = c(K, \delta)$  sufficiently large. (We use the remark following (4.3).)

Let  $N_i := |\{q \in \mathcal{Q}_i : q \subseteq B(\underline{0}, 100)\}|$  denote the number of level- $i$ -cubes that are contained in  $B(\underline{0}, 100)$ . We have

$$N_i \leq \frac{\text{vol}(B(\underline{0}, 100))}{s_i^d} \leq \frac{\text{vol}(B(\underline{0}, 100))}{s_0^d} = O(n).$$

When  $i \geq 2 \log \log n$  then

$$p_i \leq 2^{-2^{2 \log \log n}} = 2^{-\log^2 n} = o(1/n).$$

So, by the union bound, for  $i \geq 2 \log \log n$  we have

$$\mathbb{P}(\text{There is a bad level-}i \text{ cube contained in } B(0, 100)) \leq N_i p_i = o_n(1).$$

This concludes the proof of Lemma 4.9.  $\square$

The rough idea for the proof of Lemma 4.8 is to massage an (odd and appropriately Lipschitz) function that avoids bad  $(j+1)$ -cubes into one that avoids bad  $j$ -cubes without making drastic local changes, and then repeat until finally we avoid bad 0-cubes. The following lemma is the main tool we shall be using for this purpose.

**Lemma 4.10.** *For every  $d \geq 2$  and  $k \geq 1$ , there exist  $t = t(d, k)$ ,  $a_0 = a_0(d, k)$ ,  $r_0 = r_0(d, k)$ ,  $C = C(d, k) > 0$  such that the following holds. Suppose that  $0 < a < a_0$ ,  $0 < r < r_0$ , the finite set  $V \subseteq \mathbb{R}^d$  and the function  $f : S^{d-1} \rightarrow (-t, t)$  are such that  $f$  is odd and  $a$ -Lipschitz and moreover*

$$|B((1 + f(u)) \cdot u, r) \cap V| \leq k,$$

*for all  $u \in S^{d-1}$ . Then there also exists a function  $g : S^{d-1} \rightarrow \mathbb{R}$  that is odd,  $(Ca)$ -Lipschitz and satisfies*

$$|f(u) - g(u)| < Car \quad \text{and} \quad |B((1 + g(u)) \cdot u, a^C \cdot r) \cap V| = 0,$$

*for all  $u \in S^{d-1}$ .*

We find it convenient to first prove a preliminary version of the lemma with a restriction on the set  $V$  and with  $k = 1$ .

**Lemma 4.11.** *Suppose that  $0 < a < 0.01$ ,  $0 < r < 0.01/\sqrt{d}$ , the finite set  $V \subseteq [0, \infty)^d$  and the function  $f : S^{d-1} \rightarrow (-0.1, 0.1)$  are such that  $f$  is odd and  $a$ -Lipschitz and moreover*

$$|B((1 + f(u)) \cdot u, r) \cap V| \leq 1,$$

*for all  $u \in S^{d-1}$ . Then there also exists a function  $g : S^{d-1} \rightarrow \mathbb{R}$  that is odd,  $(9a)$ -Lipschitz and satisfies*

$$|f(u) - g(u)| < ar \quad \text{and} \quad |B((1 + g(u)) \cdot u, (a/2) \cdot r) \cap V| = 0,$$

*for all  $u \in S^{d-1}$ .*

*Proof.* Without loss of generality we can and do assume  $\underline{0} \notin V$ . (If we obtain the claimed  $g$  wrt.  $V \setminus \{\underline{0}\}$  then automatically  $\underline{0} \notin B((1+g(u)) \cdot u, (a/2) \cdot r)$  for all  $u$ .)

For notational convenience, let us write  $\varphi(u) := (1+f(u)) \cdot u$ . For each  $p \in V$  we define:

$$f_p(u) := b_p \cdot (\max(0, r - \|\varphi(u) - p\|) - \max(0, r - \|\varphi(-u) - p\|)),$$

where

$$b_p := \begin{cases} a & \text{if } \|p\| < 1 + f(p/\|p\|), \\ -a & \text{otherwise.} \end{cases}$$

We now set

$$g := f + \sum_{p \in V} f_p.$$

Since  $f_p(u) + f_p(-u) = 0$  for all  $p$  by construction, we clearly have  $g(u) + g(-u) = f(u) + f(-u) = 0$  for all  $u$ . So  $g$  is odd as required.

Next, we point out:

$$\text{The } f_p \text{ have disjoint supports.} \quad (4.4)$$

*Proof of (4.4).* Aiming for a contradiction, suppose  $u \in S^{d-1}$  and  $p \neq q \in V$  are such that  $f_p(u) \neq 0$  and  $f_q(u) \neq 0$ . We have either **a-1**)  $\|\varphi(u) - p\| < r$  or **a-2**)  $\|\varphi(-u) - p\| < r$ , and either **b-1**)  $\|\varphi(u) - q\| < r$  or **b-2**)  $\|\varphi(-u) - q\| < r$ . Swapping  $u$  for  $-u$  if needed, we can and do assume without loss of generality that **a-1**) holds. The condition **b-1**) cannot simultaneously hold, because otherwise  $|B(\varphi(u), r) \cap V| \geq 2$ . So **b-2**) must hold.

There is a coordinate  $1 \leq i \leq d$  for which  $|\varphi_i(u)| = (1+f(u)) \cdot |u_i| \geq 0.9/\sqrt{d}$ . As  $p \in [0, \infty)^d$ , we in fact have  $\varphi_i(u) = |\varphi_i(u)| \geq 0.9/\sqrt{d}$ , because otherwise

$$\|\varphi(u) - p\| \geq |\varphi_i(u) - p_i| \geq 0.9/\sqrt{d} > r.$$

This implies that  $\varphi_i(-u) = -|\varphi_i(u)| \leq -0.9/\sqrt{d}$  and, since  $q \in [0, \infty)^d$ , we find

$$\|\varphi(-u) - q\| \geq |\varphi_i(-u) - q_i| \geq 0.9/\sqrt{d} > r,$$

contradicting **b-2**). So  $u, p, q$  satisfying  $f_p(u) \neq 0, f_q(u) \neq 0$  cannot exist, as was to be shown.  $\square$

Having established (4.4), it immediately follows that  $|g(u) - f(u)| \leq ar$  for all  $u \in S^{d-1}$ .

Now let  $u, v \in S^{d-1}$  be arbitrary. By (4.4), there are  $p, q \in V$  such that  $g(u) = f(u) + f_p(u)$  and  $g(v) = f(v) + f_q(v)$ . It follows that

$$|g(u) - g(v)| \leq |f(u) - f(v)| + |f_p(u) - f_p(v)| + |f_q(u) - f_q(v)|. \quad (4.5)$$

(If  $p = q$  it is trivially true and if  $p \neq q$  then  $f_p(v) = f_q(u) = 0$ .) By definition of  $f_p$ :

$$\begin{aligned} |f_p(u) - f_p(v)| &\leq a \cdot (\|\varphi(u) - p\| - \|\varphi(v) - p\| + \|\varphi(-u) - p\| - \|\varphi(-v) - p\|) \\ &\leq a \cdot (\|\varphi(u) - \varphi(v)\| + \|\varphi(-u) - \varphi(-v)\|), \end{aligned} \quad (4.6)$$

where we've used the triangle inequality for the second line. We also have

$$\begin{aligned} \|\varphi(u) - \varphi(v)\| &\leq \|(1 + f(u))u - (1 + f(u))v\| + \|(1 + f(v))v - (1 + f(u))v\| \\ &= (1 + f(u)) \cdot \|u - v\| + |f(v) - f(u)| \\ &\leq 1.1 \cdot \|u - v\| + a \cdot \|u - v\| \\ &\leq 2 \cdot \|u - v\|. \end{aligned}$$

Filling this into (4.6) gives

$$|f_p(u) - f_p(v)| \leq 4a \cdot \|u - v\|.$$

Repeating the argument, the same bound applies to  $|f_q(u) - f_q(v)|$ . Combining with (4.5) and recalling that  $f$  is  $a$ -Lipschitz gives

$$|g(u) - g(v)| \leq 9a \cdot \|u - v\|.$$

In other words, we've verified that  $g$  is  $9a$ -Lipschitz.

For notational convenience, we write  $\psi(u) := (1 + g(u)) \cdot u$ . It remains to see that  $B(\psi(u), ar/2) \cap V = \emptyset$  for all  $u \in S^{d-1}$ . To this end, let  $u \in S^{d-1}$  be arbitrary. If  $u$  does not lie in the support of any  $f_p$  then  $\|\psi(u) - p\| = \|\varphi(u) - p\| \geq r$  for any  $p \in V$ . We can thus assume  $u$  lies in the support of some (unique)  $p \in V$ . For  $q \neq p$  we have

$$\begin{aligned} \|\psi(u) - q\| &\geq \|\varphi(u) - q\| - \|\psi(u) - \varphi(u)\| = \|\varphi(u) - q\| - |g(u) - f(u)| \\ &> r - ar = (1 - a)r > ar/2. \end{aligned}$$

It thus remains to consider the distance between  $\psi(u)$  and  $p$ . If  $\|\varphi(-u) - p\| < r$  then

$$\|\varphi(u) - p\| \geq \|\varphi(u) - \varphi(-u)\| - \|\varphi(-u) - p\| = 2 - \|\varphi(-u) - p\| > 2 - r.$$

Hence

$$\begin{aligned}\|\psi(u) - p\| &\geq \|\varphi(u) - p\| - \|\psi(u) - \varphi(u)\| = \|\varphi(u) - p\| - |f(u) - g(u)| \\ &> 2 - r - ar > ar/2.\end{aligned}$$

So we can assume that  $\|\varphi(u) - p\| < r$ . We denote  $v := p/\|p\|$  and  $\alpha := \angle v \underline{0} u$ . If  $\alpha \geq \pi/2$  then

$$\|\psi(u) - p\| \geq \|\psi(u)\| = (1 + g(u)) > 0.9 - ar > ar/2.$$

We can therefore assume  $0 \leq \alpha \leq \pi/2$ . By considering the distance between  $\psi(u)$  and the linear hull  $\text{span}(\{p\})$  of  $p$ , we see that

$$\|\psi(u) - p\| \geq (1 + g(u)) \cdot \sin \alpha \geq (0.9 - ar) \cdot (2/\pi) \cdot \alpha > \alpha/4,$$

(where we use that  $\sin \alpha \geq (2/\pi) \cdot \alpha$  for  $0 \leq \alpha \leq \pi/2$ ). So if  $\alpha \geq 2ar$  then  $\|\psi(u) - p\| \geq ar/2$ . We therefore assume  $0 \leq \alpha \leq 2ar$ . We have

$$\begin{aligned}\|\psi(u) - p\| &\geq \|p - \psi(v)\| - \|\psi(u) - \psi(v)\| \geq \|p - \psi(v)\| - 9a \cdot \|u - v\| \\ &= \|p - \psi(v)\| - 9a \cdot 2 \sin(\alpha/2) \geq \|p - \psi(v)\| - 9a \cdot \alpha \\ &\geq \|p - \psi(v)\| - 18a^2 r.\end{aligned}$$

(In the fourth line we use that  $\sin x \leq x$  for  $x \geq 0$ .) Because of the way we have chosen the sign of  $b_p$  in the definition of  $f_p$ , we find that

$$\begin{aligned}\|p - \psi(v)\| &= \||p\| - (1 + g(v))\| = \||p\| - (1 + f(v))\| + |f(v) - g(v)| \\ &= \|p - \varphi(v)\| + a(r - \|p - \varphi(v)\|) = ar + (1 - a)\|p - \varphi(v)\| \\ &\geq ar.\end{aligned}$$

It follows that

$$\|p - \psi(u)\| \geq ar - 18a^2 r > ar/2.$$

We have now verified that  $B(\psi(u), ar/2) \cap V = \emptyset$ , which concludes the proof of the lemma.  $\square$

We separate one more observation to facilitate the derivation of the more general Lemma 4.10 from the more restricted Lemma 4.11. Recall that an *orthant* of  $\mathbb{R}^d$  is one of the  $2^d$  subsets  $O_\sigma := \{x \in \mathbb{R}^d : x_i \sigma_i \geq 0\}$  with  $\sigma \in \{-1, +1\}^d$ .

**Lemma 4.12.** *For all  $d \geq 2, k \geq 1$  there exists a  $\ell = \ell(d, k)$  such that the following holds. Suppose  $r > 0$  and  $U, V \subseteq \mathbb{R}^d$  are such that  $V$  is finite and  $|B(u, r) \cap V| \leq k$  for all  $u \in U$ . Then we can write  $V$  as*

$$V = V_1 \cup \dots \cup V_\ell,$$

where each  $V_i$  is contained in some orthant and  $|B(u, r/4) \cap V_i| \leq 1$  for all  $u \in U$  and  $1 \leq i \leq \ell$ .

*Proof.* We set

$$W := \{v \in V : \|v - u\| < r/2 \text{ for some } u \in U\}.$$

We have  $|B(w, r/2) \cap W| \leq k$  for all  $w \in W$ , since  $B(w, r/2) \subseteq B(u, r)$  for some  $u \in U$ .

Let  $W_1 \subseteq W$  be a subset such that if  $w, w' \in W_1$  are distinct then  $\|w - w'\| \geq r/2$  and, subject to this, it has the maximum possible size. For  $2 \leq i \leq k-1$  we similarly pick

$$W_i \subseteq W \setminus (W_1 \cup \dots \cup W_{i-1}),$$

of the maximum possible size subject to the demand that  $\|w - w'\| \geq r/2$  for all distinct  $w, w' \in W_i$ . In particular

$$|B(u, r/4) \cap W_i| \leq 1.$$

(Because the diameter of  $B(u, r/4)$  is  $r/2$ .)

For each  $w \in W \setminus W_1$  we must have

$$|B(w, r/2) \cap (W \setminus W_1)| \leq k-1.$$

(Otherwise  $B(w, r/2)$  does not intersect  $W_1$  and  $w$  could have been added to  $W_1$ , contradicting maximality of  $|W_1|$ .) Repeating the argument, we find

$$|B(w, r/2) \cap (W \setminus (W_1 \cup \dots \cup W_{i-1}))| \leq k-i+1, \quad (4.7)$$

for all  $i$  and all  $w \in W \setminus (W_1 \cup \dots \cup W_{i-1})$ . In particular we have

$$W = W_1 \cup \dots \cup W_{k-1}.$$

(By (4.7) and the maximality of  $|W_{k-1}|$ , we must have  $W_{k-1} = W \setminus (W_1 \cup \dots \cup W_{k-2})$ .) Writing

$$W_k := V \setminus W,$$

we thus have  $V = W_1 \cup \dots \cup W_k$  and  $|B(u, r/4) \cap W_i| \leq 1$  for all  $1 \leq i \leq k$  and all  $u \in U$ .

We label the  $2^d$  orthants of  $\mathbb{R}^d$  arbitrarily as  $O_1, \dots, O_{2^d}$  and now define  $V_1, \dots, V_{k2^d}$  by:

$$V_{2^d \cdot (i-1) + j} := W_i \cap O_j,$$

for  $1 \leq i \leq k$  and  $1 \leq j \leq 2^d$ . Setting  $\ell := k2^d$ , we have constructed the sought decomposition  $V = V_1 \cup \dots \cup V_\ell$ .  $\square$

*Proof of Lemma 4.10.* Suppose that  $t, a, r, f : S^{d-1} \rightarrow (-t, t)$  and  $V \subseteq \mathbb{R}^d$  are such that  $f$  is odd and  $a$ -Lipschitz and  $|B((1 + f(u))u, r) \cap V| \leq k$  for all  $u$ . By the previous lemma, for some constant  $\ell = \ell(d, k)$ , we can write  $V = V_1 \cup \dots \cup V_\ell$  where each  $V_i$  is contained in some orthant and  $|B((1 + f(u))u, r/4) \cap V_i| \leq 1$  for all  $u \in S^{d-1}$ ,  $1 \leq i \leq \ell$ .

We now “iterate” Lemma 4.11. That is, we do the following. We set  $g_0 := f$ ,  $t_0 := t$ ,  $a_0 := a$ ,  $r_0 := r/4$ . Suppose that, for some  $0 \leq i \leq \ell - 1$ , we have defined  $g_i, t_i, a_i, r_i$  such that  $g_i : S^{d-1} \rightarrow (-t_i, t_i)$  is odd,  $a_i$ -Lipschitz and

$$|B((1 + g_i(u))u, r_i) \cap V_j| = 0 \text{ for } j \leq i, \text{ and}$$

$$|B((1 + g_i(u))u, r_i) \cap V_j| \leq 1 \text{ for } j > i,$$

for all  $u \in S^{d-1}$ . Provided  $t_i < 0.1$ ,  $a_i < 0.01$  and  $r_i < 0.01/\sqrt{d}$ , we can apply Lemma 4.11 (and symmetry) to  $g_i$  and  $V_{i+1}$  to obtain  $g_{i+1}$ . Setting

$$a_{i+1} = 9a_i, \quad r_{i+1} = (a_i/2) \cdot r_i, \quad \Delta_{i+1} := a_i r_i, \quad t_{i+1} = t_i + \Delta_{i+1},$$

we have that  $g_{i+1} : S^{d-1} \rightarrow (-t_{i+1}, t_{i+1})$  is odd,  $a_{i+1}$ -Lipschitz and satisfies

$$|g_{i+1}(u) - g_i(u)| \leq \Delta_{i+1}, \quad B((1 + g_{i+1}(u))u, r_{i+1}) \cap V_{i+1} = \emptyset,$$

for all  $u \in S^{d-1}$ . We have  $\Delta_{i+1} + r_{i+1} = (3/2)a_i r_i < r_i$ , which implies

$$B((1 + g_{i+1}(u))u, r_{i+1}) \subseteq B((1 + g_i(u))u, r_i) \tag{4.8}$$

for all  $u \in S^{d-1}$ . In particular

$$|B((1 + g_{i+1}(u))u, r_{i+1}) \cap V_j| = 0 \text{ for } j \leq i + 1, \text{ and}$$

$$|B((1 + g_{i+1}(u))u, r_{i+1}) \cap V_j| \leq 1 \text{ for } j > i + 1,$$

The function  $g = g_\ell$  is the one we seek. It is well defined provided the conditions of Lemma 4.11 were met each time we applied it. That is, provided  $t, t_1, \dots, t_\ell < 0.1$  and  $a, a_1, \dots, a_\ell < 0.01$  and  $r, r_1, \dots, r_\ell < 0.01/\sqrt{d}$ . We note that

$$\begin{aligned}
a_i &= 9a_{i-1} = \cdots = 9^i \cdot a, \\
r_i &= (a_{i-1}/2)r_{i-1} = \cdots = (a_{i-1} \cdots a_1 \cdot a) \cdot 2^{-i} \cdot r_0 \\
&= 9^{i(i-1)/2} \cdot 2^{-(i+2)} \cdot a^i \cdot r = 3^{i(i-1)} \cdot 2^{-(i+2)} \cdot a^i \cdot r, \\
\Delta_i &= a_{i-1}r_{i-1} = 3^{i(i-1)} \cdot 2^{-(i+3)} \cdot a^i \cdot r,
\end{aligned}$$

This gives

$$\begin{aligned}
\Delta_1 + \cdots + \Delta_i &\leq i \cdot 3^{i(i-1)} \cdot a \cdot r \leq 3^{i^2} \cdot a \cdot r, \\
t_i &= t + \Delta_1 + \cdots + \Delta_i \leq t + 3^{i^2} \cdot a \cdot r.
\end{aligned}$$

Setting  $C := 3^{\ell^2}$ , the function  $g$  is clearly well-defined if  $a < 0.01/C, r < 0.01/(C\sqrt{d}), t < 0.05$ , with plenty of room to spare.

To see that our choice of  $g$  and  $C$  is as required, we note that  $|f(u) - g(u)| \leq \Delta_1 + \cdots + \Delta_\ell \leq Car$  for all  $u \in S^{d-1}$ , that  $a_\ell < Ca$  so that  $a_\ell$ -Lipschitz implies  $Ca$ -Lipschitz, and similarly  $r_\ell > a^C r$  so that  $B((1+g(u))u, r_\ell) \supseteq B((1+g(u))u, a^C r)$  for all  $u \in S^{d-1}$ .  $\square$

*Proof of Lemma 4.8.* We apply Lemma 4.9 with parameters  $K = K(\varepsilon), \delta = \delta(\varepsilon)$  to be determined during the course of the proof. For the remainder of the proof, we assume that  $\mathcal{Z}$  is such that the conclusion of Lemma 4.9 holds. (Which happens with probability  $1 - o_n(1)$ .) Let  $C = C(d, 2^d), t = t(d, 2^d), a_0 = a_0(d, 2^d), r_0 = r_0(d, 2^d)$  be as provided by Lemma 4.10 applied in dimension  $d$  with parameter  $k = 2^d$ .

For  $0 \leq j \leq i := \lceil 2 \log \log n \rceil$ , we set

$$\mathcal{B}_j := \{p \in s_j \mathbb{Z}^d : p + [0, s_j]^d \text{ is bad and contained in } B(\underline{0}, 100)\}.$$

We construct the sought function  $h$  by reverse induction. Specifically, for  $0 \leq j \leq i$ , the function  $h_j : S^{d-1} \rightarrow \mathbb{R}$  will be odd,  $(\varepsilon \cdot C^{-j})$ -Lipschitz, and satisfy:

$$|h_j(u)| \leq s_{j+1} + s_{j+2} + \cdots + s_i, \quad (4.9)$$

and

$$B\left((1 + h_j(u)) \cdot u, 2\sqrt{d} \cdot s_j\right) \cap \mathcal{B}_j = \emptyset, \quad (4.10)$$

for all  $u \in S^{d-1}$ .

Let us point out that

$$\begin{aligned}
s_0 + s_1 + \cdots + s_i &\leq (i+1)s_i = (i+1)K^{i(i+1)/2}s_0 = e^{O((\log \log n)^2)} \cdot n^{-1/d} \\
&= n^{-1/d+o(1)}.
\end{aligned} \tag{4.11}$$

In particular demand (4.9) implies that

$$h_j(u) \in \left(-n^{-0.9/d}, n^{-0.9/d}\right),$$

for  $n$  sufficiently large. Demand (4.10) implies that  $(1 + h_0(u)) \cdot u$  lies in a good level-0 cube. In particular

$$B((1 + h_0(u)) \cdot u, \sqrt{d} \cdot s_0) \cap \mathcal{Z} \neq \emptyset.$$

Provided  $\delta$  was sufficiently small,  $\sqrt{d} \cdot s_0 = \sqrt{d} \cdot \delta \cdot \alpha < \varepsilon \cdot \alpha$ . So the function  $h_0$  can serve as the function  $h$  we seek.

As mentioned, we shall establish the existence of  $h_0$  by reverse induction. We start by defining  $h_i$ . Clearly  $h_i \equiv 0$  does the trick, as  $\mathcal{B}_i = \emptyset$ . Now suppose that we've managed to construct  $h_{j+1}$  for some  $j < i$ .

Before proceeding, let us point out that if  $h_{j+1}$  satisfies the demands, we have:

$$\left| B\left((1 + h_{j+1}(u)) \cdot u, \frac{s_{j+1}}{2}\right) \cap \mathcal{B}_j \right| \leq 2^d \quad \text{for all } u \in S^{d-1}. \tag{4.12}$$

(Because  $B((1 + h_{j+1}(u)) \cdot u, s_{j+1}/2)$  intersects at most  $2^d$  level  $j+1$  cubes. None of these can be bad, because that would contradict (4.10).) We can thus obtain  $h_j$  by invoking Lemma 4.10 with

$$k = 2^d, \quad f = h_{j+1}, \quad V = \mathcal{B}_j, \quad a = \varepsilon \cdot C^{-(j+1)}, \quad r = s_{j+1}/2.$$

(For  $n$  sufficiently large, we are allowed to apply Lemma 4.10 because  $|h_{j+1}| < t$  and  $r < r_0$  by (4.11), and  $a < a_0$  assuming without loss of generality  $\varepsilon < a_0$ .) We note that

$$Ca = \varepsilon \cdot C^{-j},$$

so that  $h_j$  is  $(\varepsilon C^{-j})$ -Lipschitz as required. We also have

$$\begin{aligned}
a^C r &= (\varepsilon \cdot C^{-(j+1)})^C \cdot (s_{j+1}/2) \\
&= (\varepsilon \cdot C^{-(j+1)})^C \cdot \frac{1}{2} \cdot K^{j+1} \cdot s_j \\
&= (\varepsilon^C/2) \cdot (K/C^C)^{j+1} \cdot s_j \\
&> 2\sqrt{d} \cdot s_j,
\end{aligned}$$

where the last line follows because we chose  $K$  sufficiently large in the beginning. So

$$B\left((1 + h_{j+1}(u)) \cdot u, 2\sqrt{d} \cdot s_j\right) \cap \mathcal{B}_j \subseteq B\left((1 + h_{j+1}(u)) \cdot u, a^C r\right) \cap \mathcal{B}_j = \emptyset,$$

for all  $u \in S^{d-1}$ , as required. Also note that  $\Delta := Car = \varepsilon C^{-j} s_{j+1}/2 < s_{j+1}$ , giving

$$\begin{aligned} |h_j(u)| &\leq |h_{j+1}(u)| + |h_j(u) - h_{j+1}(u)| \leq (s_{j+2} + \cdots + s_i) + \Delta \\ &< s_{j+1} + s_{j+2} + \cdots + s_i. \end{aligned}$$

We see that  $h_j$  satisfies the demands provided that  $h_{j+1}$  does. This concludes the proof.  $\square$

### 4.3.2 Finishing the proof of Theorem 1.7.

We fix  $\varepsilon > 0$ , small and to be determined more precisely in what follows. For the remainder of the section, we assume that  $\mathcal{X}$  is such that the conclusion of Lemma 4.8 holds (which happens with probability  $1 - o_n(1)$ ) and we let  $\alpha$  and  $h$  be as provided by that lemma. Let us write

$$A := \bigcup_{u \in S^{d-1}} B((1 + h(u)) \cdot u, \varepsilon \cdot \alpha).$$

We arbitrarily label  $\mathcal{X} \cap \pi^{-1}[A]$  as

$$\mathcal{X} \cap \pi^{-1}[A] = \{X_1, \dots, X_M\},$$

and set

$$Z_i := \pi(X_i), \quad U_i := Z_i / \|Z_i\|.$$

Perhaps superfluously, we note that the  $U_i$  lie on the  $(d-1)$ -sphere  $S^{d-1}$ . Our construction of  $h, \alpha$  has the following consequence:

**Corollary 4.13.** *Provided the constant  $\varepsilon > 0$  was chosen sufficiently small and  $n$  is sufficiently large, we have:*

$$(i) \quad \bigcup_{i=1}^M \text{cap}_{S^{d-1}}(U_i, \alpha/10) = S^{d-1}.$$

$$(ii) \quad \text{If } \text{dist}_{S^{d-1}}(U_i, U_j) > \pi - \alpha/5 \text{ then } \text{dist}_{S^d}(X_i, X_j) > \pi - \alpha.$$

*Proof of Theorem 1.7 assuming Corollary 4.13.* Part (ii) of Corollary 4.13 shows that  $G_{d-1}(\{U_1, \dots, U_M\}, \alpha/5)$  is a subgraph of  $G_d(\mathcal{X}, \alpha)$ . We thus have

$$\chi(G_d(\mathcal{X}, \alpha)) \geq \chi(G_{d-1}(\{U_1, \dots, U_M\}, \alpha/5)) \geq d + 1,$$

where the last inequality follows by part (i) of Corollary 4.13 together with Lemma 4.7.  $\square$

It remains to verify Corollary 4.13. We find it convenient to separate out a few preliminary calculations before starting its proof in earnest.

**Lemma 4.14.** *Provided the constant  $\varepsilon > 0$  was chosen sufficiently small and  $n$  is sufficiently large, we have:*

$$\|(1 + h(u)) \cdot u - (1 + h(v)) \cdot v\| \leq 2 \cdot \|u - v\|,$$

for all  $u, v \in S^{d-1}$ .

*Proof.* We have

$$\begin{aligned} \|(1 + h(u))u - (1 + h(v))v\| &\leq (1 + h(u)) \cdot \|u - v\| + |h(v) - h(u)| \\ &\leq (1 + h(u)) \cdot \|u - v\| + \varepsilon \cdot \|u - v\| \\ &= (1 + h(u) + \varepsilon) \cdot \|u - v\| \\ &\leq 2 \cdot \|u - v\|, \end{aligned}$$

where we've used that  $h$  is  $\varepsilon$ -Lipschitz and the last line follows assuming  $\varepsilon$  is sufficiently small and  $n$  sufficiently large (recalling  $|h| \leq n^{-0.9/d}$ ).  $\square$

For  $z \in \mathbb{R}^d \setminus \{0\}$ , we write:

$$\varphi(z) := \pi(-\pi^{-1}(z)).$$

That is, if  $z = \pi(x)$  with  $x \in S^d$ , then  $\varphi(z)$  corresponds to the image of the antipode of  $x$  under the stereographic projection  $\pi$ . We remark that  $\varphi$  can be written alternatively as:

$$\varphi(z) = -z/\|z\|^2. \tag{4.13}$$

(An easy way to see this is to note that  $z, \varphi(z)$  must lie on a line through the origin, and on opposite sides of the origin, and if  $z = \pi(x)$  then

$$\begin{aligned} \|z\| &= \sqrt{(x_1^2 + \dots + x_d^2)/(1 - x_{d+1})^2} = \sqrt{(1 - x_{d+1}^2)/(1 - x_{d+1})^2} \\ &= \sqrt{(1 + x_{d+1})/(1 - x_{d+1})}, \end{aligned}$$

and similarly  $\varphi(z) = \pi(-x)$  has norm  $\|\varphi(z)\| = \sqrt{(1 - x_{d+1})/(1 + x_{d+1})} = 1/\|z\|$ .

**Lemma 4.15.** *Provided the constant  $\varepsilon > 0$  was chosen sufficiently small and  $n$  is sufficiently large, we have:*

$$\|\varphi((1 + h(u)) \cdot u) + (1 + h(-u))u\| \leq \varepsilon \cdot \alpha,$$

for all  $u \in S^{d-1}$ .

*Proof.* By (4.13), we have

$$\begin{aligned} \varphi((1 + h(u)) \cdot u) &= -(1/(1 + h(u))) \cdot u \\ &= -(1 - h(u) + h(u)^2/(1 + h(u))) \cdot u \\ &= -(1 + h(-u) + h(u)^2/(1 + h(u))) \cdot u, \end{aligned}$$

using that  $h$  is odd in the last step. So

$$\begin{aligned} \|\varphi((1 + h(u)) \cdot u) + (1 + h(-u))u\| &= \|(h(u)^2/(1 + h(u))) \cdot u\| \\ &= h(u)^2/(1 + h(u)) \\ &< n^{-1.8/d}/(1 - n^{-0.9/d}) \\ &< \varepsilon \cdot \alpha, \end{aligned}$$

for  $n$  sufficiently large, using that  $\alpha = c \cdot n^{-1/d}$  in the last step.  $\square$

**Lemma 4.16.** *If  $w, z \in \mathbb{R}^d$  satisfy  $0.9999 \leq \|w\|, \|z\| \leq 1.0001$  then*

$$\|\varphi(z) - \varphi(w)\| \leq 4 \cdot \|z - w\|.$$

*Proof.* We have

$$\begin{aligned} \|\varphi(z) - \varphi(w)\| &= \|(1/\|z\|)^2 \cdot z - (1/\|w\|)^2 \cdot w\| \\ &\leq \|(1/\|z\|)^2 \cdot z - (1/\|z\|)^2 \cdot w\| + \|((1/\|z\|)^2 - (1/\|w\|)^2) \cdot w\| \\ &= (1/\|z\|)^2 \cdot \|z - w\| + |(\|w\|^2 - \|z\|^2)/(\|z\| \cdot \|w\|^2)| \cdot \|w\|. \end{aligned}$$

We now note that

$$|\|w\|^2 - \|z\|^2| = |(\|w\| - \|z\|) \cdot (\|w\| + \|z\|)| \leq \|z - w\| \cdot (\|w\| + \|z\|).$$

Feeding this into the previous inequality gives

$$\begin{aligned}\|\varphi(z) - \varphi(w)\| &\leq \left( (1/\|z\|)^2 + \|w\| \cdot (\|z\| + \|w\|) / (\|z\| \cdot \|w\|)^2 \right) \cdot \|z - w\| \\ &\leq 4 \cdot \|z - w\|,\end{aligned}$$

where we've used the bounds  $0.9999 < \|z\|, \|w\| < 1.0001$  in the last line.  $\square$

*Proof of Corollary 4.13.* Fix  $u \in S^{d-1}$ . By construction of  $h$ , we have  $B((1 + h(u))u, \varepsilon\alpha) \cap \mathcal{Z} \neq \emptyset$ . In other words, there is an  $1 \leq i \leq M$  such that  $\|(1 + h(u)) \cdot u - Z_i\| < \varepsilon\alpha$ . We have

$$\begin{aligned}\|u - U_i\| &= \left( \frac{1}{1+h(u)} \right) \cdot \|(1 + h(u)) \cdot u - (1 + h(u)) \cdot U_i\| \\ &\leq 2 \cdot (\|(1 + h(u))u - Z_i\| + \|(1 + h(u))U_i - Z_i\|) \\ &= 2 \cdot (\|(1 + h(u))u - Z_i\| + \|1 + h(u) - \|Z_i\|\|) \\ &< 4 \cdot \varepsilon \cdot \alpha.\end{aligned}\tag{4.14}$$

where we used  $h > -1/2$  (true for  $n$  sufficiently large) in the second line. Appealing to (4.1), this gives

$$\text{dist}_{S^{d-1}}(u, U_i) \leq \pi \cdot \|u - U_i\| < 4 \cdot \pi \cdot \varepsilon \cdot \alpha < \alpha/10,$$

(provided  $\varepsilon$  was chosen sufficiently small) so that part (i) follows.

Suppose  $1 \leq i < j \leq M$  are such that  $\text{dist}(U_i, U_j) > \pi - \alpha/5$ . Let us pick  $u \in S^{d-1}$  with  $\|Z_i - (1 + h(u))u\| < \varepsilon \cdot \alpha$ . By Lemma 4.14 and (4.14), we have:

$$\|(1 + h(U_i))U_i - (1 + h(u))u\| \leq 2 \cdot \|u - U_i\| \leq 8 \cdot \varepsilon \cdot \alpha,$$

It follows that

$$\begin{aligned}\|Z_i - (1 + h(U_i))U_i\| &\leq \|Z_i - (1 + h(u))u\| + \|(1 + h(U_i))U_i - (1 + h(u))u\| \\ &< 9 \cdot \varepsilon \cdot \alpha.\end{aligned}$$

Of course the same bound applies to  $\|Z_j - (1 + h(U_j))U_j\|$ . Lemma 4.16 now gives:

$$\|\varphi(Z_j) - \varphi((1 + h(U_j))U_j)\| \leq 36 \cdot \varepsilon \cdot \alpha.$$

(We are allowed to invoke Lemma 4.16 as  $\|Z_j\| - 1 < n^{-0.9/d} + \varepsilon\alpha < 0.0001$  for  $n$  sufficiently large, and similarly for  $(1 + h(U_j))U_j$ .) It follows that

$$\begin{aligned}\|Z_i - \varphi(Z_j)\| &\leq \|(1 + h(U_i))U_i - \varphi((1 + h(U_j))U_j)\| + 45 \cdot \varepsilon \cdot \alpha \\ &\leq \|(1 + h(U_i))U_i + (1 + h(-U_j))U_j\| + 46 \cdot \varepsilon \cdot \alpha,\end{aligned}\tag{4.15}$$

where we've used Lemma 4.15 in the second inequality. Applying Lemma 4.14 once again, we have

$$\|(1 + h(U_i))U_i + (1 + h(-U_j))U_j\| \leq 2 \cdot \|U_i + U_j\| \quad (4.16)$$

Combining (4.15) and (4.16) we find:

$$\|Z_i - \varphi(Z_j)\| \leq 2\|U_i + U_j\| + 46 \cdot \varepsilon \cdot \alpha.$$

By part (i) of Lemma 4.1, we have

$$\begin{aligned} \text{dist}_{S^d}(X_i, -X_j) &\leq 2 \cdot \|\pi(X_i) - \pi(-X_j)\| \\ &= 2 \cdot \|Z_i - \varphi(Z_j)\| \\ &\leq 4 \cdot \|U_i + U_j\| + 92 \cdot \varepsilon \cdot \alpha. \end{aligned}$$

To conclude the argument, we observe that  $\text{dist}(U_i, U_j) > \pi - \alpha/5$  implies that

$$\|U_i + U_j\| \leq \text{dist}(U_i, -U_j) < \alpha/5.$$

So

$$\text{dist}_{S^d}(X_i, -X_j) < (4/5 + 92 \cdot \varepsilon) \cdot \alpha < \alpha.$$

(Provided  $\varepsilon$  was chosen sufficiently small.) Equivalently,  $\text{dist}_{S^d}(X_i, X_j) > \pi - \alpha$ .  $\square$

## 4.4 Chromatic number $4, \dots, d + 2$

For  $\beta > 0$  chosen appropriately shortly, we let  $y_1, \dots, y_M \in S^d$  be a  $\beta$ -net. That is, for every  $x \in S^d$  there is some  $i$  such that  $\|x - y_i\| < \beta$ . We let  $C_i$  denote the (spherical) Voronoi cell of  $y_i$ , that is

$$C_i := \{x \in S^d : \|x - y_i\| = \min_{j=1, \dots, M} \|x - y_j\|\},$$

and we let

$$\mathcal{Y} := \{y_i : C_i \cap \{X_1, \dots, X_N\} \neq \emptyset\},$$

be the set of those  $y_i$  for which at least one of the Poisson points falls inside its cell.

**Lemma 4.17.** *For every  $\varepsilon, \mu, \alpha > 0$  there exists a  $\beta = \beta(\varepsilon, \mu, \alpha)$  such that:*

$$\mathbb{P}_\lambda(G(\mathcal{X}, \alpha) \cong G(\mathcal{Y}, \alpha)) > 1 - \varepsilon,$$

for all  $\lambda \leq \mu$ , where the error term can be chosen uniform in  $\lambda \leq \mu$ .

*Proof.* Let us write

$$\begin{aligned} E_\beta &:= \{\text{there exist } x, y \in \mathcal{X} \text{ with } \text{dist}(x, y) \in (\pi - \alpha - \pi\beta, \pi - \alpha + \pi\beta)\}, \\ F_\beta &:= \{\text{there exist distinct } x, y \in \mathcal{X} \text{ with } \text{dist}(x, y) < \pi\beta\}, \end{aligned}$$

We note that if none of these two events hold then  $G(\mathcal{X}, \alpha) \cong G(\mathcal{Y}, \alpha)$ . (We use that  $\text{dist}(x, y) \leq (\pi/2)\|x - y\|$  for all  $x, y \in S^d$  by (4.1).) Now note that

$$\lim_{\beta \searrow 0} \mathbb{P}(E_\beta) = \mathbb{P}\left(\bigcap_{\beta > 0} E_\beta\right) = \mathbb{P}(\text{there exist } x, y \in \mathcal{X} \text{ such that } \text{dist}(x, y) = \pi - \alpha) = 0.$$

Analogously,  $\lim_{\beta \searrow 0} \mathbb{P}(F_\beta) = 0$  as well. Uniformity in  $\lambda \leq \mu$  follows from the fact that we can couple the Poisson processes of intensities  $\lambda, \mu$  such that  $\mathcal{X}_\lambda \subseteq \mathcal{X}_\mu$ .  $\square$

We fix a  $\beta$ -net  $y_1, \dots, y_M \in S^d$  and let  $\mathcal{Y}$  be as above. Let us set

$$Z_i := 1_{\{y_i \in \mathcal{Y}\}} = 1_{\{C_i \cap \mathcal{X} \neq \emptyset\}}.$$

Then  $Z_1, \dots, Z_M$  are independent and  $Z_i \stackrel{d}{=} \text{Be}(p_i)$ , with

$$p_i = \mathbb{P}(C_i \cap \mathcal{X} \neq \emptyset) = 1 - e^{-\mu \cdot \nu(C_i)},$$

where  $\nu(\cdot)$  denotes the uniform measure on the sphere. For clarity, we emphasize that the random set  $\mathcal{Y}$  is completely determined by  $Z_1, \dots, Z_M$ .

Let  $A$  be the event

$$A := \{\chi(G(\mathcal{Y}, \alpha)) \geq k\}.$$

Note that  $A$  can be identified with a subset of  $\{0, 1\}^M$  (i.e.  $A = \{(Z_1, \dots, Z_M) \in B\}$  for some  $B \subseteq \{0, 1\}^M$ ) that is an up-set. So Proposition 4.4 applies. We now consider the derivative of  $\mathbb{P}_\mu(A)$  with respect to  $\mu$ . It satisfies

$$\begin{aligned} \frac{d}{d\mu} \mathbb{P}_\mu(A) &= \sum_{i=1}^M \frac{\partial}{\partial p_i} \mathbb{P}_{\bar{p}}(A) \cdot \frac{dp_i}{d\mu} \\ &= \sum_{i=1}^M \frac{\partial}{\partial p_i} \mathbb{P}_{\bar{p}}(A) \cdot \nu(C_i) \cdot e^{-\mu \cdot \nu(C_i)} \\ &= \sum_{i=1}^M \frac{\partial}{\partial p_i} \mathbb{P}_{\bar{p}}(A) \cdot \nu(C_i) \cdot (1 - p_i) \\ &\geq \frac{1}{3\mu} \cdot \sum_{i=1}^M \frac{\partial}{\partial p_i} \mathbb{P}_{\bar{p}}(A) \cdot p_i(1 - p_i), \end{aligned} \tag{4.17}$$

where the last line holds assuming  $\beta$  is chosen sufficiently small (so that  $\max_i \nu(C_i) \leq 1/\mu$  say), using that  $1 - e^{-x} \geq x/3$  for all  $0 \leq x \leq 1$ .

**Corollary 4.18.** *Fix  $\varepsilon, \delta > 0$ . There exists a  $L = L(\varepsilon, \delta) \in \mathbb{N}$  such that, for every  $\mu, \alpha > 0$  and  $k \in \mathbb{N}$  and all sufficiently small  $\beta > 0$ , one of the following holds:*

- (i)  $\mathbb{P}_\mu(A) < \varepsilon$ , or;
- (ii) *There exist indices  $i_1, \dots, i_L$  such that  $\mathbb{P}_{(1+\delta)\mu}(A|Z_{i_1} = \dots = Z_{i_L} = 1) > 1 - \varepsilon$ .*

*Proof.* We assume  $\mathbb{P}_\mu(A) \geq \varepsilon$ , otherwise there is nothing to prove.

We let  $c, K$  be as provided by Proposition 4.4 with inputs  $\varepsilon$  and  $C := 1000/\delta$ . For notational convenience, we write  $x := c\delta/100$ .

We first claim that there are indices  $i_1, \dots, i_K$  such that

$$\mathbb{P}_{(1+x)\mu}(A|Z_{i_1} = \dots = Z_{i_K} = 1) \geq \min(\mathbb{P}_\mu(A) + c, 1 - \varepsilon). \quad (4.18)$$

By obvious monotonicity,  $\mathbb{P}_{(1+x)\mu}(A|Z_{i_1} = \dots = Z_{i_K} = 1) \geq \mathbb{P}_{(1+x)\mu}(A)$ . So if  $\mathbb{P}_{(1+x)\mu}(A) > 1 - \varepsilon$  or  $\mathbb{P}_{(1+x)\mu}(A) > \mathbb{P}_\mu(A) + c$  then (4.18) is trivially true for any choice of indices  $i_1, \dots, i_K$ . We therefore assume this is not the case. By the intermediate value theorem, there is some  $\mu \leq \lambda \leq (1+x)\mu$  such that

$$\frac{d}{d\lambda} \mathbb{P}_\lambda(A) < c/(x\mu).$$

Hence, by (4.17), we have

$$\sum_{i=1}^n \frac{\partial}{\partial p_i} \mathbb{P}_{\vec{p}}(A) \cdot p_i(1 - p_i) < 3\lambda \cdot \frac{c}{x\mu} \leq \frac{3(1+x)\mu c}{x\mu} = \frac{300(1 + \frac{c\delta}{100})}{\delta} < C,$$

at that particular value of  $\lambda$ , assuming without loss of generality  $\delta < 1$  in the last step. Case (a) or (b) of Proposition 4.4 must hold. We now note that

$$\begin{aligned} \mathbb{P}_\lambda(A) &\leq \mathbb{P}_\lambda(A|Z_{i_1} = \dots = Z_{i_K} = 0) + \mathbb{P}((Z_{i_1}, \dots, Z_{i_K}) \neq (0, \dots, 0)) \\ &= \mathbb{P}_\lambda(A|Z_{i_1} = \dots = Z_{i_K} = 0) + o_\lambda(1), \end{aligned}$$

where the last equality follows provided  $\beta$  is small (so that  $\mathbb{P}(Z_i = 1) \leq \lambda\nu(C_i) = o(1)$ , uniformly in  $1 \leq i \leq M$ ). We see that, indeed, we cannot be in case (b), because that would imply  $\mathbb{P}_\lambda(A) \leq \mathbb{P}_\lambda(A) - c + o_\lambda(1)$ . So we are in case (a). As  $\mathbb{P}_{(1+x)\mu}(A|Z_{i_1} = \dots = Z_{i_K} = 1) \geq \mathbb{P}_\lambda(A|Z_{i_1} = \dots = Z_{i_K} = 1)$  by obvious monotonicity, this establishes the claim (4.18).

Now let  $A^{(1)}$  denote the event that  $A$  holds if we force the bits with indices  $i_1, \dots, i_K$  to be one. (I.e. we randomly sample all bits, and then we set the values of the selected bits to one.) In particular

$$\mathbb{P}_\lambda(A^{(1)}) = \mathbb{P}_\lambda(A | Z_{i_1} = \dots = Z_{i_K} = 1),$$

for all  $\lambda$ . To aid the reader, let us point out that  $A^{(1)}$  is an event that depends on (is determined by) the  $M - K$  independent Bernoulli random variables  $Z_j : j \in [M] \setminus \{i_1, \dots, i_K\}$ . Completely analogous to the previous, we find  $i_{K+1}, \dots, i_{2K}$  such that

$$\mathbb{P}_{(1+x)^2\mu}(A^{(1)} | Z_{i_{K+1}} = \dots = Z_{i_{2K}} = 1) \geq \min\left(\mathbb{P}_{(1+x)\mu}(A^{(1)}) + c, 1 - \varepsilon\right).$$

Repeating our reasoning several more times, we define events  $A^{(2)}, \dots, A^{(R)}$  and indices  $i_{2K+1}, \dots, i_L$ , where  $R := \lceil 1/c \rceil$  and  $L := R \cdot K$ . That is,  $A^{(j)}$  is the event that  $A$  holds if we set  $Z_{i_1} = \dots = Z_{i_{jK}} = 1$ , and by our arguments:

$$\begin{aligned} \mathbb{P}_{(1+x)^{j+1}\mu}(A^{(j+1)}) &= \mathbb{P}_{(1+x)^{j+1}\mu}(A^{(j)} | Z_{i_{jK+1}} = \dots = Z_{i_{(j+1)K}} = 1) \\ &\geq \min\left(\mathbb{P}_{(1+x)\mu}(A^{(j)}) + c, 1 - \varepsilon\right), \end{aligned}$$

for  $j = 1, \dots, R-1$ . It follows that

$$\begin{aligned} \mathbb{P}_{(1+x)^R\mu}(A | Z_{i_1} = \dots = Z_{i_L} = 1) &= \mathbb{P}_{(1+x)^R\mu}(A^{(R)}) \\ &\geq \min\left(\mathbb{P}_\mu(A) + R \cdot c, 1 - \varepsilon\right) \\ &= 1 - \varepsilon, \end{aligned}$$

where the last equality holds by choice of  $R = \lceil 1/c \rceil$ .

To conclude, we remark that

$$(1+x)^R \leq e^{xR} = e^{\lceil 1/c \rceil \cdot \frac{c\delta}{100}} \leq e^{\frac{(c+1)\delta}{100}} \leq e^{\delta/50} < 1 + \delta,$$

where the last inequality holds assuming without loss of generality that  $\delta$  was chosen sufficiently small.  $\square$

**Corollary 4.19.** *Fix  $\varepsilon, \delta > 0$ . There exist  $L = L(\varepsilon, \delta) \in \mathbb{N}$  such that, for every  $\mu, \alpha > 0$  and  $k \in \mathbb{N}$ , one of the following holds:*

- (i)  $\mathbb{P}_\mu(\chi(G(\mathcal{X}, \alpha)) \geq k) < \varepsilon$ , or;
- (ii) *There exist  $x_1, \dots, x_L \in S^d$  such that  $\mathbb{P}_{(1+\delta)\mu}(\chi(G(\mathcal{X} \cup \{x_1, \dots, x_L\}, \alpha)) \geq k) > 1 - \varepsilon$ .*

*Proof.* We notice that  $(\mathcal{Y} | Z_{i_1} = \dots = Z_{i_K} = 1) \stackrel{d}{=} \mathcal{Y} \cup \{y_{i_1}, \dots, y_{i_K}\}$ . Now notice that  $\mathcal{Y} \cup \{y_{i_1}, \dots, y_{i_K}\}$  is what we get if we follow the procedure constructing  $\mathcal{Y}$  from  $\mathcal{X}$ , but starting with  $\mathcal{X} \cup \{y_{i_1}, \dots, y_{i_K}\}$  in place of  $\mathcal{X}$ . The result follows by (a modification of) Lemma 4.17.  $\square$

We say that distinct  $y_1, \dots, y_K \in S^d$  form an  $\eta$ -near configuration with respect to  $x_1, \dots, x_K \in S^d$  if there is an orthogonal transformation  $T$  such that  $\text{dist}(T(y_i), x_i) < \eta$  for  $i = 1, \dots, K$ . (That is, we can apply an isometry of the sphere and map  $y_i$  to a point close to  $x_i$ , for each  $i$ .)

**Lemma 4.20.** *For any fixed  $K \in \mathbb{N}, \varepsilon > 0$  and  $x_1, \dots, x_K \in S^d$ , we have*

$$\mathbb{P}_\mu(\mathcal{X} \text{ contains an } \varepsilon\mu^{-1/d}\text{-near configuration}) = 1 - o_\mu(1).$$

*Proof.* We (greedily) pick orthogonal transformations  $T_1, \dots, T_L$  such that  $\text{dist}(T_i(x_j), T_{i'}(x_{j'})) > 2\varepsilon\mu^{-1/d}$  for all  $(i, j) \neq (i', j')$ . This can be done for  $L = \Omega(\mu)$ , which can be seen as follows. Suppose  $T_1, \dots, T_L$  are such that we cannot find a suitable orthogonal transformation  $T_{L+1}$ . If we pick an orthogonal transformation  $T \in O(d+1)$  uniformly at random (this makes sense as the set of all orthogonal  $(d+1) \times (d+1)$ -matrices  $O(d+1)$  can be seen as a smooth and compact manifold), then with probability one, at least one of the random points  $Y_1 = T(x_1), \dots, Y_K = T(x_K)$  falls within the set

$$A := \bigcup_{\substack{1 \leq i \leq L, \\ 1 \leq j \leq K}} \text{cap}(T_i(x_j), 2\varepsilon\mu^{-1/d}).$$

By symmetry considerations, each of  $Y_1, \dots, Y_K$  has the uniform distribution on  $S^d$  (but they are not independent). We see that

$$\begin{aligned} 1 &= \mathbb{P}(\{Y_1, \dots, Y_K\} \cap A \neq \emptyset) \leq \sum_{j=1}^K \mathbb{P}(Y_j \in A) \\ &= K\nu(A) \leq K \sum_{i=1}^L \sum_{j=1}^K \nu(\text{cap}(T_i(x_j), 2\varepsilon\mu^{-1/d})) \\ &\leq L \cdot K^2 \cdot \text{const} \cdot \varepsilon^d \cdot (1/\mu), \end{aligned} \tag{4.19}$$

where  $\text{const}$  is a constant depending only on  $d$  (the measure of a cap with opening radius  $x$  scales with  $x^d$ ). Rephrasing, (4.19) states that  $L = \Omega(\mu)$ , as claimed.

For  $1 \leq i \leq L$ , let us define the events

$$E_i := \bigcap_{j=1}^K E_{i,j}, \quad E_{i,j} := \{\mathcal{X} \cap \text{cap}(T_i(x_j), \varepsilon\mu^{-1/d}) \neq \emptyset\}.$$

Then by choice of  $T_1, \dots, T_L$ , the events  $E_1, \dots, E_L$  are independent and, by symmetry considerations, they all have the same probability. This gives

$$\mathbb{P}\left(\bigcup_{j=1}^L E_j\right) = 1 - (1 - \mathbb{P}(E_1))^L. \quad (4.20)$$

The events  $E_{1,1}, \dots, E_{1,K}$  are positively dependent, giving

$$\mathbb{P}(E_1) \geq \mathbb{P}(E_{1,1}) \cdots \mathbb{P}(E_{1,K}) = \left(1 - \exp[-\mu \cdot \nu(\text{cap}(x, \varepsilon \mu^{-1/d}))]\right)^K = \Omega(1), \quad (4.21)$$

Combining (4.20) and (4.21) with  $L = \Omega(\mu)$ , we see that  $\mathbb{P}\left(\bigcup_{j=1}^L E_j\right) = 1 - o_\mu(1)$ , which proves the lemma.  $\square$

**Corollary 4.21.** *For every  $\varepsilon, \delta, \eta > 0$  and  $3 \leq k \leq d+2$ , one of the following holds for all sufficiently large  $\mu$  and all  $\alpha > 0$ :*

- (i)  $\mathbb{P}_\mu(\chi(G(\mathcal{X}, \alpha)) \geq k) < \varepsilon$ , or;
- (ii)  $\mathbb{P}_{(1+\delta)\mu}(\chi(G(\mathcal{X}, \alpha + \eta \mu^{-1/d})) \geq k) \geq 1 - \varepsilon$ .

(Just to be clear, this holds uniformly over all  $\alpha$ . In particular  $\alpha = \alpha(n)$  is allowed to vary with  $n$ .)

*Proof.* Assume  $\mathbb{P}_\mu(\chi(G(\mathcal{X}, \alpha)) \geq k) \geq \varepsilon$ . By Corollary 4.19, there are  $x_1, \dots, x_K$  such that  $\mathbb{P}_{(1+\delta)\mu}(\chi(G(\mathcal{X} \cup \{x_1, \dots, x_K\}, \alpha)) \geq k) \geq 1 - \varepsilon$ .

Now consider the model with parameter  $(1+2\delta)\mu$ . The point set  $\mathcal{X}$  is the union of  $\mathcal{X}_1 \cup \mathcal{X}_2$  where  $\mathcal{X}_1$  is a PPP with parameter  $(1+\delta)\mu$  and  $\mathcal{X}_2$  an independent PPP with parameter  $\delta\mu$ .

By the previous lemma, with probability  $1 - o(1)$ ,  $\mathcal{X}_2$  contains at least one  $(\eta/2) \cdot \mu^{-1/d}$ -near configuration  $Y_1, \dots, Y_K$ . Let  $T$  be a corresponding (random) orthogonal transformation. That is,  $\text{dist}(T(Y_1), x_1), \dots, \text{dist}(T(Y_K), x_K) < (\eta/2) \cdot \mu^{-1/d}$ . (For definiteness, we can say that we pick  $Y_1, \dots, Y_K$  uniformly at random from all eligible  $K$ -tuples and, having chosen  $Y_1, \dots, Y_K$ , we pick  $T$  uniformly at random from all eligible transformations. Again this last step makes sense because we are picking from an open subset of a compact, smooth manifold.)

We now note that

$$\begin{aligned} G(\mathcal{X}, \alpha + \eta \mu^{-1/d}) &\supseteq G(\mathcal{X}_1 \cup \{Y_1, \dots, Y_K\}, \alpha + \eta \mu^{-1/d}) \\ &\cong G(T[\mathcal{X}_1 \cup \{Y_1, \dots, Y_K\}], \alpha + \eta \mu^{-1/d}) \\ &\supseteq G(T[\mathcal{X}_1] \cup \{x_1, \dots, x_K\}, \alpha), \end{aligned}$$

where  $\subseteq$  denotes the subgraph relation and in the last step we use the triangle inequality for the spherical distance  $\text{dist}(\cdot, \cdot)$ .

The random orthogonal transformation  $T$  is determined entirely by  $\mathcal{X}_2$  and in particular is independent of  $\mathcal{X}_1$ . It follows that  $T[\mathcal{X}_1] \stackrel{d}{=} \mathcal{X}_1$  by the symmetry properties of a constant intensity measure  $\lambda \cdot \nu(\cdot)$  on the  $d$ -sphere. This gives

$$\begin{aligned}
& \mathbb{P}_{(1+2\delta)\mu} \left( \chi \left( G(\mathcal{X}, \alpha + \eta\mu^{-1/d}) \right) \geq k \right) \\
& \geq \mathbb{P}_{(1+2\delta)\mu} \left( \chi \left( G(T[\mathcal{X}_1] \cup \{x_1, \dots, x_K\}, \alpha) \right) \geq k \right) - o(1) \\
& = \mathbb{P}_{(1+2\delta)\mu} \left( \chi \left( G(\mathcal{X}_1 \cup \{x_1, \dots, x_K\}, \alpha) \right) \geq k \right) - o(1) \\
& = \mathbb{P}_{(1+\delta)\mu} \left( \chi \left( G(\mathcal{X} \cup \{x_1, \dots, x_K\}, \alpha) \right) \geq k \right) - o(1) \\
& \geq 1 - \varepsilon - o(1).
\end{aligned}$$

(The  $o(1)$  term accounts for the event that  $\mathcal{X}_2$  does not have the required near configuration.) Adjusting the values of  $\varepsilon, \delta$ , the result follows.  $\square$

For  $n \in \mathbb{N}$  and  $4 \leq k \leq d+2$  we let  $\alpha_k(n)$  be such that

$$\mathbb{P}_n(\chi(G(\mathcal{X}, \alpha_k)) \geq k) = 1/2,$$

(This  $\alpha_k$  exists by a straightforward continuity argument.) We remark that  $\alpha_k$  is non-increasing. By (1.2) and Theorems 1.8 and 1.7, there exists a constant  $c > 0$  such that

$$\alpha_k(n) \geq cn^{-1/d}, \quad (4.22)$$

and also

$$\alpha_k(2n) > c \cdot \alpha_k(n), \quad (4.23)$$

for all  $4 \leq k \leq d+2$  and all  $n \in \mathbb{N}$ .

We fix a  $4 \leq k \leq d+2$  and  $\varepsilon, \delta > 0$  and define

$$U := \left\{ n \in \mathbb{N} : \alpha_k(n) \cdot (1 - \varepsilon/2) > \alpha_k(\lceil (1 + \delta)n \rceil) \right\}, \quad V := \left\{ n \in \mathbb{N} : \left\lfloor \frac{n}{1 + \delta} \right\rfloor \in U \right\}. \quad (4.24)$$

**Lemma 4.22.** *We have  $\frac{|U \cap [n]|}{n} \leq 100 \cdot \ln(1/c) \cdot (\delta/\varepsilon)$ , for all  $n \in \mathbb{N}$ .*

*Proof.* For some  $\ell \in \mathbb{N}$ , let  $2^\ell \leq n_1 < n_2 < \dots < n_K < 2^{\ell+1}$  be a sequence with the properties that  $n_{i+1} \geq (1 + \delta)n_i$  and  $\alpha_k(n_{i+1}) < (1 - \varepsilon/2)\alpha_k(n_i)$  and, subject to these constraints,  $K$  is as large as possible. We must have

$$\alpha_k(2^{\ell+1}) \leq (1 - \varepsilon/2)^K \alpha_k(2^\ell).$$

So, by (4.23), we have  $c < (1 - \varepsilon/2)^K$ . Rewriting, we have

$$K < \frac{\ln c}{\ln(1 - \varepsilon/2)} = \frac{\ln(1/c)}{-\ln(1 - \varepsilon/2)} \leq \frac{2 \ln(1/c)}{\varepsilon}.$$

By maximality of  $K$ , every  $n \in U \cap \{2^\ell, \dots, 2^{\ell+1} - 1\}$  must satisfy  $n_i/(1+\delta) < n < (1+\delta)n_i$  for some  $1 \leq i \leq K$ . It follows that

$$|U \cap \{2^\ell, \dots, 2^{\ell+1} - 1\}| \leq \sum_{i=1}^K 2\delta \cdot n_i \leq K \cdot 2\delta \cdot 2^{\ell+1} \leq 8 \ln(1/c) \cdot (\delta/\varepsilon) \cdot 2^\ell.$$

Now let  $n$  be arbitrary and let  $\ell \in \mathbb{N}$  be the value such that  $2^\ell \leq n < 2^{\ell+1}$ . Then

$$|U \cap [n]| \leq \sum_{i=0}^{\ell} |U \cap \{2^i, \dots, 2^{i+1} - 1\}| \leq 8 \ln(1/c) \cdot (\delta/\varepsilon) \cdot \sum_{i=0}^{\ell} 2^i \leq 16 \ln(1/c) \cdot (\delta/\varepsilon) \cdot n.$$

This proves the result.  $\square$

**Corollary 4.23.** *We have  $\frac{|V \cap [n]|}{n} \leq 1000 \cdot \ln(1/c) \cdot (\delta/\varepsilon)$ .*

*Proof.* This follows from the previous lemma and the fact that if  $u = \lfloor \frac{n}{1+\delta} \rfloor$  then  $n \in \{ \lceil u(1+\delta) \rceil, \lceil u(1+\delta) \rceil + 1 \}$  (assuming without loss of generality  $\delta < 1$ ).  $\square$

**Lemma 4.24.** *There exists  $n_0 = n_0(\varepsilon, \delta, k)$  such that for all  $n \in \{n_0, n_0+1, \dots\} \setminus (U \cup V)$  we have*

$$(i) \quad \mathbb{P}_n(\chi(G(\mathcal{X}, (1+\varepsilon)\alpha_k(n))) \geq k) > 1 - \varepsilon, \text{ and};$$

$$(ii) \quad \mathbb{P}_n(\chi(G(\mathcal{X}, (1-\varepsilon)\alpha_k(n))) \geq k) < \varepsilon.$$

*Proof.* Pick  $n \in \mathbb{N} \setminus (U \cup V)$ . Writing  $m := \lfloor \frac{n}{1+\delta} \rfloor$  for convenience, we have

$$(1+\varepsilon)\alpha_k(n) \geq (1+\varepsilon)(1-\varepsilon/2)\alpha_k(m) > \alpha_k(m) + \eta \cdot m^{-1/d},$$

where  $\eta > 0$  is a sufficiently small constant and the last inequality holds for  $n$  sufficiently large, and we use (4.22). Corollary 4.21 now shows that (i) holds for  $n$  sufficiently large.

It remains to establish (ii). Aiming for a contradiction, suppose that

$$\mathbb{P}_n(\chi(G(\mathcal{X}, (1-\varepsilon)\alpha_k(n))) \geq k) \geq \varepsilon.$$

Writing  $\ell := \lceil (1+\delta)n \rceil$ , Corollary 4.21 tells us that

$$\mathbb{P}_\ell(\chi(G(\mathcal{X}, (1-\varepsilon)\alpha_k(n) + \eta n^{-1/d})) \geq k) > 1 - \varepsilon.$$

However,  $\alpha_k(\ell) \geq (1-\varepsilon/2)\alpha_k(n)$  as  $n \notin U$  and hence

$$(1 - \varepsilon)\alpha_k(n) + \eta n^{-1/d} \leq \frac{1 - \varepsilon}{1 - \varepsilon/2} \cdot \alpha_k(\ell) + \eta n^{-1/d} < \alpha_k(\ell),$$

for  $n$  sufficiently large and we use (4.22) again, and that  $\eta$  is an appropriately chosen small constant. But now it follows that, for  $n$  sufficiently large:

$$\mathbb{P}_\ell(\chi(G(\mathcal{X}, \alpha_k(\ell))) \geq k) > 1 - \varepsilon,$$

contradicting the definition of  $\alpha_k$ . It follows that, for  $n \in \mathbb{N} \setminus (U \cup V)$  sufficiently large, we have

$$\mathbb{P}_n(\chi(G(\mathcal{X}, (1 - \varepsilon)\alpha_k(n))) \geq k) < \varepsilon,$$

proving (ii). □

*Proof of Theorem 1.10.* Fix  $4 \leq k \leq d + 2$ . We choose sequences  $\varepsilon_1, \varepsilon_2, \dots$  and  $\delta_1, \delta_2, \dots$  satisfying  $\varepsilon_i \rightarrow 0$  and  $\delta_i/\varepsilon_i \rightarrow 0$  as  $i \rightarrow \infty$ . Let  $U_i = U(k, \varepsilon_i, \delta_i)$ ,  $V_i = V(k, \varepsilon_i, \delta_i)$  be as defined in (4.24) and let  $n_i = n_0(k, \varepsilon_i, \delta_i)$  be as provided by Lemma 4.24. Without loss of generality (switching to even larger numbers if needed), we can assume that  $n_{i+1}/n_i \rightarrow \infty$  as  $i \rightarrow \infty$ .

We define

$$N_k := \bigcup_{i=1}^{\infty} \{n_i, \dots, n_{i+1} - 1\} \setminus (U_i \cup V_i).$$

Note that if  $n_i \leq n < n_{i+1}$  then

$$\begin{aligned} \frac{|[n] \setminus N_k|}{n} &\leq \frac{|U_i \cap [n]|}{n} + \frac{|V_i \cap [n]|}{n} + \frac{|U_{i-1} \cap [n_i]|}{n_i} + \frac{|V_{i-1} \cap [n_i]|}{n_i} + \frac{n_{i-1}}{n_i} \\ &\leq O(\delta_i/\varepsilon_i) + O(\delta_{i-1}/\varepsilon_{i-1}) + o(1) \\ &= o(1). \end{aligned}$$

In other words,  $N_k$  has density one. It follows that

$$N := N_4 \cap \dots \cap N_{d+2},$$

also has density one. Theorem 1.10 now follows by the previous lemma (using  $\varepsilon_i \rightarrow 0$ ). □

## 4.5 Odd cycle threshold

For this part, it will be the easiest to compare the projected Borsuk graph locally to the continuum  $AB$  percolation model. In this model, we let  $\mathcal{V}_A$  and  $\mathcal{V}_B$  be

independent Poisson point processes in  $\mathbb{R}^d$  where each has intensity  $\lambda > 0$ . We create a random graph  $(\mathcal{V}, E)$  with vertices  $\mathcal{V} = \mathcal{V}(\lambda) := \mathcal{V}_A \cup \mathcal{V}_B$  and edges

$$E := \left\{ xy \in \binom{\mathcal{V}}{2} : \|x - y\| \leq 1, x \in \mathcal{V}_A, y \in \mathcal{V}_B \right\}.$$

With a slight abuse of notation we will also write  $\mathcal{V} = \mathcal{V}(\lambda)$  for the graph and define percolation to be

$$\{\text{percolation}\} := \{\text{there exists an infinite connected component in } \mathcal{V}\}.$$

Since the law of  $(\mathcal{V}_A, \mathcal{V}_B)$  is ergodic (see e.g. Section 8.4 of [66]) and the occurrence of an infinite component is translation invariant, it follows that

$$\mathbb{P}_\lambda(\text{percolation}) \in \{0, 1\},$$

for all  $\lambda$ . We define

$$\lambda_c := \inf\{\lambda > 0 : \mathbb{P}_\lambda(\text{percolation}) > 0\}.$$

**Lemma 4.25.**  $\lambda_c \in (0, \infty)$ .

*Proof.* In the classical Boolean percolation model, we connect any two points of a PPP with constant intensity  $\lambda > 0$  if they are closer than one. As for continuum AB percolation we can define

$$\lambda_c^{\text{Bool}} := \inf\{\lambda > 0 : \mathbb{P}_\lambda(\text{percolation occurs in the Boolean model}) > 0\}.$$

It is well known (see e.g. Chapter 3 of [77]) that  $\lambda_c^{\text{Bool}} \in (0, \infty)$ . Note that  $\mathcal{V}_A \cup \mathcal{V}_B$  forms a PPP of intensity  $2\lambda$ . So if  $\lambda < \lambda_c^{\text{Bool}}/2$  then the continuum AB model certainly does not percolate. This gives

$$\lambda_c \geq \lambda_c^{\text{Bool}}/2 > 0.$$

For the upper bound we let  $L := 1/(2\sqrt{d})$  and tile  $\mathbb{R}^d$  by disjoint cubes  $Q_z := Lz + [0, L]^d$  for  $z \in \mathbb{Z}^d$ . We couple the continuum AB percolation model with the standard site percolation model on  $\mathbb{Z}^d$  by declaring  $z$  open iff.  $Q_z$  contains at least one point of  $\mathcal{V}_A$  and at least one point of  $\mathcal{V}_B$ . We have

$$\mathbb{P}(z \text{ open}) = (1 - e^{-\lambda L^d})^2,$$

and the status of a site is independent of the other sites. We can choose  $\lambda$  large enough such that  $\mathbb{P}(z \text{ open}) > p_c^{\text{site}}(\mathbb{Z}^d)$ . In particular, for  $\lambda$  sufficiently large, the site percolation model has an infinite component (with probability one). By choice of  $L$ , for  $x \in Q_z$  and  $y \in Q_{z'}$  with  $\|z - z'\| = 1$  we have  $\|x - y\|^2 \leq 1$ . So if there exists an infinite component in the coupled standard site percolation model on  $\mathbb{Z}^d$ , then the AB model has an infinite connected component too. We can conclude that  $\lambda_c < \infty$ .  $\square$

Having defined continuum  $AB$  percolation, we can now go back to the Borsuk graph. For  $-1 < t < 1$  and  $y \in S^d$  we define

$$C_t(y) := \text{cap}(y, \arccos t) = \{x \in S^d : \text{dist}(y, x) < \arccos t\} = \{x \in S^d : \langle x, y \rangle > t\}$$

and  $C_t := C_t((0, \dots, 0, -1)) = \{x \in S^d : x_{d+1} < -t\}$ . We write  $\pi_y : S^d \setminus \{-y\} \rightarrow P_y$  for the stereographic projection from  $S^d$  through  $-y \in S^d$  onto the  $d$ -dimensional plane  $P_y := \{x \in \mathbb{R}^{d+1} : \langle x, y \rangle = 0\}$ . The choice of defining it with respect to  $-y$  and not  $y$  was made, as we will use the map to approximate  $S^d$  around  $y$  with  $\mathbb{R}^d$ . Thus, the projection we have used so far is  $\pi(x) = \pi_{(0, \dots, 0, -1)}(x)$ . For  $x, y \in S^d$  we also define the map

$$g_y(x) := \frac{2\pi_y(x)}{\alpha(n)}$$

that first projects and then scales points of the sphere. We use the short form  $g := g_{(0, \dots, 0, -1)}$ . We also let

$$\rho(t, n) := \frac{2\sqrt{1-t^2}}{(1+t)\alpha(n)}.$$

Note that in the upcoming lemma,  $B(\underline{0}, \rho(t, n))$  is meant to be a  $d$ -dimensional ball that lives in the  $\{x_{d+1} = 0\}$  plane. The value that will arise in our connection threshold is now set to be

$$c_3 = c_3(d) := ((d+1)\kappa_{d+1}\lambda_c)^{1/d},$$

and because of [Lemma 4.25](#) this is a well defined positive constant (dependent only on  $d$ ).

**Lemma 4.26.** *For all  $\varepsilon > 0$  we can choose  $t_0 = t_0(\varepsilon, d)$  sufficiently close to 1 such that for all  $t > t_0$  the following holds. The points of  $g[\mathcal{X} \cap C_t]$  form an inhomogeneous Poisson point process on  $B(\underline{0}, \rho(t, n))$ . If  $\alpha = (c_3 + \varepsilon)n^{-1/d}$  we have that  $g[\mathcal{X} \cap C_t]$  stochastically dominates a homogeneous PPP with intensity  $\lambda = \lambda(\varepsilon, d) > \lambda_c$  and if  $\alpha = (c_3 - \varepsilon)n^{-1/d}$  it is stochastically dominated by a PPP with intensity  $\lambda = \lambda(\varepsilon, d) < \lambda_c$ .*

*Proof.* For  $x \in S^d$  with  $-x_{d+1} > t$  it holds that

$$\|\pi(x)\| = \frac{\sqrt{x_1^2 + \dots + x_d^2}}{1 - x_{d+1}} = \frac{\sqrt{1 - x_{d+1}^2}}{1 - x_{d+1}} \leq \frac{\sqrt{1 - t^2}}{1 + t}.$$

And thus  $\pi(C_t)$  maps into  $B(\underline{0}, \sqrt{1-t^2}/(1+t))$ . The fact that  $\pi[\mathcal{X} \cap C_t]$  is an inhomogeneous Poisson point process with intensity measure

$$\frac{n}{(d+1)\kappa_{d+1}} \left( \frac{2}{1 + \|y\|^2} \right)^d \mathbb{1}_{\{\|y\| \leq \sqrt{1-t^2}/(1+t)\}},$$

follows from [Lemma 4.3](#). As  $0 \leq \|y\|^2 \leq (1 - t^2)/(1 + t)^2$  we can bound the intensity uniformly above and below by

$$\begin{aligned} \frac{n}{(d+1)\kappa_{d+1}} 2^d &\geq \frac{n}{(d+1)\kappa_{d+1}} \left( \frac{2}{1 + \|y\|^2} \right)^d \geq \frac{n}{(d+1)\kappa_{d+1}} \left( \frac{2(1+t)^2}{(1+t)^2 + 1 - t^2} \right)^d \\ &= \frac{n}{(d+1)\kappa_{d+1}} (1+t)^d. \end{aligned}$$

By the mapping theorem (see e.g. Section 2.3 of [\[65\]](#)), our result now follows after scaling each particle with  $2/\alpha(n)$ . For the lower bound we can compute in the  $\alpha = (c_3 + \varepsilon)n^{-1/d}$  case

$$\frac{n(1+t)^d}{(d+1)\kappa_{d+1}} \frac{\alpha(n)^d}{2^d} = \left( \frac{1+t}{2} \right)^d \frac{(((d+1)\kappa_{d+1}\lambda_c)^{1/d} + \varepsilon)^d}{(d+1)\kappa_{d+1}} = \lambda_c \left( 1 + \frac{\varepsilon}{c_3} \right)^d \left( \frac{1+t}{2} \right)^d$$

and this quantity is strictly larger than  $\lambda_c$ , given that  $t$  is sufficiently close to 1. In a similar fashion we compute for the upper bound on the intensity, if we have  $\alpha = (c_3 - \varepsilon)n^{-1/d}$  that

$$\frac{n2^d}{(d+1)\kappa_{d+1}} \frac{\alpha(n)^d}{2^d} = \frac{(((d+1)\kappa_{d+1}\lambda_c)^{1/d} - \varepsilon)^d}{(d+1)\kappa_{d+1}} = \lambda_c \left( 1 - \frac{\varepsilon}{c_3} \right)^d,$$

and this quantity is less than  $\lambda_c$  in any case.  $\square$

It is sometimes easier to view the Borsuk graph as a random geometric graph on the sphere. To achieve that, we will mirror every  $X_i$  and define  $Y_i := -X_i$  and  $\mathcal{M} := \{Y_1, \dots, Y_N\}$ . Then, we connect  $X_i$  to  $Y_j$  iff  $\text{dist}(X_i, Y_j) \leq \alpha$ . We write  $G_{geo}$  for the resulting graph and for  $x \in S^d$  we let  $G_t(x)$  be the restriction of  $G_{geo}$  to  $C_t(x)$ . We sometimes also use the shorter notation  $G_t := G_t((0, \dots, 0, -1))$ . The following two lemmas help us compare the Borsuk graph with continuum  $AB$  percolation. Depending on  $\alpha$ , it will be supercritical in the first, and subcritical in the second case.

**Lemma 4.27.** *For  $\varepsilon > 0$  and  $\alpha = (c_3 + \varepsilon)n^{-1/d}$  we can choose  $t \geq t_0(\varepsilon, d)$  sufficiently close to 1 such that there exists a coupling of  $G_t$  with an continuum  $AB$  percolation model with intensity  $\lambda = \lambda(\varepsilon, t) > \lambda_c$  on  $B(\underline{0}, \rho(t, n))$  such that for any two points  $x, y$  that are adjacent in the continuum  $AB$  percolation model, it holds that  $g^{-1}(x), g^{-1}(y)$  are adjacent vertices in  $G_t$ .*

*Proof.* Both  $\mathcal{X} \cap C_t$  and  $\mathcal{M} \cap C_t$  are independent with an identical distribution. By [Lemma 4.26](#), if we project these points with  $g(\cdot)$ , each set stochastically dominates a PPP with intensity  $\lambda = \lambda(\varepsilon, t) > \lambda_c$  on  $B(\underline{0}, \rho(t, n))$ . By [Lemma 4.1](#) it holds for any  $x, y \in C_t$

$$\|g(x) - g(y)\| = \frac{2}{\alpha(n)} \|\pi(x) - \pi(y)\| \geq \frac{\text{dist}(x, y)}{\alpha(n)}.$$

Thus, if  $g(x) \in B(\underline{0}, \rho(t, n))$  has an edge to  $g(y) \in B(\underline{0}, \rho(t, n))$  in the continuum AB percolation model it holds  $\|g(x) - g(y)\| \leq 1$  and thus we obtain

$$\text{dist}(x, y) \leq \alpha(n) \|g(x) - g(y)\| \leq \alpha(n),$$

so  $x$  is a neighbor of  $y$  in  $G_t$ .  $\square$

Note that by rotational invariance we could also formulate the above for any other cap  $C_t(y)$ , part of the graph  $G_t(y)$  and projection map  $g_y(\cdot)$ . The same is true for the following.

**Lemma 4.28.** *For  $\varepsilon > 0$  and  $\alpha = (c_3 - \varepsilon)n^{-1/d}$  we can choose  $t \geq t_0(\varepsilon, d)$  sufficiently close to 1 such that there is a constant  $c = c(\varepsilon, d) > 0$  that satisfies the following. There exists a coupling of  $G_t$  with an continuum AB percolation model with  $\lambda = \lambda(\varepsilon, d) < \lambda_c$  and on  $B(\underline{0}, \rho(t, n))$  such that for any two points  $x, y$  that are adjacent in  $G_t$ , it holds that  $cg(x)$  and  $cg(y)$  are adjacent in the continuum AB percolation model.*

*Proof.* For all  $\varepsilon_1 > 0$  we obtain by Lemma 4.1 (ii) a  $t = t(\varepsilon_1)$  such that for all neighbors  $x, y \in G_t$  it holds

$$\alpha(n) = (c_3 - \varepsilon)n^{-1/d} \geq \text{dist}(x, y) > \frac{2\|\pi(x) - \pi(y)\|}{1 + \varepsilon_1}.$$

Thus by rearranging

$$\|g(x) - g(y)\| = \frac{2}{\alpha(n)} \|\pi(x) - \pi(y)\| < 1 + \varepsilon_1.$$

Note that if  $c := 1/(1 + \varepsilon_1)$ , we of course have  $\|cg(x) - cg(y)\| < 1$ . By the mapping theorem (see e.g. Section 2.3 of [65]), if we scale every point of a subcritical AB percolation model by the same scalar that is sufficiently close to 1, we obtain a new AB percolation model that is still subcritical. Therefore, if we now choose  $\varepsilon_1 = \varepsilon_1(\varepsilon, d) > 0$  small enough, we get what we wanted to show by Lemma 4.26.  $\square$

### 4.5.1 Continuum AB percolation machinery

To show an appropriate subcritical exponential decay property, we will approximate  $\mathcal{V}$  by site percolation on a discrete version  $G' = G'(\varepsilon, r_1)$ , where  $\varepsilon, r_1 > 0$  are parameters to be chosen later. The graph  $G'$  is embedded in  $\mathbb{Z}^{d+1}$  and can be viewed as two copies of  $\mathbb{Z}^d$  hovering above each other. One copy will be coupled with points of  $\mathcal{V}_A$  and the other with  $\mathcal{V}_B$ . For each  $i = (i_1, \dots, i_d) \in \mathbb{Z}^d$  we will add a site at  $(i_1, \dots, i_d, 0)$  and  $(i_1, \dots, i_d, 1)$ . We define cubes that are anchored at  $i$  with diameter  $\varepsilon$  as

$$S_i := \left\{ (x_1, \dots, x_d) : i_1\varepsilon/\sqrt{d} \leq x_1 < (i_1+1)\varepsilon/\sqrt{d}, \dots, i_d\varepsilon/\sqrt{d} \leq x_d < (i_d+1)\varepsilon/\sqrt{d} \right\}.$$

We will connect  $(i_1, \dots, i_d, 0)$  to  $(j_1, \dots, j_d, 1)$  iff

$$\sup_{u \in S_i, v \in S_j} \|u - v\| \leq r_1$$

and write  $C'_0$  for the connected component of  $\underline{0} \times \{0\}$  after performing site percolation on  $G'$ . In the following we will add  $\underline{0}$  as a deterministic point of  $\mathcal{V}_A$  (or  $\mathcal{V}_B$ , since the intensities for the two PPP's are the same, the next results will hold in either case) to  $\mathcal{V}$  and write  $C_0$  for the connected component of  $\underline{0}$  in  $\mathcal{V}$ . For two points  $z, w \in \mathcal{V}$  we write  $z \longleftrightarrow w$  if there is a path connecting  $z$  to  $w$  in  $\mathcal{V}$  and  $z \not\longleftrightarrow w$  if not. In a similar fashion if  $A \subset \mathbb{R}^d$  we write  $z \xrightarrow{A} w$  if there is a path connecting the two points while staying inside  $A$  and  $z \longleftrightarrow A$  if  $z$  is connected to some point in  $A$ .

**Lemma 4.29.** *Let  $\lambda < \lambda_c$ . Then, there exists a constant  $c = c(\lambda, d) > 0$  such that for all  $R > 0$*

$$\mathbb{P}(\underline{0} \longleftrightarrow B(\underline{0}, R)^c) \leq e^{-cR}.$$

*Proof.* We follow the ideas of Theorem 3 in Section 8.1 in [20] and aim to use a classical result on exponential decay on  $G'$ . We could have also defined  $AB$  percolation with an arbitrary radius  $r > 0$  instead of 1 and connect two points of opposite label iff the distance is less than  $r$ . Then, we can define  $\lambda_c(r)$  in terms of the radius analogously as done for  $\lambda_c$ . Note that scaling gives us the relationship

$$\lambda_c(r_1) \cdot r_1^d = \lambda_c(r_2) \cdot r_2^d,$$

for all  $r_1, r_2 > 0$ . Therefore, we can let  $r_1 > 1$  and  $\lambda_1 > 0$  be such that  $\lambda < \lambda_1 < \lambda_c(r_1) < \lambda_c$ . Further, choose  $\varepsilon > 0$  small enough such that  $1 + 2\varepsilon < r_1$ . We now let  $\mathcal{V}'_A$  and  $\mathcal{V}'_B$  be independent PPP's in  $\mathbb{R}^d$  where each has intensity  $\lambda_1 > 0$ . We consider site percolation on  $G'(\varepsilon, r_1)$  and declare  $S_i \times \{0\}$  open if  $\mathcal{V}'_A \cap S_i \neq \emptyset$  and  $S_i \times \{1\}$  if  $\mathcal{V}'_B \cap S_i \neq \emptyset$ . Thus, all sites are independently open with probability  $p_1 = 1 - e^{-\lambda_1(\varepsilon/\sqrt{d})^d}$ . We note that an infinite component in  $G'(\varepsilon, r_1)$  would imply an infinite cluster in the  $AB$  model where we connect points if they are closer than  $r_1$  to each other instead of 1 w.r.t.  $\mathcal{V}'_A, \mathcal{V}'_B$ . So it has to hold that  $p_1$  cannot be supercritical for  $G'$ , or in other words any  $p < p_1$  is subcritical. Thus, if we define the open cubes on  $G'(\varepsilon, r_1)$  w.r.t.  $\mathcal{V}_A$  and  $\mathcal{V}_B$ , we are subcritical, as  $\lambda < \lambda_1$ . Since  $1 + 2\varepsilon < r_1$ , the corresponding cube for every point of  $C_0$  with respect to  $\mathcal{V}(\lambda)$  and connection radius 1 will be part of  $C'_0$ . Thus if it holds that  $\underline{0} \longleftrightarrow B(\underline{0}, R)^c$  we will have  $|C'_0| \geq c_1 R$  for some  $c_1 = c_1(\varepsilon, d, r_1) > 0$ . This yields by Lemma 2.11 (we can apply it to our case, since a ball of radius  $n$  in  $G'$  contains no more than  $2(nr_1/\varepsilon)^d$  vertices) a constant  $c_2 = c_2(\lambda, \varepsilon, d) > 0$  with

$$\mathbb{P}(\underline{0} \longleftrightarrow B(\underline{0}, R)^c) \leq \mathbb{P}(|C'_0| \geq c_1 R) \leq e^{-c_1 c_2 R}.$$

□

**Corollary 4.30.** *For all  $\lambda < \lambda_c$  there exists a constant  $c = c(d, \lambda) > 0$  such that for all  $k \in \mathbb{N}$ , setting*

$$M_k := |\{a \in B(\underline{0}, k/2) \cap \mathcal{V}_A : a \longleftrightarrow B(\underline{0}, k)^c\}|,$$

*we have:*

$$\mathbb{P}(M_k > 0) < e^{-ck}.$$

*Proof.* By Lemma 4.29 and the Slivnyak-Mecke formula (see e.g. Corollary 3.2.3 in [89]) we obtain a constant  $c_1 = c_1(\lambda, d) > 0$  such that

$$\begin{aligned} \mathbb{P}(M_k > 0) &\leq \mathbb{E}M_k = \int_{B(\underline{0}, k/2)} \mathbb{P}(a \longleftrightarrow B(\underline{0}, k)^c \text{ w.r.t. } \mathcal{V}_A \cup \{a\}) \lambda \, da \\ &\leq \lambda \operatorname{vol}(B(\underline{0}, k/2)) e^{-c_1 k} \leq e^{-c_2 k}, \end{aligned}$$

where the last inequality holds for some constant  $c_2 = c_2(d, \lambda) > 0$ .  $\square$

For  $S \subset \mathbb{R}^d$  we will say that  $S$  is *covered by  $\mathcal{V}$*  if for every  $y \in S$  we have  $\mathcal{V}_A \cap B(y, 1/4) \neq \emptyset$  and  $\mathcal{V}_B \cap B(y, 1/4) \neq \emptyset$ . For  $k > 0, x \in \mathbb{R}^d$  we say that  $x$  is a  $k$ -*seed* if  $B(x, k)$  is covered. For a measurable  $R \subset \mathbb{R}^d$  and  $\delta > 0$  we are now adapting the intensity of  $\mathcal{V}_A$  (resp.  $\mathcal{V}_B$ ). Inside the area  $R$ , the PPP  $\mathcal{V}_A$  (resp.  $\mathcal{V}_B$ ) has intensity  $\delta$  and for  $R^c$  we use the original intensity  $\lambda$ . We write  $\mathbb{P}_{\lambda, \delta, R}$  for the corresponding measure. We also define the event

$$E_{\ell, k, R} := \left\{ R \xleftrightarrow{B(\underline{0}, 3\ell)} x \text{ for some } x \in \mathcal{V} \cap B(\ell e_1, \ell/100) \text{ and } x \text{ is a } 2k\text{-seed} \right\},$$

where  $e_1 = (1, 0, \dots, 0)$  stands for the first unit vector. We will say that  $R$  is *admissible* if for every  $x \in R$  there is a  $y \in R$  with  $\|x - y\| < 2$  and  $B(y, 1/2) \subseteq R$ . We note that if we define for some  $y \in \partial B(\underline{0}, \ell)$

$$E_{k, R}(\underline{0}, y) := \left\{ R \xleftrightarrow{B(\underline{0}, 3\ell)} x \text{ for some } x \in \mathcal{V} \cap B(y, \ell/100) \text{ and } x \text{ is a } 2k\text{-seed} \right\}, \quad (4.25)$$

by symmetry the result below also holds for this event.

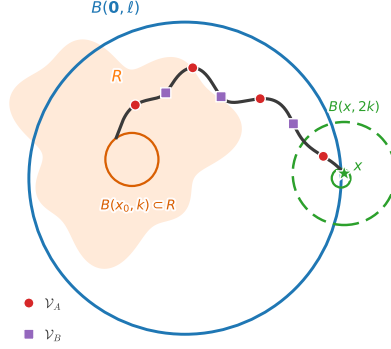
**Lemma 4.31.** *Let  $\lambda > \lambda_c$  and  $\varepsilon, \delta > 0$  be arbitrary with  $\lambda \geq \delta$ . Then, there exists a  $k = k(\lambda, \varepsilon, \delta)$  such that for all large enough  $\ell \geq \ell_0(\lambda, \varepsilon, \delta, k)$  it holds that*

$$\mathbb{P}_{\lambda, \delta, R}(E_{\ell, k, R}) > 1 - \varepsilon,$$

*for all admissible  $R$  with  $B(x, k) \subseteq R$  for some  $x \in B(\underline{0}, \ell)$ .*

The proof follows and adapts the ideas of Grimmett and Marstrand [46] and we will split it into four lemmas. We define the annulus  $A(\ell) := B(\underline{0}, \ell) \setminus B(\underline{0}, \ell - 1)$  and

$$X_{\ell, k} := \{x \in \mathcal{V} \cap A(\ell) : x \xleftrightarrow{B(\underline{0}, \ell)} B(\underline{0}, k)\}.$$

Figure 4.2: Schematic of the event  $E_{\ell,k,R}$ .

We let  $Y_{\ell,k}$  be the minimum size of a set of centers  $\{y_1, \dots, y_m\} \subset X_{\ell,k}$  such that  $X_{\ell,k} \subset \bigcup_{i=1}^m B(y_i, 100k)$  and  $\|y_i - y_j\| > 20k$  for all  $i \neq j$ .

**Lemma 4.32.** *For all  $\lambda > \lambda_c$  and  $\varepsilon > 0$  there is a  $k = k(\lambda, \varepsilon)$  such that the following holds. For all  $N \in \mathbb{N}$  we have for all  $\ell \geq \ell_0(\lambda, \varepsilon, N, k)$*

$$\mathbb{P}(Y_{\ell,k} \geq N) > 1 - \varepsilon.$$

*Proof.* We let  $\varepsilon > 0$  and since we are supercritical we can and do choose  $k = k(\lambda, \varepsilon)$  large enough such that

$$\mathbb{P}(B(\underline{0}, k) \longleftrightarrow \infty) > 1 - \varepsilon/2.$$

Then, since edges have a length of less than 1 we also have for all  $\ell > k$

$$\mathbb{P}(B(\underline{0}, k) \longleftrightarrow A(\ell)) > 1 - \varepsilon/2.$$

We now reveal  $\mathcal{V}$  in two stages. First, all points inside  $B(\underline{0}, \ell)$  and then the remaining points of  $B(\underline{0}, \ell)^c$ . We write  $S_1$  for the event that  $B(\underline{0}, k) \longleftrightarrow A(\ell)$  occurs after the first stage. Note that if for every center  $y$  of a ball counted by  $Y_{\ell,k}$  it holds that  $\mathcal{V} \cap B(y, 100k+1) \setminus B(\underline{0}, \ell) = \emptyset$ , the event  $B(\underline{0}, k) \longleftrightarrow \infty$  cannot hold. If after stage one it holds that  $Y_{\ell,k} \leq N$  for some  $N \in \mathbb{N}$ , the probability of this occurring in the second stage is lower bounded by  $\exp(-2N\lambda \text{vol}(B(\underline{0}, 100k+1)))$ , where the factor of 2 arises since we have two independent Poisson point processes with intensity  $\lambda$ . Thus for all  $N$

$$\mathbb{P}(B(\underline{0}, k) \nleftrightarrow \infty \mid S_1) \geq \exp(-2N\lambda \text{vol}(B(\underline{0}, 100k+1)))\mathbb{P}(Y_{\ell,k} \leq N \mid S_1).$$

Since we have that

$$\mathbb{P}(B(\underline{0}, k) \not\longleftrightarrow \infty \mid S_1) = 1 - \frac{\mathbb{P}(B(\underline{0}, k) \longleftrightarrow \infty)}{\mathbb{P}(B(\underline{0}, k) \longleftrightarrow A(\ell))},$$

and because the right hand side converges to 0 as  $\ell \rightarrow \infty$ , it follows that  $\mathbb{P}(Y_{\ell, k} \leq N \mid S_1) \rightarrow 0$  as  $\ell \rightarrow \infty$  too. We can conclude that for large enough  $\ell$

$$\mathbb{P}(Y_{\ell, k} \leq N) \leq \mathbb{P}(S_1^c) + \mathbb{P}(Y_{\ell, k} \leq N \mid S_1)\mathbb{P}(S_1) \leq \mathbb{P}(S_1^c) + \mathbb{P}(Y_{\ell, k} \leq N \mid S_1) < \varepsilon.$$

□

We now define the set

$$A_{\ell, k, R} := \{x \in \mathcal{V} \cap A(\ell) : x \xleftrightarrow{B(\underline{0}, \ell)} R \text{ and } x \text{ is a } 2k\text{-seed}\}.$$

**Lemma 4.33.** *For all  $\lambda > \lambda_c$  and  $\varepsilon, \delta > 0$  there exist  $k = k(\lambda, \varepsilon, \delta)$  such that for all large enough  $\ell \geq \ell_0(\lambda, \varepsilon, \delta, k)$*

$$\mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell - 2k - 2)^c}(|A_{\ell, k, B(\underline{0}, k)}| > 0) > 1 - \varepsilon.$$

*Proof.* We let  $\lambda_1 = (\lambda + \lambda_c)/2 > \lambda_c$  and  $N$  large enough to be determined in the course of the proof. Lemma 4.32 yields a  $k = k(\lambda_1, \varepsilon/2)$  and  $\ell \geq \ell_0(\lambda_1, \varepsilon/2, N, k)$  such that for continuum AB percolation with intensity  $\lambda_1$  it holds

$$\mathbb{P}_{\lambda_1}(Y_{\ell - 2k - 2, k} \geq N) > 1 - \varepsilon/2.$$

We can split  $\mathcal{V}(\lambda) \cap B(\underline{0}, \ell - 2k - 2)$  into the union of  $\mathcal{V}(\lambda_1)$  and  $\mathcal{V}(\lambda - \lambda_1)$ , where we take the union of the  $A$  and  $B$  label sets separately. We now reveal the points in two steps. First, we will reveal  $\mathcal{V}(\lambda_1)$  and then in the second stage all points of  $\mathcal{V}(\lambda - \lambda_1)$  and  $\mathcal{V}(\delta)$ . Note that if after stage one  $Y_{\ell - 2k - 2, k} \geq N$  holds, there exist at least  $N$  disjoint balls of radius  $10k$  that are centered at points of  $X_{\ell - 2k - 2, k}$ . For each such point  $x$  there is a probability that is lower bounded by some  $q := q(\lambda - \lambda_1, \delta, k) > 0$  that in the second revealment  $B(x, 9k) \cap B(\underline{0}, \ell - 2k - 2)$  is covered by  $\mathcal{V}(\lambda - \lambda_1)$  and  $B(x, 9k) \cap B(\underline{0}, \ell - 2k - 2)^c$  is covered by  $\mathcal{V}(\delta)$ . Note, for each  $x$ , this would yield a  $2k$ -seed w.r.t. the measure  $\mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell - 2k - 2)^c}$  centered at a point inside  $A(\ell)$ . Using the fact that for each  $x$ , whether  $B(x, 9k)$  is covered depends only on points inside  $B(x, 10k)$ , we obtain by writing  $S_1$  if  $Y_{\ell - 2k - 2, k} \geq N$  occurs after stage one

$$\mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell - 2k - 2)^c}(|A_{\ell, k, B(\underline{0}, k)}| > 0 \mid S_1) \geq \mathbb{P}(\text{Bi}(N, q) > 0) = 1 - (1 - q)^N.$$

Since  $q$  does not depend on  $\ell$ , we can now choose  $N$  and  $\ell$  large enough such that the right hand side is greater than  $1 - \varepsilon/2$ . As

$$\begin{aligned} \mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell - 2k - 2)^c}(|A_{\ell, k, B(\underline{0}, k)}| > 0) &\geq \mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell - 2k - 2)^c}(|A_{\ell, k, B(\underline{0}, k)}| > 0 \mid S_1)\mathbb{P}(S_1) \\ &\geq (1 - \varepsilon/2)^2 \geq 1 - \varepsilon, \end{aligned}$$

the proof is finished. □

We define an event where there is a point of  $A_{\ell,k,R}$  inside a fixed ball as

$$J_{\ell,k,R} := \{|A_{\ell,k,R} \cap B(\ell e_1, \ell/1000)| > 0\}.$$

**Lemma 4.34.** *For all  $\lambda > \lambda_c$  and  $\varepsilon, \delta > 0$  there exist  $k = k(\lambda, \varepsilon, \delta)$  such that for all large enough  $\ell \geq \ell_0(\lambda, \varepsilon, \delta, k)$*

$$\mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell-2k-2)^c}(J_{\ell,k,B(\underline{0},k)}) > 1 - \varepsilon.$$

*Proof.* Note that for  $\partial B(\underline{0}, \ell)$  and some dimension dependent constant  $c_d$ , we can place  $c_d$  many balls of radius  $\ell/1000$  with centers at the boundary such that they cover  $A(\ell)$ . We let  $B(x_i, \ell/1000)$ ,  $1 \leq i \leq c_d$  be an enumeration of it and can assume without loss of generality that  $x_1 = \ell e_1$ . For  $\varepsilon, \delta > 0$  we use Lemma 4.33 to obtain a  $k$  such that for all large enough  $\ell$

$$\mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell-2k-2)^c}(|A_{\ell,k,B(\underline{0},k)}| > 0) > 1 - \varepsilon^{c_d}.$$

If we now write  $A_i := \{|A_{\ell,k,B(\underline{0},k)} \cap B(x_i, \ell/1000)| > 0\}$  it holds that  $\cup A_i = \{|A_{\ell,k,B(\underline{0},k)}| > 0\}$  and by symmetry all  $A_i$  are identically distributed. Note that each event  $A_i$  is increasing w.r.t. adding points to either  $\mathcal{V}_A$  or  $\mathcal{V}_B$ . Therefore the decreasing events  $A_i^c$  are positively dependent, and by induction

$$\mathbb{P}\left(\bigcap_{i=1}^{c_d} A_i^c\right) \geq \prod_{i=1}^{c_d} \mathbb{P}(A_i^c).$$

Since the  $A_i$  are identically distributed, this gives

$$\mathbb{P}\left(\bigcup_{i=1}^{c_d} A_i\right) = 1 - \mathbb{P}\left(\bigcap_{i=1}^{c_d} A_i^c\right) \leq 1 - (1 - \mathbb{P}(A_1))^{c_d},$$

and hence

$$\begin{aligned} \mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell-2k-2)^c}(J_{\ell,k,B(\underline{0},k)}) &= \mathbb{P}(A_1) \\ &\geq 1 - (1 - \mathbb{P}_{\lambda, \delta, B(\underline{0}, \ell-2k-2)^c}(|A_{\ell,k,B(\underline{0},k)}| > 0))^{1/c_d} \\ &> 1 - \varepsilon. \end{aligned}$$

□

**Lemma 4.35.** *For all  $\lambda > \lambda_c$  and  $\varepsilon, \delta > 0$  with  $\lambda \geq \delta$  there exist  $k = k(\lambda, \varepsilon, \delta)$  such that for all large enough  $\ell \geq \ell_0(\lambda, \varepsilon, \delta, k)$*

$$\mathbb{P}_{\lambda, \delta, R}(J_{\ell,k,R}) > 1 - \varepsilon,$$

for all admissible  $R$  with  $B(\underline{0}, k) \subseteq R$ .

*Proof.* We let  $\lambda_1 = (\lambda + \lambda_c)/2 > \lambda_c$  and  $\varepsilon_1 > 0$  small to be chosen later. We can choose  $k, \ell_0$  as given by Lemma 4.34 such that for all  $\ell \geq \ell_0$

$$\mathbb{P}_{\lambda_1, \delta/2, B(\underline{0}, \ell-2k-2)^c}(J_{\ell, k, B(\underline{0}, k)}) > 1 - \varepsilon_1.$$

Further, we define the random set

$$S := \bigcup \{B(x, k) : x \in \mathcal{V} \cap A(\ell) \cap B(\ell e_1, \ell/1000) \text{ is a } 2k\text{-seed}\}.$$

Additionally we let  $\widehat{R} := R \cap B(\underline{0}, \ell - 2k - 2)$  and define

$$\mathcal{U} := \left\{ y \in \mathcal{V} \cap \widehat{R}^c : \inf_{w \in \widehat{R}} \|y - w\| < 1 \text{ and } y \xleftrightarrow{\widehat{R}^c \cap B(\underline{0}, \ell)} S \right\}$$

and write  $Y$  for the minimum number of balls with radius 100 that are centered at points of  $\mathcal{U}$  in a way that their union covers all points of  $\mathcal{U}$  and such that the centers are at distance of more than 10 from another. We again reveal the Poisson points in question in two steps. First, we reveal Poisson points in  $\mathcal{V}(\lambda_1) \cap B(\underline{0}, \ell - 2k - 2) \cap R^c$  and  $\mathcal{V}(\delta/2) \cap B(\underline{0}, \ell - 2k - 2)^c$ . In the second step we reveal the remaining points, so  $\mathcal{V}(\lambda_1) \cap \widehat{R}$ . Assume that after the first revealment it holds that  $Y \leq N$  for some  $N$ . We let  $\ell$  be large enough such that  $\ell - 2k - 2 \geq k$ . Then, if in the second step there are no new points of  $\mathcal{V}(\lambda_1)$  inside  $B(y, 101) \cap B(\underline{0}, \ell - 2k - 2) \cap R$  for all  $y$  counted by  $Y$ , the event  $J_{\ell, k, B(\underline{0}, k)}$  cannot hold, as  $B(\underline{0}, k) \subseteq R \cap B(\underline{0}, \ell - 2k - 2)$ . The probability of this occurring in the second step is lower bounded by  $q_1 := \exp(-2N\lambda_1 \text{vol}(B(\underline{0}, 101)))$ . Thus,

$$\varepsilon_1 > \mathbb{P}_{\lambda_1, \delta/2, B(\underline{0}, \ell-2k-2)^c}(J_{\ell, k, B(\underline{0}, k)}) \geq q_1 \cdot \mathbb{P}_{\lambda_1, \delta/2, B(\underline{0}, \ell-2k-2)^c}(Y \leq N).$$

As  $Y$  does not depend on any particles of  $R \cap B(\underline{0}, \ell - 2k - 2)$ , its distribution is identical under the measure  $\mathbb{P}_{\lambda_1, \delta/2, B(\underline{0}, \ell-2k-2)^c}$  and  $\mathbb{P}_{\lambda_1, \delta/2, B(\underline{0}, \ell-2k-2)^c \cup R}$ . This leads us to

$$\mathbb{P}_{\lambda_1, \delta/2, B(\underline{0}, \ell-2k-2)^c \cup R}(Y \leq N) \leq \frac{\varepsilon_1}{q_1}.$$

We now do two revealments with respect to  $\mathbb{P}_{\lambda, \delta/2, B(\underline{0}, \ell-2k-2)^c \cup R}$ , so the area where we reveal  $\mathcal{V}(\delta)$  is  $B(\underline{0}, \ell - 2k - 2)^c \cup R$ . Also, we can split  $\mathcal{V}(\delta) = \mathcal{V}_1(\delta/2) \cup \mathcal{V}_2(\delta/2)$  into two i.i.d. models if we take the union with respect to all  $A$  and  $B$  labels separately. The same holds for  $\mathcal{V}(\lambda) = \mathcal{V}(\lambda_1) \cup \mathcal{V}(\lambda - \lambda_1)$ . First we reveal all points of  $\mathcal{V}(\lambda_1)$  and  $\mathcal{V}_1(\delta/2)$  and in the second revealment we sprinkle with  $\mathcal{V}(\lambda - \lambda_1)$  and  $\mathcal{V}_2(\delta/2)$ . If in the second revealment we find at least one point of  $\mathcal{V}(\lambda - \lambda_1)$  inside  $R \cap B(\underline{0}, \ell - 2k - 2)$  that is connected to some particle of  $\mathcal{U}$ , the event  $J_{\ell, k, R}$  holds. For each center of a ball  $y$  counted by  $Y$  there is a chance that  $B(y, 5) \cap (B(\underline{0}, \ell - 2k - 2) \cap R^c)$  is covered with respect to  $\mathcal{V}(\lambda - \lambda_1)$  and  $B(y, 5) \cap (B(\underline{0}, \ell - 2k - 2)^c \cup R)$  is covered with respect to  $\mathcal{V}_2(\delta/2)$ . Additionally, as  $R$  is admissible there exists a  $z \in R$  such that  $B(z, 1/2) \subseteq B(y, 5) \cap R$ . So, if

in the second revealment there is a point of  $\mathcal{V}(\lambda - \lambda_1) \cup \mathcal{V}_2(\delta/2)$  inside  $B(z, 1/2)$  that is connected to  $y$ , the event  $J_{\ell,k,R}$  holds. This would be the case if the just described covering is fulfilled. The probability of all this occurring for a fixed  $y \in \mathcal{U}$  in the second revealment is lower bounded by some  $q_2 = q_2(\delta, \lambda) > 0$ . As distinct  $y$  are at distance of more than 10 to each other we obtain

$$\begin{aligned} \mathbb{P}_{\lambda, \delta/2, B(\underline{0}, \ell-2k-2)^c \cup R}(J_{\ell,k,R}) &\geq \left(1 - (1 - q_2)^N\right) \cdot \mathbb{P}_{\lambda_1, \delta/2, B(\underline{0}, \ell-2k-2)^c \cup R}(Y > N) \\ &\geq \left(1 - (1 - q_2)^N\right) \left(1 - \frac{\varepsilon_1}{q_1}\right). \end{aligned}$$

If we now choose  $\varepsilon_1 = \varepsilon_1(\delta, \lambda, \varepsilon)$  and  $N = N(\delta, \lambda, \varepsilon)$  appropriately, we can conclude that

$$\mathbb{P}_{\lambda, \delta, R}(J_{\ell,k,R}) \geq \mathbb{P}_{\lambda, \delta/2, B(\underline{0}, \ell-2k-2)^c \cup R}(J_{\ell,k,R}) > 1 - \varepsilon,$$

using that  $\lambda \geq \delta$  and that  $J_{\ell,k,R}$  remains to hold when adding points.  $\square$

*Proof of Lemma 4.31.* We let  $k(\lambda, \varepsilon, \delta)$  and  $\widehat{\ell}_0(\lambda, \varepsilon, \delta, k)$  be given by Lemma 4.35. Note that by symmetry and translation we have for all  $\widehat{\ell} \geq \widehat{\ell}_0$  and  $y \in \partial B(\underline{0}, \widehat{\ell})$

$$\mathbb{P}_{\lambda, \delta, R}(|A_{\widehat{\ell}, k, R} \cap B(y, \widehat{\ell}/1000)| > 0) > 1 - \varepsilon,$$

for all admissible  $R$  with  $B(\underline{0}, k) \subset R$ . We now let  $\ell \geq 1000\widehat{\ell}_0$  and assume  $B(x, k) \subset R$  for some  $x \in B(\underline{0}, \ell)$ . We see that for  $y = 1000\ell e_1/999$  it holds that  $\|x - y\| \in [\ell/999, 3\ell]$ , so the distance is more than  $\widehat{\ell}_0$  in any case. Additionally we have that  $B(y, \|x - y\|/1000) \subseteq B(\ell e_1, \ell/100)$ . We obtain by translation invariance

$$\begin{aligned} \mathbb{P}_{\lambda, \delta, R}(E_{\ell, k, R}) &\geq \mathbb{P}_{\lambda, \delta, R}\left(R \xleftrightarrow{B(\underline{0}, 3\ell)} z \text{ for some } 2k\text{-seed } z \in \mathcal{V} \cap B(y, \|x - y\|/1000)\right) \\ &> 1 - \varepsilon. \end{aligned}$$

$\square$

## 4.5.2 Existence of an odd cycle

From Lemma 4.1 we get to choose  $t$  sufficiently close to 1 such that for all  $x_1, x_2 \in C_t$

$$\|\pi(x_1) - \pi(x_2)\| \leq \text{dist}(x_1, x_2). \quad (4.26)$$

We fix some  $t > t_0$  that satisfies this, where  $t_0 = t_0(\varepsilon, d)$  is from Lemma 4.27. We are going to write  $\mathcal{V}_{\lambda, y}$  for the corresponding continuum AB percolation model given by Lemma 4.27 and will fix the  $\lambda$  it yields in the following. This time the projected points lie in the  $P_y = \{x \in \mathbb{R}^{d+1} : \langle x, y \rangle = 0\}$  plane. We will work with

different projections and to make it clearer in what plane certain  $d$ -dimensional balls lie we will write  $B_y(x, r)$  for the ball with radius  $r$  inside  $P_y$  around the point  $x \in P_y$ . We also let  $B(x, r) := B_{(0, \dots, 0, -1)}(x, r)$ .

### Exploration algorithm

We now define a graph  $H = H(\eta)$  on the sphere that depends on  $\eta > 0$ , which we will choose later. The goal is to define an exploration algorithm that successively declares edges of  $H$  open or closed. We let

$$m = \left\lfloor \frac{\sqrt{3}}{\sqrt{d}\eta} \right\rfloor,$$

and define  $H = \pi^{-1}(\eta\mathbb{Z}^d \cap D)$ , where  $D := [-\eta m, \eta m]^d$  and adjacencies are inherited from the nearest neighbor model. Note that this means it holds  $\sqrt{d}m\eta \leq \sqrt{3}$  and therefore for each  $x \in H$

$$\sqrt{3} \geq \|\pi(x)\| = \frac{\sqrt{1 - x_{d+1}^2}}{1 - x_{d+1}}, \quad (4.27)$$

yielding  $x_{d+1} \leq 1/2$ . Trivially, the south pole is an element of  $H$ . Additionally, we let  $A = A(d) \in \mathbb{N}$  be a constant integer that we will determine later, choose  $\lambda_c < \lambda_1 < \dots < \lambda_A < \lambda$  and define

$$\delta := \min \left\{ \lambda_1, \lambda_2 - \lambda_1, \dots, \lambda_A - \lambda_{A-1} \right\}.$$

For all  $x \in H$ , the points  $\mathcal{V}_{\lambda, x}$  that consist of the two Poisson point processes that correspond to the two labels of the  $AB$  model with intensity  $\lambda$  can be written as the union

$$\mathcal{V}_{\lambda, x} = \mathcal{V}_{\lambda_1, x} \cup \left( \bigcup_{i=2}^A \mathcal{V}_{\lambda_i - \lambda_{i-1}, x} \right) \cup \mathcal{V}_{\lambda - \lambda_A, x},$$

where we take the union of all  $A$  labels and  $B$  labels separately. Note that by [Lemma 4.27](#) each point in these continuum  $AB$  percolation models will be a point of the Borsuk graph by mapping them back with  $g_x^{-1}(\cdot)$ . For  $\varepsilon_1 > 0$  to be chosen later we let  $k(\lambda_1, \varepsilon_1, \delta), \ell_0(\lambda_1, \varepsilon_1, \delta)$  be the values from [Lemma 4.31](#). For  $x_1 = (0, \dots, 0, -1) \in H$  and an arbitrary neighbor  $y_1 \in H$  of it we start exploring to determine whether the edge  $x_1 y_1$  is open or not. We remind ourselves that  $g_{x_1}(x_1) = g(x_1) = \underline{0}$ . We declare  $x_1 y_1$  open if  $E_{k, B(\underline{0}, k)}(\underline{0}, g(y_1))$  from [Eq. \(4.25\)](#) holds with respect to  $\mathcal{V}_{\lambda_1, x_1}$ . Note that we can determine whether the event occurs by not revealing all points instantaneously. This can be done in the following way: First, we will reveal all  $\mathcal{V}_{\lambda_1, x_1} \cap B(\underline{0}, k)$ . We now write  $\mathcal{V}_{\lambda_1, x_1}(A)$  (resp.  $\mathcal{V}_{\lambda_1, x_1}(B)$ )

for all points with label  $A$  (resp. label  $B$ ). Then, for each  $a \in \mathcal{V}_{\lambda_1, x_1}(A) \cap B(\underline{0}, k)$  we can reveal all  $\mathcal{V}_{\lambda_1, x_1}(B) \cap B(a, 1)$  that are not yet revealed (so inside  $B(\underline{0}, k)^c$ ). Those points will then all be connected to  $a$ . In the same manner we will reveal for all  $b \in \mathcal{V}_{\lambda_1, x_1}(B) \cap B(\underline{0}, k)$  all  $\mathcal{V}_{\lambda_1, x_1}(A) \cap B(b, 1)$  that are not yet found. Iterating this procedure will reveal only  $\mathcal{V}_{\lambda_1, x_1}$  that lie inside  $B(\underline{0}, k)$  and points that are connected to  $B(\underline{0}, k)$ . Going back to the graph  $G_t$  on the sphere this means that we only reveal part of the points inside the ball

$$B_k := \{y \in S^d : \text{dist}(x_1, y) \leq \text{dist}(x_1, g^{-1}(ke_1))\}$$

and all other points found are connected to that set. We will never reveal area outside  $B(\underline{0}, 3\|g(y_1)\|)$  and stop exploring once either  $E_{k, B(\underline{0}, k)}(\underline{0}, g(y_1))$  is satisfied or all points we found inside  $B(\underline{0}, 3\|g(y_1)\|)$  have been used for an exploration. The next steps will be defined iteratively with the same strategy in mind as long as there exists an edge  $x_i y_i$  in  $H$  such that  $x_i$  is connected to  $x_1$  via a path of open edges and the status of  $x_i y_i$  is not determined yet. In addition, our exploration will yield a set  $R_i \subset S^d$  where we have already revealed part of the points of  $X$  and  $Y$ . We can split it into disjoint  $A_1, \dots, A_i$  such that for  $1 \leq j \leq i$  we have revealed points in  $A_j$  for  $j$  different iterations. Similarly to the initial edge, we open  $x_i y_i$  if  $E_{k, g_{x_i}(R_i)}(\underline{0}, g_{x_i}(y_i))$  occurs when we reveal  $\mathcal{V}_{\lambda_1, x_i}$  in the ‘fresh area’  $g_{x_i}(R_i^c)$  and  $\mathcal{V}_{\lambda_{j+1}-\lambda_j, x_i}$  inside  $g_{x_i}(A_j)$  for each  $1 \leq j \leq i$ . Note that by doing this, we have not encountered these points in any previous exploration. Using the same procedure as done for the first edge, we again reveal the points successively to make sure every found point is connected to  $B_{x_i}(\underline{0}, k)$ .

### Verifying requirements

In order to use [Lemma 4.31](#) to make sure edges are open with a probability close to 1, there are multiple requirements we have to check in order to make sure it is actually applicable to our setting. Firstly, for our steps in the exploration it is important that neighbors of  $H$  are sufficiently far away after projecting, as the Lemma required a distance of at least  $\ell_0$ .

**Lemma 4.36.** *For*

$$\eta = 6\ell_0\alpha(n)$$

*we have for all large enough  $n = n(d)$  that it holds for all neighboring  $x, y \in H$*

$$\ell_0 \leq \|g_x(y)\| \leq 24\ell_0.$$

*Proof.* For neighbors  $x, y \in H$  we have  $x_{d+1}, y_{d+1} \leq 1/2$  and so by [Lemma 4.1 \(i,iii\)](#) it holds that

$$\eta = \|\pi(x) - \pi(y)\| \leq 6\text{dist}(x, y) \leq 12\|\pi(x) - \pi(y)\| = 12\eta. \quad (4.28)$$

By [Lemma 4.1](#) (i) and using the symmetry of different stereographic projections we get

$$\|g_x(y)\| = \frac{2}{\alpha(n)} \|\pi_x(y)\| \geq \frac{1}{\alpha(n)} \text{dist}(x, y),$$

and the lower bound follows. Using that  $\|\pi(x) - \pi(y)\| = \eta$  we can apply [Lemma 4.1](#) (ii) and [Eq. \(4.26\)](#) to obtain for all large enough  $n = n(d)$

$$\|g_x(y)\| = \frac{2}{\alpha(n)} \|\pi_x(y)\| \leq \frac{2}{\alpha(n)} \text{dist}(x, y) \leq \frac{4}{\alpha(n)} \|\pi(x) - \pi(y)\| = 24\ell_0.$$

□

In the following, we always let  $\eta$  be as given by the lemma above. In order to determine whether we have a long connection in the Borsuk graph or not, we will project caps  $C_t(x)$ . It is important that the relative size of the cap is enough to be able to use [Lemma 4.31](#), as shown in the following. This means, we will not have to reveal points that are outside  $G_t(x_i)$  in order to determine whether  $E_{k, g_{x_i}(R_i)}(\mathbb{Q}, g_{x_i}(y_i))$  is satisfied wrt. the points we reveal.

**Lemma 4.37.** *We can choose  $n = n(t, \ell_0, d)$  large enough such that for all adjacent  $y, x \in H$  and all  $z \in \partial C_t(x)$*

$$\|g_x(z)\| \geq 3\|g_x(y)\|.$$

*Proof.* By [Lemma 4.36](#) it holds that  $24\ell_0 \geq \|g_x(y)\|$ , provided  $n = n(d)$  is large enough. Further, we have by symmetry of the different stereographic projections and [Lemma 4.26](#) for all  $z_2 \in \partial C_t$

$$\|g_x(z)\| = \|g(z_2)\| = \rho(t, n) = \frac{2\sqrt{1-t^2}}{(1+t)\alpha(n)}.$$

The claim now follows by choosing  $n = n(t, d, \ell_0)$  sufficiently large. □

We will choose  $n$  large enough such that the [Lemmas 4.36](#) and [4.37](#) above are satisfied.

**Lemma 4.38.** *For all edges  $h_1h_2$  and  $h_3h_4$  of  $H$  it holds that*

$$\text{dist}(h_1, h_2) \leq 12\text{dist}(h_3, h_4).$$

*Proof.* By [Lemma 4.1](#) we have that

$$\eta/6 = \|\pi(h_1) - \pi(h_2)\|/6 \leq \text{dist}(h_1, h_2) \leq 2\|\pi(h_1) - \pi(h_2)\| = 2\eta \leq 12\text{dist}(h_3, h_4).$$

□

Each successful exploration step that we will define after this Lemma produces a seed that gets used for the next iterations. The following makes sure that the seeds are actually large enough after the area gets distorted by the stereographic projections. The lemma implies in the beginning of each exploration step there is a  $B_{x_i}(x, k) \subseteq B_{x_i}(0, \|g_{x_i}(y_i)\|)$  such that  $g_{x_i}^{-1}(B_{x_i}(x, k))$  is connected to  $B_k$  in the Borsuk graph while staying inside  $R_i$ . Additionally,  $B_{x_i}(x, k)$  is covered with respect to  $\mathcal{V}_{\lambda, x_i}$ .

**Lemma 4.39.** *Let  $h_1h_2$  and  $h_2h_3$  be edges in  $H$ . Assume that  $x \in P_{h_1}$  and  $k \in \mathbb{N}$  are such that  $B_{h_1}(x, 2k) \subseteq B_{h_1}(g_{h_1}(h_2), 10^{-2}\|g_{h_1}(h_2)\|)$ . Then we can choose  $n = n(\ell_0, k, d)$  sufficiently large such that there is a  $y \in P_{h_2}$  with*

$$B_{h_2}(y, k) \subseteq g_{h_2}(g_{h_1}^{-1}(B_{h_1}(x, 2k))) \subseteq B_{h_2}(0, \|g_{h_2}(h_3)\|).$$

*Proof.* As  $t$  is fixed, we can and will choose  $n = n(\ell_0, k, d)$  large enough such that all areas on the sphere that we consider in this proof lie in  $C_t(h_1)$ . We let  $y = g_{h_2}(g_{h_1}^{-1}(x))$ . Then, by Eq. (4.26) and Lemma 4.1 it holds for all  $z_1 \in B_{h_2}(y, k)$  that

$$k \geq \|g_{h_2}(g_{h_1}^{-1}(x)) - z_1\| \geq \frac{1}{\alpha(n)} \text{dist}(g_{h_1}^{-1}(x), g_{h_2}^{-1}(z_1)) \geq \frac{1}{2} \|x - g_{h_1}(g_{h_2}^{-1}(z_1))\|,$$

and  $z_1 \in g_{h_2}(g_{h_1}^{-1}(B_{h_1}(x, 2k)))$  follows. For  $z_2 \in B_{h_1}(x, 2k) \subseteq B_{h_1}(g_{h_1}(h_2), 10^{-2}\|g_{h_1}(h_2)\|)$  we can bound

$$10^{-2}\|g_{h_1}(h_2)\| \geq \|z_2 - g_{h_1}(h_2)\| \geq \frac{1}{\alpha(n)} \text{dist}(g_{h_1}^{-1}(z_2), h_2) \geq \frac{1}{2} \|g_{h_2}(g_{h_1}^{-1}(z_2))\|,$$

where we utilized Lemma 4.1 again. We are now done, as it further holds by Lemma 4.38 and Eq. (4.26) that

$$\|g_{h_1}(h_2)\| \leq \frac{2}{\alpha(n)} \text{dist}(h_1, h_2) \leq \frac{24}{\alpha(n)} \text{dist}(h_2, h_3) \leq 24\|g_{h_2}(h_3)\|.$$

□

It is important that for the area that is relevant for the event, we did not encounter it in too many previous iterations. This is because if we define

$$a_i := |\{1 \leq j \leq i : g_{x_i}(A_j) \cap B_{x_i}(0, 3\|g_{x_i}(y_i)\|) \neq \emptyset\}|,$$

to be the number of revealments,  $a_i$  is uniformly bounded from above, as shown in the following two lemmas. This is needed, as we have chosen  $A(d)$  different intensities  $\lambda_i$  in the beginning. For  $x \in H$  we will write  $x^o$  for the point of  $H$  that is closest to  $-x$ . If this is not a unique point we pick a point out of the possible choice by some arbitrary deterministic rule.

**Lemma 4.40.** *Let  $r_1 > 0$  and for all  $x \in H$  define an event  $E_x$  that depends only on the points of  $\mathcal{V}_{\lambda,x}$  inside  $B_x(\underline{0}, r_1 \ell_0)$ . Then, there is a constant  $M = M(r_1, d) > 0$  such that each  $E_x$  depends only on  $E_y$  for sites  $y \in H$  that have a graph distance of less than  $M$  to  $x$  or  $x^o$ .*

*Proof.* By Lemmas 4.1 and 4.36 and Eq. (4.26) it holds for all  $z \in S^d$  with  $g_x(z) \in B_x(\underline{0}, r_1 \ell_0)$  and for all edges  $uv$  of  $H$

$$\frac{1}{\alpha(n)} \text{dist}(x, z) \leq r_1 \ell_0 \leq r_1 \|g_u(v)\| \leq \frac{2r_1}{\alpha(n)} \text{dist}(u, v).$$

Writing  $\Delta := \max\{\text{dist}(u, v) : uv \text{ is an edge in } H\}$  we can thus say that  $\text{dist}(x, z) \leq 2r_1 \Delta$ . Going back to the sphere, it follows that the region on which  $E_x$  depends on is a subset of

$$\{y \in S^d : \text{dist}(x, y) \leq 2r_1 \Delta\} \cup \{y \in S^d : \text{dist}(-x, y) \leq 2r_1 \Delta\}.$$

As a square of side length  $\ell$  in Euclidean space has a diameter of  $\ell\sqrt{d}$ , we can now further say that this is a subset of

$$\{y \in S^d : \text{dist}(x, y) \leq 2r_1 \Delta\} \cup \{y \in S^d : \text{dist}(-x, y) \leq (2r_1 + \sqrt{d})\Delta\}.$$

We follow that any  $E_y$ , where  $y$  has graph distance of more than  $2(2r_1 + \sqrt{d})$  from  $x$  and  $x^o$  is independent of  $E_x$ .  $\square$

**Lemma 4.41.** *There exists a dimension dependent integer  $A = A(d) \in \mathbb{N}$  such that for all iteration steps  $a_i < A$ .*

*Proof.* Note that by Lemma 4.36 for all  $i$  the exploration of the status of  $x_i y_i$  only depends on points that fall inside  $B_{x_i}(\underline{0}, 3\|g_{x_i}(y_i)\|) \subset B_{x_i}(\underline{0}, 100\ell_0)$ . By Lemma 4.40 there is now a constant  $c = c(d) > 0$  such that there are only  $c$  points  $x_j \in H$  where the sets  $R_i$  and  $R_j$  can intersect if  $x_j$  is an endpoint of an exploration for an edge.  $\square$

In order to use Lemma 4.31 we also have to make sure that the relevant previous area is actually admissible.

**Lemma 4.42.** *For all iteration steps,  $g_{x_i}(R_i) \cap B_{x_i}(\underline{0}, 3\|g_{x_i}(y_i)\|)$  is admissible, provided that  $n$  is large enough.*

*Proof.* Assume we finished exploring for  $x_j y_j$  and added some  $y \in B_{x_j}(\underline{0}, 3\|g_{x_j}(y_j)\|)$  to  $g_{x_j}(R_j)$ . Then, by construction, there also is a  $x \in B_{x_j}(y, 1)$  with  $B_{x_j}(x, 1) \subseteq g_{x_j}(R_j)$ . Note that by Lemma 4.40, area that gets revealed for  $x_j y_j$  will only intersect  $B_{x_i}(\underline{0}, 3\|g_{x_i}(y_i)\|)$  for  $x_i$  that have graph distance less than  $m$  from  $x_j$

or  $x_j^o$ . If  $x_i$  has distance less than  $m$  from  $x_j$  we can choose  $n$  large enough such that  $x_j \in C_t(x_i)$ . Then we obtain by using Eq. (4.26)

$$\|g_{x_i}(g_{x_j}^{-1}(x)) - g_{x_i}(g_{x_j}^{-1}(y))\| \leq \frac{2}{\alpha(n)} \text{dist}(g_{x_j}^{-1}(x), g_{x_j}^{-1}(y)) \leq 2\|x - y\| < 2.$$

Likewise, for all  $v \in \partial B_{x_j}(x, 1)$  it holds

$$1 = \|v - x\| \leq \frac{2}{\alpha(n)} \text{dist}(g_{x_j}^{-1}(v), g_{x_j}^{-1}(x)) \leq 2\|g_{x_i}(g_{x_j}^{-1}(v)) - g_{x_i}(g_{x_j}^{-1}(x))\|.$$

The same inequalities go through if  $x_i$  has a graph distance of less than  $m$  to  $x_j^o$ . It follows that  $g_{x_i}(R_i) \cap B_{x_i}(0, 3\|g_{x_i}(y_i)\|)$  is admissible.  $\square$

### Large open cluster in $H$

In the following we are going to write  $|H|$  for the number of vertices of  $H$ , which only depends on  $\eta = \eta(n, \ell_0, d)$  and the dimension.

**Lemma 4.43.** *For all  $\varepsilon_1 > 0$  we can choose  $n_0 = n_0(\varepsilon_1, \ell_0, k, d)$  large enough such that for all  $n \geq n_0$  the following holds. After our exploration procedure terminates, with a probability of more than  $1 - \varepsilon_1$ , the connected component of  $x_1$  is greater than  $(1 - \varepsilon_1)|H|$ .*

*Proof.* By the choice of  $\ell_0, k, \delta$ , the definition of our algorithm, Lemma 4.31 and the lemmas of Section 4.5.2, each edge that gets processed in our exploration is set to be open with probability of more than  $1 - \varepsilon_1$ , provided that  $n$  is sufficiently large. This lower bound holds uniformly over the revealed set  $R_i$  and also over any points we revealed. Further, by the definition of our exploration all edges that are set to be open, are connected to  $x_1$  via an open path and we will run the exploration for every edge that already has an adjacent edge that is connected to  $x_1$ . Thus, we can couple the exploration of the edges with the cluster of  $x_1$  in usual bond percolation on  $H$  with parameter  $1 - \varepsilon_1$  such that the cluster of  $x_1$  is a subset of the component our exploration yields. As  $\eta^{-1}\pi(H) = \mathbb{Z}^d \cap \{-m, \dots, m\}^d$  we can deduce the claim from Lemma 2.10 by noting that  $m \rightarrow \infty$  as  $n \rightarrow \infty$ .  $\square$

We write  $\pi_s := \pi_{(0, \dots, 0, 1)}$  for the projection through the south pole. Similarly to  $H$  we now define the mirrored version  $H^2 := \pi_s^{-1}(\eta\mathbb{Z}^d \cap D)$ , where we remember we had  $D := [-\eta m, \eta m]^d$  with  $m = \lfloor \sqrt{3}/\sqrt{d}\eta \rfloor$ . Additionally let  $D^o := [-0.9m\eta, 0.9m\eta]^d \subset D$  and

$$S := \{x \in S^d : \pi(x) \in D^o, \pi_s(x) \in D^o\}.$$

Further, we let  $b := 0.9m\eta/2$  and  $a := 1/\sqrt{2d}$ . Note that for all large enough  $n$  it holds that  $b > a$  since  $m\eta$  converges to  $\sqrt{3}/d$ . We define

$$B := \{z \in \mathbb{R}^d : a \leq |z_j| \leq b \text{ for all } j = 1, \dots, d\}$$

to obtain  $B \subset D^\circ$  and to define

$$H_B := \pi^{-1}(\eta\mathbb{Z}^d \cap B) \quad \text{and} \quad H_B^2 := \pi_s^{-1}(\eta\mathbb{Z}^d \cap B).$$

We also define the function  $I : \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{R}^d$  with

$$I(z) := \frac{z}{\|z\|^2}.$$

Note that as for  $x \in S^d$  we have

$$\frac{1 - x_{d+1}}{1 + x_{d+1}} = \frac{1}{\|\pi(x)\|^2} = \|\pi_s(x)\|^2,$$

we can easily derive the identities  $\pi_s(x) = I(\pi(x))$  and  $\pi(x) = I(\pi_s(x))$ . The two smaller graphs  $H_B$  and  $H_B^2$  are needed to deal with boundary issues in the following Lemma.

**Lemma 4.44.** *If  $n$  is large enough, the following hold:*

(i) *There is a constant  $c = c(d) > 0$  such that*

$$|H \cap S| = |H^2 \cap S| \geq |H_B| = |H_B^2| \geq c|H|.$$

(ii) *For every  $u \in H_B$  there exists  $v \in H^2 \cap S$  with*

$$\|g_u(v)\| \leq 12\sqrt{d}\ell_0,$$

*and conversely for every  $v \in H_B^2$  there exists  $u \in H \cap S$  with the same bound.*

*Proof.* It holds that

$$|\eta\mathbb{Z}^d \cap B| \geq \left( \frac{2(b-a)}{\eta} - 2 \right)^d,$$

and also  $|H| \leq (2m+1)^d \leq (3m)^d$ . Since  $m\eta$  converges to a constant as  $n \rightarrow \infty$  and  $b-a$  is also a constant depending on  $d$  once  $n$  is large, the ratio  $|\eta\mathbb{Z}^d \cap B|/|H|$  is bounded from below by some  $c = c(d) > 0$  for all large enough  $n$ . This yields  $|H_B| \geq c|H|$  and by symmetry the same bound holds for  $H_B^2$ . For the second claim take some  $u \in H_B$  and let  $p := \pi_s(u) = I(\pi(u)) \in D^\circ$ . We define  $q \in \eta\mathbb{Z}^d$  by rounding each coordinate of  $p$  to the nearest multiple of  $\eta$  and set  $v := \pi_s^{-1}(q)$ . Since  $p \in I(B) \subset D^\circ$  and  $I(B)$  is compact and strictly contained in  $D^\circ$ , for all sufficiently large  $n$  the rounding point lies in  $D^\circ$ . Since  $z = \pi(u) \in B$ , every coordinate of  $z$  satisfies  $|z_j| \leq b$ , hence

$$\inf_{x \in \partial D^\circ} \|z - x\|_\infty \geq 0.9m\eta - b = b.$$

We define  $r_0 := \min\{\|w\| : w \in I(B)\} > 0$  and also  $M := \max\{\|w\| : w \in I(B)\} < \infty$ . As  $\|p - q\| \leq (\sqrt{d}/2) \cdot \eta \rightarrow 0$  we have for all large enough  $n$  that  $\|p - q\| < r_0/2$  and  $\|p - q\| < 1$ . We now have that  $\|q\| \geq \|p\| - \|p - q\| \geq r_0/2$  and also  $\|q\| \leq \|p\| + \|p - q\| \leq M + 1$ . Thus, both  $p$  and  $q$  lie in the compact set  $K := \{x : r_0/2 \leq \|x\| \leq M + 1\}$ . Hence  $I$  is uniformly continuous on  $K$ . Since  $\|p - q\| \leq \sqrt{d}\eta/2 \rightarrow 0$ , for all sufficiently large  $n$  we have

$$\|I(p) - I(q)\|_\infty < b.$$

As  $I(p) = z$ , it follows that  $I(q) \in D^o$ . Therefore  $\pi(v) = I(q) \in D^o$ , so  $v \in S$ . As by [Lemma 4.1](#)

$$\text{dist}(u, v) \leq 2\|\pi_s(u) - \pi_s(v)\| = 2\|p - q\| \leq \sqrt{d}\eta,$$

and by using [Eq. \(4.26\)](#) it holds for all large enough  $n$

$$\|g_u(v)\| = \frac{2}{\alpha(n)}\|\pi_u(v)\| \leq \frac{2}{\alpha(n)}\text{dist}(u, v),$$

so we can conclude the proof.  $\square$

**Lemma 4.45.** *For all  $\varepsilon, \varepsilon_2 > 0$  and if  $\alpha(n) \geq (c_3 + \varepsilon)n^{-1/d}$ , we can choose  $n_0 = n_0(\varepsilon, \varepsilon_2)$  large enough such that for all  $n \geq n_0$*

$$\mathbb{P}(\text{there exists } X_i \in X \text{ that is connected to } -X_i \text{ in } G_{ge0}) > 1 - \varepsilon_2.$$

*Proof.* For  $\varepsilon_3 > 0$  we can use [Lemmas 4.43](#) and [4.44](#) to get  $\ell_0, k$  such that for all large enough  $n$ , with a probability of more than  $1 - \varepsilon_3$  the number of points of  $H$  that lie inside  $S$  that are connected to  $x_1$  by an open path is more than  $|H|(c - \varepsilon_3)$  for some  $c = c(d) > 0$ . We will write  $F$  if this event occurs. In the definition of  $H$  we have used the stereographic projection  $\pi$  that projects from the north pole and we started exploring from the south pole  $x_1$ . By symmetry we could have also used the projection from the south pole,  $\pi_s$  and start our exploration at the north pole to determine whether  $-x_1(-y_1)$  is open with respect to  $H^2$ . Note that again by symmetry, so changing the roles of  $X$  and  $Y$ , if  $F$  occurs, then there are also more than  $|H|(c - \varepsilon_3)$  points of  $H^2$  that are connected to  $-x_1$  by an open path in  $H^2$ . All  $x \in H$  that are part of an open edge  $xy$  have at least one point inside  $\{z \in S^d : \text{dist}(z, x) \leq \text{dist}(x, y)\}$  that is connected in the Borsuk graph to  $B_k$ . In the same sense if the edge  $x^2y^2$  is open in  $H^2$ , the set  $\{z \in S^d : \text{dist}(z, x^2) \leq \text{dist}(x^2, y^2)\}$  is connected to  $-B_k$ . Also, each point inside  $B_k$  (resp.  $-B_k$ ) has a mirror image inside  $-B_k$  (resp.  $B_k$ ). By [Lemma 4.44](#) for every  $x^1 \in H_B$  there is a  $x^2 \in H^2 \cap S$  with  $\|g_{x^1}(x^2)\| \leq 12\sqrt{d}\ell_0$ . We call  $x^1 \in H_B$  *good* if in addition to this distance requirement it holds that  $x^1$  is connected to  $x_1$  inside  $H$  and  $x^2 \in H^2 \cap S$  is connected to  $-x_1$  w.r.t.  $H^2$ . We also see that for  $y^2$  being adjacent to  $x^2$  and  $z$  with  $\text{dist}(z, x^2) \leq \text{dist}(x^2, y^2)$  we can bound

$$\|g_{x^1}(x^2) - g_{x^1}(z)\| \leq \frac{2}{\alpha(n)}\text{dist}(x^2, z) \leq \frac{4}{\alpha(n)}\eta = 24\ell_0.$$

For each  $x \in H$  we are going to define the event

$$E_x := \{B_x(\underline{0}, 36\sqrt{d}\ell_0) \text{ is covered w.r.t. } \mathcal{V}_{\lambda-\lambda_A, x}\}.$$

Note that if in addition to  $F$  at least one  $E_x$  occurs for a good  $x$ ,  $B_k$  is connected to  $-B_k$  and by symmetry there has to exist  $X_i \in B_k$  that is connected to  $-X_i \in (-B_k)$ . Also by [Lemma 4.41](#), we have never revealed  $\mathcal{V}_{\lambda-\lambda_A, x}$  for any  $x \in H$  during our exploration process. By [Lemmas 4.40](#) and [4.44](#) we can conclude that there is a constant  $c_2 = c_2(d) > 0$  such that we can find  $J \subset H_B$  with  $|J| \geq c_2|H|$  such that for any two  $x, y \in J$  it holds that  $E_x$  and  $E_y$  are independent. Using this independence and that under  $F$  there are less  $\varepsilon_3|H|$  points of  $H$  (resp.  $H^2$ ) that are not connected to  $x_1$  (resp.  $-x_1$ ) we obtain

$$\mathbb{P}(E_x \text{ occurs for at least one good } x \mid F) \geq 1 - \mathbb{P}(E_x^c \mid F)^{|H|(c_2-2\varepsilon_3)}.$$

As  $\mathbb{P}(E_x \mid F) > 0$  depends only on  $(\ell_0, \lambda, d)$ , the lemma follows by letting  $n$  be large enough and choosing  $\varepsilon_3$  appropriately, together with the fact that  $|H| \rightarrow \infty$  with  $n \rightarrow \infty$ .  $\square$

**Lemma 4.46.** *For all  $\varepsilon, \varepsilon_2 > 0$  and if  $\alpha(n) \geq (c_3 + \varepsilon)n^{-1/d}$ , we can choose  $n_0 = n_0(\varepsilon, \varepsilon_2)$  large enough such that for all  $n \geq n_0$*

$$\mathbb{P}(\chi(G(\mathcal{X}, \alpha)) \geq 3) \geq 1 - \varepsilon_2.$$

*Proof.* Assume that there exists  $X_i \in X$  that is connected to  $-X_i$  in  $G_{geo}$ . Then, there is a path  $x_0, y_0, x_1, y_1, \dots, x_k, y_k$  with  $x_0 = X_i, y_k = -X_i$  and  $x_j \sim y_j \sim x_{j+1}$  for all  $0 \leq j \leq k-1$  and also  $x_k \sim y_k$ . Note that in the Borsuk graph this implies the existence of the path  $x_0, -y_0, x_1, -y_1, \dots, x_k, -y_k$ . Thus,  $X_i$  is connected to itself via an odd cycle. Since no odd cycle can be colored with just two alternating colors, the statement we are proving follows directly from [Lemma 4.45](#).  $\square$

### 4.5.3 Bipartite regime

We will write  $\mathcal{W}_{\lambda, x}$  for the AB model as given by [Lemma 4.28](#) if we use  $c \cdot g_x(\cdot)$  to project the points from  $C_t(x)$  (where  $c > 0$  is from the lemma and by symmetry the identical claim also holds in the cases where we have a different projection).

**Lemma 4.47.** *For all  $\varepsilon > 0$  and if  $\alpha(n) \leq (c_3 - \varepsilon)n^{-1/d}$*

$$\mathbb{P}(\chi(G(\mathcal{X}, \alpha)) \leq 2) = 1 - o(1).$$

*Proof.* We let  $t = t(\varepsilon)$  be as provided by [Lemma 4.28](#). For each  $X \in \mathcal{X}$  define

$$E_X := \{\text{the component of } X \text{ in } G_{geo} \text{ is not contained inside } C_t(X)\}$$

and also for each  $x \in S^d$

$$J_x := \{\exists y \in \mathcal{W}_{\lambda,x} \cap B_x(\mathbb{Q}, \rho(t, n)/4) : y \longleftrightarrow B_x(\mathbb{Q}, \rho(t, n)/2)^c\}.$$

We note that by [Lemma 4.28](#) it holds that  $E_X \subseteq J_X$ . We express the total number of occurrences of  $E_X$  as

$$N := \sum_{X \in \mathcal{X}} \mathbb{1}_{E_X}.$$

Since  $\mathcal{X}$  is a Poisson point process on  $S^d$  with intensity  $n/((d+1)\kappa_{d+1})$  we can write by using the Mecke formula (see e.g. Theorem 3.2.5 of [89]) and [Corollary 4.30](#)

$$\mathbb{E}N = \frac{n}{(d+1)\kappa_{d+1}} \int_{S^d} \mathbb{P}(E_u \text{ occurs w.r.t. } \mathcal{X} \cup \{u\}) d\sigma(u),$$

where  $d\sigma$  is the surface measure on  $S^d$ . Note that since  $\rho(t, n)/4 > 2$  for all large enough  $n$ , the probability of  $E_u$  occurring is unchanged if we add a deterministic point of either label  $A$  or  $B$  to the center of  $B_u(\mathbb{Q}, \rho(t, n)/4)$ . Therefore, by [Corollary 4.30](#) we obtain a constant  $c = c(\varepsilon) > 0$  such that for all  $u \in S^d$

$$\mathbb{P}(E_u \text{ occurs w.r.t. } \mathcal{X} \cup \{u\}) \leq \mathbb{P}(J_u) \leq e^{-c\rho(t, n)}.$$

We conclude that  $\mathbb{E}N = o(1)$  as  $n \rightarrow \infty$ . If  $\mathbb{E}N = 0$ , almost surely for all  $X \in \mathcal{X}$  it holds that the component of  $X$  in  $G_{geo}$  is contained in  $C_t(X)$ . If this is the case, for each  $G_t(X)$  we can assign a color to the points of  $\mathcal{X}$  and another color to the points of  $\mathcal{M}$ , since on  $C_t(X)$  these sets are disjoint, and all particles are now connected to a different color than themselves. Since any two coloring of  $G_{geo}$  implies a two coloring of  $G(\mathcal{X}, \alpha)$ , we can conclude the proof.  $\square$

*Proof of Theorem 1.8.* [Theorem 1.8](#) follows directly by combining [Lemma 4.46](#) and [4.47](#).  $\square$

## 4.6 Empty graph

For the Borsuk graph  $G(n, \alpha)$  on  $n$  vertices we write  $I_{ij} = \mathbb{1}_{\{X_i \text{ is connected to } X_j\}}$ . Then, we can also express the total number of edges as

$$W_n := \sum_{i < j} I_{ij}$$

and the connection probability as

$$p_n := \mathbb{P}(I_{ij} = 1) = \mathbb{P}(U \in \text{cap}((0, \dots, 0, 1), \alpha)),$$

where  $U \stackrel{d}{=} \text{Unif}(S^d)$ .

**Lemma 4.48.** *If  $\alpha = \alpha(n) \rightarrow 0$  as  $n \rightarrow \infty$  then*

$$p_n = (1 + o_n(1)) \cdot c_d \cdot \alpha^d,$$

where

$$c_d := \frac{\Gamma((d+3)/2)}{(d+1) \cdot \sqrt{\pi} \cdot \Gamma((d+2)/2)}$$

*Proof.* Writing  $Z := \pi(U)$ , parts (i) and (ii) of Lemma 4.1 tell us that

$$\mathbb{P}(Z \in B(\underline{0}, \alpha/2)) \leq p_n \leq \mathbb{P}(Z \in B(\underline{0}, (1+\varepsilon) \cdot \alpha/2)),$$

for  $\varepsilon > 0$  an arbitrary fixed small constant and  $n$  sufficiently large (using  $\alpha(n) \rightarrow 0$ ). Invoking Lemma 4.3, this gives

$$\frac{\left(\frac{2}{1+(\alpha/2)^2}\right)^d}{(d+1)\kappa_{d+1}} \cdot \kappa_d(\alpha/2)^d \leq p_n \leq \frac{2^d}{(d+1)\kappa_{d+1}} \cdot \kappa_d((1+\varepsilon)\alpha/2)^d.$$

Sending  $\varepsilon \searrow 0$  and using that  $\alpha \rightarrow 0$ , we see that indeed

$$p_n = (1 + o_n(1)) \cdot \frac{\kappa_d}{(d+1)\kappa_{d+1}} \cdot \alpha^d = (1 + o_n(1)) \cdot \frac{\Gamma((d+3)/2)}{(d+1) \cdot \sqrt{\pi} \cdot \Gamma((d+2)/2)} \cdot \alpha(n)^d,$$

where we have used (2.1) for the final identity.  $\square$

**Lemma 4.49.** *If  $n^2\alpha(n)^d \rightarrow \nu \in (0, \infty)$  it holds that*

$$W_n \xrightarrow{\mathcal{L}} \text{Po}((c_d/2) \cdot \nu).$$

*Proof.* We are going to show that for all  $m \in \mathbb{N}$

$$\lim_{n \rightarrow \infty} \mathbb{E}(W_n)_m = (c_d\nu/2)^m.$$

The proof is then done by Lemma 4.5. We can write

$$\mathbb{E}(W_n)_m = \sum_{\substack{e_1, \dots, e_m \\ \text{distinct}}} \mathbb{P}(I_{e_1} = 1, \dots, I_{e_m} = 1), \quad (4.29)$$

where the sum is over all ordered  $m$ -tuples of distinct candidate edges. We write  $2 \leq s \leq 2m$  for the number of unique vertices that occur in  $(e_1, \dots, e_m)$  in the sum above. For  $s = 2m$  there are  $(n)_{2m}/2^m$  possible choices of  $(e_1, \dots, e_m)$ . This holds since  $(n)_{2m}$  gives the number of ordered lists of length  $2m$  out of  $n$  vertices, but an edge is an unordered pair we count each of the  $m$  edges twice. If all vertices

are different it holds by independence that  $\mathbb{P}(I_{e_1} = 1, \dots, I_{e_m} = 1) = p_n^m$ . Thus, the  $s = 2m$  part of the sum is equal to

$$\frac{(n)_{2m}}{2^m} p_n^m = n^{2m} \prod_{i=0}^{2m-1} \left(1 - \frac{i}{n}\right) \cdot \left(\frac{p_n^m}{2^m}\right) = \left(\frac{n^2 p_n}{2}\right)^m (1 + o(1)),$$

and the right hand side converges to  $(c\nu/2)^m$  by assumption and [Lemma 4.48](#). We will now fix  $e = (e_1, \dots, e_m)$  that corresponds to some  $s$  with  $2 \leq s \leq 2m - 1$ . For that, we will write  $c = c(e)$  for the number of connected components w.r.t. the graph  $G = G(e)$  with structure  $e$ . We will write  $F = F(e)$  for an arbitrary spanning forest of  $G$ . For each tree we can start at one arbitrary vertex as the root and reveal a point that is a candidate for an edge. We can iterate this and in each revealment we choose a point that is already part of the tree and check whether an unrevealed point that was a candidate for a connection falls into the necessary cap. Then, in each step the probability of an edge being present equals  $p_n$ . Note that there are  $e(F) = s - c$  edges in total. Therefore by independence

$$\mathbb{P}(\text{all edges are present in } G) \leq \mathbb{P}(\text{all edges are present in } F) = p_n^{e(F)}.$$

We further claim to have  $s \geq 2c + 1$ . This can be seen as follows. First,  $s \geq 2c$ , as every component has at least two vertices. If  $s = 2c$  would be true every component would have exactly two vertices and one edge, leading to  $s = 2m$ . But we assumed  $s < 2m$  here, leading to the contradiction. We conclude that  $p_n^{e(F)} \leq p_n^{s/2+1/2}$ . We now want to compute an upper bound on the possible number of edges  $(e_1, \dots, e_m)$  that correspond to a fixed  $s$ . First, we choose an ordered list of  $s$  distinct vertex labels. There are less than  $(n)_s \leq n^s$  possible ways of doing that. We can now form  $\binom{s}{2}$  many unordered pairs out of them. There are  $\binom{s}{2}^m$  many ways of forming an ordered  $m$  tuple made out of these pairs. We note that this construction over-counts (the list of  $s$  vertex labels is ordered and we allowed for repetitions in the  $m$  tuple), but the bound is good enough for our needs. Finally, we sum over all possible  $c \leq s/2$  to conclude with our derivation above that each fixed  $2 \leq s \leq 2m - 1$  has a contribution for [Eq. \(4.29\)](#) of less than

$$n^s \binom{s}{2}^m \cdot \frac{s}{2} \cdot p_n^{s/2+1/2} = \frac{s}{2} \binom{s}{2}^m (n^2 p_n)^{s/2+1/2} \cdot n^{-1}.$$

As  $s$  and  $m$  are fixed and  $n^2 p_n \rightarrow c_d \nu$  by our assumption and [Lemma 4.48](#), the parts that do not correspond to  $s = 2m$  vanish with  $n \rightarrow \infty$ .  $\square$

*Proof of Theorem 1.9.* [Theorem 1.9](#) follows from [Lemma 4.49](#) by noting that for  $X \sim \text{Po}(\mu)$  it holds that  $\mathbb{P}(X = 0) = e^{-\mu}$  and that  $W_n = 0$  implies  $\chi(G(n, \alpha)) = 1$ .  $\square$

## 4.7 Discussion and suggestions for further work.

Theorem 1.10 showed that the property of having chromatic number  $\geq k$  has a sharp threshold “for almost all  $n$ ”. Naturally we conjecture that this last addition is not needed:

**Conjecture 4.50.** For every  $d \geq 2$  and  $4 \leq k \leq d + 2$ , the property of having chromatic number at least  $k$  has a sharp threshold.

The proof of Theorem 1.10 leveraged the semi-sharpness of the threshold to show that the value of  $\alpha$  where the probability of chromatic number  $\geq k$  is exactly  $1/2$  as a function of  $n$  can only “jump” relatively rarely. If one could argue that instead of rarely, it does not jump at all (for large  $n$ ) then one would have proved the conjecture.

We offer the following more detailed conjectures stating also the form of the threshold that most researchers in the field will probably expect:

**Conjecture 4.51.** For every  $d \geq 2$  there exists a constant  $c = c(d)$  such that, for every fixed  $\varepsilon > 0$  and sequence  $\alpha = \alpha(n)$ :

$$\lim_{n \rightarrow \infty} \mathbb{P}(\chi(G(n, \alpha)) \geq d + 2) = \begin{cases} 1 & \text{if } \alpha > (1 + \varepsilon) \cdot c \cdot (\ln n/n)^{1/d}, \\ 0 & \text{if } \alpha < (1 - \varepsilon) \cdot c \cdot (\ln n/n)^{1/d}. \end{cases}$$

**Conjecture 4.52.** For each  $d \geq 2$ , there exist constants  $c_4 = c_4(d), \dots, c_{d+1} = c_{d+1}(d)$  such that

$$c_3 < c_4 < \dots < c_{d+1},$$

where  $c_3$  is as provided by Theorem 1.8 and moreover, for every fixed  $\varepsilon > 0$  and  $4 \leq k \leq d + 1$ , and every sequence  $\alpha = \alpha(n)$ :

$$\lim_{n \rightarrow \infty} \mathbb{P}(\chi(G(n, \alpha)) \geq k) = \begin{cases} 1 & \text{if } \alpha > (1 + \varepsilon) \cdot c_k \cdot n^{-1/d}, \\ 0 & \text{if } \alpha < (1 - \varepsilon) \cdot c_k \cdot n^{-1/d}. \end{cases}$$

The latter conjecture would in particular answer question 2 of Kahle and Martinez-Figueroa ([62], Section 4).

The proof of Theorem 1.8 (establishing the existence of  $c_3$ ) hinged on the fact that chromatic number  $\geq 3$  is the same as having an odd cycle and a comparison with continuum AB percolation. One might hope that, for  $4 \leq k \leq d + 1$ , the constant  $c_k$  will correspond to the critical intensity for some sort of “higher order percolation event” in the continuum AB percolation model. It is however not so clear (at least to the authors) what might be a useful property or characterization of graphs with chromatic number  $\geq k$  that we could exploit in place of existence of odd cycles.

For Conjecture 4.51 the relevant regime is where the average degree is of logarithmic order. Here we can imagine that a more careful analysis along the lines of the arguments in [62] might work. In particular, perhaps there is an (approximate) correspondence between chromatic number  $\geq d + 2$  and some kind of “coverage of the sphere by random caps” event.

Lovász’ pioneering paper [69] gives a general lower bound for the chromatic number : if the neighbourhood complex of  $G$  is  $k$ -connected then  $\chi(G) \geq k + 3$ . (See e.g. [73] for the relevant definitions and much more background.) It can be shown that the neighbourhood complex of the subgraph of the random Borsuk graph that we construct in the proof of Theorem 1.7 is in fact homotopy equivalent to  $S^{d-1}$  and hence  $(d - 2)$ -connected (see e.g. Theorem 4.3.2 in [73]). We have chosen not to follow this route in our proof and instead invoke Lemma 4.7, which is quicker. It does however suggest the following conjecture. For  $n$  fixed, we can imagine increasing  $\alpha$  from zero to  $\pi$  while keeping the random points fixed. This results in what might be called the “random Borsuk graph *process*” : edges are included one-by-one in order of decreasing geodesic distance between their endpoints. Analogously to the *hitting time* for Erdős-Rényi random graph and *hitting radius* for random geometric graphs, we can define the *hitting angle*  $\alpha(n, \mathcal{P})$  for a given monotone property  $\mathcal{P}$  (preserved under the addition of edges but not necessarily vertices) as the (random) value of  $\alpha$  where the random Borsuk graph process first attains the property  $\mathcal{P}$ . In other words it equals  $\pi$  minus the geodesic length of the edge being added the moment  $\mathcal{P}$  starts holding. We offer the following conjecture.

**Conjecture 4.53.** For  $k = 4, \dots, d + 2$  let us write  $\mathcal{P}$  for the property of having chromatic number  $\geq k$  and  $\mathcal{Q}$  for the property of containing a subgraph whose neighbourhood complex is homotopy equivalent to  $S^{k-2}$ . Then we have

$$\mathbb{P}(\alpha(n, \mathcal{P}) = \alpha(n, \mathcal{Q})) = 1 - o_n(1).$$

We point out that for  $k = 3$  the assertion in the above conjecture in fact holds. (Having chromatic number at least three is equivalent to containing an odd cycle. The neighbourhood complex of an odd cycle other than  $K_3$  is homotopy equivalent to  $S^1$ , while the neighbourhood complex of any bipartite graph with more than one vertex is disconnected, so in particular not homotopy equivalent to  $S^1$ .)



# CHAPTER 5

## THE TYPICAL POISSON–VORONOI CELL IN HIGH DIMENSIONS

This chapter is based on joint work with Zakhar Kabluchko and Tobias Müller. At the time of writing, the manuscript is online as a preprint [56]. We actually obtained slightly more results than what was advertised in the introduction. We also offer the following less precise result on the width of the typical cell:

**Proposition 5.1.** *We have, for every fixed  $\varepsilon > 0$  and  $\lambda = \lambda(d) > 0$ :*

$$\mathbb{P}\left(\frac{2\sqrt{5}}{2 + \sqrt{5}} - \varepsilon \leq \frac{\text{width}(\mathcal{V}_{\text{typ}})}{\sqrt[d]{\lambda \cdot \text{vol}(B)}} \leq \frac{3}{2} + \varepsilon\right) = 1 - o_d(1).$$

To clarify, we emphasize that in the above result we choose  $\varepsilon > 0$  fixed as the dimension increases towards infinity. Note that  $2\sqrt{5}/(2 + \sqrt{5}) \approx 1.0557 > 1$ . So Proposition 5.1 shows that the width is both bounded away by a multiplicative constant from twice the inradius (a trivial lower bound) and bounded away by a multiplicative constant from the diameter (a trivial upper bound). A well-known folklore result states that  $\mathbb{E} \text{vol}(\mathcal{V}_{\text{typ}}) = 1/\lambda$ , in any dimension. (This can for instance be seen from (1.1) together with some relatively straightforward considerations.) Alishahi and Sharifitabar [7] have shown that  $\text{Var}(\text{vol}(\mathcal{V}_{\text{typ}})) = o_d(1)$  as the dimension  $d \rightarrow \infty$  and the intensity  $\lambda$  is kept constant, which in particular implies that the volume of  $\mathcal{V}_{\text{typ}}$  tends to  $1/\lambda$  in probability as  $d \rightarrow \infty$ . In the light of (1.1), this can be informally paraphrased as: “in high dimension, almost all Voronoi cells have volume arbitrarily close to  $1/\lambda$ ”. The result of Alishahi and Sharifitabar was later extended by Yao [102], who showed that the

variance of the volume of the intersection of  $\mathcal{V}_{\text{typ}}$  with any measurable set is bounded by the variance of the volume of  $\mathcal{V}_{\text{typ}}$ . Additional results in [7] imply that if  $r = \sqrt[d]{1/\lambda \cdot \text{vol}(B)}$  is such that  $\text{vol}(B(\underline{0}, r)) = 1/\lambda$  then almost all mass of  $\mathcal{V}_{\text{typ}}$  is contained in  $B(\underline{0}, (1+\varepsilon)r)$  and almost all mass of  $B(\underline{0}, (1-\varepsilon)r)$  is contained in  $\mathcal{V}_{\text{typ}}$ . By this we mean that both the ratios  $\text{vol}(\mathcal{V}_{\text{typ}} \cap B(\underline{0}, (1-\varepsilon)r)) / \text{vol}(B(\underline{0}, (1-\varepsilon)r))$  and  $\text{vol}(\mathcal{V}_{\text{typ}} \cap B(\underline{0}, (1+\varepsilon)r)) / \text{vol}(\mathcal{V}_{\text{typ}})$  tend to one in probability as the dimension  $d \rightarrow \infty$ . This can be interpreted as credence for the idea that the typical cell is somehow “ball like”, at least as far as the volume is concerned. On a similar note, Hörmann et al. [55] investigated the asymptotics of the expected number of  $k$ -faces of  $\mathcal{V}_{\text{typ}}$  as the dimension goes to infinity. Based on their findings, they mention (page 13, paragraph following Theorem 3.20) “... we roughly speaking see that the typical Poisson-Voronoi cells are approximately spherical in the mean ...”.

In contrast, our results on diameter, mean width and outradius all seem to support the idea the typical cell is somehow close to a ball of the same volume, while the (proof of the) results on the inradius and width suggest the typical cell behaves rather differently from a ball of the same volume. Our results for instance imply that the Hausdorff distance between  $\mathcal{V}_{\text{typ}}$  and any ball is large (i.e. at least of the same order as the diameter of  $\mathcal{V}_{\text{typ}}$ ). We spell out the short argument demonstrating this in [Section 5.3](#).

During the course of the proofs of the above results, we will derive and heavily rely on the following observation that is of independent interest: with probability  $1 - o_d(1)$ , all vertices of  $\mathcal{V}_{\text{typ}}$  have norm close to  $\sqrt[d]{\lambda \cdot \text{vol}(B)}$ . This generalizes as follows. The union  $\bigcup \mathcal{F}_k(P)$  of all  $k$ -faces is sometimes also called the  $k$ -skeleton of  $P$ .

**Proposition 5.2.** *Let  $\varepsilon > 0$  be fixed and  $k = k(d)$  be such that  $d - k \rightarrow \infty$ . For any  $\lambda = \lambda(d) > 0$  we have:*

$$\mathbb{P}\left(\bigcup \mathcal{F}_k(\mathcal{V}_{\text{typ}}) \subseteq B(\underline{0}, (1+\varepsilon)r) \setminus B(\underline{0}, (1-\varepsilon)r)\right) \xrightarrow{d \rightarrow \infty} 1,$$

where  $r = r(d, \lambda) := \sqrt[d]{\lambda \cdot \text{vol}(B)}$ .

This last proposition tells us that, with probability  $1 - o_d(1)$ , all faces of non-constant co-dimension are contained in an annulus around  $\underline{0}$  of width  $o\left(\sqrt[d]{\lambda \text{vol}(B)}\right)$  and (inner) radius  $(1 + o_d(1)) \cdot \sqrt[d]{\lambda \text{vol}(B)}$ . From this it can be seen that, with probability  $1 - o_d(1)$ , every face of non-constant co-dimension is “microscopic” (i.e. of diameter  $o\left(\sqrt[d]{\lambda \text{vol}(B)}\right)$ ).

**Corollary 5.3.** *Let  $k = k(d)$  be such that  $d - k \rightarrow \infty$ . For any  $\lambda = \lambda(d) > 0$  we have:*

$$\frac{\max_{F \in \mathcal{F}_k(\mathcal{V}_{typ})} \text{diam}(F)}{\sqrt[d]{\lambda \cdot \text{vol}(B)}} \xrightarrow[d \rightarrow \infty]{\mathbb{P}} 0.$$

(For completeness, we spell out the short derivation of Corollary 5.3 from Proposition 5.2 in Section 5.2.6.)

Theorem 1.1 tells us that Proposition 5.2 does not extend to facets, i.e. faces of co-dimension one. (If all facets of a polytope  $P$  with  $\underline{0} \in P$  lie in the annulus from Proposition 5.2 then  $\text{outr}(P)/\text{inr}(P)$  must be close to one.) What is more, Corollary 5.3 does not extend to facets. As part of the proof of Proposition 5.1 we will exhibit the existence (with probability  $1 - o_d(1)$ ) of facets that have distance roughly  $1/2 \cdot \sqrt[d]{\lambda \cdot \text{vol}(B)}$  to the origin. Since, with probability  $1 - o_d(1)$ , all vertices have distance roughly  $\sqrt[d]{\lambda \cdot \text{vol}(B)}$  to the origin, some pair of vertices on such a face will have distance  $\Omega\left(\sqrt[d]{\lambda \cdot \text{vol}(B)}\right)$ . Although we will not prove this explicitly here, similar remarks apply to faces of any constant co-dimension.

The constant in the right hand side of Theorem 1.3 is equals  $k!$  times the  $k$ -dimensional volume of a regular  $k$ -simplex inscribed in  $S^{k-1}$ . Each edge of such a simplex has length  $\ell_k := \left(\frac{2(k+1)}{k}\right)^{1/2}$ . For  $\varepsilon > 0$ , we will say that  $z_1, \dots, z_k \in \mathbb{R}^d$  is an  $\varepsilon$ -near regular simplex if  $\ell_k - \varepsilon < \|z_i - z_j\| < \ell_k + \varepsilon$  for all  $0 \leq i < j \leq k$  where  $z_0 := \underline{0}$  denotes the origin. We say that  $z_1, \dots, z_k \in \mathcal{Z}$  define a  $(d-k)$ -face of  $\mathcal{V}_{typ}$  if there is a face  $F \in \mathcal{F}_{d-k}(\mathcal{V}_{typ})$  such that

$$F = \mathcal{V}_{typ} \cap \bigcap_{i=1}^k \{x \in \mathbb{R}^d : \|x\| = \|x - z_i\|\}.$$

Let  $M_{k,\varepsilon}$  denote the number of  $(d-k)$ -faces of  $\mathcal{V}_{typ}$  that are defined by a  $k$ -set  $\{z_1, \dots, z_k\} \subseteq \mathcal{Z}$  that forms an  $\varepsilon$ -near regular simplex. The following observation tells us that “almost all” faces of constant co-dimension are defined by near-regular simplices.

**Proposition 5.4.** *For  $k \in \mathbb{N}$  and  $\varepsilon > 0$  fixed and any  $\lambda = \lambda(d) > 0$  we have that*

$$\frac{M_{k,\varepsilon}}{f_{d-k}(\mathcal{V}_{typ})} \xrightarrow[d \rightarrow \infty]{\mathbb{P}} 1.$$

## 5.1 Notation and preliminaries

We briefly recall the definitions of the main parameters we'll be studying in the present chapter. The *diameter* of a set  $A \subseteq \mathbb{R}^d$  is

$$\text{diam}(A) := \sup_{a,b \in A} \|a - b\|.$$

The *inradius* and *outradius* are defined respectively by

$$\begin{aligned}\text{inr}(A) &:= \sup\{r > 0 : \exists p \in \mathbb{R}^d \text{ such that } B(p, r) \subseteq A\}, \\ \text{outr}(A) &:= \inf\{r > 0 : \exists p \in \mathbb{R}^d \text{ such that } B(p, r) \supseteq A\}.\end{aligned}$$

We remark that (in the case when  $A$  is the typical cell of a Poisson-Voronoi tessellation) when defining the inradius, respectively outradius, some authors (e.g. [28]) take the sup, respectively inf, of the radius of all balls centered on the origin  $\underline{0}$  that are contained in  $A$ , respectively contain  $A$ .

We denote by  $S^{d-1} \subseteq \mathbb{R}^d$  the  $(d-1)$ -dimensional unit sphere (in ambient  $d$ -dimensional space). The *width in the direction*  $u \in S^{d-1}$  is defined by

$$w(u, A) := \sup_{a \in A} u^t a - \inf_{a \in A} u^t a.$$

Here and in the rest of the chapter  $v^t w$  denotes the inner product of the vectors  $v, w \in \mathbb{R}^d$ . The (ordinary) *width* is

$$\text{width}(A) := \inf_{u \in S^{d-1}} w(u, A).$$

The *mean width*, as its name suggests, is the average of  $w(u, A)$  over all  $u \in S^{d-1}$ . One of several equivalent ways to define this more formally is by setting

$$\text{meanw}(A) := \mathbb{E}w(U, A),$$

where  $U$  is a point chosen uniformly at random on  $S^{d-1}$ .

For  $x \in \mathbb{R}^d \setminus \{\underline{0}\}$  we will denote by

$$H_x := \{y \in \mathbb{R}^d : x^t y \leq x^t x / 2\}, \quad (5.1)$$

for the set of all points that are at least as close to the origin  $\underline{0}$  as they are to  $x$ .

We note that the typical cell of the Poisson-Voronoi tessellation can be written as

$$\mathcal{V}_{\text{typ}} = \bigcap_{z \in \mathcal{Z}} H_z, \quad (5.2)$$

where  $\mathcal{Z}$  denotes the Poisson point process that generates the typical cell  $\mathcal{V}_{\text{typ}}$ .

In the proofs below, we will repeatedly make use of the dissection  $\mathcal{Q}_\delta$  of  $\mathbb{R}^d$  into axis parallel cubes of side length  $\delta/\sqrt{d}$  (and hence diameter  $\delta$ ) in the obvious way, where  $\delta > 0$  will be a constant chosen independently of the dimension. In more detail:

$$\mathcal{Q}_\delta := \left\{ \left[ \frac{i_1 \delta}{\sqrt{d}}, \frac{(i_1 + 1) \delta}{\sqrt{d}} \right) \times \cdots \times \left[ \frac{i_d \delta}{\sqrt{d}}, \frac{(i_d + 1) \delta}{\sqrt{d}} \right) : i_1, \dots, i_d \in \mathbb{Z} \right\}.$$

We point out that if

$$\mathcal{Q}_{\delta,R} := \{q \in \mathcal{Q}_\delta : q \subseteq B(\underline{0}, R)\},$$

and

$$N_{\delta,R} := |\mathcal{Q}_{\delta,R}|, \quad (5.3)$$

is the number of cubes in the dissection that are contained in the ball of radius  $R$  around the origin, then

$$N_{\delta,R} \leq \frac{\text{vol}(B(\underline{0}, R))}{(\delta/\sqrt{d})^d} = \frac{\kappa_d R^d}{(\delta/\sqrt{d})^d} = (R/\delta)^d \cdot \exp[O(d)], \quad (5.4)$$

where the asymptotics is for  $d \rightarrow \infty$  and the term  $O(d)$  depends only on  $d$  and not  $R$  or  $\delta$ . (Here we've used that  $\kappa_d = d^{-d/2} \cdot \exp[O(d)]$ , as can for instance be seen from Stirling's approximation to the Gamma function and the exact expression (2.1) for  $\kappa_d$ .)

## 5.2 Proofs

We recall (Lemma 2.3) that a dilation of an intensity  $\lambda$  Poisson point process on  $\mathbb{R}^d$  by a factor of  $\rho > 0$  yields a Poisson process of intensity  $\rho^{-d}\lambda$  whose typical cell is just the typical cell of the original process rescaled by  $\rho$ . In particular, for the proofs of our results we can take any value of  $\lambda$  we like (and the result will follow for all choices  $\lambda$ ).

We find it convenient to take

$$\lambda := 1/\kappa_d.$$

(This way the numerator in our main results equals one; and also the expected number of Poisson points in a ball of radius  $s$  is simply  $s^d$ ). We will be using this choice of  $\lambda$  throughout the remainder of the chapter, without stating it explicitly every time. In this chapter, we will denote by  $\mathcal{Z} \subseteq \mathbb{R}^d$  the Poisson point process of intensity  $\lambda = 1/\kappa_d$  on  $\mathbb{R}^d$ .

### 5.2.1 All vertices have norm approximately one

**Lemma 5.5.** *For every fixed  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , all vertices of  $\mathcal{V}_{\text{typ}}$  have norm  $> 1 - \varepsilon$ .*

*Proof.* We use the elementary observation that if  $p \in \mathbb{R}^d$  is a vertex of  $\mathcal{V}_{\text{typ}}$  then there are  $d$  points  $z_1, \dots, z_d \in \mathcal{Z}$  of the Poisson process such that  $z_1, \dots, z_d \in \partial B(p, \|p\|)$  and in addition  $\mathcal{Z} \cap B(p, \|p\|) = \emptyset$ . (This is because a vertex of  $\mathcal{V}_{\text{typ}}$  must lie on the common boundary of  $d + 1$  Voronoi cells, one of which must be  $\mathcal{V}_{\text{typ}}$  itself. This translates to  $p$  being equidistant to  $\underline{0}$  and  $d$  points of the Poisson

point process  $\mathcal{Z}$ , and in addition no point of  $\mathcal{Z}$  can be closer to  $p$  than the common distance – which must be  $\|p\|$ .)

We let  $\delta = \delta(\varepsilon) < \varepsilon/1000$  be a small constant. Any vertex  $v$  of  $\mathcal{V}_{\text{typ}}$  with norm  $\leq 1 - \varepsilon$  is contained in one of the cubes of  $\mathcal{Q}_\delta$  as defined above. That cube itself is completely contained in  $B(\underline{0}, 1 - \varepsilon + \delta) \subseteq B$ . Let  $c$  be one of the corners of the cube containing  $v$ . By the elementary observation above,  $B(c, 1 - \varepsilon + 2\delta)$  contains at least  $d$  points of the Poisson process. Hence, if  $E$  denotes the event that there exists a vertex of  $\mathcal{V}_{\text{typ}}$  with norm  $\leq 1 - \varepsilon$  then by the union bound:

$$\mathbb{P}(E) \leq N_{\delta,1} \cdot \mathbb{P}(\text{Po}((1 - \varepsilon + 2\delta)^d) \geq d),$$

(The first term is an upper bound on the number of cubes considered and the second term is the probability the ball around a corner of a given cube has  $\geq d$  Poisson points in it.)

By (5.4) we have  $N_{\delta,1} = e^{O(d)}$ . On the other hand, using the version of the Chernoff bound we presented as Lemma 2.1, we see that

$$\mathbb{P}(\text{Po}((1 - \varepsilon + 2\delta)^d) \geq d) \leq \exp \left[ -(1 - \varepsilon + 2\delta)^d H \left( \frac{d}{(1 - \varepsilon + 2\delta)^d} \right) \right] = e^{-\Omega(d^2)}.$$

Combining these bounds, we see that  $\mathbb{P}(E) = o_d(1)$ , as claimed in the lemma statement.  $\square$

**Lemma 5.6.** *For every fixed  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , every vertex of  $\mathcal{V}_{\text{typ}}$  has norm  $< 1 + \varepsilon$ .*

*Proof.* As in the proof of the previous lemma, we use that if  $p \in \mathbb{R}^d$  is a vertex of  $\mathcal{V}_{\text{typ}}$  then there are  $d$  points  $z_1, \dots, z_d \in \mathcal{Z}$  of the Poisson process such that  $z_1, \dots, z_d \in \partial B(p, \|p\|)$  and in addition  $\mathcal{Z} \cap B(p, \|p\|) = \emptyset$ .

We fix some  $\delta = \delta(\varepsilon) \in (0, \min(\varepsilon/2, 1))$ . If  $p \in B(\underline{0}, 2) \setminus B(\underline{0}, 1 + \varepsilon)$  is a vertex of  $\mathcal{V}_{\text{typ}}$  then there is some cube  $q \in \mathcal{Q}_\delta$  satisfying that  $q \subseteq B(\underline{0}, 3)$  and  $B(c, 1 + \varepsilon - \delta) \cap \mathcal{Z} = \emptyset$  for any corner  $c$  of  $q$ . Similarly, if for some  $n \geq 2$  a point  $p \in B(\underline{0}, n + 1) \setminus B(\underline{0}, n)$  is a vertex of  $\mathcal{V}_{\text{typ}}$  then there is some cube  $q \in \mathcal{Q}_\delta$  satisfying that  $q \subseteq B(\underline{0}, n + 2)$  and  $B(c, n - \delta) \cap \mathcal{Z} = \emptyset$  for  $c$  any corner of  $q$ .

It follows that if  $E$  is the event that  $\mathcal{V}_{\text{typ}}$  has at least one vertex outside  $B(\underline{0}, 1 + \varepsilon)$  then

$$\begin{aligned} \mathbb{P}(E) &\leq N_{\delta,3} \cdot \mathbb{P}(\text{Po}((1 + \varepsilon - \delta)^d) = 0) + \sum_{n \geq 2} N_{\delta,n+2} \cdot \mathbb{P}(\text{Po}((n - \delta)^d) = 0) \\ &= e^{O(d) - (1 + \varepsilon - \delta)^d} + \exp[O(d)] \cdot \sum_{n \geq 2} (n + 2)^d \cdot e^{-(n - \delta)^d} \\ &= o_d(1), \end{aligned}$$

using (5.4) above.  $\square$

### 5.2.2 The outradius, diameter and mean width

Since  $\mathcal{V}_{\text{typ}}$  is the convex hull of its vertices, Lemma 5.6 implies that  $\mathcal{V}_{\text{typ}} \subseteq B(0, 1 + \varepsilon)$  (for any fixed  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ ). In particular:

**Corollary 5.7.** *For every  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have  $\text{meanw}(\mathcal{V}_{\text{typ}}) \leq \text{diam}(\mathcal{V}_{\text{typ}}) < 2 + \varepsilon$  and  $\text{outr}(\mathcal{V}_{\text{typ}}) < 1 + \varepsilon$ .*

We need to prove matching lower bounds. The following observation does the trick for the diameter and outradius.

**Lemma 5.8.** *For every  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have  $(1 - \varepsilon, 0, \dots, 0) \in \mathcal{V}_{\text{typ}}$  and  $(\varepsilon - 1, 0, \dots, 0) \in \mathcal{V}_{\text{typ}}$ .*

*Proof.* By symmetry and the union bound, we just need to show that  $\mathbb{P}((1 - \varepsilon, 0, \dots, 0) \notin \mathcal{V}_{\text{typ}}) = o_d(1)$ . We simply note that

$$\begin{aligned} \mathbb{P}((1 - \varepsilon, 0, \dots, 0) \in \mathcal{V}_{\text{typ}}) &= \mathbb{P}(\mathcal{Z} \cap B((1 - \varepsilon, 0, \dots, 0), 1 - \varepsilon) = \emptyset) \\ &= \mathbb{P}(\text{Po}((1 - \varepsilon)^d) = 0) \\ &= e^{-(1 - \varepsilon)^d} \\ &= 1 - o_d(1). \end{aligned}$$

□

**Corollary 5.9.** *For every  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have  $\text{diam}(\mathcal{V}_{\text{typ}}) > 2 - \varepsilon$  and  $\text{outr}(\mathcal{V}_{\text{typ}}) > 1 - \varepsilon$ .*

For the mean width the lower bound also follows quite easily.

**Lemma 5.10.** *For every  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have  $\text{meanw}(\mathcal{V}_{\text{typ}}) > 2 - \varepsilon$ .*

*Proof.* We pick  $\delta = \delta(\varepsilon) > 0$  a small constant to be determined. We define

$$\mathcal{X} := \{u \in S^{d-1} : w(u, \mathcal{V}_{\text{typ}}) < 2 - \delta\}, \quad X := \sigma(\mathcal{X}) / \sigma(S^{d-1}),$$

where  $\sigma$  denotes the Haar measure on  $S^{d-1}$  (the “ $(d - 1)$ -dimensional area”). Put differently,  $X$  is the fraction of  $S^{d-1}$  that is covered by directions in which the width is  $< 2 - \delta$ . We have

$$\text{meanw}(\mathcal{V}_{\text{typ}}) \geq (1 - X) \cdot (2 - \delta).$$

Next, we observe that

$$\begin{aligned}
\mathbb{E}X &= \mathbb{E} \left( \frac{1}{\sigma(S^{d-1})} \int_{S^{d-1}} 1_{\{w(u, \mathcal{V}_{\text{typ}}) < 2-\delta\}} \sigma(du) \right) \\
&= \frac{1}{\sigma(S^{d-1})} \int_{S^{d-1}} \mathbb{E} (1_{\{w(u, \mathcal{V}_{\text{typ}}) < 2-\delta\}}) \sigma(du) \\
&= \frac{1}{\sigma(S^{d-1})} \int_{S^{d-1}} \mathbb{P}(w(u, \mathcal{V}_{\text{typ}}) < 2-\delta) \sigma(du) \\
&= \frac{1}{\sigma(S^{d-1})} \int_{S^{d-1}} \mathbb{P}(w(e_1, \mathcal{V}_{\text{typ}}) < 2-\delta) \sigma(du) \\
&= \mathbb{P}(w(e_1, \mathcal{V}_{\text{typ}}) < 2-\delta).
\end{aligned}$$

where  $e_1 = (1, 0, \dots, 0)$  denotes the first standard basis vector, we used Fubini for non-negative integrands in the second equality and symmetry in the fourth. Applying Lemma 5.8:

$$\mathbb{E}X = \mathbb{P}(w(e_1, \mathcal{V}_{\text{typ}}) < 2-\delta) \leq 1 - \mathbb{P}((1-\delta/2, 0, \dots, 0), (\delta/2-1, 0, \dots, 0) \in \mathcal{V}_{\text{typ}}) = o_d(1).$$

Markov's inequality thus gives

$$\mathbb{P}(X > \delta) \leq \mathbb{E}X/\delta = o_d(1).$$

It follows that, with probability  $1 - o_d(1)$ ,  $\text{meanw}(\mathcal{V}_{\text{typ}}) > (2-\delta) \cdot (1-\delta) > 2-\varepsilon$ , where the last inequality holds having chosen  $\delta$  appropriately.  $\square$

### 5.2.3 The inradius

A lower bound on the inradius of  $\mathcal{V}_{\text{typ}}$  is given by the following lemma.

**Lemma 5.11.** *For every  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have  $B(\underline{0}, \frac{1}{2} - \varepsilon) \subseteq \mathcal{V}_{\text{typ}}$ .*

*Proof.* We simply note that if  $\mathcal{Z} \cap B(\underline{0}, 1-2\varepsilon) = \emptyset$  then  $B(\underline{0}, 1/2 - \varepsilon) \subseteq \mathcal{V}_{\text{typ}}$ , and hence

$$\mathbb{P}(B(\underline{0}, 1/2 - \varepsilon) \subseteq \mathcal{V}_{\text{typ}}) \geq \mathbb{P}(\text{Po}((1-2\varepsilon)^d) = 0) = e^{-(1-2\varepsilon)^d} = 1 - o_d(1).$$

$\square$

To derive a matching upper bound we need to show that *no* ball of radius  $\frac{1}{2} + \varepsilon$  is contained in  $\mathcal{V}_{\text{typ}}$  (with probability  $1 - o_d(1)$ ). Even if  $B(\underline{0}, \frac{1}{2} + \varepsilon) \not\subseteq \mathcal{V}_{\text{typ}}$  there are in principle still infinitely many balls of the same radius but with a different center that need to be excluded. In order to deal with this issue, we will make use of the following observation.

**Lemma 5.12.** *For every  $p \in \mathbb{R}^d$  and  $r > 0$  we have*

$$\mathbb{P}(B(p, r) \subseteq \mathcal{V}_{\text{typ}}) \leq \sqrt{\mathbb{P}(B(\underline{0}, r) \subseteq \mathcal{V}_{\text{typ}})}.$$

*Proof.* By symmetry, we can take  $p$  on the positive  $x_1$ -axis without loss of generality. We set

$$A := B(\underline{0}, 2r), \quad A^+ := A \cap \{x \in \mathbb{R}^d : x_1 > 0\}.$$

We note that  $A$  can be written alternatively as  $A = \{x \in \mathbb{R}^d : B(\underline{0}, r) \not\subseteq H_x\}$ , where  $H_x$  is as defined by (5.1). Appealing to the observation (5.2), we have

$$\mathbb{P}(B(\underline{0}, r) \subseteq \mathcal{V}_{\text{typ}}) = \mathbb{P}(\mathcal{Z} \cap A = \emptyset).$$

We claim that if  $\mathcal{Z} \cap A^+ \neq \emptyset$  then  $B(p, r) \not\subseteq \mathcal{V}_{\text{typ}}$ . To see this, let  $x \in A^+$  be arbitrary. Then there exists a  $y \in B(\underline{0}, r) \setminus H_x$ . That is,  $y \in B(\underline{0}, r)$  and  $x^t y > x^t x/2$ . But then  $w := y + p \in B(p, r)$  and

$$x^t w = x^t y + x^t p > x^t y > x^t x/2,$$

where the first inequality holds by choice of  $p$  and  $A^+$  ( $p$  is on the positive  $x_1$ -axis and  $x$ 's first coordinate is positive). This establishes that  $B(p, r) \setminus H_x \neq \emptyset$  for all  $x \in A^+$ . In other words, we've proven the claim that  $\{\mathcal{Z} \cap A^+ \neq \emptyset\} \subseteq \{B(p, r) \not\subseteq \mathcal{V}_{\text{typ}}\}$ .

This gives

$$\mathbb{P}(B(p, r) \subseteq \mathcal{V}_{\text{typ}}) \leq \mathbb{P}(\mathcal{Z} \cap A^+ = \emptyset) = e^{-\lambda \cdot \text{vol}(A^+)} = e^{-\lambda \cdot \text{vol}(A)/2} = \sqrt{\mathbb{P}(\mathcal{Z} \cap A = \emptyset)}.$$

The lemma follows.  $\square$

**Lemma 5.13.** *For every  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have  $\text{inr}(\mathcal{V}_{\text{typ}}) \leq \frac{1}{2} + \varepsilon$ .*

*Proof.* Again we let  $\delta = \delta(\varepsilon) > 0$  be an appropriately chosen small constant. As we've already established that  $\mathcal{V}_{\text{typ}} \subseteq B(\underline{0}, 1 + \varepsilon)$ , with probability  $1 - o_d(1)$ , it suffices to show that there is no  $p \in B(\underline{0}, 1/2)$  such that  $B(p, 1/2 + \varepsilon) \subseteq \mathcal{V}_{\text{typ}}$ . Each such  $p$  lies inside some cube  $q \in \mathcal{Q}_{\delta, 1}$ , and any corner  $c$  of  $q$  must satisfy that  $B(c, 1/2 + \varepsilon - \delta) \subseteq \mathcal{V}_{\text{typ}}$ .

Applying the previous lemma we find

$$\begin{aligned} \mathbb{P}(\text{inr}(\mathcal{V}_{\text{typ}}) > 1/2 + \varepsilon) &\leq \mathbb{P}(\mathcal{V}_{\text{typ}} \not\subseteq B(\underline{0}, 1 + \varepsilon)) \\ &\quad + N_{\delta, 1} \cdot \sqrt{\mathbb{P}(B(\underline{0}, 1/2 + \varepsilon - \delta) \subseteq \mathcal{V}_{\text{typ}})} \\ &= o_d(1) + e^{O(d)} \cdot \sqrt{\mathbb{P}(\mathcal{Z} \cap B(\underline{0}, 1 + 2\varepsilon - 2\delta) = \emptyset)} \\ &= o_d(1) + e^{O(d)} \cdot e^{-(1+2\varepsilon-2\delta)^d/2} \\ &= o_d(1), \end{aligned}$$

having chosen  $0 < \delta < \varepsilon$  appropriately.  $\square$

## 5.2.4 Upper bound for the width

**Lemma 5.14.** *For every  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have  $\text{width}(\mathcal{V}_{\text{typ}}) < 3/2 + \varepsilon$ .*

*Proof.* As noted before, it follows from Lemma 5.6 that  $\mathcal{V}_{\text{typ}} \subseteq B(\underline{0}, 1 + \varepsilon)$  with probability  $1 - o_d(1)$ . Therefore we also have, with probability  $1 - o_d(1)$ , that

$$\mathcal{V}_{\text{typ}} \subseteq B(\underline{0}, 1 + \varepsilon) \cap H_z,$$

for all  $z \in \mathcal{Z}$  – where  $H_z$  is as defined by (5.1) above and we use (5.2). This gives that, with probability  $1 - o_d(1)$ :

$$\begin{aligned} \text{width}(\mathcal{V}_{\text{typ}}) &\leq \inf_{z \in \mathcal{Z}} \text{width}(B(\underline{0}, 1 + \varepsilon) \cap H_z) \\ &= \inf_{z \in \mathcal{Z}} \inf_{u \in S^{d-1}} w(u, B(\underline{0}, 1 + \varepsilon) \cap H_z) \\ &\leq \inf_{z \in \mathcal{Z}} w\left(\frac{z}{\|z\|}, B(\underline{0}, 1 + \varepsilon) \cap H_z\right) \\ &\leq 1 + \varepsilon + \inf_{z \in \mathcal{Z}} \|z\|/2. \end{aligned}$$

Since

$$\mathbb{P}(\mathcal{Z} \cap B(\underline{0}, 1 + \varepsilon) \neq \emptyset) = 1 - e^{-(1+\varepsilon)^d} = 1 - o_d(1),$$

it follows that, with probability  $1 - o_d(1)$ , we have  $\inf_{z \in \mathcal{Z}} \|z\| < 1 + \varepsilon$ . In other words, with probability  $1 - o_d(1)$ ,  $\text{width}(\mathcal{V}_{\text{typ}}) \leq (3/2) \cdot (1 + \varepsilon)$ . Adjusting the value of  $\varepsilon$ , the result follows.  $\square$

## 5.2.5 Lower bound on the width

Recall that the *polar* of a set  $A \subseteq \mathbb{R}^d$  is defined by

$$A^\circ := \{y \in \mathbb{R}^d : a^t y \leq 1 \text{ for all } a \in A\}.$$

(For background and an overview of properties of the polar set, see e.g. [25].) We will find it convenient to switch attention to the polar  $\mathcal{V}_{\text{typ}}^\circ$  of  $\mathcal{V}_{\text{typ}}$ . Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$  be the map defined by  $x \mapsto (2/\|x\|^2)x$  for  $x \neq \underline{0}$  and  $\underline{0} \mapsto \underline{0}$ . We let the point set  $\mathcal{Y} \subseteq \mathbb{R}^d$  be defined by

$$\mathcal{Y} := f[\mathcal{Z}],$$

where  $\mathcal{Z}$  as usual is the Poisson point process of constant intensity  $\lambda = 1/\kappa_d$  that we used to define the typical cell  $\mathcal{V}_{\text{typ}}$ . By the mapping theorem (2.3)  $\mathcal{Y}$  is also a

Poisson point process on  $\mathbb{R}^d \setminus \{\underline{0}\}$ . Its intensity function can easily be worked out to be  $\text{const} \cdot \|x\|^{-2d}$ , but we shall not be needing that. Convex hulls of Poisson processes with power-law intensity and their duals were studied in [58–61].

We will rely on the following well-known observation. For a proof, see for instance the footnote on page 17 of [58].

**Proposition 5.15.** *Almost surely,  $\mathcal{V}_{typ}^\circ = \text{conv}(\mathcal{Y})$ .*

Although we shall not need this fact below, let us remark that  $\underline{0}$  belongs to the interior of  $\text{conv}(\mathcal{Y})$  almost surely (see Corollary 4.2 on page 1040 of [60]).

We define for  $u \in S^{d-1}$  and  $A \subseteq \mathbb{R}^d$ :

$$\varphi(u, A) := \sup\{u^t a : a \in A\}, \quad \psi(u, A) := \sup\{\lambda \geq 0 : \lambda u \in A\}.$$

We remark that we can write

$$w(u, A) = \varphi(u, A) + \varphi(-u, A).$$

Another well-known, elementary observation is the following. It for instance occurs as equation (14.42) in [89].

**Lemma 5.16.** *For  $A \subseteq \mathbb{R}^d$  compact and convex with  $\underline{0} \in \text{int } A$  and all  $u \in S^{d-1}$  we have*

$$\varphi(u, A) = \frac{1}{\psi(u, A^\circ)}.$$

**Lemma 5.17.** *Let  $0 < r < 2$  be fixed. With probability  $1 - o_d(1)$ , any two distinct  $Y_1, Y_2 \in \mathcal{Y}$  with  $\|Y_1\| > r, \|Y_2\| > r$  satisfy*

$$\alpha < \angle Y_1 \underline{0} Y_2 < \pi - \alpha,$$

where  $\alpha := \arcsin(r^2/4) < \pi/2$ .

*Proof.* Let us write

$$\begin{aligned} X &:= \left| \left\{ (Y_1, Y_2) \in \mathcal{Y}^2 : \begin{array}{l} \|Y_1\|, \|Y_2\| > r \text{ and } Y_1 \neq Y_2 \text{ and} \\ \angle Y_1 \underline{0} Y_2 \leq \alpha \text{ or } \angle Y_1 \underline{0} Y_2 \geq \pi - \alpha \end{array} \right\} \right| \\ &= \left| \left\{ (Z_1, Z_2) \in \mathcal{Z}^2 : \begin{array}{l} \|Z_1\|, \|Z_2\| < 2/r \text{ and } Z_1 \neq Z_2 \text{ and} \\ \angle Z_1 \underline{0} Z_2 \leq \alpha \text{ or } \angle Z_1 \underline{0} Z_2 \geq \pi - \alpha \end{array} \right\} \right|. \end{aligned}$$

Using the Mecke formula (Lemma 2.5), we find

$$\begin{aligned}
\mathbb{E}X &= \lambda^2 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} 1 \left\{ \|z_1\|, \|z_2\| < 2/r \text{ and } \right. \\
&\quad \left. \angle z_1 \underline{0} z_2 \leq \alpha \text{ or } \angle z_1 \underline{0} z_2 \geq \pi - \alpha \right\} d z_1 d z_2 \\
&= \lambda^2 \cdot \text{vol}(B(\underline{0}, 2/r))^2 \cdot \mathbb{P}(\angle U_1 \underline{0} U_2 \leq \alpha \text{ or } \angle U_1 \underline{0} U_2 \geq \pi - \alpha) \\
&= 2 \cdot (4/r^2)^d \cdot \mathbb{P}(\angle U_1 \underline{0} v \leq \alpha) \\
&= o((4/r^2)^d \cdot \sin^d \alpha) \\
&= o_d(1),
\end{aligned}$$

where  $U_1, U_2$  denote points chosen uniformly at random on  $S^{d-1}$  and  $v \neq \underline{0}$  is an arbitrary but fixed point in  $\mathbb{R}^d$ . (The second line follows by symmetry considerations; the third line uses the choice of  $\lambda$  and more symmetry considerations, and; we applied Lemma 2.6 in the fourth line.)  $\square$

**Lemma 5.18.** *For every  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have  $\mathcal{Y} \subseteq B(\underline{0}, 2 + \varepsilon)$ .*

*Proof.* This follows from the fact that

$$|\mathcal{Y} \setminus B(\underline{0}, 2 + \varepsilon)| = |\mathcal{Z} \cap \text{cl}(B(\underline{0}, 2/(2 + \varepsilon)))| \stackrel{d}{=} \text{Po}\left((2/(2 + \varepsilon))^d\right),$$

where  $\text{cl}(\cdot)$  denotes topological closure. We see that

$$\mathbb{P}(\mathcal{Y} \not\subseteq B(\underline{0}, 2 + \varepsilon)) \leq \left(\frac{2}{2 + \varepsilon}\right)^d = o_d(1).$$

$\square$

For the remainder of this section, we set

$$r := \frac{4}{\sqrt{5}}, \quad \alpha := \arcsin(r^2/4). \quad (5.5)$$

(Note that  $r < 2$  and  $\alpha < \pi/2$ .) We separate out the following observation as a lemma.

**Lemma 5.19.** *With  $r, \alpha$  as given by (5.5), we have  $\cos(\alpha/2) = r/2$ .*

*Proof.* By the double angle formula

$$4/5 = r^2/4 = \sin(\alpha) = 2 \sin(\alpha/2) \cos(\alpha/2) = 2 \cos(\alpha/2) \sqrt{1 - \cos^2(\alpha/2)}.$$

We see that we must have either  $\cos(\alpha/2) = 1/\sqrt{5}$  or  $\cos(\alpha/2) = 2/\sqrt{5}$ . We can exclude the first possibility, as  $\cos(\alpha/2) = 1/\sqrt{5} < 1/\sqrt{2}$  would imply that  $\alpha/2 > \pi/4$ , which in turn would imply that  $\alpha > \pi/2$ , but clearly  $\alpha = \arcsin(4/5) \in (0, \pi/2)$ . It follows that  $\cos(\alpha/2) = 2/\sqrt{5} = r/2$ , as stated by the lemma.  $\square$

**Lemma 5.20.** *For all  $\varepsilon > 0$  there is a  $\delta = \delta(\varepsilon) > 0$  such that the following holds. If  $V \subseteq \mathbb{R}^d$  and  $v \in \mathbb{R}^d$  are such that*

(i)  $V \subseteq B(0, 2 + \delta)$ , and;

(ii)  $\|v\| > r + \varepsilon$ , and;

(iii)  $\angle v0w \geq \alpha/2$  for all  $w \in V$  with  $\|w\| > r$ ,

then  $v \notin \text{conv}(V)$ .

*Proof.* Applying a suitable isometry, we can assume without loss of generality that  $v = (x, 0, \dots, 0)$  with  $x > r + \varepsilon$ . We consider the projection  $\pi : \mathbb{R}^d \rightarrow \mathbb{R}$  onto the first coordinate.

For every  $w \in V$  with  $\|w\| \leq r$  we of course have  $\pi(w) \leq r$  as well. For  $w \in V$  with  $\|w\| > r$  we have

$$\pi(w) = \cos(\angle v0w) \cdot \|w\| \leq \cos(\alpha/2) \cdot (2 + \delta) = r \cdot (1 + \delta/2),$$

where we've used Lemma 5.19 and the specific choice of  $r, \alpha$ . Having chosen  $\delta := 2\varepsilon/r$  appropriately, we see that

$$\pi[\text{conv}(V)] \subseteq (-\infty, r + \varepsilon].$$

So  $x = \pi(v) \notin \pi[\text{conv}(V)]$  and in particular  $v \notin \text{conv}(V)$  as well.  $\square$

**Lemma 5.21.** *For all  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have*

$$\sup_{u \in S^{d-1}} (\psi(u, \mathcal{V}_{\text{typ}}^\circ) + \psi(-u, \mathcal{V}_{\text{typ}}^\circ)) \leq 2 + r + \varepsilon.$$

*Proof.* Let  $\varepsilon > 0$  be arbitrary and let  $\delta = \delta(\varepsilon)$  be as provided by Lemma 5.20. We assume that  $\mathcal{Y}$  is such that the conclusions of Lemmas 5.17 and 5.18 hold (which happens with probability  $1 - o_d(1)$ ), with  $\min(\delta, \varepsilon)$  taking the role of  $\varepsilon$  in Lemma 5.18 and  $r, \alpha$  as specified by (5.5).

Aiming for a contradiction, suppose that there is some  $u \in S^{d-1}$  such that both  $\psi(u, \mathcal{V}_{\text{typ}}^\circ)$  and  $\psi(-u, \mathcal{V}_{\text{typ}}^\circ)$  are  $> r + \varepsilon$ . In other words, we can find  $\lambda, \mu > r + \varepsilon$  such that the points  $v := \lambda u, w := -\mu u$  satisfy  $v, w \in \mathcal{V}_{\text{typ}}^\circ = \text{conv}(\mathcal{Y})$ . There must be a  $y \in \mathcal{Y}$  such that  $\|y\| > r$  and  $\angle y0v \leq \alpha/2$ , because otherwise by Lemma 5.20 (with  $V = \mathcal{Y}$ ) would imply that  $v \notin \mathcal{V}_{\text{typ}}^\circ$ , contradicting the choice of  $v$ . Analogously, there is a  $z \in \mathcal{Z}$  with  $\|z\| > r$  and  $\angle z0w \leq \alpha/2$ .

We now point out that  $\angle v0w = \pi$ , which gives  $\angle y0z \geq \pi - \alpha$ , contradicting our assumption that the conclusion of Lemma 5.17 holds. It follows that, for every  $u \in S^{d-1}$  at least one of  $\psi(u, \mathcal{V}_{\text{typ}}^\circ) \leq r + \varepsilon$  or  $\psi(-u, \mathcal{V}_{\text{typ}}^\circ) \leq r + \varepsilon$  holds. Hence,

$$\sup_{u \in S^{d-1}} (\psi(u, \mathcal{V}_{\text{typ}}^\circ) + \psi(-u, \mathcal{V}_{\text{typ}}^\circ)) \leq (2 + \varepsilon) + (r + \varepsilon) = 2 + r + 2\varepsilon.$$

Adjusting the value of  $\varepsilon$ , the lemma follows.  $\square$

**Corollary 5.22.** *For all  $\varepsilon > 0$ , with probability  $1 - o_d(1)$ , we have*

$$\text{width}(\mathcal{V}_{\text{typ}}) > \frac{4}{2+r} - \varepsilon.$$

*Proof.* Applying Lemma 5.16 (and recalling that  $\mathcal{V}_{\text{typ}}$  is almost surely a polytope with  $\underline{0} \in \text{int}(\mathcal{V}_{\text{typ}})$ ), almost surely, for every  $u \in S^{d-1}$  we have

$$\begin{aligned} w(u, \mathcal{V}_{\text{typ}}) &= \frac{1}{\psi(u, \mathcal{V}_{\text{typ}}^\circ)} + \frac{1}{\psi(-u, \mathcal{V}_{\text{typ}}^\circ)} \\ &= \frac{1}{2} \cdot \left( \frac{2}{\psi(u, \mathcal{V}_{\text{typ}}^\circ)} + \frac{2}{\psi(-u, \mathcal{V}_{\text{typ}}^\circ)} \right) \\ &\geq \frac{\frac{1}{2} (\psi(u, \mathcal{V}_{\text{typ}}^\circ) + \psi(-u, \mathcal{V}_{\text{typ}}^\circ))}{4} \\ &= \frac{1}{\psi(u, \mathcal{V}_{\text{typ}}^\circ) + \psi(-u, \mathcal{V}_{\text{typ}}^\circ)}, \end{aligned}$$

using the convexity of  $x \mapsto 2/x$  (or the arithmetic-harmonic inequality). Hence

$$\text{width}(\mathcal{V}_{\text{typ}}) = \inf_{u \in S^{d-1}} w(u, \mathcal{V}_{\text{typ}}) \geq \frac{4}{\sup_{u \in S^{d-1}} (\psi(u, \mathcal{V}_{\text{typ}}^\circ) + \psi(-u, \mathcal{V}_{\text{typ}}^\circ))}.$$

The result thus follows from Lemma 5.21, adjusting the value of  $\varepsilon$ .  $\square$

### 5.2.6 Faces of large co-dimension.

Having already obtained Lemma 5.6, in order to prove Proposition 5.2 it suffices to show:

**Lemma 5.23.** *Let  $\varepsilon > 0$  be fixed, and let  $k = k(d)$  satisfy  $d - k \rightarrow \infty$ . With probability  $1 - o_d(1)$ , we have  $\min \{\|x\| : x \in \bigcup \mathcal{F}_k(\mathcal{V}_{\text{typ}})\} > 1 - \varepsilon$ .*

The proof is essentially the same as that of Lemma 5.5. For completeness we spell it out.

*Proof.* We use the observation that if  $x \in F$  for some  $F \in \mathcal{F}_k(\mathcal{V}_{\text{typ}})$  then there must be  $z_1, \dots, z_{d-k} \in \mathcal{Z}$  such that  $z_1, \dots, z_{d-k} \in \partial B(x, \|x\|)$ . If there exists any such  $x \in B(\underline{0}, 1 - \varepsilon)$  then there exists a cube  $q \in \mathcal{Q}_\delta$  with  $x \in q \subseteq B(\underline{0}, 1 - \varepsilon + \delta)$ , where  $\delta = \delta(\varepsilon) > 0$  is a constant, to be chosen sufficiently small. If  $c$  is any corner of  $q$  then  $B(c, 1 - \varepsilon + 2\delta)$  contains at least  $d - k$  points of  $\mathcal{Z}$ . We arrive at:

$$\begin{aligned} \mathbb{P}(\min \{\|x\| : x \in \bigcup \mathcal{F}_k(\mathcal{V}_{\text{typ}})\} \leq 1 - \varepsilon) &\leq N_{1,\delta} \cdot \mathbb{P}(\text{Po}((1 - \varepsilon + 2\delta)^d) \geq d - k) \\ &\leq e^{O(d)} \cdot e^{-\Omega(d(d-k))} = o_d(1), \end{aligned}$$

again using (5.4) and Lemma 2.1.  $\square$

Corollary 5.3 follows immediately from the following observation and Proposition 5.2.

**Lemma 5.24.** *If  $F \subseteq B(\underline{0}, 1 + \varepsilon) \setminus B(\underline{0}, 1 - \varepsilon)$  is convex then  $\text{diam}(F) \leq 4\sqrt{\varepsilon}$ .*

*Proof.* Let  $p, q \in F$  be two arbitrary points. The midpoint  $m := (p + q)/2$  also lies on  $F$ . We write  $\alpha := \angle p\underline{0}m, \beta := \angle q\underline{0}m$ . Since  $m$  is the middle of the line segment between  $p$  and  $q$ , we have  $\alpha + \beta = \pi$ . One of  $\alpha, \beta$  is  $\geq \pi/2$ . Without loss of generality it is  $\alpha$ . The cosine rule gives:

$$\|p\|^2 = \|m\|^2 + \|p - m\|^2 - 2\|m\|\|p - m\|\cos\alpha \geq \|m\|^2 + \|p - m\|^2.$$

The assumptions on  $\varepsilon, F$  now give:

$$\|p - m\|^2 \leq \|p\|^2 - \|m\|^2 \leq (1 + \varepsilon)^2 - (1 - \varepsilon)^2 = 4\varepsilon.$$

Hence

$$\|p - q\| = 2\|p - m\| \leq 4\sqrt{\varepsilon}.$$

Since  $p, q \in F$  were arbitrary, this shows that  $\text{diam}(F) \leq 4\sqrt{\varepsilon}$ .  $\square$

### 5.2.7 Faces of constant co-dimension

In this section, we will prove Theorems 1.2 and 1.3 and Proposition 5.4. Before we can start in earnest, we need some more definitions and preliminary observations.

For the remainder of the section, we fix  $k \in \mathbb{N}$ . For notational convenience, we'll write:

$$v_k := \left( \frac{(k+1)^{k+1}}{k^k} \right)^{1/2}, \quad \ell_k := \left( \frac{2(k+1)}{k} \right)^{1/2}.$$

As mentioned earlier,  $v_k$  equals  $k!$  times the volume and  $\ell_k$  equals side-length of a regular simplex inscribed in  $S^{k-1}$ . A short verification is standard and omitted. (Computing  $v_k$  for instance occurs as an exercise with a difficulty rating of 2 in Section 13 of [72] and the value of  $v_k$  is simply stated in Section 49 of [95], without further elaboration. Determining  $\ell_k$  is even easier, at least in the opinion of the authors.)

Also for notational convenience, we write for  $u_1, \dots, u_k \in \mathbb{R}^k$ :

$$D(u_1, \dots, u_k) := |\det(u_1 | \dots | u_k)|, \quad (5.6)$$

where  $(u_1 | \dots | u_k)$  denotes the  $k \times k$  matrix whose columns are  $u_1, \dots, u_k$ . (Note we consider  $k$  vectors in  $k$ -dimensional Euclidean space.) We denote by  $P(u_1, \dots, u_k)$  the *parallelepiped spanned* by  $u_1, \dots, u_k \in \mathbb{R}^k$ . That is:

$$P(u_1, \dots, u_k) := \{\lambda_1 u_1 + \dots + \lambda_k u_k : 0 \leq \lambda_1 \leq 1, \dots, 0 \leq \lambda_k \leq 1\}.$$

A standard elementary fact states that:

$$\begin{aligned} D(u_1, \dots, u_k) &= \text{vol}(P(u_1, \dots, u_k)) \\ &= k! \cdot \text{vol}(\text{conv}(\{0, u_1, \dots, u_k\})). \end{aligned} \quad (5.7)$$

(This last identity is for instance stated as equation (7.6) in [89].)

For notational convenience, for  $x_1, \dots, x_k \in \mathbb{R}^d$ , we write

$$\rho(x_1, \dots, x_k) := \inf \{r > 0 : \exists x \text{ such that } 0, x_1, \dots, x_k \in \partial B(x, r)\}.$$

(Note that now the ambient dimension  $d$  is not necessarily the same as the number of points  $k$ .) We collect some observations on  $\rho$  needed in the sequel.

**Lemma 5.25.**

- (i) If  $x_1, \dots, x_k \in \mathbb{R}^d$  are linearly independent, then  $0 < \rho(x_1, \dots, x_k) < \infty$  and there is a unique open ball  $B$  of radius  $\rho(x_1, \dots, x_k)$  such that  $0, x_1, \dots, x_k \in \partial B$ .
- (ii) The map  $(x_1, \dots, x_k) \mapsto \rho(x_1, \dots, x_k)$  is continuous on the set  $I_{k,d} \subseteq \mathbb{R}^{kd}$  given by

$$I_{k,d} := \{(x_1, \dots, x_k) \in \mathbb{R}^d \times \dots \times \mathbb{R}^d : x_1, \dots, x_k \text{ are linearly independent}\}.$$

*Proof.* A ball that has  $0, x_1, \dots, x_k$  on its boundary has to be of the form  $B(x, \|x\|)$  with  $x$  satisfying

$$\|x\| = \|x - x_1\| = \dots = \|x - x_k\|.$$

Squaring and rewriting the squared norm in terms of the inner product gives

$$\langle x, x \rangle = \langle x - x_i, x - x_i \rangle = \langle x, x \rangle - 2\langle x_i, x \rangle + \langle x_i, x_i \rangle \quad (i = 1, \dots, k).$$

Reorganising gives:

$$\langle x_i, x \rangle = \|x_i\|^2/2 \quad (i = 1, \dots, k).$$

Writing  $A$  for the  $k \times d$  matrix whose  $i$ -th row is  $x_i^t$  and  $b \in \mathbb{R}^k$  for the vector whose  $i$ -th entry equals  $\|x_i\|^2/2$ , we see that  $x$  must satisfy

$$Ax = b, \quad (5.8)$$

and any  $x$  satisfying this equation defines a ball of the sought form. If  $x_1, \dots, x_k$  are linearly independent then the  $k \times k$  matrix  $AA^t$  is non-singular. In particular  $(AA^t)^{-1}$  is well defined. So if  $x_1, \dots, x_k$  are linearly independent then

$$x := A^t(AA^t)^{-1}b \quad (5.9)$$

solves (5.8). Note that  $x$  is a linear combination of the columns  $A^t$ . In other words, it lies in the linear hull  $\mathcal{L}(\{x_1, \dots, x_k\})$  of  $x_1, \dots, x_k$ .

Now let  $y$  be an arbitrary solution of (5.8). We have  $A(y - x) = 0$ , so that  $\langle y - x, x_i \rangle = 0$  for  $i = 1, \dots, k$ . But then we also have  $\langle y - x, x \rangle = 0$ . Pythagoras's theorem tells us that

$$\|y\|^2 = \|x\|^2 + \|y - x\|^2.$$

So  $x$  is the (unique) solution with the smallest possible norm. This proves (i).

To see (ii) we note that, using the Leibniz formula for the determinant and Cramer's rule for the matrix inverse, each individual coordinate of  $x = x(x_1, \dots, x_k)$  can be written as  $P(x_1, \dots, x_k)/Q(x_1, \dots, x_k)$  where  $P, Q$  are polynomials in the coordinates of  $x_1, \dots, x_k$  and specifically  $Q = \det(AA^t)$ . The polynomial  $Q$  is non-zero on  $I_{kd}$ , so that  $x$  is a continuous function of  $x_1, \dots, x_k$  on  $I_{kd}$ . So  $\rho(x_1, \dots, x_k) = \|x\|$  is also continuous on  $I_{k,d}$ .  $\square$

The unique ball that has  $0, x_1, \dots, x_k$  on its boundary, guaranteed to exist when  $x_1, \dots, x_k$  are linearly independent, will be denoted by  $B(x_1, \dots, x_k)$ . We shall only consider  $B(x_1, \dots, x_k)$  for  $x_1, \dots, x_k$  distinct elements of the Poisson point process  $\mathcal{Z}$ . As long as  $k \leq d$ , almost surely any set of  $k$  distinct points of  $\mathcal{Z}$  is linearly independent.

Another observation we'll use is that for  $\mu > 0$  and  $u_1, \dots, u_k \in \mathbb{R}^k$  we have

$$\rho(\mu u_1, \dots, \mu u_k) = \mu \rho(u_1, \dots, u_k), \quad D(\mu u_1, \dots, \mu u_k) = \mu^k D(u_1, \dots, u_k). \quad (5.10)$$

We set:

$$\mathcal{X}(r) := \left\{ (z_1, \dots, z_k) \in \mathcal{Z}^k : \begin{array}{l} z_1, \dots, z_k \text{ are distinct, and} \\ \rho(z_1, \dots, z_k) \leq r \end{array} \right\}, \quad X(r) := |\mathcal{X}(r)|, \quad (5.11)$$

Applying the Mecke formula we obtain

$$\mathbb{E}X(r) = \lambda^k \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} 1_{\{\rho(x_1, \dots, x_k) \leq r\}} dx_1 \cdots dx_k. \quad (5.12)$$

The following approximations will be reused a few times.

**Lemma 5.26.**

- (i) For  $r > 0$  fixed and  $d \rightarrow \infty$ , we have  $\sqrt[d]{\mathbb{E}X(r)} = r^k v_k + o_d(1)$ .
- (ii) For  $r > 1/\sqrt{2}$  fixed and  $d \rightarrow \infty$ , we have  $\mathbb{E}X(r)^2 = (1 + o_d(1)) \cdot (\mathbb{E}X(r))^2$ .

The proof of Lemma 5.26 makes use of the following observations.

**Lemma 5.27.**

- (i) For every  $k \in \mathbb{N}$ , we have

$$\max_{\substack{u_1, \dots, u_k \in \mathbb{R}^k \\ \rho(u_1, \dots, u_k) \leq 1}} D(u_1, \dots, u_k) = v_k.$$

Moreover, the maximum is attained by  $u_1, \dots, u_k \in \mathbb{R}^k$  if and only if  $\underline{0}, u_1, \dots, u_k$  form the vertices of a regular simplex of side-length  $\ell_k$ .

- (ii) For every  $k, \ell \in \mathbb{N}$ , we have

$$\max_{\substack{u_1, \dots, u_{k+\ell} \in \mathbb{R}^{k+\ell}, \\ \rho(u_1, \dots, u_k) \leq 1, \\ \rho(u_{k+1}, \dots, u_{k+\ell}) \leq 1}} D(u_1, \dots, u_{k+\ell}) \leq v_k \cdot v_\ell.$$

Before we get to the proof of Lemma 5.26, we first deal with Lemma 5.27. We make use of a “folklore” fact about simplices inscribed in the unit ball.

**Proposition 5.28.** *Among all simplices inscribed into the unit sphere, only the regular simplices maximize the volume. More precisely, for  $w_0, \dots, w_k \in S^{k-1}$  we have*

$$\text{vol}(\text{conv}(\{w_0, \dots, w_k\})) \leq v_k/k!,$$

with equality if and only if  $\|w_i - w_j\| = \ell_k$  for all  $0 \leq i < j \leq k$ .

This result can be found in the literature. It is for instance derived in [90] and (independently) [93]. We believe it is significantly older, but have not managed to find an earlier, explicit reference. In both [90] and [93] the proposition is a consequence of a more general result. There is a much shorter and simpler proof of Proposition 5.28 however. See for instance the start of the proof of Lemma 13.2.2 in [72].

*Proof of Lemma 5.27.* The relations (5.10) show that when computing the sought maximum we can restrict attention to  $k$ -tuples  $u_1, \dots, u_k$  such that  $\rho(u_1, \dots, u_k) = 1$ .

Let  $u_1, \dots, u_k \in \mathbb{R}^k$  be such that  $\rho(u_1, \dots, u_k) = 1$ . Let  $B$  be the (unique) ball of radius one such that  $\underline{0}, u_1, \dots, u_k \in \partial B$  and let  $c$  denote its center. If

we set  $w_i = u_i - c$  for  $i = 0, \dots, k$  where  $u_0 := \underline{0}$ , then  $w_0, \dots, w_k \in S^{k-1}$ . In fact, this procedure provides a one-to-one correspondence between  $(k+1)$ -tuples  $w_0, \dots, w_k \in S^{k-1}$  and  $k$ -tuples  $u_1, \dots, u_k \in \mathbb{R}^k$  satisfying  $\rho(u_1, \dots, u_k) = 1$ . We also have

$$\begin{aligned} D(u_1, \dots, u_k) &= k! \cdot \text{vol}(\text{conv}(\{\underline{0}, u_1, \dots, u_k\})) \\ &= k! \cdot \text{vol}(\text{conv}(\{w_0, \dots, w_k\})). \end{aligned}$$

Part (i) now follows immediately from Proposition 5.28.

For the proof of part (ii) we use the first interpretation of the absolute value of the determinant given in (5.7). The volume of the parallelepiped  $P(u_1, \dots, u_{k+\ell})$  can be written as

$$\begin{aligned} D(u_1, \dots, u_{k+\ell}) &= \prod_{i=1}^{k+\ell} \text{dist}(u_i, \mathcal{L}(\{u_1, \dots, u_{i-1}\})) \\ &\leq \left( \prod_{i=1}^k \text{dist}(u_i, \mathcal{L}(\{u_1, \dots, u_{i-1}\})) \right) \cdot \left( \prod_{j=1}^{\ell} \text{dist}(u_{k+j}, \mathcal{L}(\{u_{k+1}, \dots, u_{k+j-1}\})) \right) \end{aligned}$$

where  $\text{dist}(x, A) := \inf_{y \in A} \|x - y\|$ . Writing  $v_{k,\ell}$  for the maximum in the LHS in Part (ii) of the lemma statement, we find:

$$\begin{aligned} v_{k,\ell} &\leq \left( \max_{\substack{u_1, \dots, u_k \in \mathbb{R}^{k+\ell} \\ \rho(u_1, \dots, u_k) \leq 1}} \prod_{i=1}^k \text{dist}(u_i, \mathcal{L}(\{u_1, \dots, u_{i-1}\})) \right) \\ &\quad \cdot \left( \max_{\substack{w_1, \dots, w_\ell \in \mathbb{R}^{k+\ell} \\ \rho(w_1, \dots, w_\ell) \leq 1}} \prod_{i=1}^{\ell} \text{dist}(w_i, \mathcal{L}(\{w_1, \dots, w_{i-1}\})) \right) \\ &= \left( \max_{\substack{u_1, \dots, u_k \in \mathbb{R}^k \\ \rho(u_1, \dots, u_k) \leq 1}} \prod_{i=1}^k \text{dist}(u_i, \mathcal{L}(\{u_1, \dots, u_{i-1}\})) \right) \\ &\quad \cdot \left( \max_{\substack{w_1, \dots, w_\ell \in \mathbb{R}^\ell \\ \rho(w_1, \dots, w_\ell) \leq 1}} \prod_{i=1}^{\ell} \text{dist}(w_i, \mathcal{L}(\{w_1, \dots, w_{i-1}\})) \right) \\ &= v_k \cdot v_\ell, \end{aligned}$$

where in the penultimate line we use that if we apply an orthogonal transformation  $T$  mapping  $u_1, \dots, u_k \in \mathbb{R}^{k+\ell}$  to  $Tu_1, \dots, Tu_k \in \mathbb{R}^k \times \{0\}^\ell$  then  $\rho(Tu_1, \dots, Tu_k) =$

$\rho(u_1, \dots, u_k)$  and  $\text{dist}(Tu_i, \mathcal{L}(\{Tu_1, \dots, Tu_{i-1}\})) = \text{dist}(u_i, \mathcal{L}(\{u_1, \dots, u_{i-1}\}))$ .  $\square$

*Proof of Lemma 5.26.* Applying the linear Blaschke-Petkantschin formula (see e.g. [89], Theorem 7.2.1) to (5.12), we have:

$$\begin{aligned} \mathbb{E}X(r) &= \lambda^k \cdot \frac{(d+1) \cdots (d-k+2)}{(k+1) \cdots 2} \cdot \frac{\kappa_{d+1} \cdots \kappa_{d-k+2}}{\kappa_{k+1} \cdots \kappa_2} \cdot \\ &\quad \int_{\mathbb{R}^k} \cdots \int_{\mathbb{R}^k} 1_{\{\rho(u_1, \dots, u_k) \leq r\}} \cdot D(u_1, \dots, u_k)^{d-k} \, du_1 \cdots du_k, \end{aligned} \quad (5.13)$$

(To obtain this identity from Theorem 7.2.1 of [89], we use the identities  $\omega_i = (i+1)\kappa_{i+1}$  and that  $\rho(\cdot)$  is invariant under orthogonal transformations. That is  $\rho(Tx_1, \dots, Tx_k) = \rho(x_1, \dots, x_k)$  for any orthogonal transformation  $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$  and any  $x_1, \dots, x_k$ . Note that in [89], the inner integral in the RHS of (7.7) does not depend on the choice of  $L \in G(d, q)$  and  $\nu_q(\cdot)$  denotes the uniform measure on  $G(d, q)$ , the space of all linear subspaces of dimension  $q$ .)

The choice  $\lambda = 1/\kappa_d$  implies that the constant in the RHS of (5.13) equals

$$\frac{1}{(k+1)!} \cdot \frac{1}{\prod_{i=2}^{k+1} \kappa_i} \cdot \frac{\kappa_{d+1}}{\kappa_d} \cdots \frac{\kappa_{d-k+2}}{\kappa_d} = d^{O(1)}, \quad (5.14)$$

using (2.1) and that  $\Gamma(t+\alpha)/\Gamma(t) = (1+o_t(1)) \cdot t^\alpha$  if  $\alpha \in \mathbb{R}$  is fixed and  $t \rightarrow \infty$ , as can for instance be seen from Stirling's approximation to the Gamma function.

If  $\|u_i\| > 2r$  for some  $1 \leq i \leq k$  then  $\rho(u_1, \dots, u_k) > r$ . Hence

$$\mathbb{E}X(r) = d^{O(1)} \cdot \int_{B_{\mathbb{R}^k}(0, 2r)} \cdots \int_{B_{\mathbb{R}^k}(0, 2r)} 1_{\{\rho(u_1, \dots, u_k) \leq r\}} \cdot D(u_1, \dots, u_k)^{d-k} \, du_1 \cdots du_k. \quad (5.15)$$

By Lemma 5.27 and (5.15), (5.10), we have

$$\mathbb{E}X(r) \leq d^{O(1)} \cdot (\kappa_k(2r)^k)^k \cdot (r^k v_k)^{d-k}.$$

Taking  $d$ -th roots, we find

$$\sqrt[d]{\mathbb{E}X(r)} \leq (1 + o_d(1)) \cdot r^k v_k = r^k v_k + o_d(1).$$

It remains to derive a lower bound of the same form. To this end, we fix an arbitrarily small  $\eta > 0$  and let  $A \subseteq \mathbb{R}^{k^2}$  be defined by

$$A := \left\{ (u_1, \dots, u_k) \in \mathbb{R}^k \times \cdots \times \mathbb{R}^k : \begin{array}{l} \rho(u_1, \dots, u_k) < r, \text{ and} \\ D(u_1, \dots, u_k) > (1 - \eta)r^k v_k \end{array} \right\}.$$

We have

$$\mathbb{E}X(r) \geq d^{O(1)} \cdot \text{vol}_{k^2}(A) \cdot ((1-\eta)r^k v_k)^{d-k}. \quad (5.16)$$

The set  $A$  is non-empty. (If  $0, u_1, \dots, u_k$  form a regular simplex of side-length  $(1-\eta/2)^{1/k} r \ell_k$ , then  $(u_1, \dots, u_k) \in A$  by (5.10) and Lemma 5.27.) The set  $A$  is also open. (To see this, note that  $D$  is continuous, so that  $\{D > (1-\eta)r^k v_k\}$  is open. Now note that  $\rho$  is continuous on  $\{D > (1-\eta)r^k v_k\}$ .) Being non-empty and open implies that  $A$  has positive  $k^2$ -dimensional volume. So, taking the  $d$ -th root of (5.16) gives

$$\sqrt[d]{\mathbb{E}X(r)} \geq (1 + o_d(1)) \cdot (1-\eta)r^k v_k.$$

Since  $\eta > 0$  was arbitrary this in fact gives  $\sqrt[d]{\mathbb{E}X(r)} \geq r^k v_k + o_d(1)$ , finishing the proof of (i).

We now turn our attention to the second moment of  $X(r)$ . We remark that

$$X(r)^2 = \left| \left\{ (z_1, \dots, z_k, w_1, \dots, w_k) \in \mathcal{Z}^{2k} : \begin{array}{l} z_1, \dots, z_k \text{ are distinct, and} \\ w_1, \dots, w_k \text{ are distinct, and} \\ \rho(z_1, \dots, z_k) \leq r, \text{ and} \\ \rho(w_1, \dots, w_k) \leq r. \end{array} \right\} \right|. \quad (5.17)$$

We define for  $\ell = 0, \dots, k$ :

$$X_\ell := \left| \left\{ (z_1, \dots, z_{k+\ell}) \in \mathcal{Z}^{k+\ell} : \begin{array}{l} z_1, \dots, z_{k+\ell} \text{ are distinct, and} \\ \rho(z_1, \dots, z_k) \leq r, \text{ and} \\ \rho(z_1, \dots, z_{k-\ell}, z_{k+1}, \dots, z_{k+\ell}) \leq r. \end{array} \right\} \right|.$$

Accounting for the ways in which  $(z_1, \dots, z_k)$  and  $(w_1, \dots, w_k)$  in (5.17) might overlap and using symmetry, we find

$$\mathbb{E}X(r)^2 = \sum_{\ell=0}^k (k-\ell)! \cdot \binom{k}{\ell}^2 \cdot \mathbb{E}X_\ell. \quad (5.18)$$

We will bound  $\mathbb{E}X_\ell$  separately for each  $\ell$ . The easiest value is of course  $\ell = 0$ , as  $X_0 = X(r)$ . In particular, Part (i) tells us that  $\mathbb{E}X_0 = \mathbb{E}X(r) = (r^k v_k + o_d(1))^d$ . The next simplest value is  $\ell = k$ . Now the Mecke formula gives

$$\begin{aligned} \mathbb{E}X_k &= \lambda^{2k} \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} 1_{\{\rho(x_1, \dots, x_k), \rho(x_{k+1}, \dots, x_{2k}) \leq r\}} \, dx_1 \cdots dx_{2k} \\ &= \left( \lambda^k \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} 1_{\{\rho(x_1, \dots, x_k) \leq r\}} \, dx_1 \cdots dx_k \right)^2 \\ &= \mathbb{E}X(r)^2. \end{aligned} \quad (5.19)$$

For the remaining values  $0 < \ell < k$  a little more work is needed. Arguing as in the proof of Lemma 5.26, we have:

$$\begin{aligned}
\mathbb{E}X_\ell &= \lambda^{k+\ell} \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} 1_{\{\rho(x_1, \dots, x_k) \leq r, \rho(x_1, \dots, x_{k-\ell}, x_{k+1}, \dots, x_{k+\ell}) \leq r\}} \mathrm{d}x_1 \dots \mathrm{d}x_{k+\ell} \\
&\leq \lambda^{k+\ell} \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} 1_{\{\rho(x_1, \dots, x_k) \leq r, \rho(x_{k+1}, \dots, x_{k+\ell}) \leq r\}} \mathrm{d}x_1 \dots \mathrm{d}x_{k+\ell} \\
&= d^{O(1)} \cdot \int_{B_{\mathbb{R}^{k+\ell}}(\underline{0}, 2)} \cdots \int_{B_{\mathbb{R}^{k+\ell}}(\underline{0}, 2)} 1_{\{\rho(u_1, \dots, u_k) \leq r, \rho(u_{k+1}, \dots, u_{k+\ell}) \leq r\}} \\
&\quad \cdot D(u_1, \dots, u_{k+\ell})^{d-(k+\ell)} \mathrm{d}u_1 \dots \mathrm{d}u_{k+\ell} \\
&\leq d^{O(1)} \cdot (\kappa_{k+\ell} 2^{k+\ell})^{k+\ell} \cdot (r^{k+\ell} v_k v_\ell)^{d-(k+\ell)} \\
&= (r^{k+\ell} v_k v_\ell + o_d(1))^d,
\end{aligned} \tag{5.20}$$

where we used Part (ii) of Lemma 5.27 in the penultimate line.

We have  $v_1 = 2$  and for  $\ell \geq 2$ :

$$\begin{aligned}
\left( \frac{v_\ell}{v_{\ell-1}} \right)^2 &= \frac{(\ell+1)^{\ell+1} (\ell-1)^{\ell-1}}{\ell^\ell} = \frac{(\ell+1)^2}{\ell} \cdot \left( \frac{(\ell+1)(\ell-1)}{\ell^2} \right)^{(\ell-1)/2} \\
&= \frac{\ell^2 + 2\ell + 1}{\ell} \cdot \left( 1 - \frac{1}{\ell^2} \right)^{\ell-1} \geq (\ell+2) \cdot \left( 1 - \frac{1}{\ell^2} \right)^\ell \\
&\geq (\ell+2) \cdot \left( 1 - \frac{1}{\ell} \right) \geq 4 \cdot (1 - 1/2) = 2.
\end{aligned}$$

So if  $r > 1/\sqrt{2}$  then  $1 < rv_1 < r^2 v_2 < \dots < r^k v_k$ . Thus, by Part (i) and (5.20), provided  $r > 1/\sqrt{2}$ :

$$\mathbb{E}X_\ell = o(\mathbb{E}X(r)^2) \quad (\ell = 0, \dots, k-1).$$

Part (ii) now follows by combining with (5.18) and (5.19).  $\square$

*Proof of Theorem 1.3.* We let  $\varepsilon > 0$  be arbitrary and we let  $\delta = \delta(\varepsilon, k) > 0$  be a small constant to be chosen more precisely in the remainder of the proof. We note that if  $z_1, \dots, z_k \in \mathcal{Z}$  define a  $(d-k)$ -face  $F$  then every  $x \in F$  must satisfy  $\|x\| \geq \rho(z_1, \dots, z_k)$  as  $B(x, \|x\|)$  contains  $\underline{0}, z_1, \dots, z_k$  on its boundary. Since every face contains a vertex, Lemma 5.6 implies that, with probability  $1 - o_d(1)$ , the only  $k$ -tuples  $z_1, \dots, z_k \in \mathcal{Z}$  that define a  $(d-k)$ -face must satisfy  $\rho(z_1, \dots, z_k) < 1 + \delta$ . We have:

$$\mathbb{P}(f_{d-k}(\mathcal{V}_{\text{typ}}) \leq X(1 + \delta)) = 1 - o_d(1). \tag{5.21}$$

Provided  $\delta = \delta(\varepsilon, k)$  was chosen sufficiently small, Markov's inequality tells us that

$$\mathbb{P}(X(1+\delta) > (1+\varepsilon)^d v_k^d) \leq \frac{\mathbb{E}X(1+\delta)}{(1+\varepsilon)^d v_k^d} = \left( \frac{(1+\delta)^k v_k + o_d(1)}{(1+\varepsilon)v_k} \right)^d = o_d(1).$$

Combining with (5.21), this gives

$$\begin{aligned} \mathbb{P}\left(\sqrt[d]{f_{d-k}(\mathcal{V}_{\text{typ}})} > (1+\varepsilon)v_k\right) &\leq \mathbb{P}(f_{d-k}(\mathcal{V}_{\text{typ}}) > X(1+\delta)) \\ &\quad + \mathbb{P}(X(1+\delta) > (1+\varepsilon)^d v_k^d) \\ &= o_d(1). \end{aligned} \quad (5.22)$$

It remains to derive an upper bound on  $\mathbb{P}\left(\sqrt[d]{f_{d-k}(\mathcal{V}_{\text{typ}})} < (1-\varepsilon)v_k\right)$  that tends to zero with  $d$ . For this purpose, we define

$$Y := \left| \left\{ (z_1, \dots, z_k) \in \mathcal{Z}^k : \begin{array}{l} z_1, \dots, z_k \text{ are distinct, and} \\ \rho(z_1, \dots, z_k) \leq 1-\delta, \text{ and} \\ B(z_1, \dots, z_k) \cap \mathcal{Z} = \emptyset \end{array} \right\} \right|. \quad (5.23)$$

Clearly each  $k$ -tuple counted by  $Y$  defines a  $(d-k)$ -face of  $\mathcal{V}_{\text{typ}}$ . As each  $k$ -tuple has  $k!$  re-orderings, we have

$$f_{d-k}(\mathcal{V}_{\text{typ}}) \geq \frac{1}{k!} \cdot Y.$$

(We also use that, almost surely, no two distinct  $k$ -sets of points in  $\mathcal{Z}$  define the same face.) Using the Mecke formula once again we have

$$\mathbb{E}Y = \lambda^k \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} 1_{\{\rho(x_1, \dots, x_k) \leq 1-\delta\}} \cdot \mathbb{P}(B(x_1, \dots, x_k) \cap \mathcal{Z} = \emptyset) \, dx_1 \dots dx_k.$$

Comparing to (5.12) we see that  $e^{-(1-\delta)^d} \cdot \mathbb{E}X(1-\delta) \leq \mathbb{E}Y \leq \mathbb{E}X(1-\delta)$ . In other words

$$\mathbb{E}Y = (1 + o_d(1)) \cdot \mathbb{E}X(1-\delta). \quad (5.24)$$

Since  $Y \leq X(1-\delta)$ , provided  $\delta = \delta(\varepsilon, k)$  was chosen sufficiently small, Part (ii) of Lemma 5.26 now gives

$$\mathbb{E}Y^2 \leq \mathbb{E}X(1-\delta)^2 = (1 + o_d(1)) \cdot (\mathbb{E}Y)^2.$$

By (5.24) and Part (i) of Lemma 5.26, provided  $\delta = \delta(\varepsilon, k)$  is chosen sufficiently small, we have

$$k! \cdot (1 - \varepsilon)^d v_k^d < \frac{1}{2} \mathbb{E}Y,$$

for  $d$  sufficiently large. We can now apply Chebyshev's inequality to obtain:

$$\begin{aligned} \mathbb{P}\left(\sqrt[d]{f_{d-k}(\mathcal{V}_{\text{typ}})} < (1 - \varepsilon)v_k\right) &\leq \mathbb{P}(Y < k! \cdot (1 - \varepsilon)^d v_k^d) \\ &\leq \mathbb{P}(|Y - \mathbb{E}Y| > \tfrac{1}{2}\mathbb{E}Y) \\ &\leq 4 \text{Var } Y / (\mathbb{E}Y)^2 \\ &= o_d(1). \end{aligned}$$

Together with (5.22) this establishes Theorem 1.3.  $\square$

*Proof of Theorem 1.2.* Recall that if  $F \in \mathcal{F}_{d-k}(\mathcal{V}_{\text{typ}})$  is defined by  $z_1, \dots, z_k \in \mathcal{Z}$  then  $\|x\| \geq \rho(z_1, \dots, z_k)$  for all  $x \in F$ . So by Lemma 5.6, with probability  $1 - o_d(1)$ , any face defined by some  $z_1, \dots, z_k \in \mathcal{Z}$  with  $\rho(z_1, \dots, z_k) \geq 1 - \varepsilon$  is contained in the annulus  $B(\mathbf{0}, 1 + \varepsilon) \setminus B(\mathbf{0}, 1 - \varepsilon)$ . By Theorem 1.1, with probability  $1 - o_d(1)$ ,  $\mathcal{V}_{\text{typ}}$  has diameter at most  $2 + \varepsilon$ . Applying Lemma 5.24 we obtain that, with probability  $1 - o_d(1)$ :

$$\sum_{F \in \mathcal{F}_{d-k}(\mathcal{V}_{\text{typ}})} \text{diam}(F) \leq 4\sqrt{\varepsilon} \cdot f_{d-k}(\mathcal{V}_{\text{typ}}) + (2 + \varepsilon) \cdot X(1 - \varepsilon).$$

We note that

$$\begin{aligned} \mathbb{P}(X(1 - \varepsilon) > \varepsilon f_{d-k}(\mathcal{V}_{\text{typ}})) &\leq \mathbb{P}(X(1 - \varepsilon) > \varepsilon(1 - \varepsilon/2)^d v_k^d) \\ &\quad + \mathbb{P}(f_{d-k}(\mathcal{V}_{\text{typ}}) < (1 - \varepsilon/2)^d v_k^d). \end{aligned}$$

The first term in the RHS is  $o_d(1)$  by Lemma 5.26 and Markov's inequality. The second term is  $o_d(1)$  as well, by Theorem 1.3. We arrive at:

$$\mathbb{P}\left(\frac{1}{f_{d-k}(\mathcal{V}_{\text{typ}})} \sum_{F \in \mathcal{F}_{d-k}(\mathcal{V}_{\text{typ}})} \text{diam}(F) > 4\sqrt{\varepsilon} + \varepsilon(2 + \varepsilon)\right) = o_d(1).$$

The theorem follows.  $\square$

It remains to prove Proposition 5.4. We need one more preparatory lemma.

**Lemma 5.29.** *For every  $\varepsilon > 0$  and  $k \in \mathbb{N}$  there exist  $\delta > 0$  and  $c < v_k$  such that the following holds. For all  $u_1, \dots, u_k \in \mathbb{R}^k$  that are not  $\varepsilon$ -near regular, we have  $\rho(u_1, \dots, u_k) > 1 + \delta$  or  $D(u_1, \dots, u_k) \leq c$ , or both.*

*Proof.* Let us set

$$g(u_1, \dots, u_k) := \sum_{i=1}^k (\|u_i\| - \ell_k)^2 + \sum_{1 \leq i < j \leq k} (\|u_i - u_j\| - \ell_k)^2,$$

$$W := \left\{ (u_1, \dots, u_k) \in \mathbb{R}^k \times \dots \times \mathbb{R}^k : \begin{array}{l} D(u_1, \dots, u_k) \geq v_k/2, \text{ and,} \\ \rho(u_1, \dots, u_k) \leq 1, \text{ and,} \\ g(u_1, \dots, u_k) \geq \varepsilon^2/4 \end{array} \right\},$$

and

$$c' := \max \left( \sup_{(u_1, \dots, u_k) \in W} D(u_1, \dots, u_k), \frac{v_k}{2} \right).$$

We claim that  $c' < v_k$ . If  $W = \emptyset$  then this is clearly true, so suppose  $W$  is non-empty. Since  $D, g$  are continuous, the set  $W_0 := \{D \geq v_k/2, g \geq \varepsilon^2/4\}$  is closed. By Lemma 5.25,  $\rho$  is continuous on  $W_0$ . So  $W$  is closed too. It is also bounded, since  $\rho \leq 1$  implies  $\|u_1\|, \dots, \|u_k\| \leq 2$ . We see that  $W$  is compact, and hence the supremum  $c'' := \sup_{(u_1, \dots, u_k) \in W} D(u_1, \dots, u_k)$  is attained by some  $(w_1, \dots, w_k) \in W$ . Lemma 5.27 tells us that  $c'' < v_k$  as  $g(w_1, \dots, w_k) \neq 0$  implies that  $0, w_1, \dots, w_k$  is not a regular simplex of side length  $\ell_k$ . So  $c' = \max(v_k/2, c'') < v_k$  as claimed.

We now choose a small  $\delta = \delta(\varepsilon, k) > 0$ , small enough so that  $(1+\delta)^k c' < v_k$  and  $(\ell_k + \varepsilon)/(1+\delta) > \ell_k + \varepsilon/2$ . Let  $u_1, \dots, u_k$  be such that  $\rho(u_1, \dots, u_k) \leq 1 + \delta$  and  $u_1, \dots, u_k$  is not  $\varepsilon$ -near regular, but otherwise arbitrary. Setting  $w_i := u_i/(1+\delta)$  we see that  $\rho(w_1, \dots, w_k) \leq 1$  by (5.10). Moreover, if  $i, j$  are such that  $\|u_i - u_j\| \geq \ell_k + \varepsilon$  then  $\|w_i - w_j\| \geq \ell_k + \varepsilon/2$ . Similarly, if  $\|u_i\| \geq \ell_k + \varepsilon$  then  $\|w_i\| \geq \ell_k + \varepsilon/2$ . We also have  $\|w_i\| \leq \|u_i\|, \|w_i - w_j\| \leq \|u_i - u_j\|$ . It follows that  $g(w_1, \dots, w_k) \geq \varepsilon^2/4$ . Appealing to (5.10), we find that

$$D(u_1, \dots, u_k) = (1+\delta)^k D(w_1, \dots, w_k) \leq (1+\delta)^k c' < v_k.$$

This shows that the lemma holds with the choice  $c := (1+\delta)^k c'$ .  $\square$

*Proof of Proposition 5.4.* We fix  $\varepsilon > 0$ , let  $\delta, c$  be as provided by Lemma 5.29 and define

$$N := \left| \left\{ (z_1, \dots, z_k) \in \mathcal{Z}^k : \begin{array}{l} \rho(z_1, \dots, z_k) \leq 1 + \delta, \text{ and} \\ z_1, \dots, z_k \text{ not } \varepsilon\text{-near regular} \end{array} \right\} \right|.$$

Arguing as in the proof of Lemma 5.26, we find:

$$\begin{aligned}
\mathbb{E}N &= \lambda^k \int_{\mathbb{R}^d} \cdots \int_{\mathbb{R}^d} 1_{\left\{ \begin{array}{l} \rho(x_1, \dots, x_k) \leq 1+\delta, \text{ and} \\ x_1, \dots, x_k \text{ not } \varepsilon\text{-near regular} \end{array} \right\}} dx_1 \dots dx_k \\
&= d^{O(1)} \int_{\mathbb{R}^k} \cdots \int_{\mathbb{R}^k} 1_{\left\{ \begin{array}{l} \rho(u_1, \dots, u_k) \leq 1+\delta, \text{ and} \\ u_1, \dots, u_k \text{ not } \varepsilon\text{-near regular} \end{array} \right\}} \cdot D(u_1, \dots, u_k)^{d-k} du_1 \dots du_k \\
&\leq d^{O(1)} \cdot (\kappa_k 3^k)^k \cdot c^{d-k} \\
&= (c + o_d(1))^d.
\end{aligned}$$

where  $c$  is as provided by Lemma 5.29, which we've applied in the penultimate line. For every fixed  $\eta > 0$  we have

$$\begin{aligned}
\mathbb{P}(N \geq \eta f_{d-k}(\mathcal{V}_{\text{typ}})) &\leq \mathbb{P}\left(N \geq \eta \left(\frac{c + v_k}{2}\right)^d\right) + \mathbb{P}\left(f_{d-k}(\mathcal{V}_{\text{typ}}) \leq \left(\frac{c + v_k}{2}\right)^d\right) \\
&= o_d(1),
\end{aligned}$$

by Theorem 1.3 and Markov's inequality. This proves  $N/f_{d-k}(\mathcal{V}_{\text{typ}}) \xrightarrow[d \rightarrow \infty]{\mathbb{P}} 0$ .

To conclude the proof, we remind the reader that as explained in the start of the proof of Theorem 1.3, with probability  $1 - o_d(1)$ , every  $(d - k)$ -face of  $\mathcal{V}_{\text{typ}}$  is defined by some  $z_1, \dots, z_k$  with  $\rho(z_1, \dots, z_k) \leq 1 + \delta$ . It follows that, with probability  $1 - o_d(1)$ :

$$f_{d-k}(\mathcal{V}_{\text{typ}}) - M_{k,\varepsilon} \leq N.$$

In other words

$$1 - \frac{N}{f_{d-k}(\mathcal{V}_{\text{typ}})} \leq \frac{M_{k,\varepsilon}}{f_{d-k}(\mathcal{V}_{\text{typ}})} \leq 1,$$

with probability  $1 - o_d(1)$ . The proposition follows.  $\square$

### 5.3 Discussion and suggestions for further work

We have shown that the inradius, outradius, diameter and mean width of the typical Poisson–Voronoi cell, after normalization and when the dimension tends to infinity, all tend in probability to explicit constants. For the width we've shown non-matching upper and lower bounds. We would like to mention that while finalizing the paper corresponding to this chapter, we learned that K. Alishahi has shown in his PhD thesis [6] that the ratio  $\text{outr}(\mathcal{V}_{\text{typ}})/\text{inr}(\mathcal{V}_{\text{typ}})$  tends to 2 in probability as  $d \rightarrow \infty$ , using different methods to the ones we employ. Alishahi's thesis is written in Persian and does not appear to be available online at the time of writing.

As a side product we also obtain that, with probability  $1 - o_d(1)$ , the Hausdorff distance between the typical cell  $\mathcal{V}_{\text{typ}}$  and any ball is at least a constant times the diameter of  $\mathcal{V}_{\text{typ}}$ . For completeness let us first recall that the *Hausdorff distance* between sets  $X, Y \subseteq \mathbb{R}^d$  is given by:

$$d_H(X, Y) := \max \left( \sup_{x \in X} \inf_{y \in Y} \|x - y\|, \sup_{y \in Y} \inf_{x \in X} \|x - y\| \right).$$

We note that the Hausdorff distance satisfies

$$2 \cdot d_H(X, Y) \geq |\text{diam}(X) - \text{diam}(Y)|, |\text{width}(X) - \text{width}(Y)|.$$

If  $Y$  is a ball then  $\text{diam}(Y) = \text{width}(Y)$ . So for  $X \subseteq \mathbb{R}^d$  arbitrary and  $Y \subseteq \mathbb{R}^d$  a ball we must have

$$\text{diam}(X) - 2 \cdot d_H(X, Y) \leq \text{diam}(Y) = \text{width}(Y) \leq \text{width}(X) + 2 \cdot d_H(X, Y),$$

giving

$$d_H(X, Y) \geq (\text{diam}(X) - \text{width}(X))/4.$$

Hence, by Theorem 1.1 and Proposition 5.1, with probability  $1 - o_d(1)$ , the typical cell  $X = \mathcal{V}_{\text{typ}}$  satisfies:

$$d_H(\mathcal{V}_{\text{typ}}, Y) \geq (1/8 - o_d(1)) \cdot \sqrt[d]{\lambda \cdot \text{vol}(B)} = (1/16 - o_d(1)) \cdot \text{diam}(\mathcal{V}_{\text{typ}}),$$

for every ball  $Y$ .

As already stated in the introduction, we could not prove the following natural conjecture.

**Conjecture 5.30.** There is a constant  $c$  such that

$$\frac{\text{width}(\mathcal{V}_{\text{typ}})}{\sqrt[d]{\lambda \cdot \text{vol}(B)}} \xrightarrow[d \rightarrow \infty]{\mathbb{P}} c.$$

The lower bound on the width can probably be improved via a more technical variation on our argument, but it seems unlikely to us that it will give a sharp result without significant new ideas. The argument giving the upper bound exhibits a direction  $u \in S^{d-1}$  such that  $w(u, \mathcal{V}_{\text{typ}})$  is small. The direction  $u$  is chosen perpendicular to one of the facets of  $\mathcal{V}_{\text{typ}}$ . A priori we see no reason to believe a direction of this type should minimize  $w(u, \mathcal{V}_{\text{typ}})$  over all possible directions. Let us however point out that if  $u$  is the direction that minimizes  $w(u, \mathcal{V}_{\text{typ}})$  then each of the two supporting hyperplanes perpendicular to  $u$  must contain some face and

the sum of the dimensions of these two faces must be at least  $d - 1$ . (Otherwise, a small perturbation of  $u$  will yield a direction with even smaller width.) Put differently, the union of the two supporting hyperplanes perpendicular to  $u$  contains at least  $d + 1$  vertices of  $\mathcal{V}_{\text{typ}}$ . We have not been able to turn this observation into an argument that gives better bounds than the ones provided by Proposition 5.1, but perhaps other teams will be able to succeed in doing that.

An important ingredient in our proofs was the observation that, with probability  $1 - o_d(1)$ , all vertices of the typical cell have approximately the same norm (namely  $1 \pm o_d(1)$  under the scaling  $\lambda = 1/\kappa_d$  used throughout the chapter). As can for instance be seen from Theorem 1.2 in [55], the expected number of vertices of  $\mathcal{V}_{\text{typ}}$  is  $\exp[(d/2) \cdot \ln d \cdot (1 + o_d(1))]$ . It seems natural to compare the behaviour of the typical cell to that of the convex hull of the same number of points taken i.i.d. uniformly at random on  $S^{d-1}$ . The latter set-up has been studied by Bonnet and O'Reilly [22]. A relevant result in [22] is Theorem 3.4, which tells us the convex hull of  $n = \exp[(d/2) \cdot \ln d \cdot (1 + o_d(1))]$  i.i.d. uniform points on  $S^{d-1}$  has the property that, with probability  $1 - o_d(1)$ , each of its facets has distance  $1 \pm o_d(1)$  to the origin. This is rather different from the behaviour of  $\mathcal{V}_{\text{typ}}$ . In the proof of Lemma 5.14 we have exhibited the existence (with probability  $1 - o_d(1)$ ) of some facet with distance  $1/2 \pm o_d(1)$  to the origin. A straightforward adaptation of the argument shows that for all  $1/2 \leq r \leq 1$  there exist (with probability  $1 - o_d(1)$ ) some facet of  $\mathcal{V}_{\text{typ}}$  whose distance to the origin is  $r \pm o_d(1)$ . So, in some sense the typical Poisson–Voronoi cell is less “symmetric” or “regular” than the convex hull of  $n$  i.i.d. points on the unit sphere, if  $n$  is taken comparable to the (expected) number of vertices of  $\mathcal{V}_{\text{typ}}$ .

Another natural direction for further research is to try and obtain more precise results on the behaviour of the faces of  $\mathcal{V}_{\text{typ}}$ . E.g. give more detailed quantitative bounds on the diameter of (almost all)  $k$ -faces.

As the reader may have noticed – and if not can readily check – all the error probabilities  $o_d(1)$  obtained in our proofs were in fact exponentially small in  $d$  or much smaller in many cases, with the exception of the lower bound in Proposition 5.1. We’ve always considered the probability that the considered quantity (inradius, outradius, etc.) differs by a constant  $\varepsilon > 0$  from the target value. A natural set of follow-up questions is to see how fast we can let  $\varepsilon$  tend to zero as  $d \rightarrow \infty$ . In the same vein, it would be natural to find or bound the variances or perhaps even find a normalization that yields non-trivial limiting distributions.

## CHAPTER 6

# CONCLUSION

This thesis studied how high-dimensional geometry affects random spatial structures, and in particular how it leads to threshold phenomena. After the right normalization, many geometric quantities concentrate strongly and become almost deterministic. The location of a global transition is often determined, at least to first order, by a small number of parameters, such as an expected degree, an intensity scale, or an angular window. Once such a parameter reaches its critical scale, the behavior of the model changes sharply.

Two chapters of the thesis are set in the Poisson–Voronoi tessellation generated by a homogeneous Poisson point process in  $\mathbb{R}^d$ . This setting gives both a natural random polytope, namely the typical cell, and natural proximity graphs, in particular the Delaunay graph and the Gabriel graph. For Bernoulli site percolation on these graphs, a main high-dimensional input is that the degree of a typical vertex grows exponentially in  $d$ . For the Delaunay graph this comes with explicit asymptotics, while for the Gabriel graph there is an exact formula. We show that this local statistic determines the percolation threshold: the critical probability satisfies

$$p_c \mathbb{E}D \rightarrow 1$$

as  $d \rightarrow \infty$ . The proofs are based on a branching-process type exploration of clusters. The main difficulty is that the graphs are not independent or tree-like objects; they come from the Poisson–Voronoi tessellation and therefore contain geometric dependencies. A large part of the work is to control these dependencies well enough so that they do not change the behavior.

The Poisson–Voronoi tessellation also has a rigid high-dimensional structure at the level of the typical cell. After the natural rescaling of lengths, or equivalently after choosing the intensity so that the expected number of Poisson points in the unit ball is one, several basic geometric quantities concentrate. The outer radius is close to a deterministic constant, while the inradius is close to half of that value.

The diameter and mean width also stabilize on the same macroscopic scale. This gives a two-scale picture of the cell: extreme points are typically found near the outer radial scale, while the largest inscribed ball is smaller by roughly a factor of two.

At the same time, the cell does not become smooth or close to a Euclidean ball in a strong metric sense. The cell remains genuinely polyhedral. In fixed codimension, the number of faces grows exponentially with the dimension, and asymptotically almost all of these faces are generated by nuclei that are close to regular simplex position. Faces of growing codimension behave differently: they become microscopic compared with the macroscopic scale and concentrate in a thin annulus. Thus high dimension makes the coarse geometry of the cell almost deterministic, while the fine polyhedral structure remains visible and can be described quantitatively.

The other part of the thesis studies coloring thresholds in the random Borsuk graph on the sphere. In this model the vertices are random points on  $S^d$ , and two vertices are connected when the corresponding points are nearly antipodal. The parameter  $\alpha$  controls how close to antipodal the points must be. The threshold question here concerns the chromatic number. For very small  $\alpha$ , edges appear in a sparse regime governed by the scale of  $n^2\alpha^d$ . In this range, the transition from having no edges to having edges can be easily identified.

The more interesting coloring obstructions appear in the thermodynamic regime

$$\alpha \asymp n^{-1/d}.$$

In this regime, we identify a sharp threshold for the loss of bipartiteness, or equivalently for the event  $\chi \geq 3$ , at a dimension-dependent constant multiple of  $n^{-1/d}$ . We also show that, at another constant multiple of the same scale, the chromatic number is at least  $d+1$  with high probability. More generally, for each fixed

$$k \in \{4, \dots, d+2\},$$

we obtain sharp-threshold behavior for the property  $\{\chi \geq k\}$  along a set of values of  $n$  of asymptotic density one. One key idea we use is that, after rescaling, the local structure in the thermodynamic regime is well approximated by a continuum AB percolation model. This approximation captures the appearance of odd cycles and the coloring constraints that follow from them.

Taken together, these results show a common pattern. Concentration helps identify the right normalization and often makes the leading constants accessible. At the same time, the rapid growth of local complexity creates enough combinatorial room for sharp transitions to occur.

There are several natural directions for further work. For percolation on Poisson–Voronoi-type graphs, one important next step is to understand the critical window and the behavior inside it. This includes questions about mean-field critical exponents, for example through the triangle condition, as well as sharper

asymptotics for  $p_c(d)$  beyond the leading term. For random Borsuk graphs, it would be useful to make the threshold locations more explicit, including constants where possible, to study regimes in which  $k$  varies with  $n$  or  $d$ , and to determine whether the density-one restriction can be removed. For the typical cell, natural questions include sharper asymptotics for the width and possible universality results beyond the Poisson setting.



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# SAMENVATTING

## Poisson–Voronoi-geometrie in hoge dimensies en drempelverschijnselen

Dit proefschrift gaat over stochastische meetkundige structuren en hun faseovergangen. Onze meetkundige intuïtie in twee- en driedimensionale ruimten schiet in hoge dimensies vaak tekort. Bijvoorbeeld, twee onafhankelijk en uniform gekozen richtingen staan vrijwel loodrecht op elkaar, het grootste deel van het volume van een bol ligt dicht bij de rand en veel stochastische waarden concentreren zich, in de juiste schaal, sterk rond een deterministische waarde. Tegelijk neemt de combinatorische complexiteit snel toe: er zijn veel mogelijke burens en lokale configuraties. Juist die combinatie leidt vaak tot faseovergangen: een kleine verandering in een parameter kan dan een abrupte omslag in het gedrag van het model teweegbrengen.

In elk van de drie hoofdstukken beginnen we met een stochastische puntenverzameling: een homogeen poissonproces in de euclidische ruimte  $\mathbb{R}^d$ , of onafhankelijk en uniform gekozen punten op de eenheidsbol  $\mathbb{S}^d$ . Uit deze punten construeren we met een lokale meetkundige regel een tessellatie of een graaf. In de twee hoofdstukken over Poisson–Voronoi-geometrie laten we de dimensie  $d$  naar oneindig gaan. In het hoofdstuk over Borsuk-grafen blijft  $d$  vast en laten we in plaats daarvan het aantal punten  $n$  groeien.

**Percolatie.** Het eerste deel gaat over percolatie op de Delaunay-graaf van een Poisson–Voronoi-tessellatie en op de verwante Gabriel-graaf. Elke knoop wordt onafhankelijk met kans  $p$  open verklaard. Percolatie treedt op als er een oneindige samenhangende component van open knopen bestaat. De kritieke kans  $p_c(d)$  scheidt het subkritische regime, waarin geen oneindige component ontstaat, van het superkritische regime, waarin dat met positieve kans wel gebeurt. Door de schaalinvariantie van beide modellen hangt  $p_c(d)$  niet af van de intensiteit van het Poisson-puntproces, maar wel van de dimensie.

Hoewel deze grafen sterke meetkundige afhankelijkheden vertonen, lijkt de verkenning van een open component rond het kritieke punt in hoge dimensies steeds meer op een vertakkingsproces. Op de relevante schaal spelen cycli en de

onderlinge beïnvloeding van verschillende verkenningsspaden een steeds kleinere rol. In overeenstemming met deze heuristiek blijkt de kritieke kans asymptotisch gelijk te zijn aan de inverse van de verwachte graad van een typische knoop. Concreet geldt

$$p_c^{\text{Del}}(d) = \frac{e}{d2^d}(1 + o(1)), \quad p_c^{\text{Gab}}(d) = 2^{-d}(1 + o(1)).$$

De bewijzen steunen op lokale meetkundige schattingen, op de koppeling van verkenningsprocessen aan vertakkingsprocessen en op tellingen van mogelijke lange open paden. De kern is dat de meetkundige afhankelijkheden in de onderliggende grafen de asymptotische hoofddterm niet veranderen.

**Borsuk-grafen.** Het tweede hoofdstuk bestudeert de random Borsuk-graaf. Op de  $d$ -dimensionale bol worden  $n$  punten onafhankelijk en uniform gekozen. Tussen twee punten wordt een kant getrokken als hun sferische afstand groter is dan  $\pi - \alpha$ ; de punten zijn dan bijna antipodaal. Centraal staat het chromatische getal: het kleinste aantal kleuren dat nodig is om de knopen zo te kleuren dat geen twee aangrenzende knopen dezelfde kleur krijgen.

De verschillende overgangen van het chromatische getal worden onderzocht. De overgang van een lege graaf naar een graaf met ten minste één kant vindt plaats op de schaal  $\alpha \asymp n^{-2/d}$  en is niet scherp. Daarentegen heeft het ontstaan van een oneven cyclus—en daarmee het verlies van tweekleurbaarheid—een scherpe drempel bij  $\alpha = c_3 n^{-1/d}$ . De constante  $c_3$  wordt bepaald door de kritieke intensiteit van een continuïmpercolatiemodel met twee typen punten, A en B. Voor een andere constante  $c = c(d)$  zijn bij  $\alpha = cn^{-1/d}$  met hoge waarschijnlijkheid minstens  $d + 1$  kleuren nodig. Bovendien wordt voor elk  $k \in \{4, \dots, d + 2\}$  scherp drempelgedrag aangetoond wanneer  $n$  wordt beperkt tot een deelverzameling van  $\mathbb{N}$  met asymptotische dichtheid één. Voor  $4 \leq k \leq d + 1$  is de relevante schaal  $n^{-1/d}$ ; voor  $k = d + 2$  is dat  $(\log n/n)^{1/d}$ .

**De typische cel.** Het derde hoofdstuk keert terug naar de Poisson–Voronoi-tessellatie en bestudeert de typische cel. Concreet voegen we de oorsprong als extra punt toe aan een homogeen Poisson-puntproces en beschouwen we de bijbehorende Voronoi-cel. Deze constructie levert de typische cel, die in statistische zin representatief is voor de cellen van de tessellatie.

Na een natuurlijke herschaling convergeren de straal van de kleinste omsluitende bol, de straal van de grootste ingeschreven bol, de diameter en de gemiddelde breedte in kans naar respectievelijk 1,  $1/2$ , 2 en 2. Voor de minimale breedte worden asymptotische onder- en bovengrenzen verkregen. Daarnaast worden de aantallen en de meetkundige eigenschappen van de lagerdimensionale delen van de cel onderzocht. Voor elke vaste codimensie wordt de exponentiële groeisnelheid van hun aantal expliciet bepaald. Op de macroscopische schaal van de cel is de diameter van bijna al deze delen verwaarloosbaar. Bovendien worden ze bepaald door punten die samen met de oorsprong bijna een regelmatige simplex vormen.

Deze sterke concentratie betekent niet dat de cel als geheel bolvormig wordt.

Met hoge waarschijnlijkheid is de Hausdorff-afstand van de typische cel tot elke bol minstens een vaste fractie van de diameter van de cel. De polyedrische structuur blijft dus essentieel, ook al worden verschillende globale grootheden in hoge dimensies vrijwel deterministisch. Na de herschaling tekenen zich twee karakteristieke radiale schalen af: de hoekpunten liggen asymptotisch op afstand 1 van de oorsprong, terwijl steunhypervlakken al op afstand  $1/2$  voorkomen.

De bewijzen maken gebruik van technieken uit de stochastische meetkunde, de kansrekening in hoge dimensies en de theorie van random grafen. Daarbij spelen onder meer concentratie-ongelijkheden, koppelingen met vertakkingprocessen, percolatieargumenten en algemene stellingen over scherpe faseovergangen voor monotone eigenschappen een belangrijke rol.

De drie delen vertonen hetzelfde onderliggende patroon. Wanneer de dimensie of het aantal punten toeneemt, worden bepaalde globale grootheden vrijwel deterministisch, terwijl de lokale combinatorische complexiteit juist groeit. Die combinatie ligt ten grondslag aan de uiteenlopende faseovergangen die in dit proefschrift worden onderzocht.



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